

Foreword

The Urban and Regional Planning Act No. 3 of 2015 provides that all Local authorities in Zambia are mandated to develop the IDPs of their areas. To this end, Ndola City is non-exceptional and has therefore developed its Integrated Development Plan (IDP) that shall serve as a strategic framework in guiding planning, budgeting, implementation, monitoring and evaluation of all development agenda in the city's jurisdiction. In order to provide democratic and accountable governance for the local communities, the city consulted both its internal and external stakeholders in the IDP development and formulation process. The key and priority projects identified for implementation in the IDP emanate from the grass roots/local communities and other relevant stakeholders through various public communication and participation platforms. The Ndola City IDP is therefore based on informed priority development goals, emerging socio-economic trends, the ever increasing demand and social cry of the people for better services and other related issues that provides a framework in which the City can ensure effective and efficient implementation of development agenda.

It is important to note that the City operates in an ever-changing and dynamic environment: socially, politically, technologically, culturally, economically, and is therefore, not immune to such changes either positively or negatively. The dynamism in environments at all levels: local, provincial, national and global constantly pose the City with new challenges and new demands. Similarly, the needs and priorities of the people within the City area also change from time to time and therefore, we need to ensure that the IDP is reviewed to allow inclusion of such needs as and when they arise. Consequently, it is imperative to further take note that some of the identified needs as captured during the consultative or public communication and participation platforms do fall within the mandate of the city and other relevant players at all levels. In order to make sure that the needs of local communities are properly met, Ndola city has ensured that its planning efforts are aligned to the Eighth National Development Plan and Vision 2030, and it is therefore a stepping stone towards advancing and realizing the goals of the National Development Plan.

Moreover, to enable a close co-ordination and integration amongst projects, programs and activities, the City will continue to work closely with all stakeholders and local community at large. This is aimed at achieving sustainable development within the City. The city is committed to adhere to good governance principles; participation, efficiency, effectiveness, accountability, transparency, equity, fairness and the rule of law.

In conclusion allow me to state that the successful implementation of the IDP can be achieved through a coordinated approach and partnerships with all spheres of government, communities,

civil society and the private sector. It is indeed true that working together, we can do more as a friendly City and without leaving anyone behind.



Jones Kalyati

His Worship the Mayor

Acknowledgment

The Ndola City Integrated Development Plan (2022- 2032) was developed through a wide consultation process of all key stakeholders from both within and outside the district including our local communities. On behalf of Ndola City, I would like to extend our appreciation to all institutions (government line ministries, Council, FBOs, NGOs, private, business community, Civic leaders, and Ward Development Committees etc.) and individuals who provided valuable contributions towards the formulation of the Integrated Development Plan.

I would like also to state that it was going to be impossible without the technical and financial backstopping from the Common Wealth in partnership with Local Government Association of Zambia to produce such an important document. The Ministry of Local Government through the Department of Physical Planning as well provided timeless policy guidance on the preparation of the planning document and I thank them for this very important role offered to Ndola City.

Special thanks go to the IDP task force that worked tirelessly to ensure the document is produced in line with the stipulated guidelines. The document therefore must be accorded the importance it deserves and it will guide development in the district and that all players to draw their plans and budgets based on this document.



Joseph Phiri

District Commissioner (DC)

Acknowledgement

The Ndola Integrated Development Plan for 2022-2032 is a culmination of months of planning and working process that involved Government leaders, officials, the private sector, community and faith based organizations and Ndola City Council staff, in particular the Department of City Planning.

Representatives from various heads of Government departments in Ndola participated in a series of meetings in which documents were reviewed and discussions were held, after which, reports were generated.

Special thanks goes to each individual and organization that volunteered their time and expertise to ensure that an Integrated Development Plan which responds to the needs of Ndola is developed.

With the invaluable input, it is expected that the Integrated Development Plan for 2022 to 2032 would boost the city's efforts to regain its position as a commercial and distributive Centre in Zambia.



Shilla C. Songolo (Mrs.)

Town Clerk

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KEY ACRONYMS

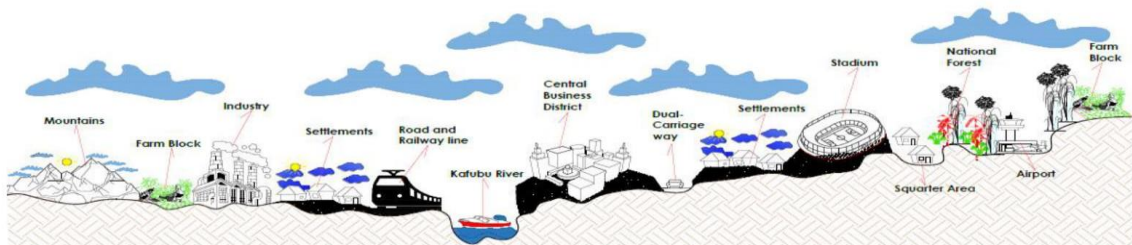
NCC	Ndola City Council
GRZ	Government of the Republic of Zambia
IDP	Integrated Development Plan
URP Act	Urban and Regional Planning Act
LGA	Local Government Act
CSO	Central Statistics Office
PRP	Poverty Reduction Paper
SP	Strategic Plan
DDP	District Development Plan
SB	Sector Budgets
SDGs	Sustainable Development Goals
8NDP	Eighth National Development Plan
LGAZ	Local Government Association of Zambia
MLG	Ministry of Local Government
EU	European Union
DFID	Department for International Development
UK	United Kingdom
ZAFFICO	Zambia Forestry & Forest Industries Cooperation
MVR	Main Valuation Roll
CBD	Central Business District
IMR	Infant Mortality Rate
DANIDA	Danish International Development Agency
ZMW	Zambian Kwacha/ Zero Mode Waveguide
DHMT	District Health Management Team
MTN	Mobile Telecommunications Network
ZAMTEL	Zambia Telecommunications
ZAMPOST	Zambia Postal Services
ZANIS	Zambia National Information Service
FEDEX	Federal Express
DHL	Dalsey Hillblom & Lynn

CBS	Copperbelt Broadcasting Services
UCZ	United Church of Zambia
ZNBC	Zambia National Broadcasting Cooperation
PUSH	Program for Urban Self Help
TDRC	Tropical Diseases Research Centre
SMKIA	Simon Mwansa Kapwepwe International Airport
HIV	Human Immunodeficiency Virus
AIDS	Acquired Immunodeficiency Syndrome
FHH	Female Headed Households
HKO	Household Keeping Orphans
DHH	Disabled Headed Households
UY	Unemployed Youth
AHH	Aged Headed Household
TIHH	Terminally Ill Headed Household
CHH	Child Headed Household
SCT	Social Cash Transfer
PWAS	Public Welfare Assistance Scheme
MSC	Minimum Standard of Care
GBV	Gender Based Violence
SEEVCA	Service Efficiency Effectiveness for Vulnerable Children and Adolescents
MYCD	Ministry of Youth and Child Development
CP	Child Protection

Ndola District at Glance



Transact Work Cross Section – Ndola



PART ONE BACKGROUND TO THE IDP

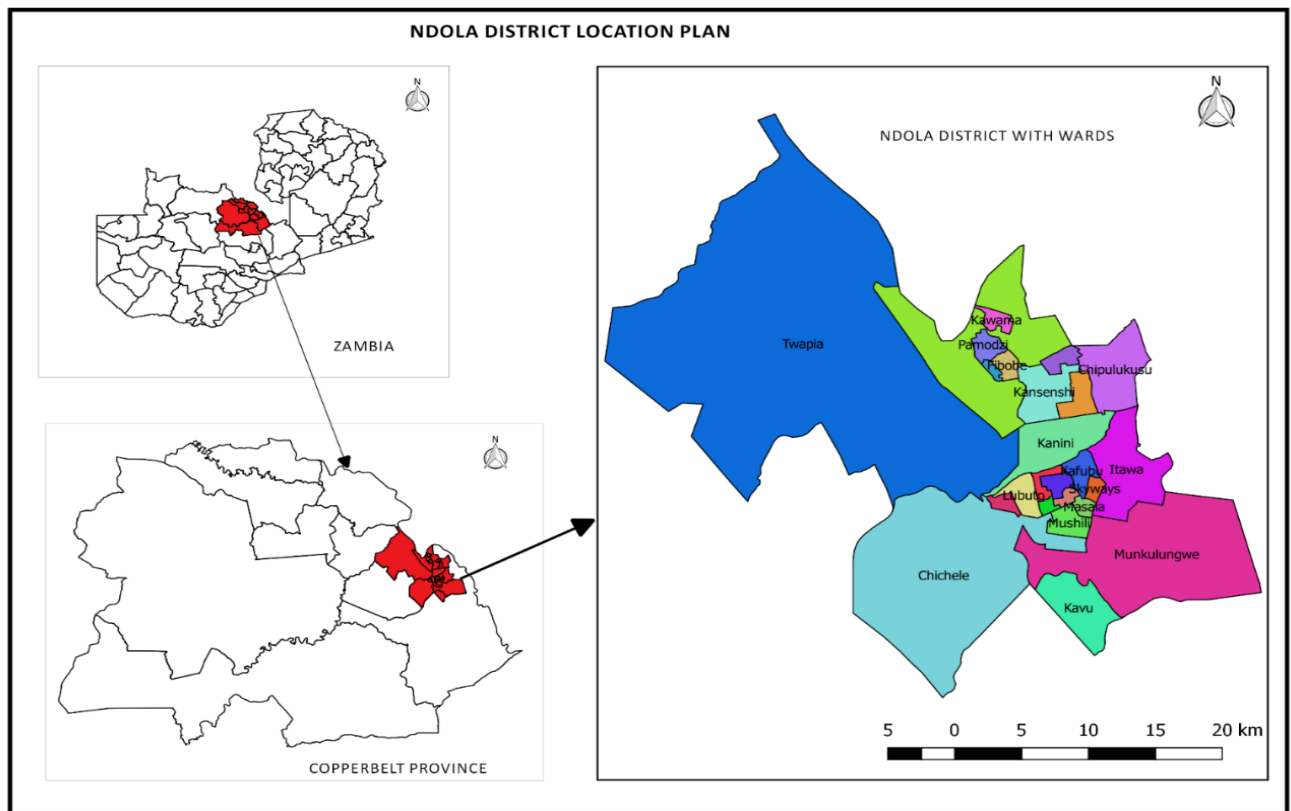
1.0 Introduction and Background

Ndola is the third largest city of Zambia, covering some 110, 300 hectares with a population of over 451,246 (Central Statistics Office – CSO 2010). During the slave trade era, Ndola was a slave trading post and was declared a Village Management Board in 1920, a Municipality in 1932 before attaining City status in 1967.

Ndola lies between latitudes 120° and 160° south and between 250° and 30° east of the Equator. It has borders with three (3) districts namely; Masaiti, Luanshya and Kitwe and shares an international boundary with the Democratic Republic of Congo (DRC) to the north as indicated in **Figure 1** with a total coverage area of 1,108 km².

Ndola falls within the gently sloping Copperbelt pen plain at an attitude ranging from 1250 to 1455 above sea level. The District is characterized by undulating terrain of less than 10 slopes and is dissected by the Mwambashi, Muliashi, Kafubu and Luansobe Rivers, the principle drainage lines to the Kafue River. Though there are highly isolated outcrops at Dola, Nakaputa and Kaloko hills, the average elevation is 1200 m above sea level.

Figure 1: Ndola District Location Map



Source: Ndola City Council, 2021

The Central Government System is composed of all government departmental heads, under the headship of the District Commissioner (DC) who co-ordinates all district developmental activities. To discharge these functions, the District Commissioner and the Town Clerk co-chairs the District Development Coordinating Committee (DDCC), whose composition encompasses district government departments, the Council, major companies in the district, Community Based Organizations (CBOs), Non-governmental Organizations (NGOs) and many other stakeholders. Currently, the DDCC membership stands at forty-six (46) members.

Local Government system: The local government system is composed of the elected councilors headed by his Worship the Mayor. These collectively constitute the council.

1.1 Background

This document represents the first generation of Integrated Development Plan (IDP) for Ndola District which will be operational for 10 years (2022 to 2032). This is a departure from District Development and Strategic plans (IDP Guidelines, 2018). The ultimate objective within each cycle remains the improved implementation of the said dispensation's ten year strategies, as well as ensuring improved responsiveness to community needs over time. The IDP is based on

lessons learned from the previous plans as well as changing environments. The IDP focuses on the following:

- Alignment with national and provincial issues of importance
- Strengthening the analysis principles of strategic planning processes
- Initiating a zonal planning process that involves communities from planning, implementation, and Monitoring and Evaluation of developmental programs
- More emphasis is placed on area-based interventions – addressing the needs of specific communities. This will also facilitate better feedback to communities in terms of programmes and projects that bridge the gap between the state of the nation address, state of the province address, and community needs and inputs
- Ensure better coordination through a programmatic approach and focused budgeting process; and
- Strengthening performance management and monitoring systems.

It is important that the IDP developed by Ndola District correlates with National and District intent. The aim of this document is to develop and coordinate a coherent plan to improve the quality of life for all the people living in the District and also reflecting on issues of National and District importance. One of the key objectives is therefore, to ensure that there is and existence of alignment between National and District priorities, policies and strategies and the District's response to these requirements.

1.2 The Legal Framework

In accordance with the provisions of **Urban and Regional Planning Act No 3 of 2015 (URP)** under **SECTION (19)** Ndola District presented to the Ordinary Council Sitting for Approval and subsequent securing of a minute to prepare an IDP for the District. The **ORDINARY COUNCIL MINUTE NUMBER: OC/315/05/17** approved the IDP preparation to allow for a multi-sectorial approach to planning in the district while encouraging seamless approach to planning system through integration of all development system components in Ndola while harmonising national, regional and provincial influence. This brings about balance in accelerating national development efforts 'towards the Vision 2030 of becoming a middle income nation without leaving anyone behind' under the primary goal of creating a diversified and resilient economy for sustained growth and social – economic transformation driven by among other factors Agriculture (ibid).

The strategic location of Ndola during the booming copper mining era of the 1960s and 70s, quickly established the District as the Country's leading Commercial, industrial and distributive Centre. Attributed mainly to the Government's economic liberalization and privatization policies of the 1990s, Ndola experienced a period of economic slump, however currently the economy is gradually coming up and this is evidenced by a number of business houses and infrastructure which have opened up and also been built up in the recent past respectively.

The District has the potential to regain its position as the most important industrial District of Zambia. The single greatest threat to the local economy however, is the northward shift of mining activities which is threatening to erode Ndola's position as the premier service provision Centre in the province. As alluded to earlier on, development in Ndola just like elsewhere in Zambia was shaped by planning documents (Poverty Reduction Paper, Strategic Plan, District Development Plan, Sector Budgets etc.) that promoted Sectorial Approach to Planning. Contrary to this, Zambia started promoting Decentralization in 2002 yet tolling Centralized way of doing public and economic administration.

The Vision 2030 is the National Long Term Vision and the first long range plan, that was launched in December 2006, following a broad national consultation process. This expresses the aspirations of Zambians to be realised by the year 2030, while the concurrent National Development five-year plan and annual budgets set out shorter-term targets for the achievement of these aspirations. The current National Development Plan is the Eighth National Development Plan.

In April, 2022, Zambia developed the Eighth National Development Plan (8NDP) which sets out Zambia's strategic direction, development priorities and implementation strategies for the period 2022 to 2026. This Plan is a successor to the Seventh National Development Plan (7NDP) that was implemented over the period 2017 to 2021 and the fourth in the series of National Development Plans (NDPs) towards the national Vision 2030 in which Zambians aspire to live in a prosperous middle-income nation. As the penultimate Plan, the 8NDP will play a pivotal role as a building block towards the attainment of the national vision (MoFNP, 8NDP, 2022).

The Plan is a critical vehicle to support the attainment of Zambia's international and regional commitments under various frameworks, including in the last decade of action towards the

realization of the Sustainable Development Goals (SDGs) and delivery on the Africa We Want, espoused in the African Union Agenda 2063. Others are the Southern African Development Community (SADC) Regional Indicative Strategic Development Plan (RISDP) and Common Market for Eastern and Southern Africa (COMESA). (MoFNP. 8NDP, 2022).

PART TWO: PLANNING SURVEY AND ISSUES REPORT

2.0. Demographic Analysis

2.1. Population and Population Growth

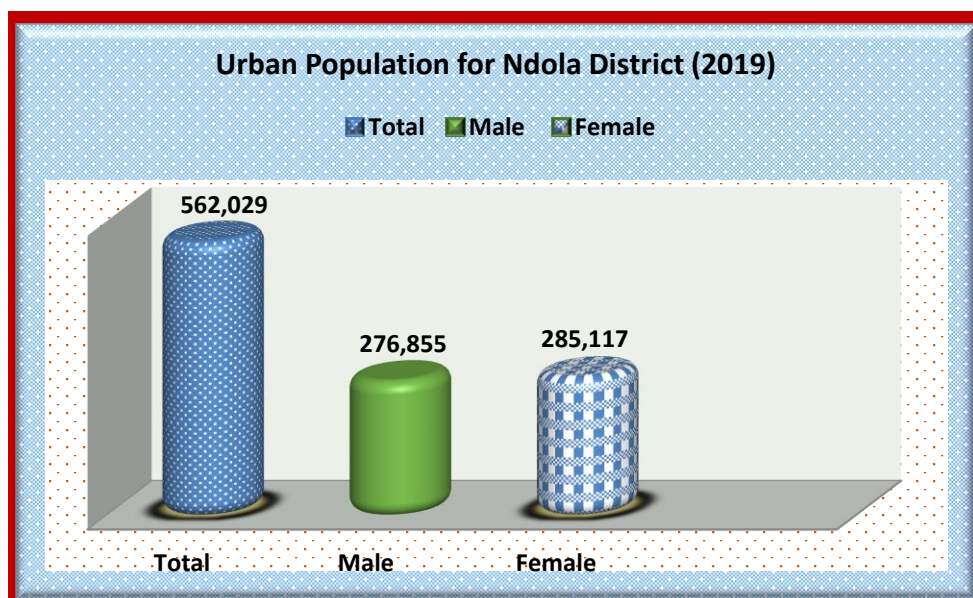
The population of Ndola district has steadily grown from 159,786 inhabitants in 1969, 281,315 in 1980, 334,531 in 1990, 374,757 in 2000, 451,246 in 2010 to 574, 437 in 2019 at a rate of 1.9%. The 2019 national population figure represents 22.9% of the Copper-belt Provincial population, which now stands at 1, 972, 3173. Out of the district population figure, 49.4% are male with females commanding 50.6%. However, it is important to note that the population of urban and rural areas of Ndola is different.

2.1.1 Urban and Rural Population of Ndola District

2.1.1.1 Urbanization

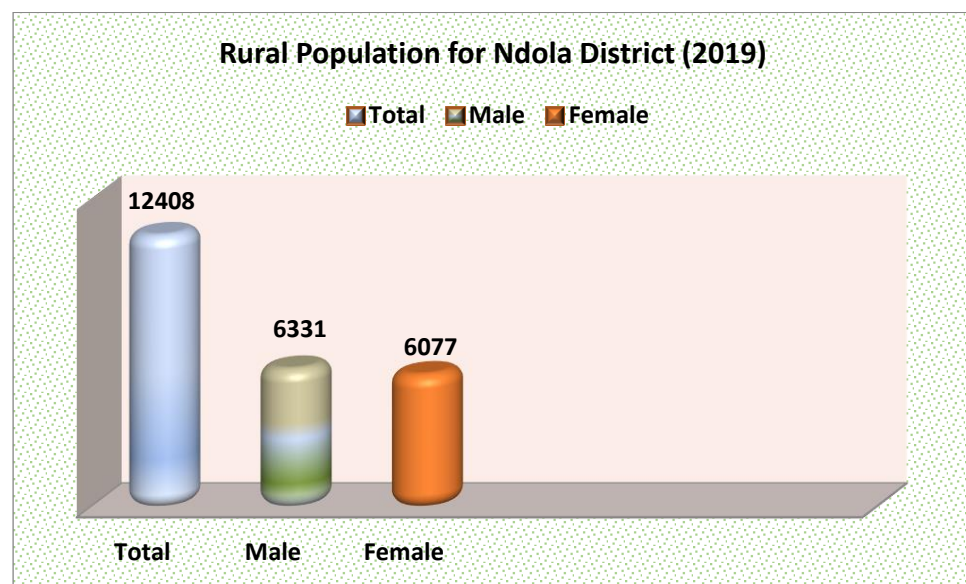
Urbanization is one key factor contributing to the Population growth rate in the Districts. According to the 2019 population projection (CSO 2019), the percentage of the total population living in urban areas will increase gradually in the next 35 years, rising from 40.6 per cent in 2010 to 46.1 per cent by 2035. Over time the District has recorded gradual increase in Economic growth. This has been and will continue to be a source of inward migration due to availability of High standard social amenities such as Universities, Colleges, Secondary schools and Hospital. Ndola is one of the most urbanised districts in the Copper belt province with the urban population of 562,029 where 12,408 are in the rural parts of the district. It can be observed in Figures 2 and 3 respectively, that the number of females out numbers that of males in urban areas. But this picture is different in the rural area where there are more males than females

Figure 2.1: Urban population of Ndola District (2019)



CSO, 2019 population data

Figure 2.2: Rural Population of Ndola District (2019)

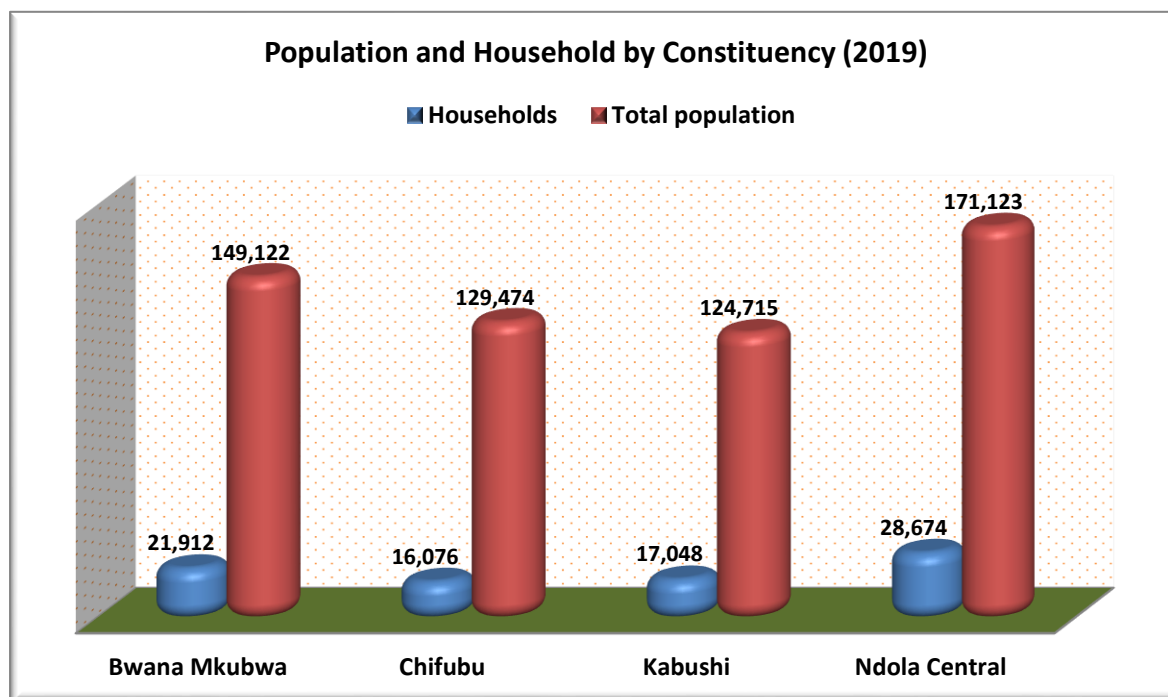


CSO, 2019 population data

2.1.1.2 Population and Housing by Constituency

It is estimated from the 2019 projections that Ndola district has 85,707 households out of which 28,674 are in Ndola Central Constituency with the least being 16,073 households in Chifubu Constituency. Furthermore, Ndola Central Constituency has the largest population in the district with 171,123 and Kabushi Constituency with the lowest population of 124,715 inhabitants as shown in Figure 4 below.

Figure 2.3: Population and Housing by Constituency

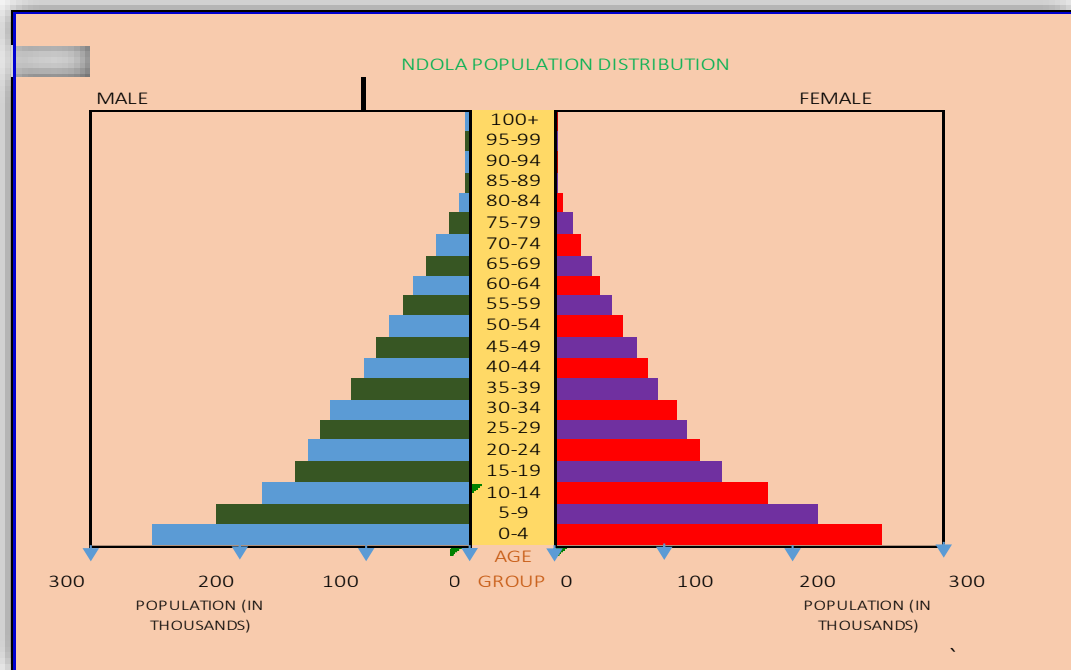


CSO, 2019 population data

2.1.2. Population Characteristics

Ndola District population distribution resembles that of the national distribution where 20.8% of its population is between 15- 24 years and 36.7% is between 25 and 35 years. The district has a pyramid with a wider base indicating a larger number of people in young age group due to a number of factors such as high birth rate and high fertility rate and narrow top of the pyramid attributed to factors such as high death rate and low life expectancies. Therefore, such a pyramid suggests a growing population. It further explains that from the total population of 574,437 according to 2019 CSO projection report based on gender, there are more females than male where 290,104 represents that of female and 284,333 represent the male population. The population pyramid for Ndola district is depicted in Figure 5 as shown below.

Figure 2.2.1: Population Pyramid for Ndola District

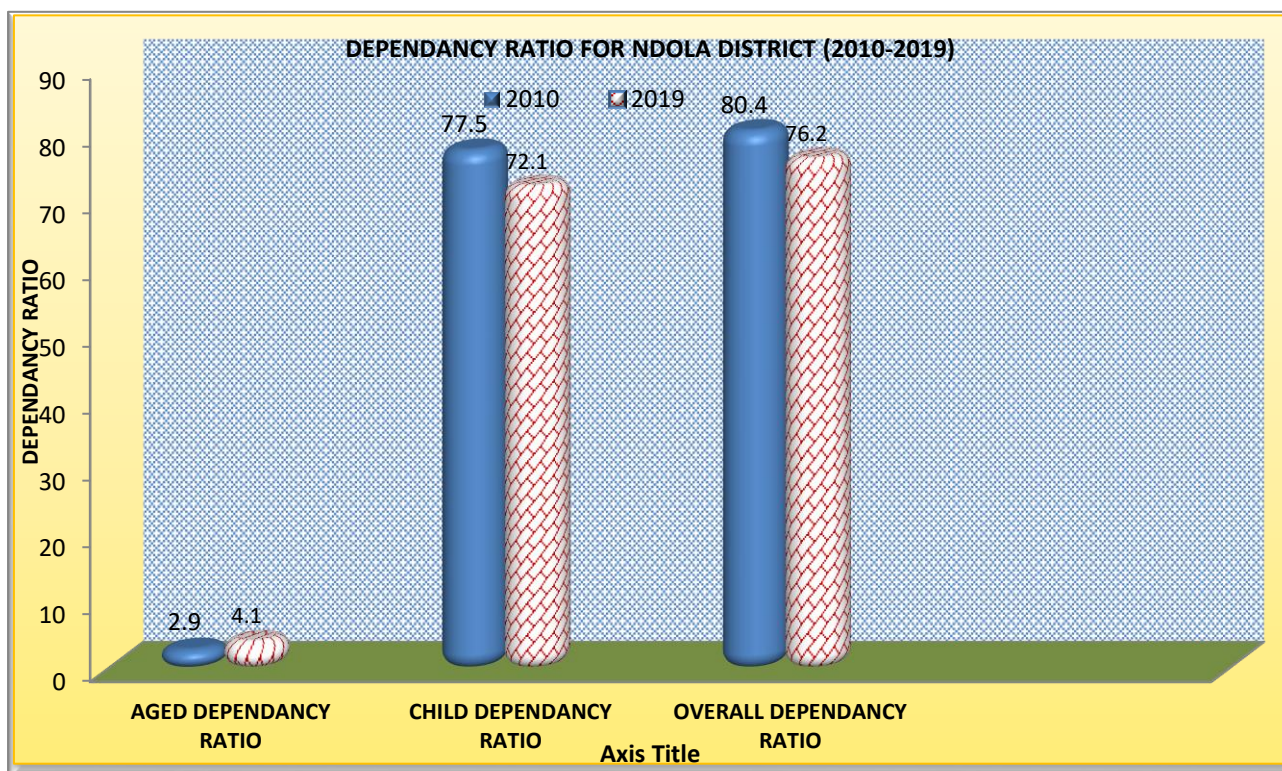


CSO, 2019 population data

Dependency Ratios

The overall dependency ratio for the District is 76 persons, meaning that for every 100 persons about 76 are dependent on those who are economically productive capabilities. Figure 6 below shows the overall dependency ratios for children and the aged for periods 2010 and 2019.

Figure 2.2.2: Dependency ratio for Ndola District (2000 – 2019)



CSO, 2019 population data

2.3 Projected Population and Likely Characteristics

According to the 2010 Census of Population and housing, Ndola district Population stood at 451,246. This is composed of 223, 020 Males and 228,226 females. The Population grew at an annual rate of 1.9 %. The projected information is depicted below in both line graph and bar chart for both male and female on growth rate of the district for the period 1950-2035. The districts total population is expected to grow to 926,000 for the year 2035 on an annual growth rate of 1.9% as shown in Figures 2.2.3 and 2.2.4 respectively.

Figure 2.2.3: Ndola District Projected Population

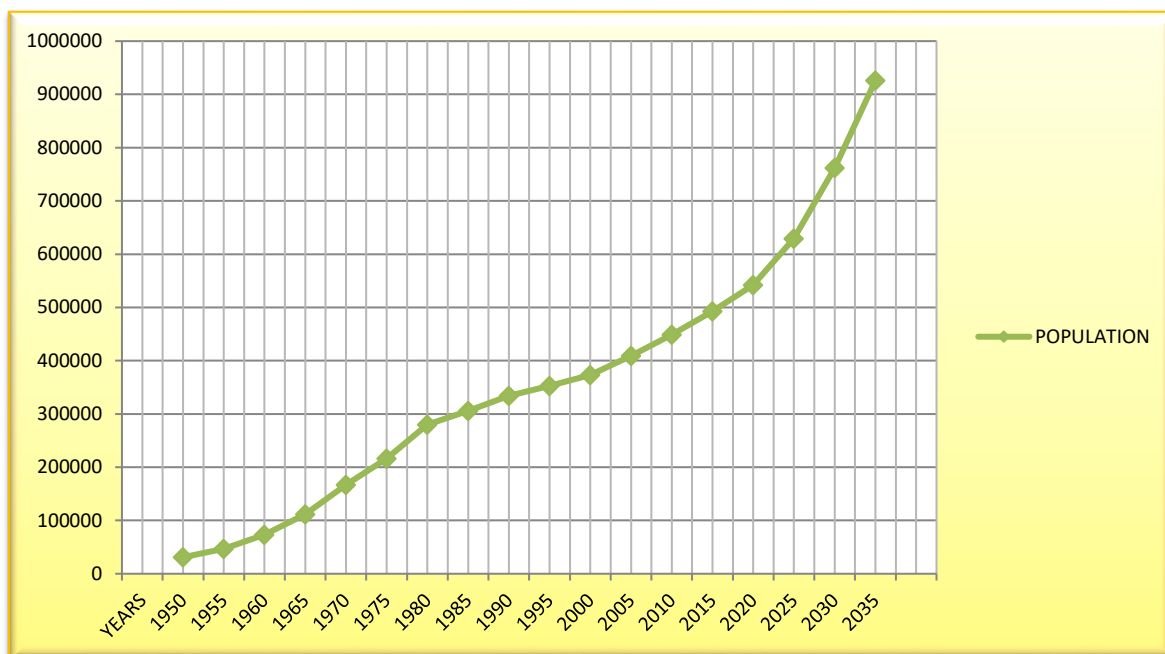
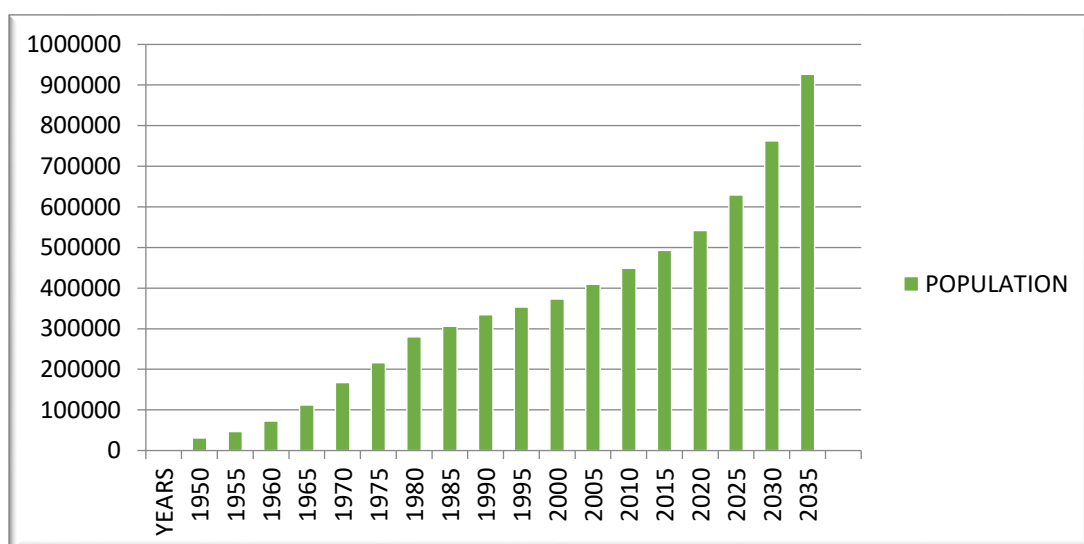


Figure 2.2.4: Projected Population for Ndola District



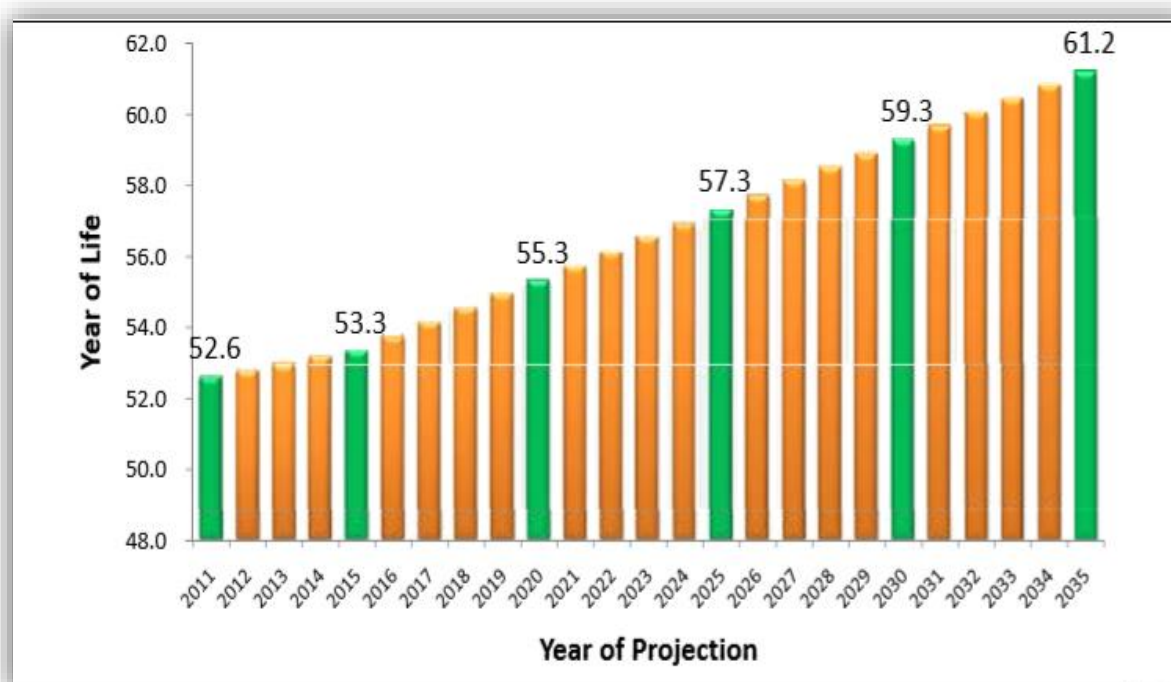
Source: CSO (2010)

2.3.1 Life Expectance for Ndola District

Life expectancy for Ndola stands at 53 years; this is based on data collected during the 2010 Census and the projected figure of population and housing. Life expectancy refers to the average number of years a newly born baby is expected to live, if the current existing mortality conditions were to prevail for a long time. According to the 2010 Census of Population and Housing Population Projections the District will have a gradual increase in life expectancy. Life expectancy at birth is also expected to increase during the projection period for both males and females. Overall life expectancy at birth is expected to rise by 8.6% years during the projection period from 2011 to 2035. The increase in life expectancy at birth is assumed to

result from overall declines in both childhood and adult mortality. The District has seen an increase in health facilities as well as a general improvement in health Care aimed at reducing the High infectious disease burden. The projected life expectancy for Ndola District is depicted in Figure 2.2.5 below.

Figure 2.2.5: Projected Life Expectancy for Ndola District (2011-2035)

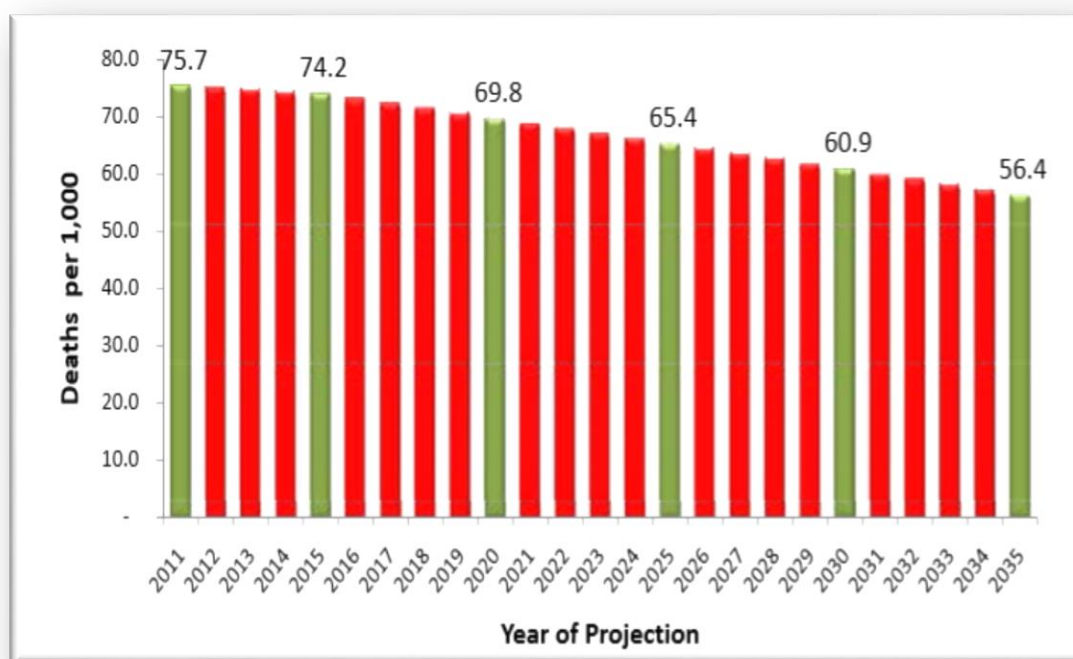


2.3.2 Infant mortality

The infant mortality rates (IMR) for the district is projected to decline by a quarter (25%) during the period 2011-2035 dropping from 75.7 to 56.4 deaths per thousand live births. Historically IMR has had downward trends which has declined by 24% in both rural and urban areas between 1980 and 2019, from 106 per thousand live births in rural areas and 89 per thousand live births in urban areas in 1980, to 80.5 per thousand live births and 67.6 per thousand live births in rural and urban areas respectively in 2019. The District has been implementing interventions aimed at a higher rate of reduction in IMR including the provision of essential primary health-care services for children across a range of care that spans from pregnancy to child birth and after delivery leading to care for children in the crucial early years of life. However, there has been an increase in the number of health facilities built which are providing a full range of health services including Antenatal and post-natal services thereby reducing unsafe child births (home deliveries). The District health office has embarked on upgrading of health facilities to offer antenatal and post-natal health services thereby increasing the districts

capacity to meet demand of the growing population. The provision of less than five year care services such as immunisation and vaccinations have been scaled up. In addition, sensitization on the importance of antenatal and postnatal services remains a key intervention in reduction of IMR. The graph below shows projected IMR rates for the period 2011-2035 for Ndola District showing a downward trend.

Figure 2.2.6: Ndola District Projected Infant Mortality Rate (2011 -2035)

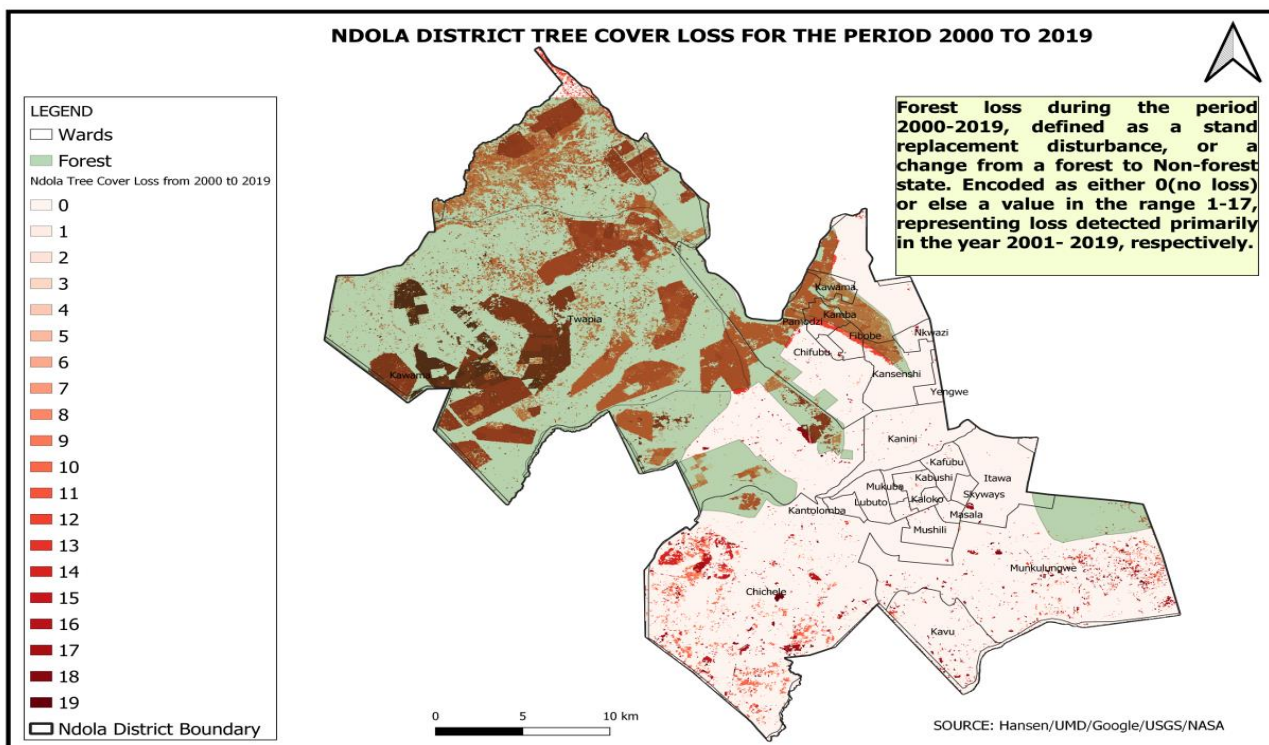


Source; Ndola District Health Office (2019)

2.4. Impact on Existing Population Trends on Land Use and Spatial Development Patterns

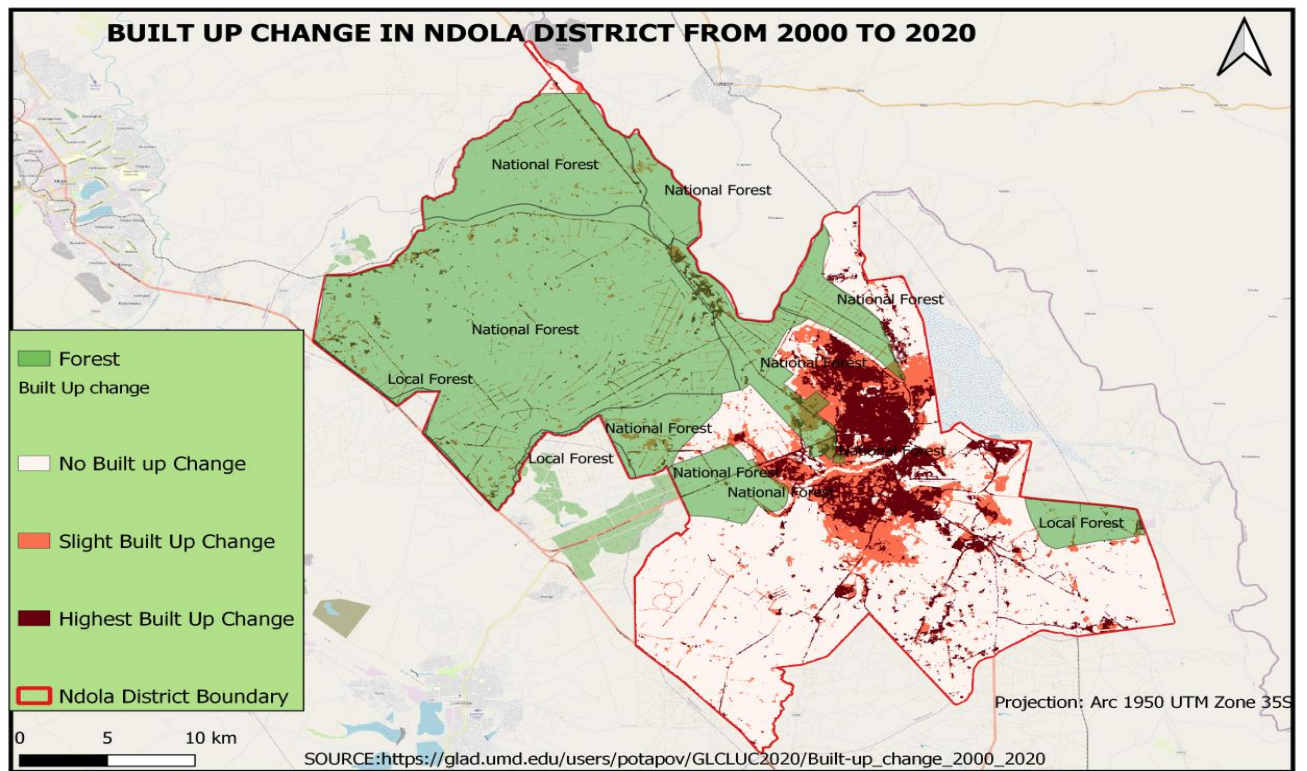
With the expansion of population in Ndola District due to several factors such as a drop in mortality rates, and high fertility rate as observed from the 2019 to 2031 demographic trends, human activities are likely to expand in all directions of the district. The rapid growth is leading to uncontrolled urbanisation which is causing overcrowding, destitution and crime. This pressure has led to over use of arable land and forest reserves where people are now changing the land use from farming to residential to accommodate the ever increasing demand for land causing massive deforestation. As it can be observed in *Figure 11 and 12* below showing the tree cover loss for the period 2000 to 2019, and the built up change from 2000 to 2020 respectively, showing a rapid expansion of human activities towards forest reserves and land meant for agriculture.

Figure 11: Tree cover loss for the period 2000 to 2019



Source: Ndola City Council GIS Unit (2021)

Figure 12: Built up change from 2000 to 2020



Source: Ndola City Council GIS Unit (2021)

3.0 SECTOR ANALYSIS

3.1. Economy

3.1.1. Review - economic policies and Plans

In order to stimulate the key sectors of the economy the Government has revised and formulated National policies which are aimed at stimulating growth in the key sectors to enhance economic development. The Eighth National Development Plan is one of the key planning national development plans in which the country's focus is to attain economic transformation that will be marked by advancements in industrialisation and economic diversification for sustained growth driven by agriculture, mining, manufacturing and tourism.

The government has also formulated policies to enhance economic growth through the promotion of value addition on locally produced materials through the establishment of value chain and support of the Small Medium Enterprises (SME) under the Citizen Economic Empowerment Commission. In order to stimulate growth in the Economic sector, the government has established the Economic Multi-Facility Economic Zones (EMFZ) in various districts under the Industrial Development Policy. The EMFZ will provide a platform for value chain and establish market linkages between the suppliers and manufacturer

3.1.1.1 Existing state of Development

Ndola has a diversified economy with the major economic activities ranging from Cement, Manufacturing, Construction, Transportation, Wholesale and trade and Forestry. The district has other economic activities such quarrying, hospitality industry and financial services.

3.1.1.2 Mining and Quarrying

The mining industry was one of the major drivers of the economy until the privatization of the Zambia Consolidated Copper Mine in 1991. This saw the closure of the Bwana Mkubwa Mine and industries supporting the mining industry. Currently mining activities borders on production of Lime, this is mainly due to the presence of the huge dolomitic limestone deposit in the Itawa springs which provides an important resource limestone for lime and cement production. The presence of resource has stimulated investment in lime and cement production as seen in the number of new companies investing in lime and cement production.

3.1.1.3 Forestry

Ndola district lies entirely on state land whose total area is 110,800 hectares (1,108 km²) of which 37.6 percent (41,418 hectares) is classified as medium to high potential for agriculture. Forest Reserves were established for production of timber, poles, fuel wood, non- wood forest products and protection of water catchment areas such as the Lamba head waters commonly known as the Kafue River Basin (**The Forest Acts No. 4 of 2015**). Ndola District has a total of 49,088 hectares of forest reserves of which 19,000 hectares is covered by ZAFFICO pine and eucalyptus plantations. The remaining portion is covered by the Miombo Forest characterised by species of general *Brachystegia*, *Julbernadia* and *Isoberlinia* (A.E.G Storrs 1995).

These reserves have continued being the source of timber and poles for our industrial and domestic utilization. Due to excessive demand of the resource, there has been over exploitation of pine trees. However, in order to have a sustainable supply, the company replants trees in Ndola on an average of 1000 hectares per annum. To contribute towards plantation

establishment and management, the Forestry Department in the district has established a total of 100 hectares of pine trees located in Ndola and Chichele National Forests. The department also encourages private sector participation in plantation establishment in order to contribute towards meeting the high demand for timber and poles. This is in line with the keep Zambia Clean, Green and Healthy Campaigns.

3.1.1.4 Agriculture

The district is in region III and normally receives between 1,000 - 1,500mm of rainfall annually. The soils are mostly sandy loams. This is suitable for the production of a broad range of crops, fish and livestock. The major field crops grown in the District are maize (both rain-fed and irrigated), soya beans, groundnuts, sweet potatoes, dry beans and cassava. The vegetables that are commonly grown include tomatoes, egg plants, cabbage, Chinese cabbage, spinach, lettuce, green beans and rape, while fruits include bananas, citrus, avocados, guavas and mangoes. The production trends from the recent past show that soya beans is an upcoming cash crop owing to the fact that a number of agro processing industries within the district are using it as a raw material in the production of cooking oil and animal feed. These are Olympic milling, Global industries and Golden Lay.

The main types of livestock reared in the district are cattle, pigs, goats, sheep, poultry. Minsundu and Sakania are predominantly the main dairy farming areas whereas rearing of Piggery is evenly distributed across the district. The village chickens are mostly concentrated in the peri-urban whereas Broiler chicken rearing is mainly concentrated in Ndola Central.

Fish farming is highly concentrated in Sakania vet camp. However, some ponds have not been stocked due to low water levels. Sakania, Kavu and Ndola Central have potential to increase fish production from the current production levels. The main species being reared in the district are the three spotted, red breasted, green headed and Nile tilapia breams. The Market potential within the district is quite viable. There is also a viable international market for processed fish in the Congo DR. To facilitate growing the fish sector in the district, major inputs like feed and fingerlings have to be readily available. Therefore, there is potential to attract investment in production and processing of food for fish and fingerings

3.1.1.5 Manufacturing

The manufacturing sector is one of the top performing and also a priority sector in the development of the district. Ndola's current base in this sector comprises of companies into food processing which includes beverages production, oil refinery, agro-processing and the main activity being cement processing. However, the economic down turn of 1991 brought about by the decline in the copper prices which was followed by the privatization of government owned companies resulted in the closure of a number of manufacturing companies and industries supporting the mining sector. This led to loss of employment and hence loss of income to stimulate growth in the manufacturing sector. The current state of most of the former industries is an eye sore as they are abandoned and are in deplorable state. Currently Ndola has been recording an upsurge in the number of manufacturing industries in agro processing and food production. This has been mainly due to readily available raw materials from the local farmers as well as neighbouring districts.

3.1.1.6 Tourism

The tourism sector still remains one of the key industries that haven't been fully exploited in the district yet it has potential and opportunities to attract local and foreign investment which can bring about social and economic development in the district.

Ndola district boasts of a number of natural, cultural and heritage sites which have potential for tourism investment. However, most of these sites still lack basic infrastructure to attract tourists. Some of these sites include the Dag Hammarskjold memorial site which has 7 hectare piece of land available for investment in hotel or shopping infrastructure, the Itawa Springs for creating an ecological garden and the Copperbelt Museum. The completion of the New Simon Mwansa Kapwepwe airport will further boost the tourism, manufacturing and the hospitality industry.

3.2.2. Quality of Service

- The City of Ndola is linked to other towns with a good roads network. However, the roads within the townships and in the industrial areas are in a deplorable condition
- There is a constant supply of electricity. However high electricity tariffs have a bearing on the cost of production
- Processing and manufacturing sectors do not have modern machinery to convert raw materials into much valuable goods and services.
- Bulk storage facilities are in a deplorable state
- Sawmills operate as make shift without proper basic infrastructure and modern equipment
- Inadequate infrastructure for agriculture extension services such as veterinary, deep tanks and breeding centres
- Inadequate veterinary services to cater for the meat processing industry
- Unchecked dumping of solid waste by sawmills, processing and manufacturing industries
- Bad state of tourist facilities in the communities
- High interest rate discriminate SMEs from accessing loans

3.2.2.1. Future demand for services and facilities

- The current population estimates for Ndola district stands at 574,437 and its projected to be 760,000 by the year the 2031(CSO). The Change in the population will result in increase in the demand for goods and services. This will consequently put pressure on the manufacturing industries and service providers to increase their output to meet the demand for goods and services.

3.2.2.2. Existing and Proposed Investment Development programs

- Increase investment in the manufacturing and processing industries especially in cement production will also stimulate growth in other sectors such as the construction industry and hence create employment opportunities in the sector and also accelerate infrastructure development in the district.
- The current supply of agricultural and livestock products in the district do not meet the consumption requirements of the district population. Ndola district heavily relies on the surrounding district to meet the demands for agricultural and livestock product. Increased investment in agricultural services will not only increase productivity to meet

the demand but will also create employment opportunities for labour force in the district. Investment in agriculture will also stimulate growth agriculture sub sectors such stock feed production, agro business, dairy processing and meat processing

- Increased investment quarrying will increase the production output especially in cement production thereby reduce the price of the commodity which consequently reduce the cost of building.

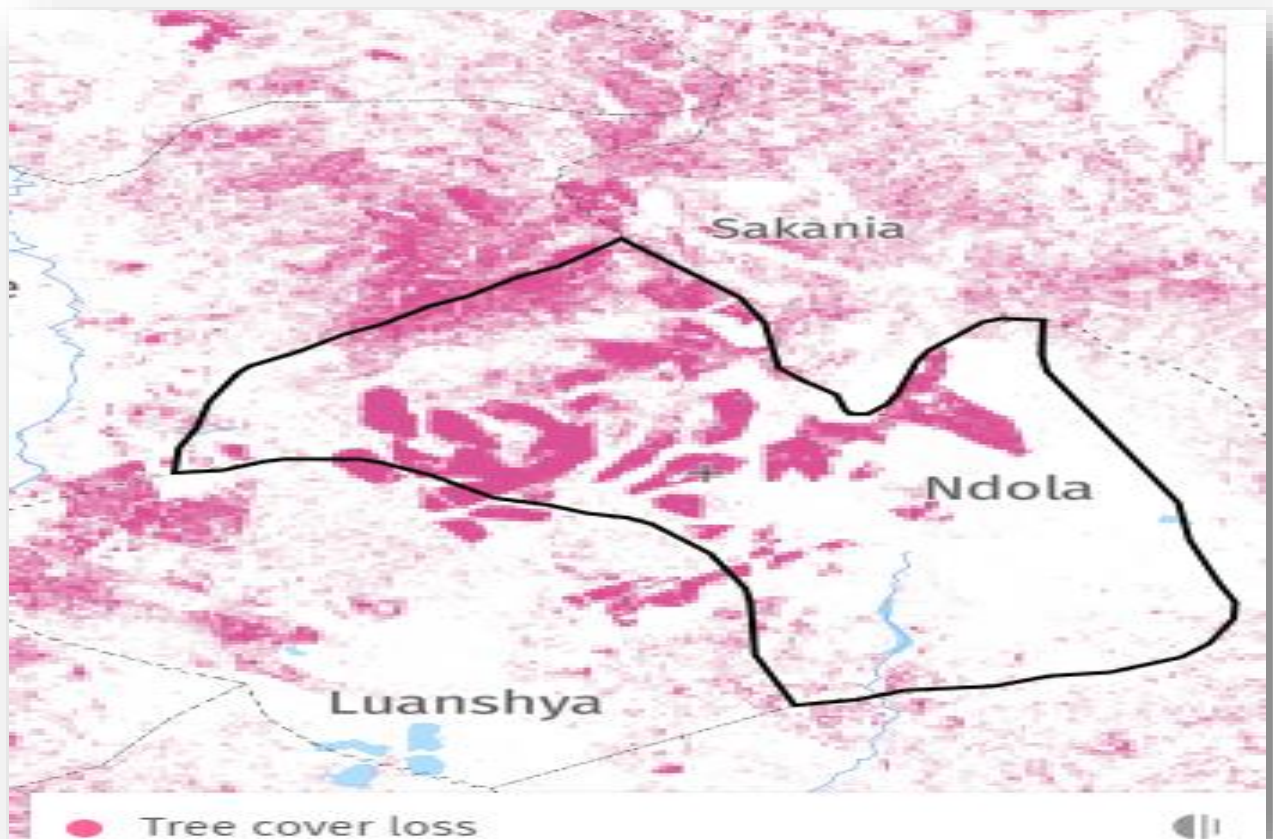
3.2.3. The impact of existing trends on land use and population distribution pattern

The operation of Manufacturing and processing industries is generally characterized by utilization of large pieces of land. Cement processing industries usually occupies larger pieces of land and therefore operation involves extensive excavation of land and this leads to larger pieces of land being degraded altered. The high demand of cement and lime entails further expansion of the production capacity to meet the demand of the commodity. Therefore, to meet the demand for the commodity will mean setting up new plants or expansion of the existing plants. This will mean displacing the population and disturbing flora and fauna sitting on the raw material. This will also necessitate alteration of the land use to mining and quarrying activities which will further cause land degradation making the land inhabitable.

3.2.4. Environment and Climate Change Analysis

In 2010, Ndola had 23.8kha of tree cover, extending over 26% of its land area. In 2021, it lost 476ha of tree cover, equivalent to 226kt of CO₂ emissions. The loss of tree cover will therefore have a negative effect on the environment and will result in climate change.

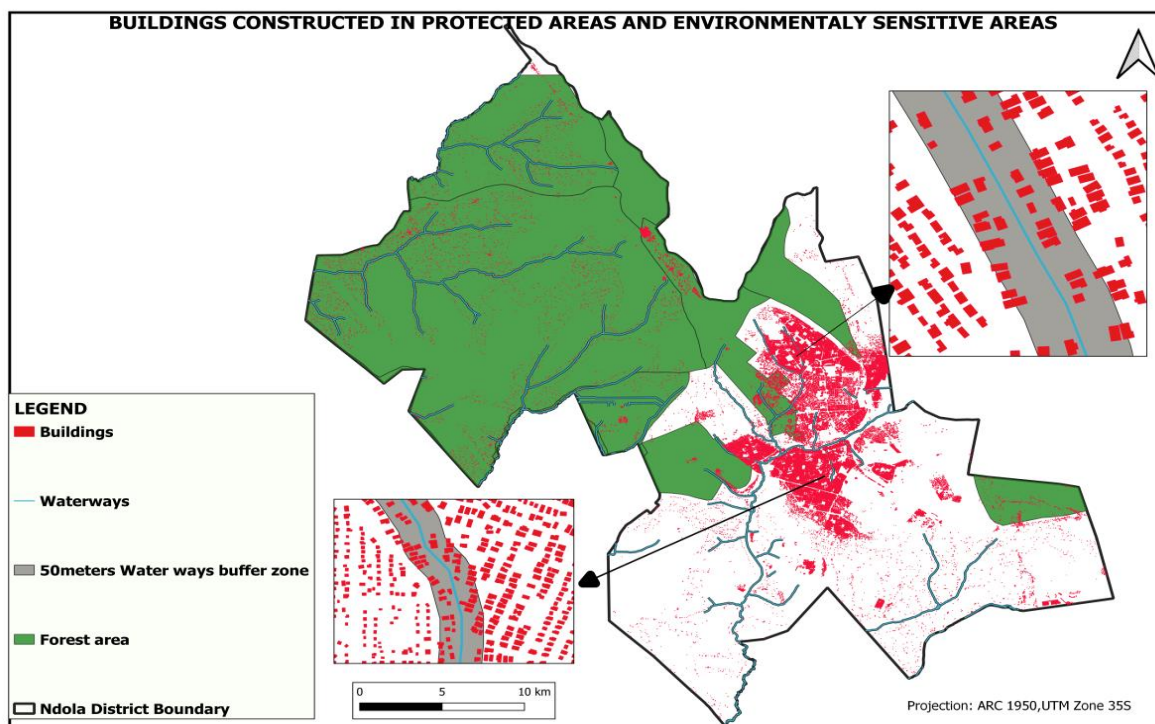
Figure 13: Tree Cover loss in Ndola



Source: <https://www.globalforestwatch.org/>

The rapid population growth has led to an increase in demand for land for infrastructure development, wood extraction, agriculture expansion hence urban sprawl and encroachment on environmentally sensitive areas.

Figure 13: Buildings constructed in protected areas and environmentally sensitive areas



Source: Ndola City Council GIS Unit (2021)

3.2.5. The impact of the existing trends on the Environment and Climate

Though, the district has recorded an increase in agricultural activities, it is mainly practiced on a small scale and mainly in the peri-urban areas of the district. The increase in agricultural activities has been necessitated by the increase in demand of agriculture and livestock produce in the district. The call for diversification of the economy in the district to agriculture has also stimulated investment in the value addition to agricultural produce and hence the increase in demand for agriculture produce as raw materials. The current trend for the demand of agricultural produce is likely to continue due to the projected increase in the district population. This will result in the degazetting of forest reserves and forest plantation to be used for farming activities to increase production to meet the demand. This will have a negative effect on the environment and the climate as agricultural activities in the district are one of the reasons for deforestation and change in land use. The use of fertiliser and pesticides to increase productivity has a harmful effect as it pollutes water and threatens the aquatic ecosystem.

The presence of the limestone which is one of the raw materials needed in the production of cement lime and cement has attracted a lot of investment in cement processing in the district. Investment in cement process has been brought about by the increased demand of the commodity for construction purpose for both the public and private infrastructure across the country and neighbouring countries. Cement is an extremely important construction material used for housing and infrastructure development and key for economic growth. The demand for cement is directly associated to economic growth. The country is currently undergoing rapid infrastructure development which has resulted in an increase in demand for the commodity. However, cement production is one of the major contributors to global warming and climate change. The emissions from cement manufacturing process result in global warming, ozone depletion, acid rain, bio-diversity loss and reduced crop productivity.

3.2.5.1. Impact of Environment and Climate Change issues on the Economic sector

- The effects of environmental and climate change have had detrimental effects on the growth of the economy in the district. The low rainfall pattern the country as experienced in the past years has resulted in shortage of energy as the country is largely dependent on hydroelectricity as a source of energy. This has resulted in losses in the manufacturing industry and increased production cost by using alternative sources of energy.
- Environment and Climate Change has a direct effect on the amount of rainfall and the annual temperature. Decline in the amount of rainfall due to climate change has resulted in the decrease in the agriculture yields recorded in the past year which has led to food security issues. Though currently the demand for agriculture and livestock products has been increased, the current output cannot meet the district demand for consumption of the products. This can be attributed to poor rainfall pattern and high temperature to support crop production and rearing of livestock.

3.2.5.2 Issues Arising Relating to Gender Groups and Vulnerable Groups

- Ndola district has a total labour force of 143,678 of which 75 % are in active employment. However, 68% of the actively employed are skilled and semi-skilled and usually work in the manufacturing sector, construction industry, mining and quarrying industry, trade and the service industry (CSO Population Census 2010). The population actively employed in the formal sector comprises of 25% women and 78% men. Most of the unemployed population are women who are engaged in small scale agriculture and informal businesses to sustain their livelihood. This indicates that still a larger section of the population of women are not actively involved in major economic activities such as manufacturing, mining and timber processing.
- Women have unequal access to opportunities compared to their male counterparts in terms of ownership to factors of production. Women are in the informal sector and small scale farming sectors which are linked to their lower education levels.
- The district has a disproportionate ratio in terms of property ownership such as land between male and females which favours males.
- Most of the women in the informal sector do not have the entrepreneurship skill to enable them expand their business and compete favourably with the male counterparts.
- Difficulties in accessing finance to support. Most financial institutions are not willing to give loans to small businesses which appear to be non-profitable.

3.2.5.3. Underlying Factors contributing to the Issues identified

- The main contributing factor to the disparity in ownership of factors of production by the women and the vulnerable are mainly due to cultural practices which hindered women to have access to factors of production such as land and capital.
- Limited access to these factors also limits women participation in key economic activities such as manufacturing, mining and quarrying and processing as these require substantial operating capital. Women and the vulnerable are limited to informal business such as market trading which require less operation capital.

3.2.5.4. Summary-Core Issues

- Long hours without power supply due to load shading

- Lack of or inadequate infrastructure to support the thriving manufacturing and agriculture sector.
- Lack of access to finance to support local business

3.3. Environment

The district has a specific mandate to oversee and implement strategies that limit the aspect of environmental pollution. Management of activities and projects from as minimal as individual levels to industrial level have impacts that lead to alteration of land, water, and air resources and contribute to issues pertaining to solid waste disposal, effluent discharge and environmental degradation.

3.3.1. Review of environment Policies and Plans

Vision 2030: The Vision 2030 is a long-term plan that expresses the aspirations of the Zambian people to live in a strong and dynamic, middle-income industrial nation that provides opportunities for improving the well-being of all.

The plan underlines that Zambia's comparative advantage lies in its endowment of renewable and natural resources which can be productively harnessed. Overall good climate and soils make organic and climate-smart agriculture development and processing poised for growth.

Proactively, Vision 2030 falls in line with the United Nations Environmental Program (UNEP) which stretches out six criteria for environmental management and these include; (i) Strong linkage of environment with socio-economic developmental goals; (ii) Decoupling of socio-economic development from escalating resource use and environmental degradation; (iii) Coverage of critical issues of environmental sustainability such as important irreversible changes in the global environment; (iv) Take into account current global environmental goals and targets, (v) Scientifically credible and verifiable; and (vi) Progress must be "trackable".

The 8th National Development Plan (8NDP) 2022 - 2026; The Eighth National Development Plan is the country's blue-print unearthing a development plan projected within a five-year period. Building on the success and lessons learnt from the implementation of the previous NDPs, and informed by the developmental challenges facing the country, the theme for the 8NDP is "Socio-Economic Transformation for Improved Livelihoods." The overall aim is to improve the efficiency and competitiveness of the economy to sustainably lift the living standards of the people.

In pursuit of the economic transformation agenda, it is imperative that development pathways are sustainable. This entails the sustainable utilisation of natural resources which are the basis for wealth creation, as well as building resilience to the adverse effects of climate change.

Thus, the 8NDP prioritises measures aimed at promoting green growth and environmental sustainability, enhancing mitigation and adaptation to climate change while promoting sustainable management of the environment, ecosystems and natural resources. The Government will also enact legislation on climate change to strengthen the coordination framework and enhance coherence between adaptation and disaster risk reduction, among others

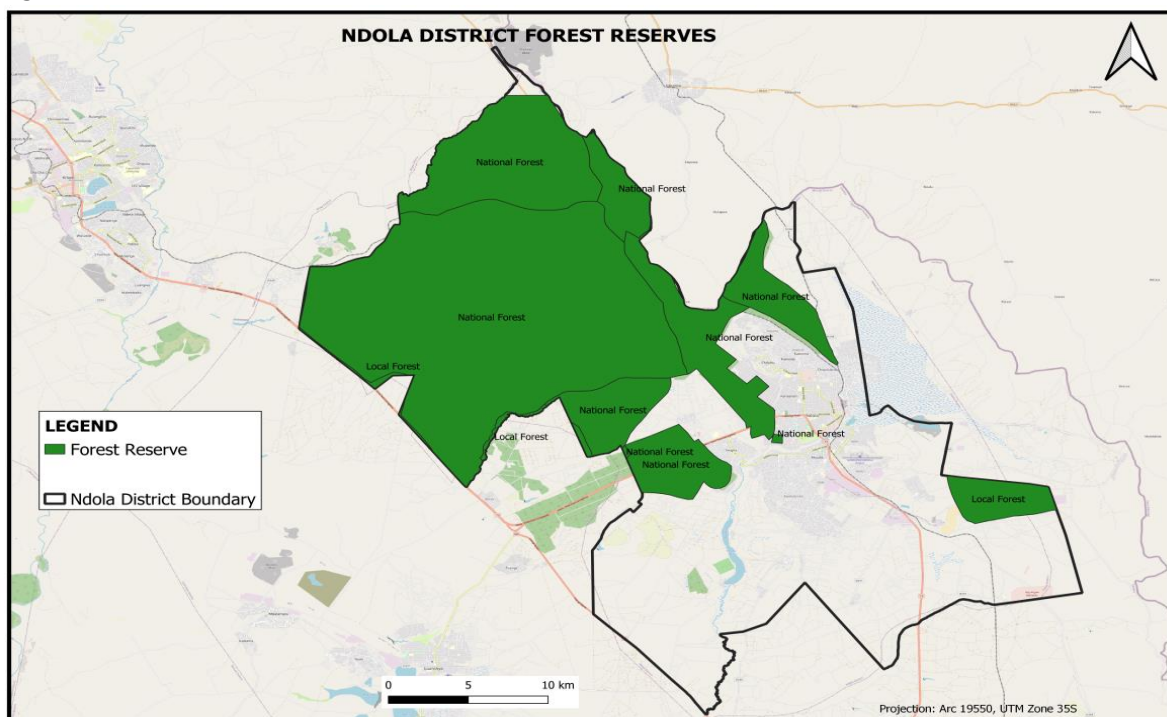
At district level, the outcome indicators involve areas of project implementation that address climate change adaptation, climate change mitigation, and disaster risk reduction.

Climate Change Adaptation; Climate change is a major threat to the environment and natural resources, which we need for the sustainable development of our country. The effects of climate change will undermine the very foundation of socioeconomic development and will increase inequality and poverty.

At district level, there is an establishment of a multi-stakeholder process whose strategies increase public awareness of climate change and gender equality issues to stimulate interest, participation and consultation with different stakeholders in developing climate change policies, and to increase capacity of those national actors to continue their involvement.

Sustainable Forest Management Strengthened; Forest Reserves were established for production of timber, poles, fuel wood, non- wood forest products and protection of water catchment areas such as the Lamba head waters commonly known as the Kafue River Basin (**The Forest Acts No. 4 of 2015**). Ndola District has a total of 49,088ha of forest reserves of which 19,000ha is covered by ZAFFICO pine and eucalyptus plantations. The remaining portion is covered by the Miombo Forest characterised by species of genera *Brachystegia*, *Julbernadia* and *Isoberlinia* (A.E.G Storrs 1995).

Figure 14: Forest Reserves



Source: Ndola City Council GIS Section (2021)

These reserves have continued to provide the source of timber and poles for our industrial and domestic utilization. Due to excessive demand of the resource, there has been over exploitation of pine trees. However, in order to have a sustainable supply, the company replants trees in Ndola on an average of 1000 hectares per annum. To contribute towards plantation establishment and management the Forestry Department in the district has established a total of 100ha of pine trees located in Ndola and Chichele National Forests. The department also

encourages private sector participation in plantation establishment in order to contribute towards meeting the high demand for timber and poles. This is in line with the keep Zambia Clean, Green and Healthy Campaigns.

Safeguarding Water Resources; The district is manned in protecting water resources within Ndola district. An assessment of the level of environmental knowledge held by local water-users in Ndola demonstrates a coherent understanding of the safety of their water sources, the quality of these, the threats to them, and the fundamentals of how their local hydrology works, all of which is contrary to the perspective of key informants who are involved in water supply. On developmental projects, the local authority ensures that projects undertaken do not hinder flows of any natural water systems such as the case of the cancellation of title to a piece of land in Ndola's Hillcrest area, which was earmarked for the development of a filling station because if the developer is given a go ahead to construct the filling station, the development will not only lead to the contamination of ground water but also disturb the flow of water into the Kafubu River (WRM Act No. 21 of 2011).

Solid waste management in Ndola; Ndola District has a population of approximately 600, 194 (CSO 2020). Services provided include provision of waste bins in public places, street cleaning, market cleaning, waste collection and management of the disposal site.

Implementation of the Environmental Management Act No. of 2011; The Environmental Management Act No of 2011 replaced the repealed Environmental pollution and Protection Control Act (EPPCA) of 1990. This is the main environmental law in Zambia with superiority over other legislation on environmental matters. The Act continues and renames the Zambia Environmental Management Agency (ZEMA) which was established under the EPPCA.

Every project in the district as required by the Act demands that a developer whose project will have a significant impact on the environment must carry out an environmental impact assessment which must be approved by ZEMA prior to the commencement of the development. According to the guidelines provided in environmental impact assessment (SI No. 28, 1997), this project qualifies for an environmental project brief. (EPB)

The Environmental Management (Licensing) Regulations, 2013(Statutory instrument No. 122 of 2013), promulgated under this Act provides for control of pollution through licensing of waste generation, storage and disposal and discharge of effluent among others. Accordingly, the project will need to comply with regulations on discharge of emissions to the air and transportation of waste.

3.3.2. Existing state of development

The most convincing measure of environment-development integration would be an increase in positive outcomes on the ground. The fact is, however, that such outcomes are not routinely measured. We only have glimpses of progress – many of them noted in the foregoing sections – and for most of these we would want to know more if we were intent on bringing the twin endeavors of development and environmental management together to realize their synergies. A quick review, based in part on the four-part development framework of assets, risks, powers and time frames (suggested in Box 3), shows partial progress in all areas:

3.3.2.1. Availability of Service

3.3.2.1.1 Street Cleaning

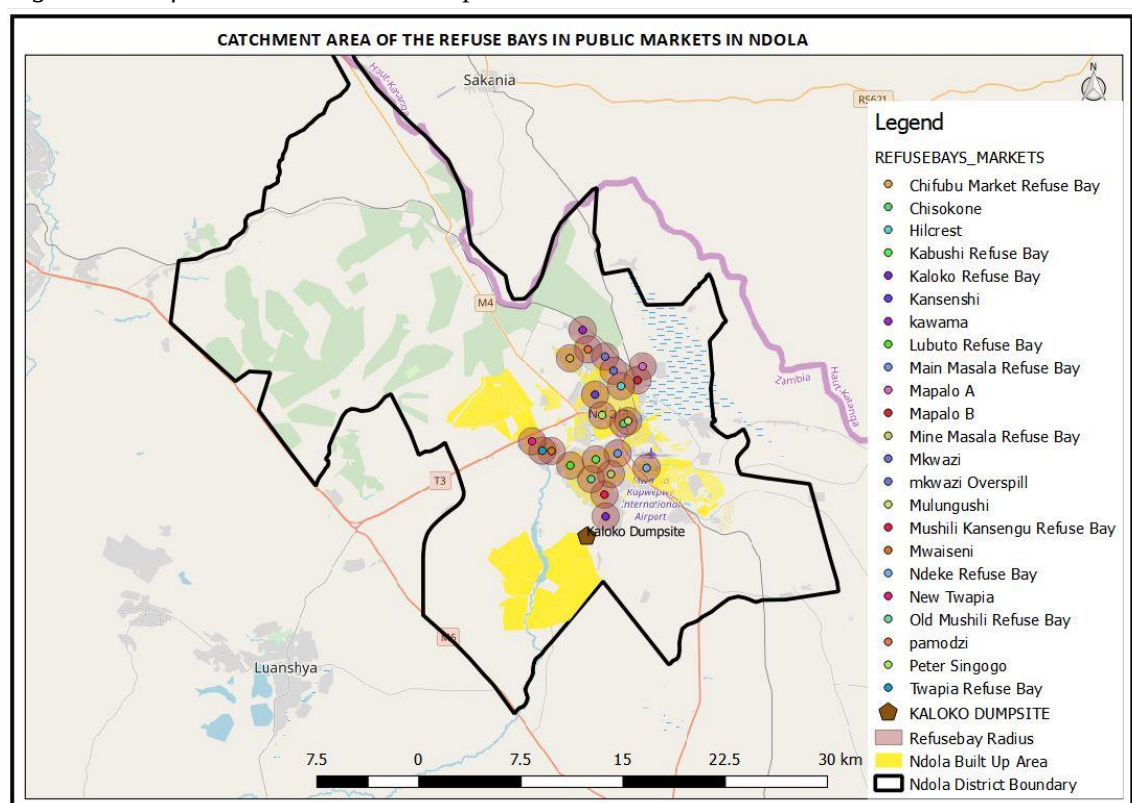
This service is currently available in the central business district. An assigned number of street sweepers are used to clean the streets in the central business district on a daily basis. Major roads outside the Central Business District are swept at least once in a month. Provision of bins in the town centre for town goers to throw little quantities of waste has greatly improved the cleanliness. However, the bins are being abused by owners of premises in town who throw huge amounts of waste and vandals who uprooted bins. The Council has since embarked on a sensitization campaign and some vandals who vandalised the bins in January were jailed. The expansion of this service to areas outside the Central Business District is dependent on availability of manpower and refuse trucks to collect the sweepings.

3.3.2.1.2 Waste Collection

The District has a waste generation rate of 0.5kg per capital per day and approximately 13, 276.5 tons per month. This represented by a composition of organic waste- 70%, paper plastics- 20%, metal- 5%, other 5%. Only a total of 5,624 tons is collected by both the council and private companies representing an efficiency rate of 42.4%. The amount collected by the council represents 47.9% of the total collected. The Council collects refuse from residential and commercial areas at a charge.

Waste from public places that include two legal and one illegal intercity bus stations and 30 markets refuse bays, namely; Kawama market, Pamodzi market, Chifubu market, Nkwazi market, Nkwazi overspill market, Riverside market, Kansenshi market, Hillcrest market, New Twapia market, Old Twapia market, Mwaiseni market, Northrise market, Peter Singogo market, Chisokone market, Mulungushi market, Chipuluskusu mapalo A, Chipulukusu mapalo B, Ndeke market, Kaloko market, Mine masala market, Main Masala market, Kapalalala market, Kabushi market, Old mushili market, Mushili market, Mushili Kansengu market, Lubuto market, Kansenshi market and Kayele market. Illegally dumped waste is usually collected for free when the author of the nuisance cannot be traced. If the author is known before or after collection of the illegally dumped waste, the cost of removal of the waste is borne by such a person.

Figure 15: Refuse Collection and Dump Site



Source: Ndola City Council Public Health Department (2021)

Waste Collection Equipment

The Council currently has the following equipment for refuse collection:

- Refuse compactor truck
- Tipper trucks
- Frontend Loaders
- Bull Dozer
- Skip Loaders

Most of the equipment above needs repair works. Furthermore, the above listed equipment is still insufficient for the council to provide efficient services to its residents. A brief background is that in the 80s, the Council had twenty-three (23) refuse trucks, (7) seven tipper trucks, (1) one front end loader and (2) two bulldozers for solid waste management when the population of the city was relatively small.

Involvement of the private sector in waste collection is being done through implementation of Solid Waste Regulation and Management Act No 20 of 2018 which requires the Zoning of the city and granting franchise contracts to private waste collection companies in accordance to section 37 of this same act. With the zoning of the city, all owners or tenants of properties are to subscribe for waste collection to specific companies awarded a contract through public tender.

3.3.2.1.3 Market Cleaning

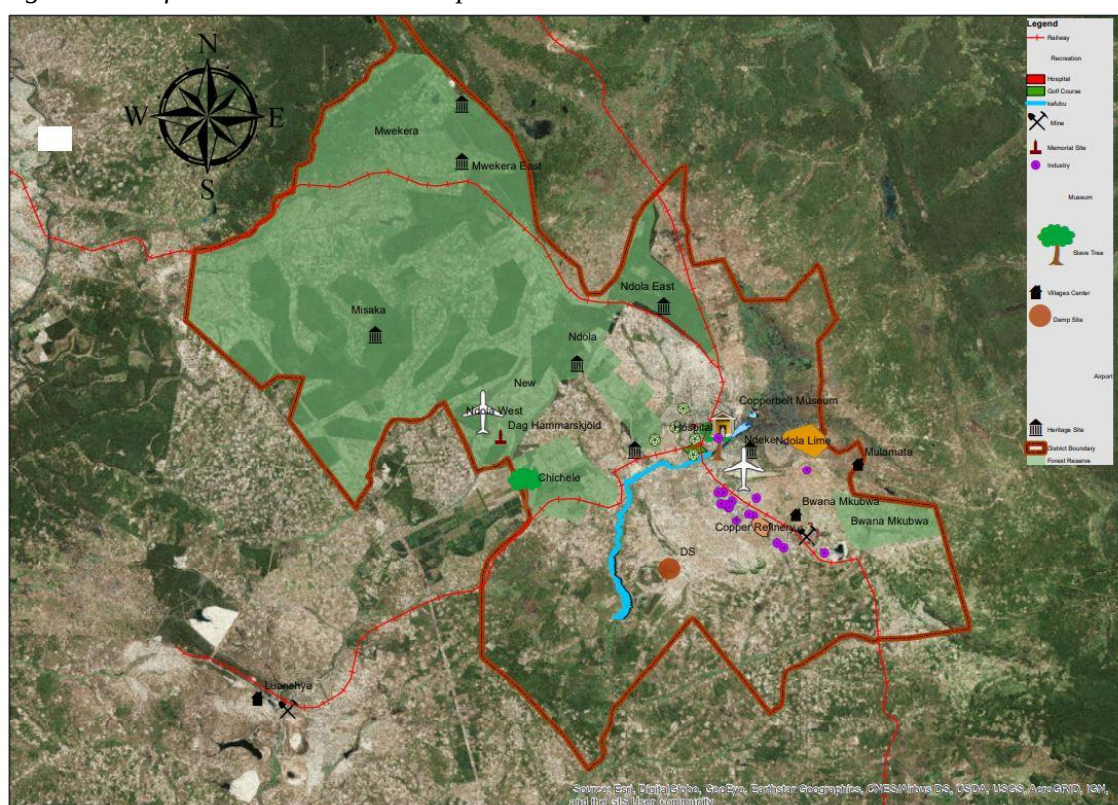
Markets are cleaned on a daily basis. The sweepings are either stored in waste bins, market refuses bays or communal dumps pending collection. Communal dumps are being phased out

because residents near the markets and proprietors of private properties also illegally dump their waste there which tends to be unhygienic and requires machine loading. On the other hand, the bin storage system works well in preventing illegal dumping, promoting hygiene and imparting a sense of responsibility on the waste generators. However, the workability of this system is dependent on the number of waste bins and frequency of collection which is dependent on the availability of refuse trucks. Markets on the bin storage system are Chifubu, Nkwazi, Pamodzi, Overspill, Kansenshi, Northrise, Lubuto, Mushili Kansengu, Kabushi and Mine Masala. Markets on the communal dump system include Main Masala, Twapia, Kawama, Peter Singogo, Mulungushi, Chisokone, Ndeke, Old Mushili and Kaloko

3.3.2.1.4 Final Disposal of Waste

The District has one disposal site, the Kaloko dumpsite where all the waste which is collected by the Council and private sector is dumped. The site is a disused quarry site with controlled dumping. The dumpsite is maintained by a bull dozer which pushes the waste into the ditch. The maintenance is affected by the type of vehicles accessing the site. Smaller vehicles do not easily reach the required areas for dumping. This is further worsened by frequent breakdowns of the bull dozer. In the absence of a bulldozer a frontend loader is used to push light loads of waste in order to just clear the waste.

Figure 16: Refuse Collection and Dump Site



Kaloko Dump-site location – Source: Ndola City Council GIS Unit (2021)

3.3.2.1.5 Effects of Poor Solid Waste Management on the Environment

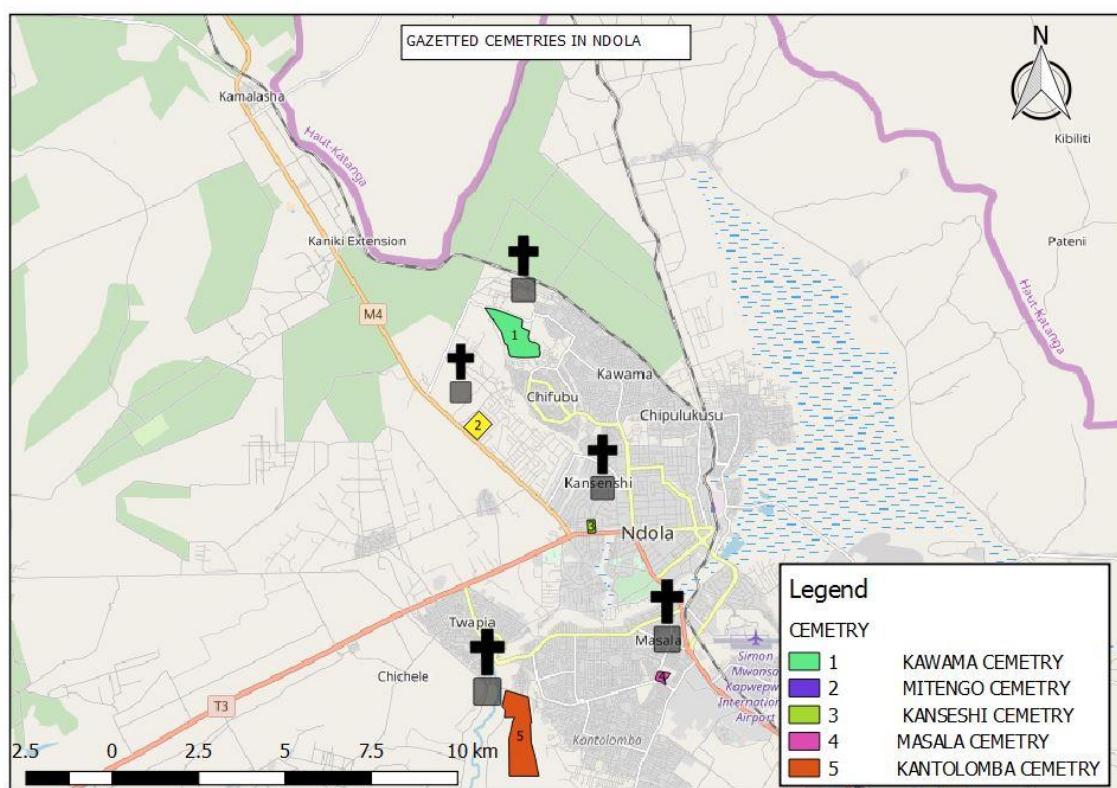
Due to inadequate collections contributed by increased frequency of equipment breakdowns and non-payment for the service, there has been increase of illegal dumping sites emerging within the district of Ndola. Consequently, diseases associated with poor sanitation have been

occurring. This scenario poses economic, environmental and health threats to the city. Environmental degradation has been seen in increased offensive odours in townships due to purification of organic waste, fly and rodent infestation, unsightly aesthetic of the residential surrounding and open spaces in the city. Such conditions do not attract investors and the environment is unattractive for sustainable development.

3.3.2.1.6 Cemetery

Ndola District is currently managing four gazetted cemeteries namely: Kasenshi, Mitengo, Kawama and Kantolomba cemeteries. Among the gazetted cemeteries, Kasenshi is the oldest and it's a high-cost fee paying cemetery. It houses Common Wealth graves, Jewish sites, Muslim site and the general public. Mitengo cemetery is a medium cost fee paying cemetery and it also houses Muslim Section and the General Public. Kawama and Kantolomba are the low-cost fee-paying cemetery. All destitute, unclaimed bodies and still born are buried from these two cemeteries.

Figure 17 Map of Gazetted Cemeteries in Ndola



Source: Ndola City Council GIS Unit (2021)

Environmental Education

It is clear that there is improved environmental awareness amongst the general public, largely attributed to the District's long and rich history of environmental education, as well as some media activity. More routine information is needed, especially along river banks and industrial environment links and economic information. Environmental education has helped reduce environmental risks of development activity, especially through the 'Keep Zambia Clean

Campaign.’ However, the production of timber (saw millers) continues to be destructive to the environment, and urban development continues to pollute aquifers with poor provision of sanitation.

Assessment of proposed developmental projects

The district is tasked with ensuring that any project that is to take place should be assessed to ensure that it does not cause any harm to the environment and pose a risk to human health. In order to achieve this, Environmental Impact Assessments are conducted on proposed projects.

Enhancing Environmental Management compliance

As part of the measures to ensure compliance of environmental policies within the district, frequent monitoring is done on industries, water resources, protected areas as well as temporal business activities. This usually entails the environmentalist conducts inspections as a follow up to the submitted Environmental Project Brief (EPB) and check if there is compliance to the mitigation measures.

3.3.2.2. Quality of Service

Percentage of number of industries inspected.

Ndola has a total number of 1073 industries. A total inspection pertaining to environmental compliance has been done on 412 industries representing 39.7%. 63.7% projects the number of industries not inspected out of which some are non-functional and others going on temporal closures due to the surfacing of the covid-19 pandemic.

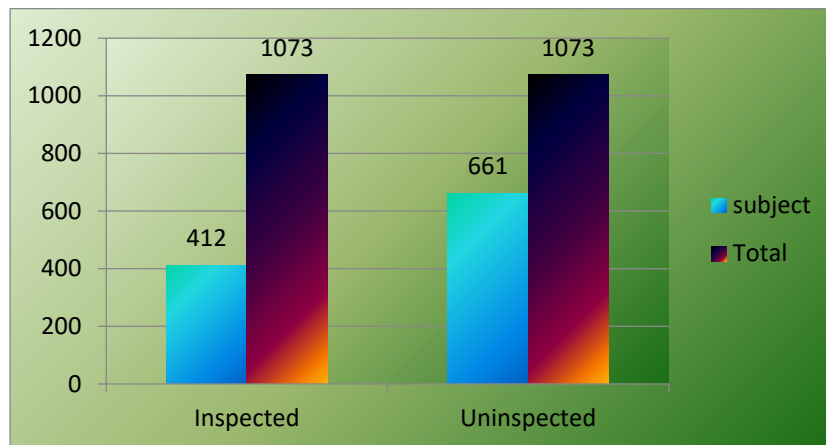
The number of inspected industries (**412**) and divide it by the total number of industries (**1073**) in the District and examining. This gives us the percentage of inspected as industries (**39.7 %**). Clearly, the lower you can get this number, the better.

This information is represented in the figures below.

Table showing the inspected industries in Ndola

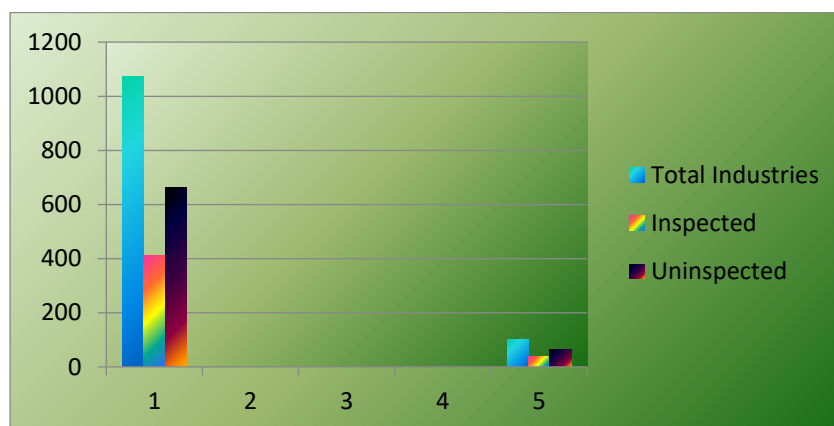
Industries	Substantial figure	Total	Percentage
Inspected	412	1073	39.7%
Uninspected	661	1073	63.7%

Bar graph showing the inspected industries in Ndola District



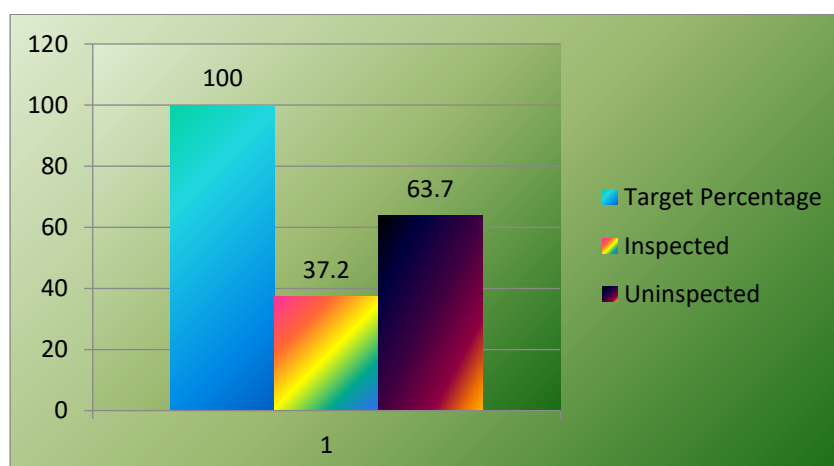
Source: Ndola City Council, Inspectorate Unit (2021)

Bar graph showing the inspected industries in Ndola District



Source: Ndola City Council, Inspectorate Unit (2021)

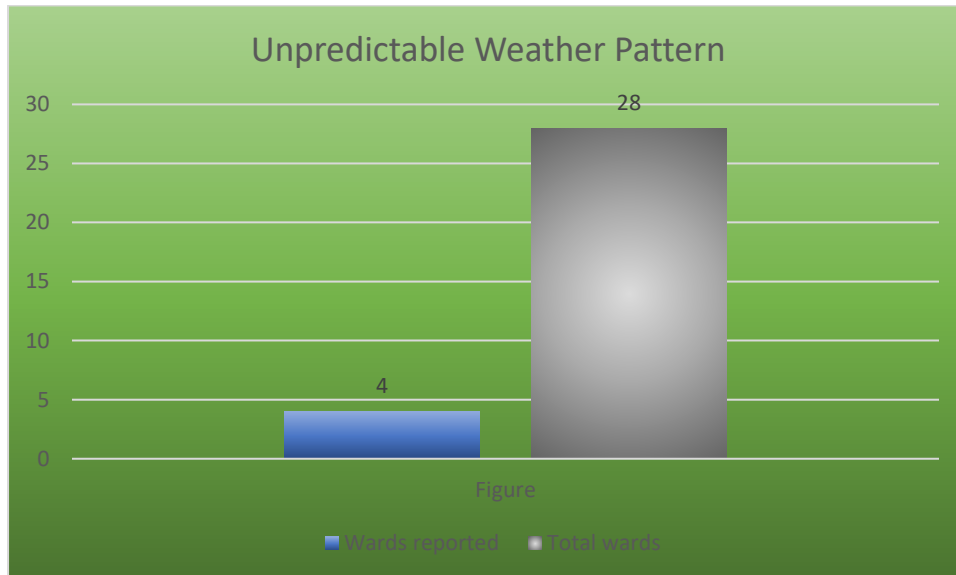
Bar graph showing the percentage expression of inspected industries in Ndola District



3.3.3. Issues Arising from the Public Participation Process

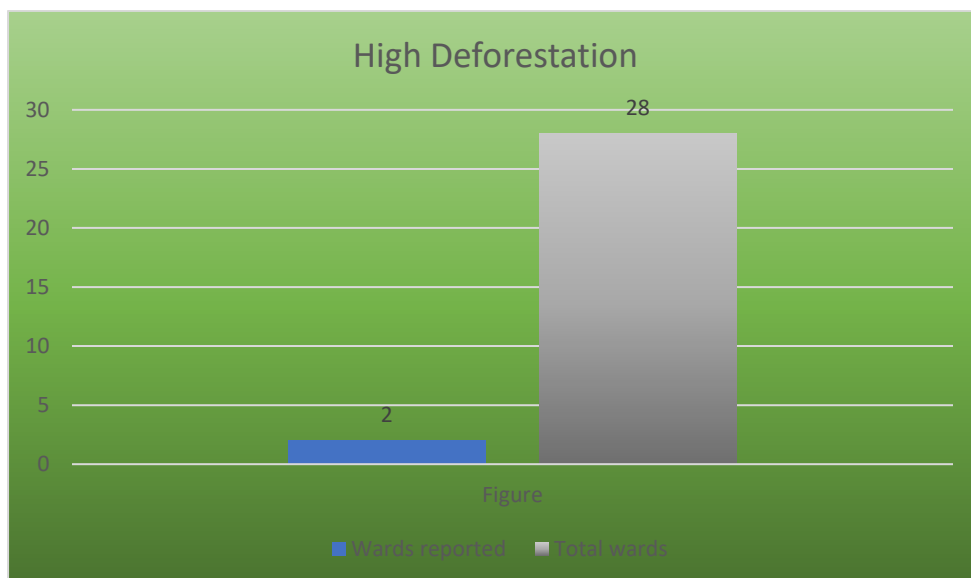
Unpredictable weather patterns were at 14%.

Variability of the weather pattern is going to become the new normal. This means that there will be heavy rainfall in one year, drought in the next, and then heavy rainfall in the following year. This weather pattern disadvantages farmers. Unpredictable weather can disrupt food availability, reduce access to food, and **affects** food quality. For example, projected increases in temperature, changes in precipitation patterns, changes in extreme **weather** events, and reductions in water availability may all result in reduced **agricultural productivity**



Source: Field Survey, 2020

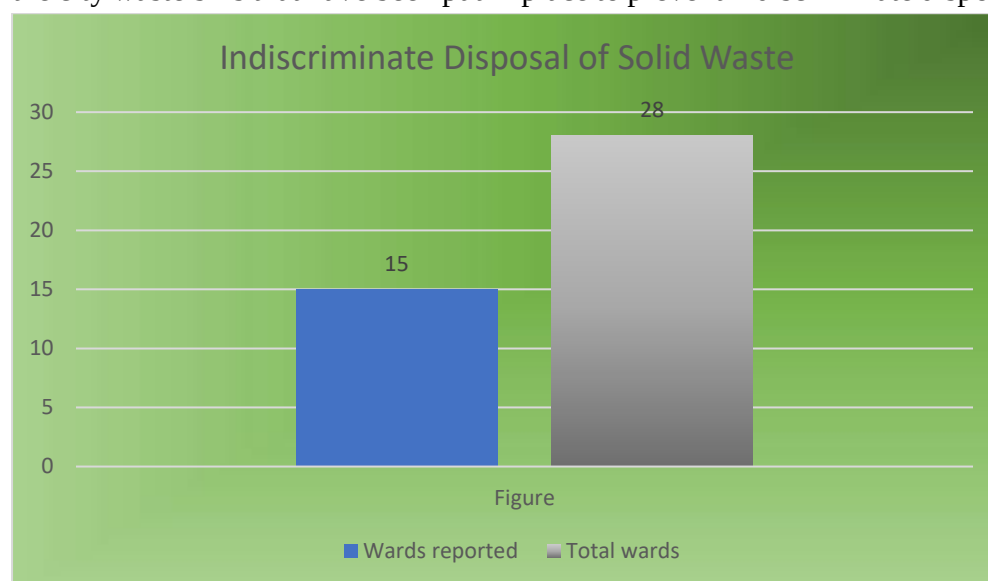
Deforestation is high, this submission was rated at 7% from all the 28 wards and



Source: Field Survey 2020

Indiscriminate disposal of solid waste was rated 53%.

Indiscriminate waste disposal still remains a challenge for the council. Despite putting a number of measures to control waste disposal, there still remains a hindrance in meeting the current set out target of achieving zero-waste disposal due to a rising issues of vandalism of the city waste bins that have been put in place to prevent indiscriminate disposal of solid waste.



Source: Field Survey 2020

Ndola district has in the past years experienced the above mentioned hazards and the following are the areas that have been affected:

- **Chipulukusu Area:** this is an area which has a high-density population, mainly the houses found in this area have weak foundations making it very easy to collapse during the rainy season, normally Chipulukusu area becomes a victim of flash floods during the rainy season (a flash flood occurs when water temporarily covers an area that it usually does not) leaving the people vulnerable due to the collapsing of houses caused by the said floods. In the same area, there has been epidemic hazards like Malaria, Cholera and HIV/AIDS.
- **Kabushi Area:** this an area with high density population, due to the high population it has, it has been privileged with drainages which in most cases people turn them into rubbish pits causing drainage blockages during the rainy season, because of this culture that has trended in the area for a long period of time, Kabushi area hasn't been spared of epidemics like cholera and malaria.
- **Twapia Area:** In this area, most of the drainages are blocked as is the case in Kabushi, and Residents of Twapia become vulnerable to Malaria and Cholera during the rainy season.
- **Bwana Mukubwa:** is normally affected with floods during the rainy season, due to the floods experienced in the area, houses with weak foundations collapse causing displacement of the people in the area.
- **Chichele Area:** this is an area whose main activity is farming and the main hazard experienced in this area is pests' infestation, following the assessment that has been

done by Ministry Agriculture, it has shown that this area is mainly affected with army worms.

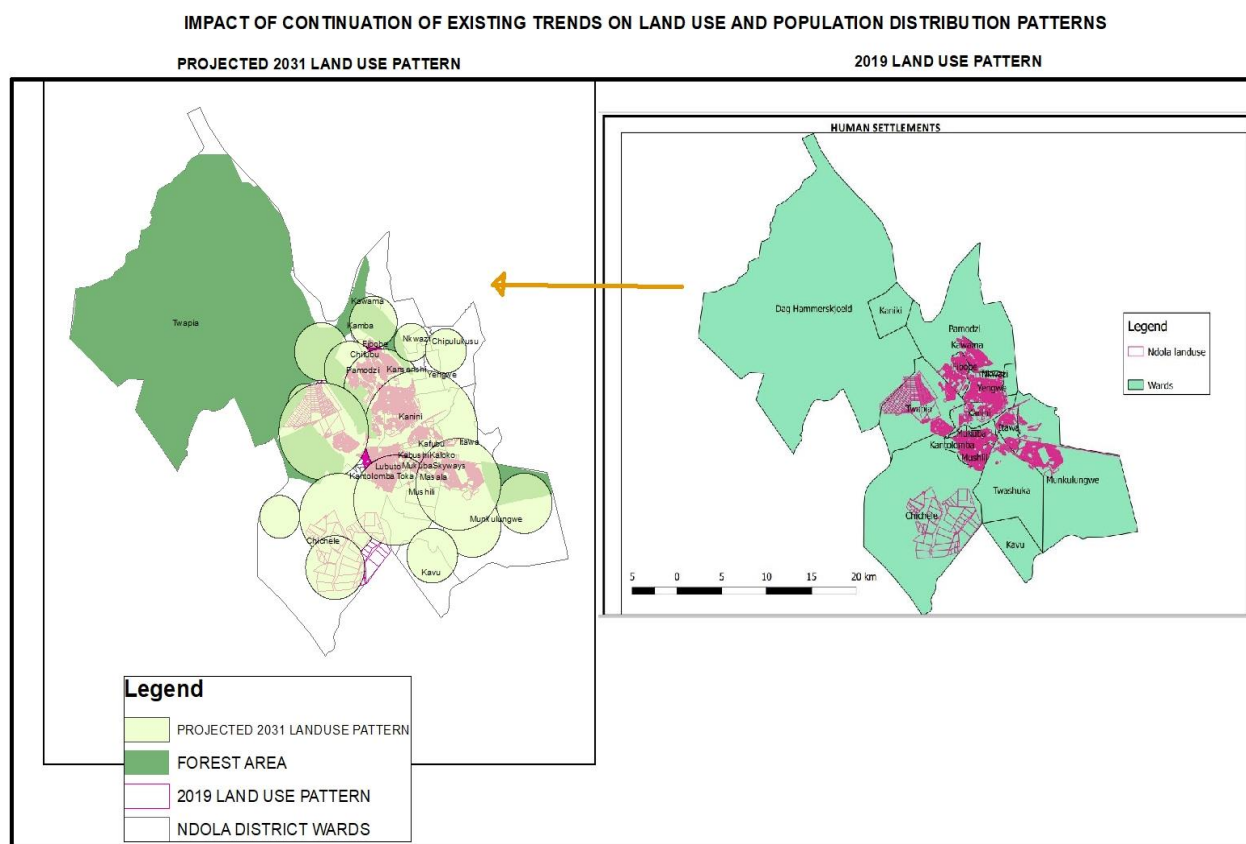
Despite the above discussed hazards, Disaster Management and Mitigation Unit is doing its best to ensure that as a District we have a sustainable safe-net for the protection of the citizenry, its assets and the environment from disaster and develop coordination and monitoring disaster risk management programmes in order to minimize loss of life, damage to property and the environment.

3.3.4. Impact of Changes Anticipated Over the Next Ten Years

3.3.4.1. Future Demand for Services and Facilities

With expected increase in total urban population, the district in the next 10 years is expected to continue recording gradual increase in economic growth. Continued de-gazetting of forest reserves is anticipated in order to meet the increased demand for land as a result of population increase. This pressure will lead to over use of arable land and forest reserves where people will continue to change the land use from farming to residential to accommodate the ever-increasing demand for land causing massive deforestation. Therefore, Economic growth anticipated as more land is made available to developers for development. Population Change

As it can be observed in figure below; a comparison of years 2019 to 2031 showing the rapid expansion of human activities towards forest reserves and land meant for agriculture.



3.3.4.1. Existing and Proposed Investment and Development Programmes

3.3.4.1.1 Protection of Environmental Sensitive Areas

Ndola district has a rich biophysical environment characterised by soils, minerals, climate, air and water. Added to this is the land resource which supports various economic activities. Population growth, industrial and commercial activities and competing land uses have increased pressure on land thereby threatening the availability and accessibility to ecological resources and making the district vulnerable to a number of hazards. Some areas in the district have therefore been identified as environmentally sensitive/hazard areas.

3.3.4.1.2 Environmental Sensitive areas

The following are identified environmental sensitive areas:

i. Wetlands

Wetlands are areas of marsh fen, peat land or water, whether natural or artificial permanent or temporary with water that is static or flowing, fresh or blackish or salt, including areas of marine water of which at low tide does not exceed six meters (MTENR, 2007). This definition excludes open waters that are several meters deep, but include permanent swamps, floodplains, and dambos. Wetlands are important for agriculture, aquaculture, flood control, water purification and micro climate regulation. However, wetland in the district is among key freshwater resource areas facing increasing pressure from human activities which include agriculture, mining, human settlements and climate change. It is therefore important that any developmental plan in the district should ensure that wetland is protected.

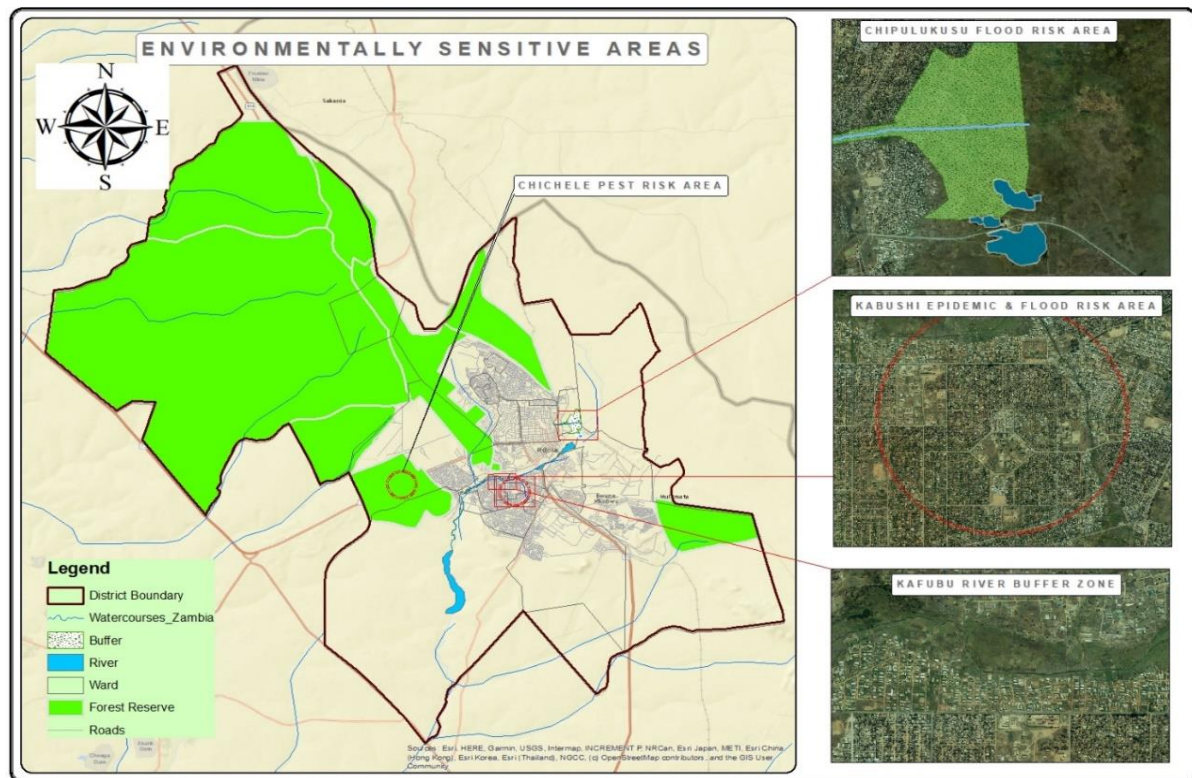
3.3.4.1.3 Kafubu River

Kafubu River is the main river that cuts through Ndola town and main source of water for the district. The river stretches about 80.7Km before joining the Kafulafuta river with a catchment of about 1,375km. The river has many tributaries of which Ndola, Mushili, Kanini, Kansenshi, Misishi, Itawa spring and Twapia fall within Ndola. The Kafubu catchment area has many inhabitants that depend on the water resource. Some communities within the catchment area grow maize, vegetable and other crops. Others rear cattle, chickens, pigs and other animals. Apart from Kafubu water and Sewerage Company (KWSCO) abstracting water from the Kafubu River, the community also abstract water from the river for domestic use.

The Kafubu River is being threatened in many ways ranging from the Kafubu weed, siltation and encroachments especially from development of residential housing units. In some cases raw sewer is directly discharged into the river and its tributaries thereby compromising the water quality for domestic and commercial use.

Therefore, it is imperative that all developmental plans in the district ensure that the Kafubu catchment area is protected. Fifty meters away from the river bank should not be encroached.

Environmentally Sensitive Sites



Source: Ndola City Council GIS Unit (2020)

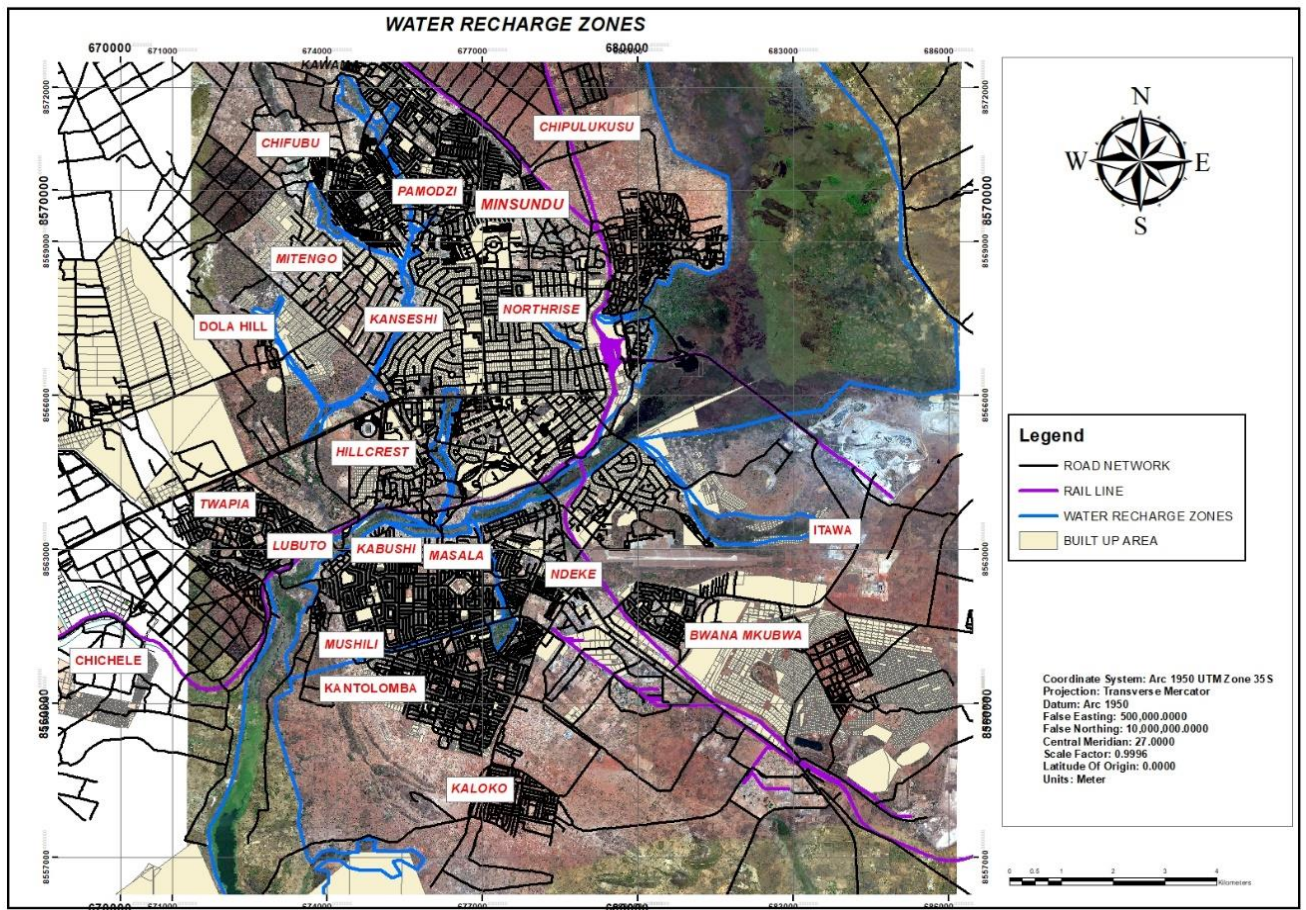
3.3.4.1.4 Misundu Water Recharge Zone

Preliminary assessments indicate that the Misundu catchment area and the Itawa dambo are water recharge zones in Ndola district. Much of the Misundu area is covered by small holding farms. Within these small holding farms lies the Misundu well field which hosts 11 boreholes for Kafubu Water and Sewerage Company (KWSCO). The well field is heavily encroached mainly due to land use change where most of the small holding farms are being turned into medium cost residential settlements. This has threatened the ground water resource in terms of quality and availability. This means that much of the water supplied by KWSCO to the communities within the district will not be fit for human consumption.

The potential environmental hazards arising from the land use change include the following:

- Underground water pollution through seepage from pit latrines and sewer soakaways.
- Loss of vegetation covers to support water recharge system.
- Depletion of ground water. Some of KWSCO's boreholes are showing signs of drying up.
- Land degradation.

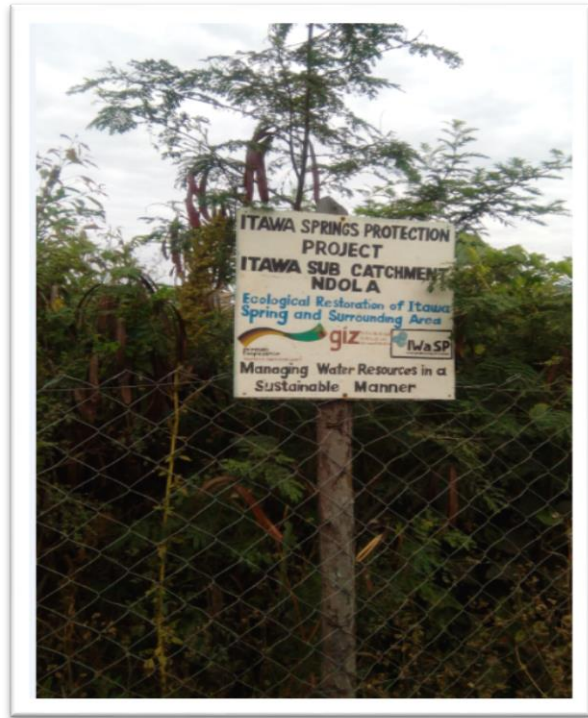
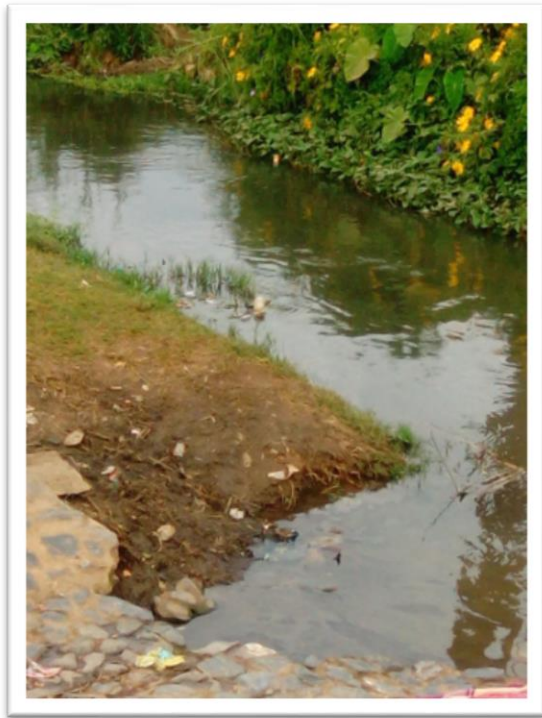
In order to protect the Misundu Water Recharge zone it is highly recommended that the land use change (conversion of small holding farm into medium cost residential) should not be permitted further and the Misundu well field should not be altered.



Source: Ndola City Council GIS Unit (2021)

3.3.4.1.5 Itawa Springs

The Itawa Spring has major economic, social and ecological importance as a water resource for domestic and industrial purposes to the city of Ndola. The water spring has its origin from groundwater which rises at an impermeable barrier in the subsurface. Where the rising groundwater level meets the surface, it causes the surface flow – the spring. The spring however is not a single spring but a much larger system that includes multiple points where the water flows from the ground and an elongated zone on either side of the stream where water continuously seeps through the soil into the stream that flows east towards the Itawa Dambo.



The environment around the spring has been altered by industrial development and by people settling close to the spring. The issues related to settlements and changes in land use around the Itawa spring have put water supply, quality and integrity of the slopes around the spring under threat. Sanitation issues around the area affect the water quality of the stream in that pit latrines are mostly used in Mapalo community which in combination with a very shallow water table creates potential pathways for contamination.

It is therefore important that the Itawa spring is protected from all potential threats

3.3.4.1.6 Afforestation/Reforestation

In a quest to conserve nature and alleviate the impact of climate change, the district at local level continues to plan for tree planting projects. In the long run, afforestation will continue to take place within the district. In support of these projects, the district has so far implemented two activities of tree planting in 2021. The first project was launched through the partnership of Limestone Resources Limited, Plant a Million and Ndola City Council. Tree seedlings were planted along Airport road. The second project was launched with the partnership of Rotary Club/Ndola City Council. Tree seedlings planting activities were initiated in Itawa – corner of Dag Hammaerskjold Drive and Mwasumina Avenue, next to the border wall fence with ZAFFICO flats, opposite Oryx Filling Station, and continued along Kafubu River near Kafubu Water and Sanitation Company Pump Station as indicated above.

S/NO	DATE	ACTIVITY	QUANTITY	LOCATION	ENGANGED PARTNERS
01	18.02.2021	Tree planting	500 trees -125 Jacaranda -125 Flamboyant -250 Mahogany	Itawa – TAZARA Railway, Airport road	Limestone Resources Ltd, Plant A Million
02	27.02.2021	Tree planting	1000 trees	Banks of Kafubu river, Airport road	Rotaract club, WARMA, ZAFFICO, KWSC

Summary of tree planting activities

3.3.4.1.7 Continued Climate Change Management

Changes in climate pose challenges to Zambia's ongoing efforts to combat poverty, reduce food insecurity and sustainably manage natural resources. Climate projections in Zambia include increases in temperature and more extreme weather (intense precipitation, floods and droughts) and increased rainfall variability. Climate changes will have key impacts on agriculture, water resources, human health, ecosystems, and energy and infrastructure.

At district level, there is an establishment of a multi-stakeholder process whose strategies increase public awareness of climate change and gender equality issues to stimulate interest, participation and consultation with different stakeholders in developing climate change policies, and to increase capacity of those national actors to continue their involvement.

3.3.4.1.8 Sustainable Industrial Waste Management

Copperbelt province has experienced an increase in lack of environmentally friendly, sustainable and affordable waste management which has led to the wide spread open dumping and open burning of solid waste. In another study that included Zambia and reported in the Africa Review Report on waste management [7], it was concluded that poor waste management practices in particular the widespread dumping of wastes in water bodies and uncontrolled dump sites aggravates the problems of generally low sanitation levels across the African continent.

A Site Waste Management Plan (SWMP) is a tool that is and will be used to manage waste by industries on-site.” A SWMP provides a framework which can help contractors or project managers to forecast and record the amount and type of construction waste that is likely to be generated on a project, as well as assist in setting up appropriate management actions that reduce the amount of waste that is destined for to landfills, WRAP, [17]. The main aim of a construction SWMP is to improve materials resource efficiency by following the waste hierarchy in employing reuse, recovery and recycling to cut down illegal dumping by recording waste removal processes.

The strategic plan also involves liaising with specific departments of Safety, Health and Environment in all industries within the district in order to meet compliance with statutory measures regarding industrial waste management and disposal.

3.4. Impact of Existing Trends on Land Use and Population Distribution Patterns

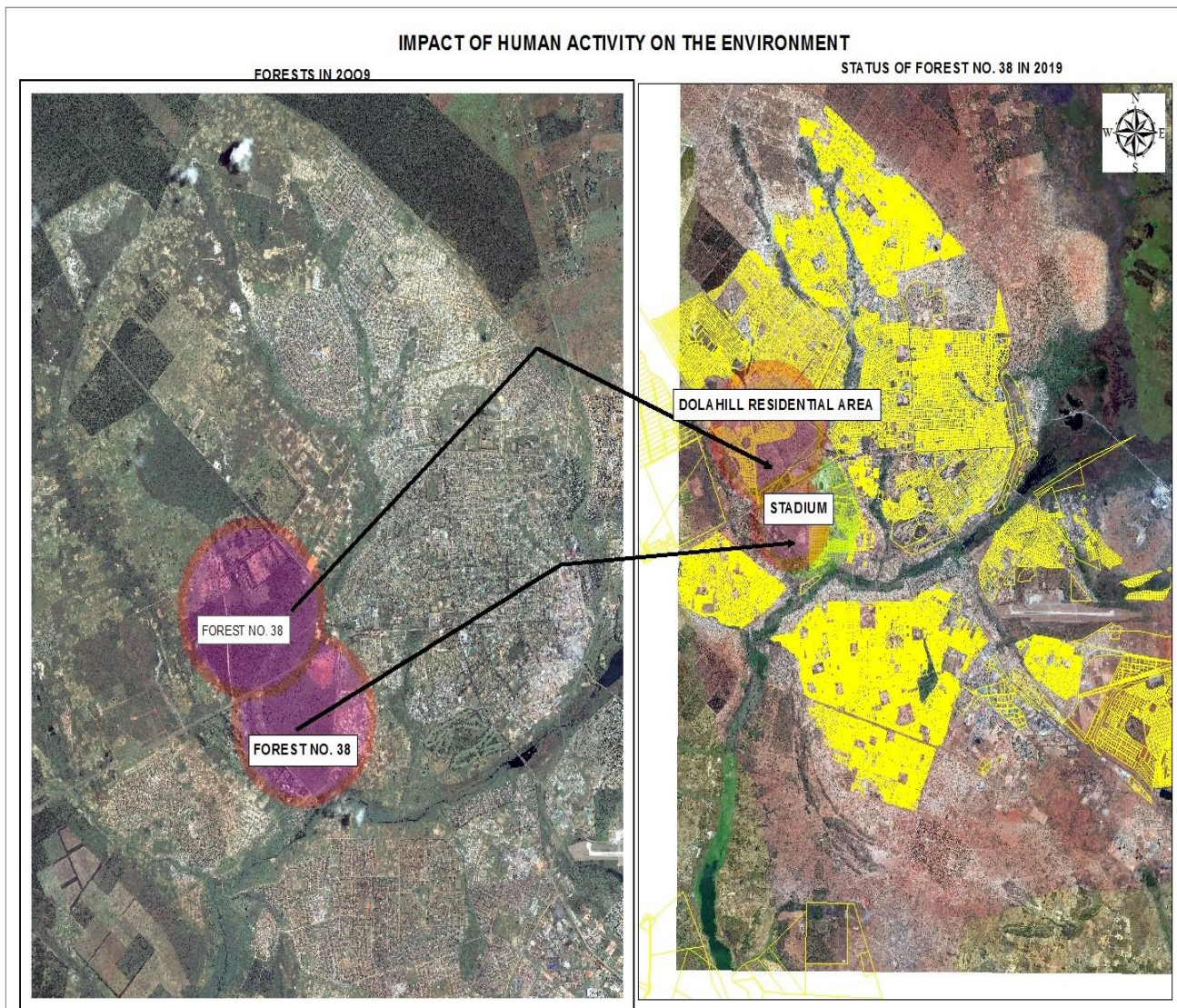
According to the 2010 census, the population was 451 246. This population is expected to be more than double by 2030 implying that there will be increased demand for services and facilities. Infrastructure services such communication, are provided by infrastructure assets, such as power stations, and networks such as fibre optic cables. Infrastructure services are consumed by households, business and government and third sector organisations.

The number of people has a direct impact on demand for infrastructure services, as people use infrastructure to communicate, stay warm, do washing etc. Mostly in their homes (although e.g., travel and mobile communication, are not based at home). A greater number of people means more or bigger households, both of which will drive up demand for infrastructure services, although not necessarily by the same amount, and more people making journeys and using mobile communications.

Therefore, the pace of population growth influences the infrastructure investment required to deal with the variation in demand. A small increase in population growth can often be addressed by making incremental improvements to existing infrastructure or by accepting higher congestion. Larger and relatively faster changes in population growth require new and bigger investment. Finally, the higher the population, the greater the demands placed on infrastructure will tend to be. So, having an understanding of the possible paths for the future population is a key input into developing scenarios for possible future infrastructure demand.

With the increase in population and Human activity, an increase in demand for land arises hence the need to excise forest reserves into other uses. As indicated in the figure below where

forest reserves are de-gazetted into residential areas to meet the increased demand for land.



Source: Ndola City Council GIS Unit (2021)

3.4.1. Environment and Climate Change Analysis

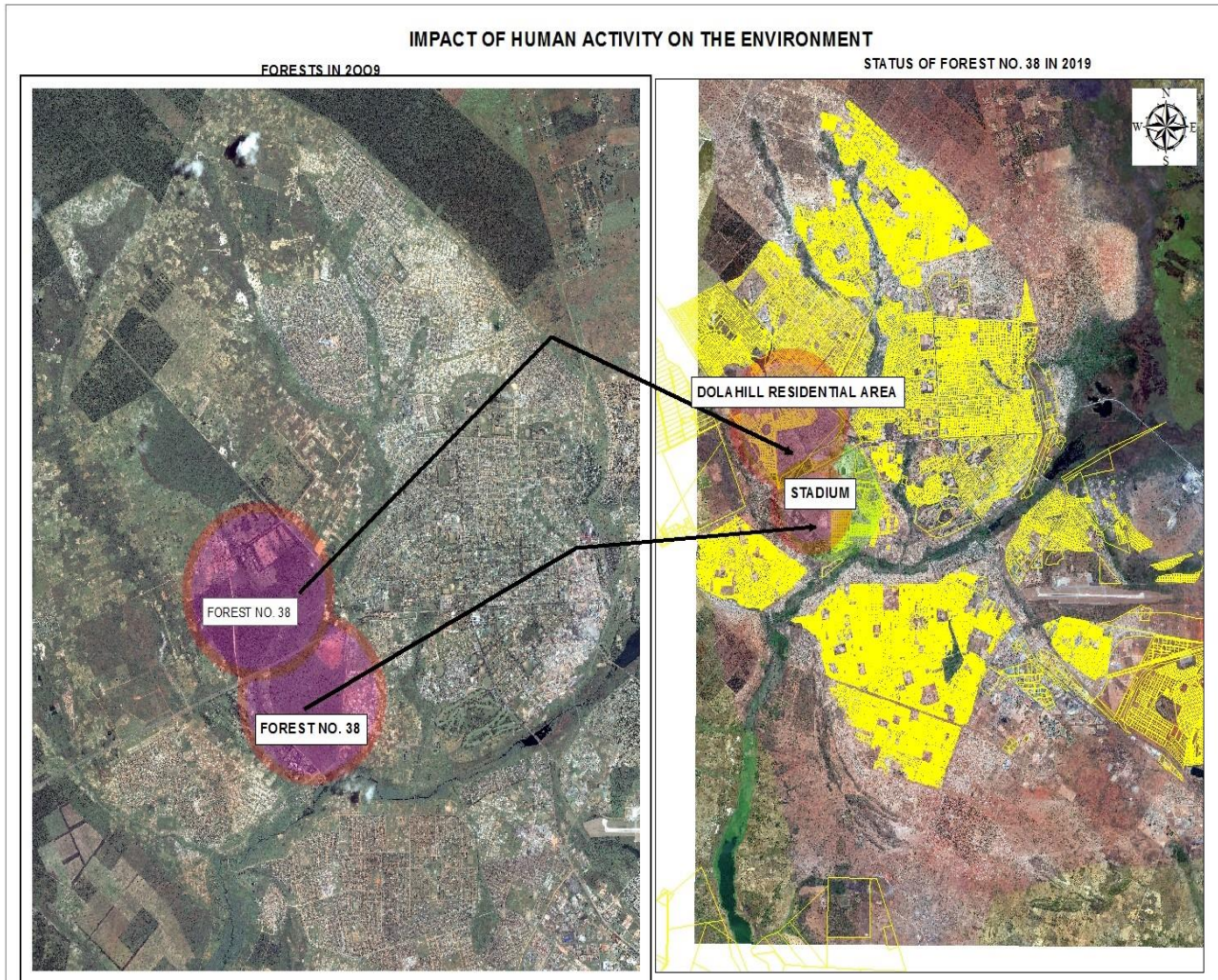
It is widely recognized that climate change constitutes a significant and serious threat to sustainable development of any country, including Zambia. Evident shows that Zambia has over the past years experienced a number of climate related hazards including droughts and dry spells, seasonal flash floods, and extreme temperatures. Some of these, especially droughts and floods have increased in frequency and intensity over the last two decades and have adversely impacted on food and water security, energy and livelihoods of communities. Temperatures also indicate a rising trend with potential for increased heat stress, land degradation and desertification. Such impacts are likely to compound the daunting economic and social challenges the country already faces, (MNDP, 2016).

Copperbelt province and Ndola district in particular has not been spared to climate related hazards mentioned above. Impacts like droughts and dry spells have threaten the food security in the district while floods and temperature rise has given rise to many health issues for the community

3.4.2. The Impact of Existing Trends on the Environment and Climate Change

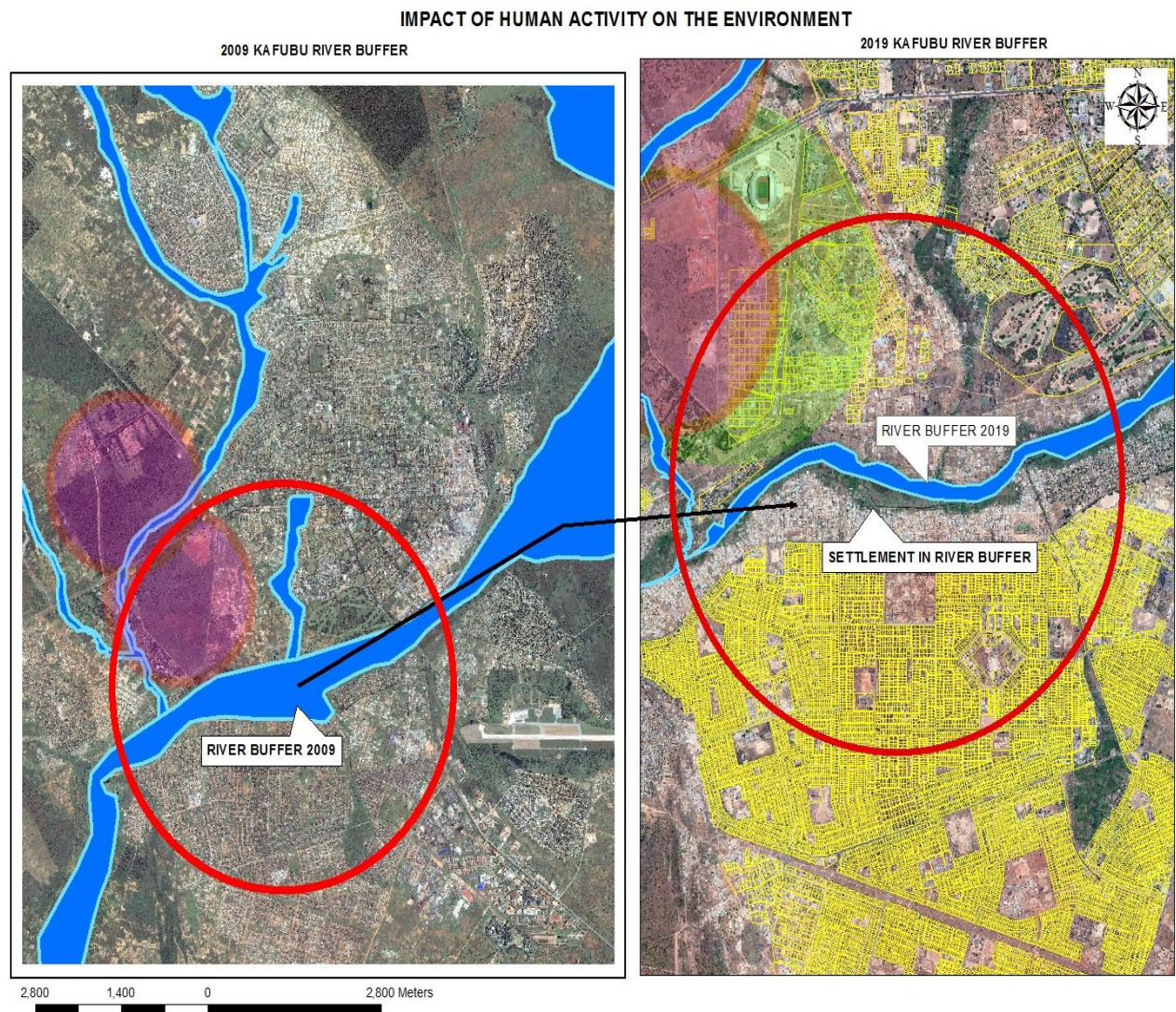
Ndola District has transformed environmentally in so many ways due to climate change. In most cases this has been necessitated by human activities as well as economic development which heavily depend on natural resources and environment itself. The maps below give an insight on how the environment has been affected due to aforementioned in the preceding text

Spatial Trend in the Built Environment



Source: Ndola City Council GIS Unit (2021)

Spatial Trends on the River Course



Source: Ndola City Council GIS Unit (2021)

3.4.2.1. The Impact of Environmental and Climate Change Issues on the Sector

The United Nations identified thirteen major environmental risks resulting from the impact of climate change on biodiversity globally (CCFR, 2014). However, it is just as much as imperative to address how climate change impacts the district.

Increased risk from pests, disease and invasive non-native species

Pest invasions come from a combination of ecological and climatic factors, such as weather patterns, mono cropping, the introduction of new species, or pest migration routes. Because the army worm jumps across borders, it has not only been reported in Zambia but also in neighboring countries such as Malawi and Zimbabwe.

In Zambia, the 2017/18 pest invasion caused about 30-40% damage on crops.

Reduced water quantity

Many factors of the economy are run by use of hydroelectricity energy which effectuates from the Kariba Dam. With regards to this, Ndola as a district, relies on Kariba's hydropower project for half of their electricity needs. A reduction of water means that like many, the district scales back electricity usage.

Impacts on energy

In a warmer climate, people will use more electricity for air conditioning and less natural gas, oil, and wood for heating. If the nation's climate warms by 5.8°C, the demand for energy used for cooling is expected to increase by about 5-20%, while the demand for energy used for heating is expected to decrease by about 3-15%.

Impacts on agriculture

Agriculture and fisheries are highly dependent on the climate. Increases in temperature and carbon dioxide (CO₂) can increase some crop yields in some places. But to realize these benefits, nutrient levels, soil moisture, water availability, and other conditions must also be met. Changes in the frequency and severity of droughts and floods could pose challenges for farmers and ranchers and threaten food safety. Overall, climate change could make it more difficult to grow crops, raise animals, and catch fish in the same ways and same places as we have done in the past. The effects of climate change also need to be considered along with other evolving factors that affect agricultural production, such as changes in farming practices and technology.

Changes in species migration patterns

Many animals, especially birds, migrate to warmer climates in winter. Changing climates can therefore impact migration patterns. This may present both risks and opportunities to migratory species. Observations suggest that patterns are already changing, in terms of geography and timing, behaviour which itself might be considered an adaptation to climate change. However, concern is that it may impact breeding patterns and successes.

Impact of climate mitigation programmes

This consequence is indirectly related to climate change, resulting from lead to a change in rainfall patterns, and indirectly impact the use of fertilisers in agriculture. The key challenge is to reduce pollution at its source

3.4.2.2. Issues Arising Relating to Gender Groups and Vulnerable Groups

Gender considerations are an obligation in development programming and implementation in relation to sustainable development and environmental compliance. The National gender policy of 2014 aim at “ensuring the attainment of gender equality in the development process by redressing the existing gender imbalances. For Equitable and Sustainable development to be attained gender equality must be achieved. However, gender inequalities are still very high between males and females in the District and the country at large. According to the 2010 Census of Population and Housing Report, Ndola District had a total population of 451,246 with a distribution of 223,020 being male and 228, 226 being female. The population of females has been slightly more than males.

Unequal Access to Resources

Women have unequal access to opportunities compared to their male counterparts in terms of access to jobs, women are in the lowest paying sectors and non-technical jobs, which are linked to their lower education levels. This is as a result of high illiteracy levels among women compared to men. In Ndola school attendance between boys and girls is at 93% and 86% respectively. According to the 2010 Census unemployment was higher among women compared to men with rates of 24.20% - 20.80%. Employment rates indicate 63.2 % for men compared to 37.7 for women. The district has a disproportionate ratio in terms of property ownership such as land between male and females which favours males.

Poverty

Several studies conducted project a direct link between poverty and its effects towards the environment. Out of the many baseline studies assessed, one of the biggest ways that the environment is being affected by poverty is through deforestation. Ndola is surrounded by vast forest cover managed by ZAFFICO. Part of the forests face encroachment from the impoverished communities unaware of the errant, harmful ways in which they use natural resources, such as forest wood and soil, are continuing the destructive cycle that spirals the environment further downward.

Due to the existing gender inequalities the Impact of poverty is more devastating among women compared to men. This is as result of significant differences in opportunities available between the sexes. According to the Living Conditions Monitoring Survey (2010), overall poverty is marginally higher among female populations at 80 per cent compared to the male population at 78 percent.

Decision Making

In Ndola decision making have generally been dominated by men. Women have been under-represented at all levels of national decision-making especially in government, parliament, the private sector, special committees, religious groupings, boards and other institutions in the community. Gender equality can only be archived through women's participation and in decision making at all levels, starting in the home and extending to the highest levels of government. The fact that gender inequalities continue to exist in all sectors at all levels, has negative implications on national development.

Traditional & Culture Beliefs

In Ndola, gender inequalities are perpetuated through cultural roles across ethnic groups that are determined and by one's sex, socialization and culture. Cultural norms, values and taboos that lead to the perpetuation of gender imbalances are transmitted through the socialization process. For instance, girls are socialized to be wives and mothers who are generally submissive while boys are socialized to be husbands, fathers and leaders. Girls and women are overburdened than boys and men in the sharing of roles particularly at the household and community levels.

Gender based Violence (GBV)

Gender Based Violence (GBV) continues to be a huge problem in the district. This is despite various legislature and policies such as the Anti-Gender Based Violence (GBV) Act No. 1 of 2011. The 2007 Zambia Demographic Health Survey (ZDHS) indicates that one in five women has reported having experienced sexual violence at some point in her life. The majority

of GBV cases are violence against women. These are mainly in the form of physical violence, Sexual assault.

Legal reforms and Gender

Inequalities in Ndola not only imitate from traditional, cultural and social norms but also from the dual structure of statutory law and customary law. Rights, which are supposed to be protected under statutory law, are not necessarily observed and women endure unfair treatment in terms of child marriage, unequal distribution of property, etc. Meanwhile, there have been some positive developments at the policy level, including the establishment of an independent Ministry of Gender, introduction of specific national and local authority level gender policies and revision of certain provisions of the Constitution, which epitomise gender inequality (currently being deliberated).

Environment and Climate Change

Climate change as a development subject matter is critical for the District where the majority of the population are female small holder farmers and they rely on agriculture for livelihoods. The gender differences between men and women imply that their vulnerabilities differ. Since women are already in a disadvantaged position, effects of climate change threaten to further increase the inequality.

3.4.3. Summary

Unequal Access to Resources

Access to and secure tenure over land is also closely linked to natural resource access and management, such as water and forest resources, with benefits for sustainable ecosystems. While women that manage land may want to adopt sustainable management approaches, if they do not have their name on the land title, they may not be able to access loans to invest in technology and inputs.

Poverty

Poverty often causes people to put relatively more pressure on the environment which results in larger families (due to high death rates and insecurity), improper human waste disposal leading to unhealthy living conditions, more pressure on fragile land to meet their needs, overexploitation of natural resources and more deforestation. Insufficient knowledge about agricultural practices can also lead to a decline in crop yield and productivity etc.

Decision Making

Facilitating access to decision-making power and maintaining cultural identity is most likely be a driving factor towards equal access to resources. Securing land rights for Indigenous women and men contributes to reduced deforestation rates and is a cost-effective measure for climate change mitigation.

Traditional & Culture Beliefs

The importance of secure and equal land rights in sustainable ecosystem management cannot be understated. The Intergovernmental Panel on Climate Change (IPCC), reaffirmed that strengthening women's access to and control over land improves sustainable management efforts, such as by increasing investments and willingness to engage in conservation activities, including tree planning and sustainable soil management. These are some of key performance

plans adopted by the local authority to cut out gender and vulnerability constraints in resource management.

Legal reforms and Gender

The rights to sell manage or control the economic output from land should be liberally met by all social groups. Insecure land rights are a huge barrier for women in participating in or leading sustainable management efforts, as they may not have decision-making power over how land is used and managed if they do not own it. These are some of the marginal links we try to merge, especially in rural areas.

3.4.4. Factors Contributing to the Issues Identified

Electricity Load shedding

It is a known fact that energy drives the economy of the world by enabling processes and activities that support economic development. Constrained energy or electricity supply eventually cripples government operations and productive systems thus freezing the economy. However, these events usually do not last for a very long time. The deficit of electricity supply in Ndola often results in high rate of electric power outages, material theft and use of wood fuel which leads to an increase in deforestation.

Shortage in fuel supply

Petroleum plays a crucial role in the running of Zambia's economy as most sectors use petroleum as energy requirement. All petroleum products in the country are all imported which makes petroleum a significant component of the country's import costs. The petroleum industry consists of TAZAMA Pipelines, Indeni Refinery and Ndola Fuel Terminal. A shortfall in petroleum in the country affects products' and services' delivery. A follow-up factor is the hiking of transportation costs which also contributes to increase in prices of products and services.

Disease pandemics (covid-19)

Health calamities, like other calamities are great revellers. The covid-19 pandemic is emblematic of this. During its early stage, the International Monetary Fund (IMF) warned that the economic fallout will be worse than the Great Financial Recession of 2008. The IMF projected that the Zambian economy will experience negative growth this year, shrinking by at least 2.6% (Ng'andu, 2020). The pandemic brought about difficulty in access to basic needs such as food. There was scarcity in moving goods between towns. Imports and exports were blocked as borders were closed. With all these constraints, the government have embraced it and life should continue as the new normal.

Absolute poverty

One in every two Zambians lives below the poverty line, with a substantial percentage living just above it and risk of slipping into poverty (IMF 2019). Even in the midst of high economic growth, poverty in Zambia remains high and inequality worsens. Social cash transfer schemes have proven to be the most effective tool for poverty alleviation for countries such as Zambia. Of eight million Zambians living in poverty, only 700,000 are on social cash transfer programme. Currently Zambia runs other programmes like Farmer Input Support.

Delayed supply of agricultural inputs (fertilizer)

Farmer Input Support Programme in 2020 was allocated 5% of the national budget. It is often imperative that agricultural inputs are delivered on time to cater for the oncoming farming period. But oftentimes, there has been a delay in delivering inputs like fertilizer to farmers and delay often lead to periods of disaster, of which it is often the most vulnerable that are hardest hit and find it most difficult to recover. A delay in obtaining fertilizer by farmers means the farming calendar gets affected and thereby, affecting resultant food supply.

Unequal access to resources

Some evidence indicates that many older people are routinely denied access to resources and services as a result of age-based discrimination (UNICEF & UN Women, 2013; UNDESA, 2013; Cain, 2012). Studies have found that there are more children on average in the poorest households and thus they are more exposed to economic inequalities than adults (UNICEF & UN Women, 2013).

An increasing body of evidence indicates that people with disabilities often face inequalities in all areas of life (comparatively lower educational attainment, higher unemployment rates, worse living conditions, and higher poverty rates, for example) (Mitra et al., 2013; UNICEF & UN Women, 2013; UNDESA, 2013).

Erratic water supply

In Ndola District, limited access to adequate water and sanitation is a key developmental challenge particularly for rapidly expanding peri-urban areas. From the district public consultation submissions on the quality of services offered on water and sanitation, it is evident enough that communities feel that water is supplied for less hours and in some areas, which are already serviced, there is literally no water that runs through the pipes hence it being pointed out as the major contributor to disease outbreaks.

3.4.5 Core Issues

- Unpredictable weather pattern
- High deforestation
- Indiscriminate disposal of solid waste
- Increased risk from pests, disease and invasive non-native species
- Reduced water quantity
- Impacts on energy
- Impacts on agriculture
- Changes in species migration patterns
- Impact of climate mitigation programmes

3.5. Institutional framework

3.5.1. Reviewed Institutional Policies and plans

State Police; The Ministry of Home Affairs is a government institution charged with the responsibility of providing and maintaining internal security in order to promote sustainable socio-economic development for the people of Zambia. The Zambia police Act Chapter 107 of the laws of Zambia provides for the organisation functions and discipline of the Zambia police force and of special constables and for matters incidental thereto. The institutional mandate is

to provide and promote quality internal security in order to create a safe, secure and peaceful environment for sustainable socio-economic development for all internal security in order to promote sustainable socio-economic development for the people.

Zambia correctional service; Zambia correctional service is there to effectively and efficiently provide human custodial and correctional services to inmates and increase industrial and Agricultural production in order to contribute to the wellbeing and reform inmates and maintenance of inter security. In line with the Zambia correctional service chapter 97 of the laws of Zambia (prisons act) charged with a mandate of providing human custody and quality correctional services in order to promote public safety and to contribute to the social economic development of the country.

Ministry of justice; the ministry is charged with the responsibility of facilitating the administration of justice and promoting the observance of the rule of law. According to government gazette notice no. 183 of 2012, the ministry is responsible for the functions and statutory bodies and institutions

The judiciary administrative act no. 23 of 2016 an act to revise the law relating to the administration of the judiciary; make provision for the appointment of the judicial officers and staff of the judiciary. Repeal and replace the judicature administration act 1994 and provide matters connected with or incidental to the foregoing.

The functions of judiciary include, administration of justice through resolving dispute between individual or between individual and the state. One of the major functions of judiciary is to interpret and apply laws to specific cases. This interprets and applies to specific case. This interpretation of law is done by Judges, Magistrate and Commissioners.

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To enhance security in the district, the Ndola District has embarked on the construction of different infrastructure projects such as the construct of Mushili Police Station in Bwanamkubwa Constituency and the expansion of Mapalo Police Post in Mapalo Ward of Ndola central constituency.

In Ndola District, the Judiciary has been constructing Courts e.g. the Fast Track Court located opposite Ndola High Court of and has also embarked on the rehabilitation of court premises.

3.5.2. Existing state of development

Ndola district is geographically divided into both rural and urban areas. there are seven police stations in the district namely; Ndola central which has six police posts, Masala which has six

police posts, Kansenshi which has three police posts, Chifubu which has three police posts, Twapia has two police posts, Kafulafuta and Sakania have no police posts.

Table 1: Infrastructure (Staff houses and water availability)

Court name	District	Number of staff			AV number cases/month	Court accommodation			Staff accommodation	
		Required								
Ndola Magistrate Court	Ndola	22	17	05	600	09	05	04	0	0
					Cri-83	Ch-12	05	07		
		35	32	03	civ-35	Cr-12	04	-		
						Off-04				
Totals		22	17	05	600	09	05	04	0	0

Name of Police Station/Post	Number of Staff Houses	Type of Water Supply
Ndola Central	269	Piped water from KWSC and boreholes
Kansenshi	286	Piped from KWSC
Masala	165	Piped from KWSC
Chifubu	145	Piped from KWSC
Sakania Grade B	08	Bore hole
Twapia	27	Piped water from KWSC and borehole
Kafulafuta	08	Bore hole

Prison Infrastructure in the district

The infrastructure in the correctional service in Ndola is tabulated in Table 2 below;

Table 2: Prison Infrastructure in the district

Name of Prison	Location and Type Of Prison	No. of Cells	Personnel	No. of Staff Houses
Kansenshi State Prison	Kansenshi and State Prison	6 Dormitories	83	47
Ndola Remand Prison	Peter Singogo (Kanini) and Remand Prison	3 Dormitories	30	9

Magistrates Court: The functions of judiciary include, administration of justice through resolving dispute between individual or between individual and the state. One of the major functions of judiciary is to interpret and apply laws to specific cases. This interprets and applies

to specific case. This interpretation of law is done by Judges, Magistrate and Commissioner

3.5.3. Availability of service

3.5.3.1 Zambia Police staffing levels

The current staff is inadequate as the population of Ndola is expanding at a fast rate. The establishment capacity is low and hence the need to recruit officers to fill some vacancies and training those in need of training.

Table 3: Ndola District Staffing Levels and transport availability

Name of Police Post/Station	Location	Number of Cells	Number of Staff	Number of vehicles
Ndola Central	Town Centre	1	108	1
Sakania Grade B	Sakania	2	22	-
Chifubu Police	Chifubu	3	78	1
Kansenshi	Kansenshi	3	77	-
Masala	Masala	2	88	1
Twapia	Twapia	3	34	-
Kafulafuta	Kafulafuta	2	24	-
TOTAL		16	431	3

Zambia Police Victims Support Unit

The counselling that is on demand mostly is that of spouse abuse, sexual, emotional, and physical abuse, property grabbing and HIV/AIDS related counselling. It has proved that such counselling is effective looking at the number of cases reported and victims who have benefited.

Table 4: Number of Clients Counseled

Name of Police Station	Location	No. of Trained Counselors	Number of Clients counseled in Previous years				
			2002	2003	2004	2005	2006
Ndola Central	Town Centre	03	1200	1800	2000	4600	5300
Kansenshi	Kansenshi	02	800	1600	2000	1200	1000
Chifubu	Chifubu	02	600	2000	1800	4320	2700
Masala	Masala	03	1300	2800	3000	4000	6000
TOTAL		10	3,900	8,200	8,800	14,120	15,000

Crime Pattern

Kafulafuta and Sakania police stations have been declared rural areas. The most prevalent cases in these include murder, stock theft, rape, assault/wounding, burglary and theft, defilement and GBV. As for the urban stations the most common crimes are breaking in, assaults, thefts, rape and occasional robberies. Most crimes are contained by available officers who are committed to carry out their duties.

Table 5: Records of Crimes in Ndola District - 2015 to 2019

OFFENCE	2015	2016	2017	2018	2019
Breaking (General)	259	301	360	410	480
Theft	448	203	540	585	602
6Assault/Wounding	553	143	597	573	573
Burglary	173	55	298	208	340
Murder	19	21	41	35	31
Malicious damage	6	11	46	64	38
Threatening violence	1	4	17	24	11
Stock theft	6	5	9	17	10
Forgery	4	6	5	2	7
Robbery	3	4	3	10	10
House breaking and theft	141	29	169	94	127
Aggravated robbery	23	15	18	38	23
Theft of motor vehicle	22	5	26	10	7
Criminal trespass	29	3	9	15	24
Theft by servant/agent	16	63	92	81	96
Obtaining by false pretence	94	82	95	38	101
Defilement	43	9	39	43	7
Indecent assault on female	6	3	2	3	5
Escape from lawful custody	0	5	20	22	15
Rape	0	3	10	7	4

Source: Zambia Police Copperbelt Division (2021)

Magistrates Court

The Judiciary is composed of High Court, Industrial Relations Court, Magistrate Court, Small Claims Court and Local Court and Gender Based Violence Court. In Ndola precisely, there is

only one high Court, Industrial Relations Court, Magistrate Court, Small Claims Court, Local Courts and Gender Based Violence Court.

The composition adjudication includes 6 judges, 6 Local Court Magistrates and 11 Magistrates. The prevalent cases in court are Divorce Cases, Gender Based Violence cases and Small Claims Cases. The table below shows this tabulation.

Table 6: Number of Court Cases

COURT	NUMBER OF CASES PER YEAR					
	2015	2016	2017	2018	2019	2020
HIGH COURT	339	442	452	459	340	11
MAGISTRATE	411	469	536	626	623	29
LOCAL COURT	1557	1287	1118	928	1131	72
SMALL CLAIM COURT	891	969	906	857	706	21
GENDER BASED VIOLENCE			4	207	170	11
INDUSTRIAL RELATIONS COURT	121	132	144	103	133	4

Source: The Judiciary, Copperbelt Region (2021)

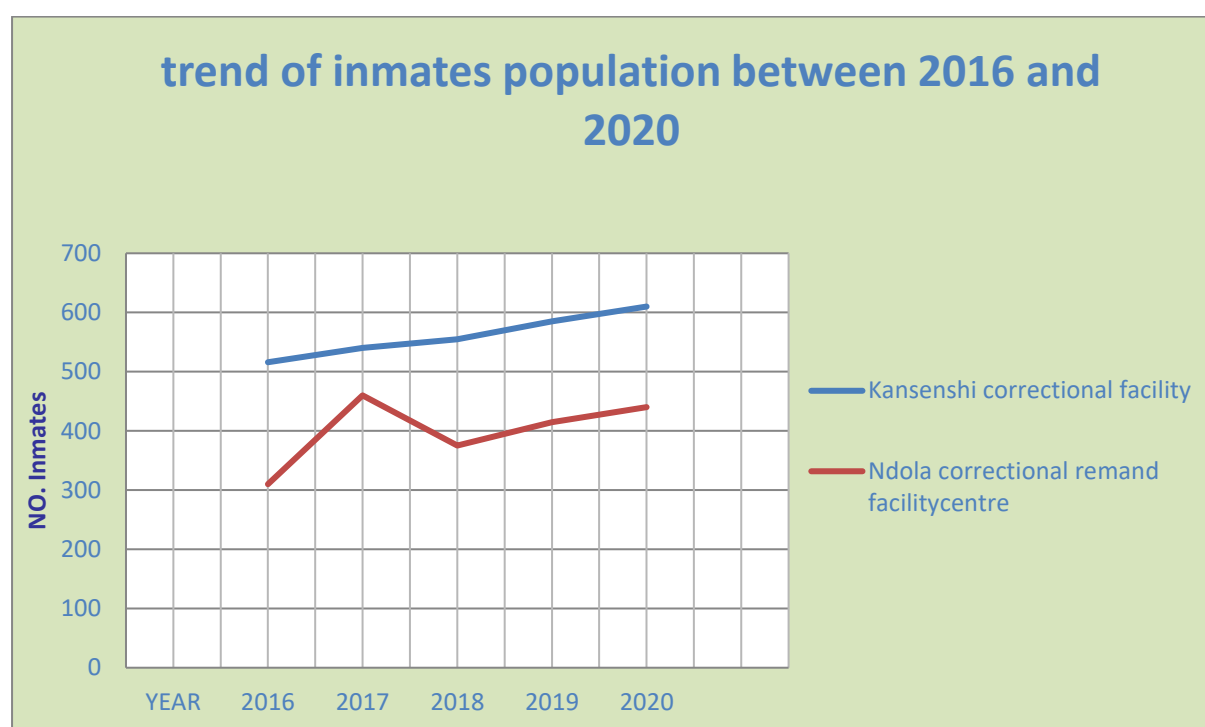
Zambia Correctional Service

Apart from providing the custody services, Zambia correctional service in the district is also providing a variety of services to the inmates to meet a range of basic needs including medical care counselling, psychiatric, spiritual and religious care, recreation and crisis intervention.

Agriculture is a major activity in the Prison Service and many inmates participate in this program as a way of acquiring skills they would be able to use upon their release. Agricultural farm produce can be used as food for prisoners. This would improve the nutritional value of their diets and afford them three meals as opposed to one meal per day. The Prison Service's vision states that it aims to be self-sustaining in agriculture to eliminate hunger in prisons and reduce dependence on the National Treasury. To this end it continues to organize activities that allow inmates to produce their own food

Below is the figure showing the trend of inmates in the two correctional facilities for the period 2016 to 2020. It can be seen that the number of inmates has been increasing for the period of study showing the upward trend.

Table 7: Trends of Inmates Population



Source: Zambia Correctional Services (2021)

3.5.4. Issues arising to gender groups and vulnerable groups

Gender-Based Violence (GBV): Common forms of gender based violence in Ndola in the three institutions include rape, defilements, incest, wife battering, and general mistreatment of women. Violence arises due to issues of infidelity, dissatisfaction, refusal to have sex, financial issues or when a woman does something without the consent of the husband. Though violence against women is more common than against men, men also experience violence and do not report due to pride and shame.

HIV/AIDS in prisons: Apart from tuberculosis, HIV/AIDS and other sexually transmitted diseases are the most dangerous diseases prisoners have to contend with, a situation that is exacerbated by overcrowding in prisons. The 2005 Human Rights Commission's Report for Central Province confirmed that HIV/AIDS and tuberculosis (TB) pose a great danger to the lives of prisoners.

3.5.5 Summary

The three institutions in Ndola District namely judiciary, Zambia police and correctional service has a mandate of providing and maintaining internal security in order to promote sustainable socio-economic development for the people of Zambia and to facilitate the administration of justice and promoting the observance of the rule of law. To this effect, the institutions has been facing a number of challenges where the major ones includes inadequate resources and infrastructure as well as human resource as these come with a huge challenge in the execution of duty.

3.5.6. Factors contributing to the issues identified

- Most infrastructures for the three government institution are in the state of disrepair and need urgent remedial steps before the situation becomes completely helpless.

- The two prisons in the district are overcrowded posing a serious hazard to inmates and officers in this time of Tuberculosis.
- Inadequate funding
- Prison staff is not adequate to effectively man prison day to day operations. In addition, housing for staff is not sufficient to accommodate all prison officers as most of them are sharing one house to two or three officers.
- HIV/AIDS cut across all sectors, prisons too have not been spared. Firstly the prisons service has lost staff robbing of valuable manpower and the prevalence of the scourge exists in the prison among inmates that come with the sickness from outside before they are committed to prison. The concerned authorities in collaboration with other concerned partners are carrying out the sensitization campaigns in the prisons.

3.5.6.1 Core Issues

- Inadequate space in correctional facilities
- It takes time for cases to be disposed off
- Inadequate resources for correctional facilities
- Not adequate nutrition in correctional facilities
- There are no visitors' shelters correctional facilities
- There less no female juvenile dormitories.
- Inadequate transport in police stations/police posts
- Less police stations/police posts to cutter for the ever growing population
- No local courts in most areas of the district making community members to move long distances in search of justice.

3.6. Education

3.6.1. Review of education policies and plans

According to the principle Education document, Educating Our Future, (1996), Education is viewed as a right for each individual. It is also a means for enhancing the well-being and quality of life for the entire society. The Government's role in education arises from its overall concern to protect the rights of individuals, promote social well-being and achieve a good quality of life for every person through an all-embracing economic development. Therefore, the Government seeks to create, promote and support the conditions within which education can realize its potential in society.

It is for this reason that the Ministry of Education National Strategic Plan aims at addressing the needs of the Zambian people in the field of education. In the overall development of the country, education has been recognized as the greatest asset and contributor to national development and that an educated population will be a leading force in development and in poverty reduction levels.

The National Education strategic plan has been based mostly on three key documents, Educating Our Future (1996), the Poverty Reduction Paper and Report on the Restructuring and Decentralization of the Ministry of Education. The overall vision of the ministry is ***“to provide quality education for all, which is accessible, inclusive, equitable and relevant to individuals, national, global needs and value system”***.

To ensure that the overall vision of the ministry at national level is implemented, the Provincial Office interprets it in simpler terms that can easily be applied and implemented in the province, which is “***A Province with Excellent Results, Clean and green schools with no one left behind***”.

A province with excellent results- this aspect of the vision entails that performance at all levels should improve. The targets for the various levels are as follows:

Reading levels from grade 1 to 4 should improve from 60% in 2018 to 90% in 2021,

Grade 7 performance should improve from 80% in 2018 to 100 in 2021,

Grade 9, performance should improve from 70% in 2018 to 100% in 2021 and;

Grade 12 performance should also improve from 70% in 2018 to 100% in 2021.

Clean and green schools - to ensure that all schools in the province have clean, green and conducive learning environment. All schools should have adequate sanitation facilities accessible by all learners and teachers.

No one left behind- to ensure that all pupils have equal and equitable access to have an education.

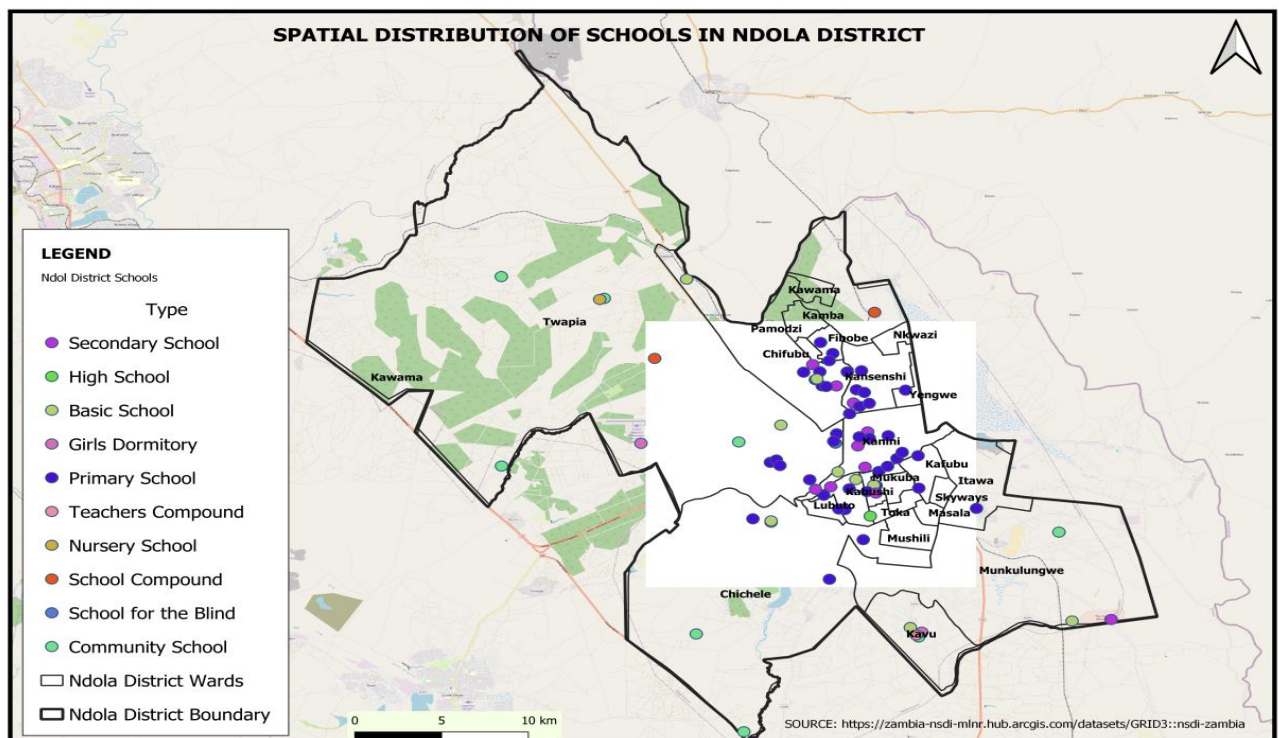
Through the Ministry of General Education, the government has embraced the mandate of the provision of education. In so doing, Government provides infrastructure, teaching and learning materials and human resource. At district level, the education sector has four functions; Early Childhood Education, Primary Education, Adult Literacy and Secondary Education. For the purposes of decentralization, Early Childhood Education, Primary Education, Adult Literacy have been devolved to the Local Authorities together with the administrative function at the district level. Meanwhile, Secondary Education and Standards department have remained in the mainstream and will continue to report directly to the line Ministry. Due to the fact that Government public schools cannot manage to accommodate all the school-age going children, other players, such as private and community schools, have come on board to help supplement Government’s effort in the provision of education.

3.6.2. Existing State of Development

3.6.2.1. Availability of Services

Access to education has been necessitated by the provision of free education from grade 1 to 12. The schools are also provided with textbooks for all subjects in all the grades for public and community schools only.

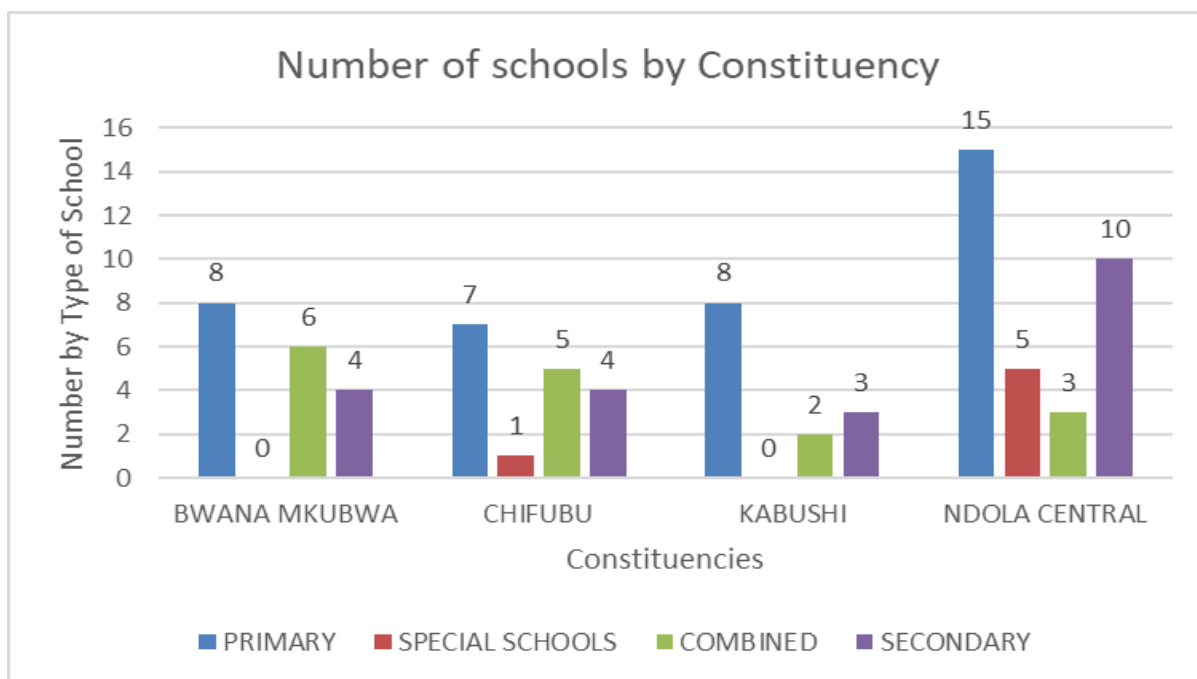
Figure 12: Spatial Distribution of Schools in Ndola District



Source: Ndola City Council GIS Section (2021)

Currently, Ndola district has a total of 238 schools. These are run by various sectors ranging from public, community and private. In this regard, 81 are public schools; 86 are run by the community and 71 by private sector. The public schools are further disaggregated into 43 primaries - 6 of which are special schools, 16 combined, 15 secondary and 7 Grant Aided schools. The table below shows the distribution of public schools by constituency:

Chart 1: Public Schools by Constituency



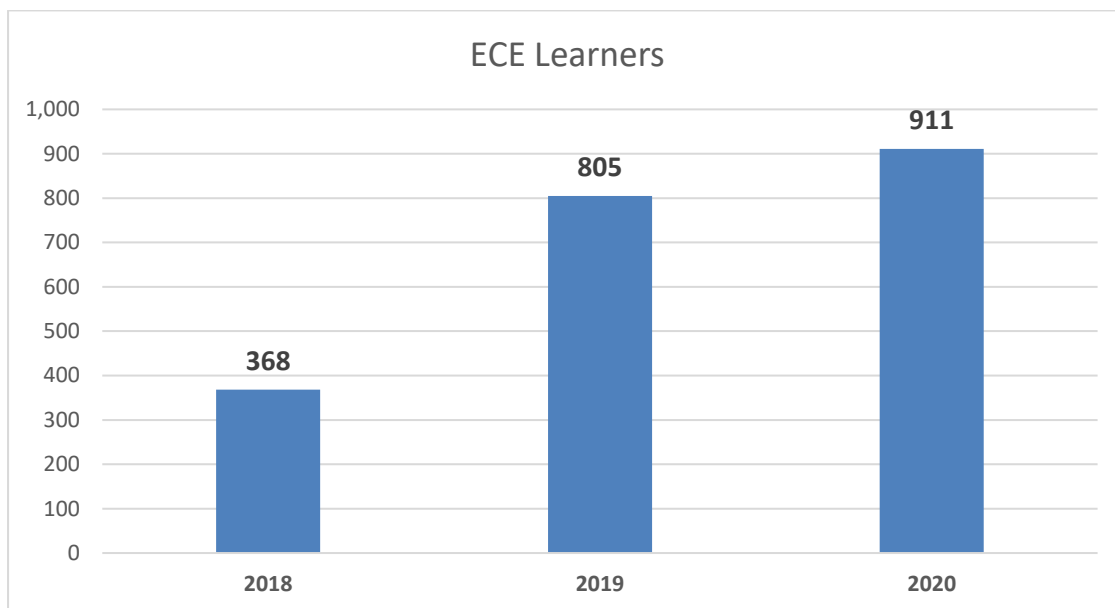
Source: DEBS Ndola Office (2021)

These schools are able to accommodate 79,371 learners at primary 21,981 in Combined and 45,086 at secondary schools. Besides these, there are also 23,085 and 17,204 in Community and Private Schools respectively.

The 16 Combined schools were opened in already existing primary schools. For them to cater for the secondary section, the schools had to allocate some classrooms that belong to the primary section thereby reducing access at primary level. This entails that these combined schools have to construct more classrooms to cater for the secondary section so that the primary section is not disturbed. In addition, there is need for qualified teachers to handle the secondary section in combined schools.

The other service being provided is the Early Childhood Education (ECE). There are 28 schools that provide ECE center services. The enrolment trends show that each year the enrolment figures keep on increasing. The table below shows the trends at Early Childhood Education (ECE), primary and secondary levels.

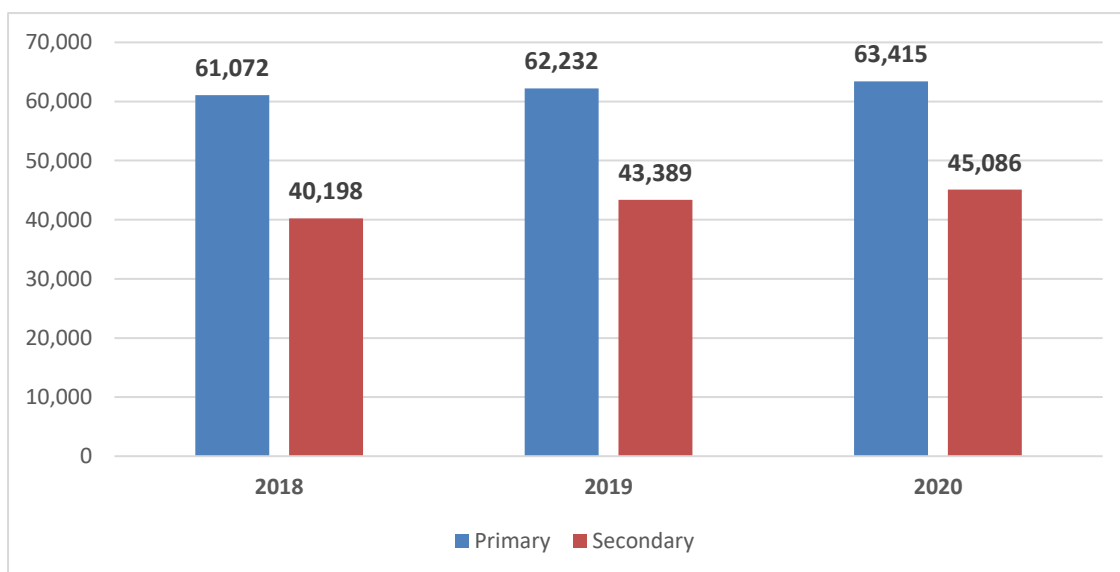
Chart 2: Enrolment Trend at ECE Level.



Source: DEBS Ndola Office (2021)

The enrolment trends at ECE level greatly increased from 2017 to 2018 and marginally in 2018. This is so because in 2017 the department was still laying the groundwork for the establishment of ECE centres.

Chart 3: Enrolment Trend in Primary and Secondary Schools



Source: DEBS Ndola Office (2021)

The trend of pupil enrolment at primary and secondary school levels show that it is increasing each year. Primary school enrolments increasing from 61,072 in 2018 to 62,232 in 2019 representing an increment of 1.89%. then from 2019 to 2020 the enrolment levels changed

from 63,232 to 63,415 respectively, representing an increase of 1.9%. at secondary sector there was an increase from 40,198 in 2018 to 43,389 in 2019, which is a 7.94%. while the change from 2019 to 2020 was from 43,389 to 45,089, representing a 3.91%.

In order to accommodate these learners, the sector has 405, with 8 incomplete, classrooms in the 22 secondary schools and 501, with 19 incomplete classrooms in the 37 primary schools. The 16 combined schools which run from grades 1 to 12 have 362 classrooms with 21 incomplete. Since these combined schools were not initially catered for in the classrooms available in these schools, the sector is expected to construct 9 classrooms for the secondary section so that the primary section is not disturbed. Hence there is need for 144 more classrooms in combined schools. The other way to increase access at primary section is by turning the existing and established community schools into public schools and expanding them by building more classrooms. To ensure that education is provided to all, the sector has 6 special schools with 27 classrooms.

In terms of access to education, about 96.6% of children in Ndola are in school. Following Government policy of leaving no one behind, more schools are required to cater for the remaining 3.4% of children that are out of school.

Granted, not all school-age going children are able to access education due to vulnerabilities of different aspects. The sector policy indicates that **The Ministry of Education will take positive action to ensure that the education system caters satisfactorily for the poor and vulnerable.** It is in this vein that the sector, with the help from Government, helps such children by providing learning materials and school fees and also with the help of partners and well-wishers.

One other challenge that hampers the access to education for most of the children is distance to the nearest schools. The standard spacing between schools is 5 kilometres or less. But in some areas in Bwana Mkubwa, Ndola Central and Chifubu constituencies schools are far apart, making children walk long distances to access school while others just drop out. In addition, some newly developed areas in these constituencies are increasing in population thereby increasing the need for more classrooms. This challenge can be curbed by constructing 722 and 596 classrooms at primary and secondary sectors respectively.

Tertiary Education

There are 14 colleges and 5 Universities in Ndola. These offer courses ranging from education, ICT, theology, technical, nursing and medical. Of all these colleges and Universities, only 2 colleges and 1 University are public while the rest are private. This indicates that the cost of

acquiring a tertiary education locally is expensive for the learners. Otherwise, the learners have to go to other towns or provinces to access tertiary education.

3.6.2.2. Quality of Service

The recommended class size for primary and secondary schools in Zambia is 40 pupils per class. The current scenario shows that the classrooms are overcrowded. The table below shows the standard class size, the actual obtaining in the district and the shortfall.

Table 8: Class size in the District

Education Level	Standard Class size	District Actual Scenario	Deficit/Shortfall
Primary	40 pupils per class	73 pupils per class	722
Secondary	40 pupils per class	111 pupils per class	722

Source: DEBS Ndola Office (2021)

These classrooms are necessary in order to accommodate all the grade externals whose numbers are ever increasing.

The district has 2,128 teachers at primary section and 1,228 teachers at secondary section. The pupil-teacher ratio is 1 to 29 at primary school and 1 to 37 at secondary. The ratio at primary level seems to be fair mathematically but, on the ground, some schools are in dire need for teachers than others. At secondary section, teachers have subject areas which they teach some of which need more teachers while others less, depending on whether they are optional or compulsory subjects. In order to improve quality of education provision and reduce the teacher pupil ratio, qualified teachers need to be recruited especially in science and mathematics.

In order to comply with the policy of ‘leaving no one behind’, the district has 6 special schools to cater for children that have special learning needs ranging from intellectual, physical, mental and hearing impairments. There are also 10 other centres in primary and secondary schools that offer special education. Others travel long distances to reach these schools. So there is need to construct 8 boarding facilities in special schools to cater for children who stay far from the special schools.

Quality of education is also enhanced when there is adequate furniture in classrooms. For instance, in primary schools, they are supposed to use double seater desks and single seater desks in secondary schools. The following table shows the desks available and the deficit:

Table 9: Desks available and the deficit

		Available	Required	Deficit
<i>Primary</i>	Double Seater	10,427	25,714	15,287
<i>Combined</i>	Double Seater	8,812	10,929	2,117
	Single Seater	85	12,542	12,457
<i>Secondary</i>	Double Seater	7,763	0	0
	Single Seater	4,341	19,799	15,458

Source: DEBS Ndola Office (2021)

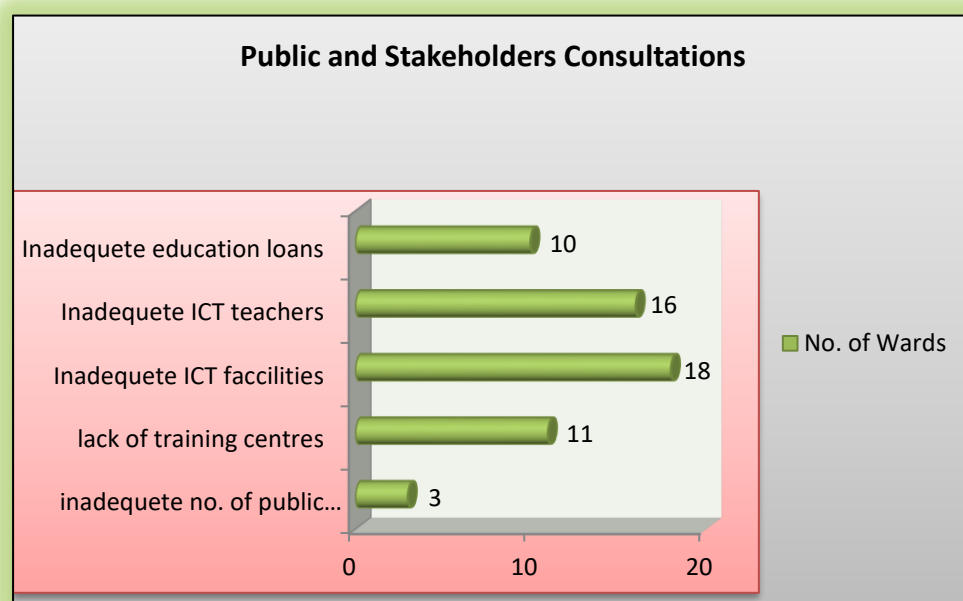
The table indicates that there is a deficit of 17,404 double seater desks and 27,915 single seater desks altogether.

3.6.3. Issues Arising from Public Partnership Process

3.6.3.1. Availability of Services Non-Formal Education (Adult literacy)

The sector has 14 adult centers in public schools. Adult Literacy offers learners the opportunity to proceed from the lower levels into formal education. These learners can proceed into Grade 5 to Grade 7 and beyond, i.e., Secondary and Tertiary levels of education. There is also another non formal education (adult literacy) program implemented by the department of community development meant to impart knowledge by teaching learners how to read, write and solve simple arithmetic. The program encourages the practice of functional literacy by incorporating other income generating activities during their learning period. Assessment of learners is undertaken to measure their understanding prior to graduation.

Figure 15: Public and stakeholder consultation



Source: Field Survey Data (2020)

Figure 15 shows the issues which were raised from the public and stakeholders consultations with regards to challenges that are being faced in the education sector in the district. As can be seen in the figure above, majority of the schools in the district have inadequate ICT facilities and teachers. In particular, it was reported that the district has limited opportunities for education loans, inadequate number of public schools, and lack of training centres. Overall, the findings imply that the district does not have enough ICT facilities and teachers.

3.5.4. Impact of Changes Anticipated Over the Next 10 Years

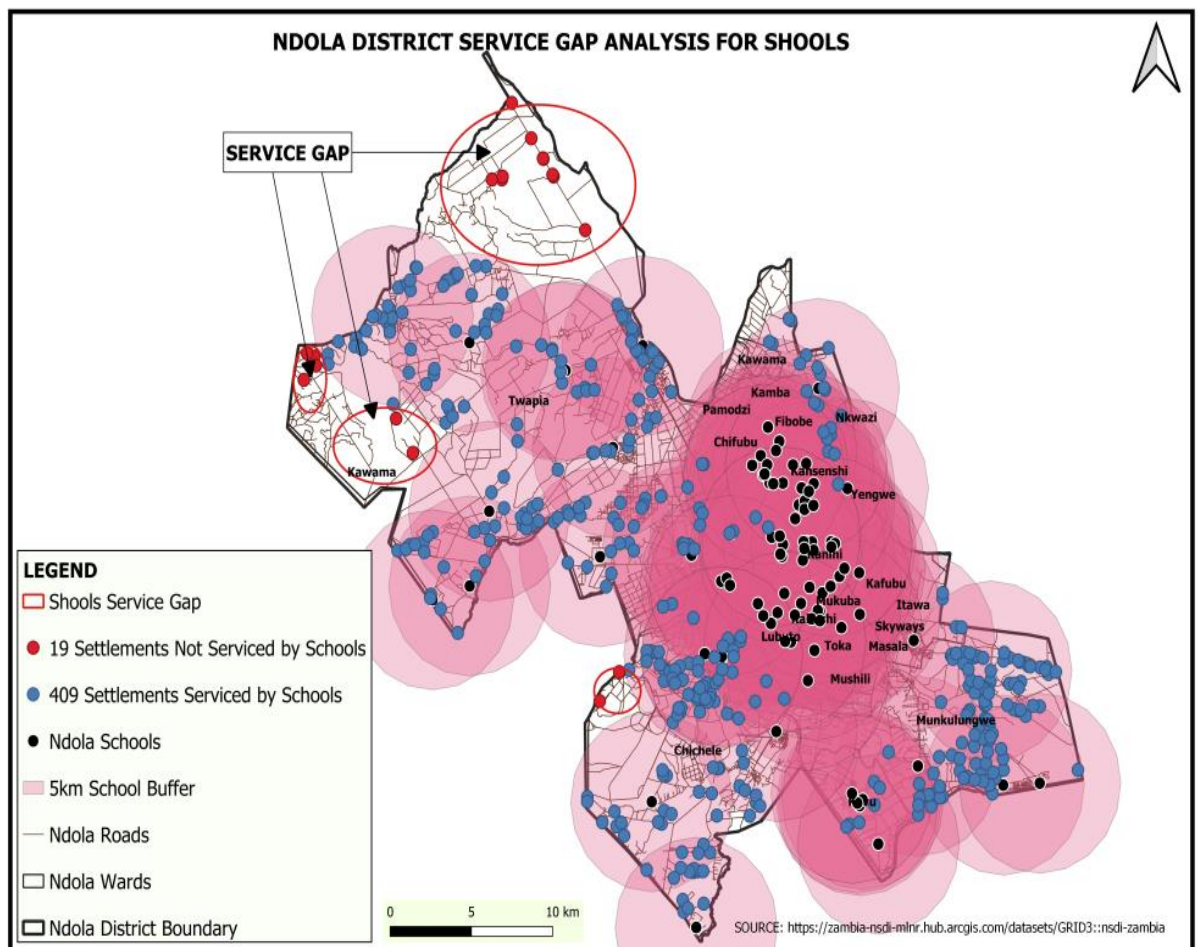
3.5.4.1. Future Demand for Services and Facilities

In the next 10 years, the district is likely to experience increased population which will require increased number of classrooms to cater for the increase in number of school going children both at primary and secondary levels. Some lands reserved for school construction is prone to encroachment by the community. There is need to put billboards to indicate that the area is designated for a school construction. If possible, such areas should be fenced.

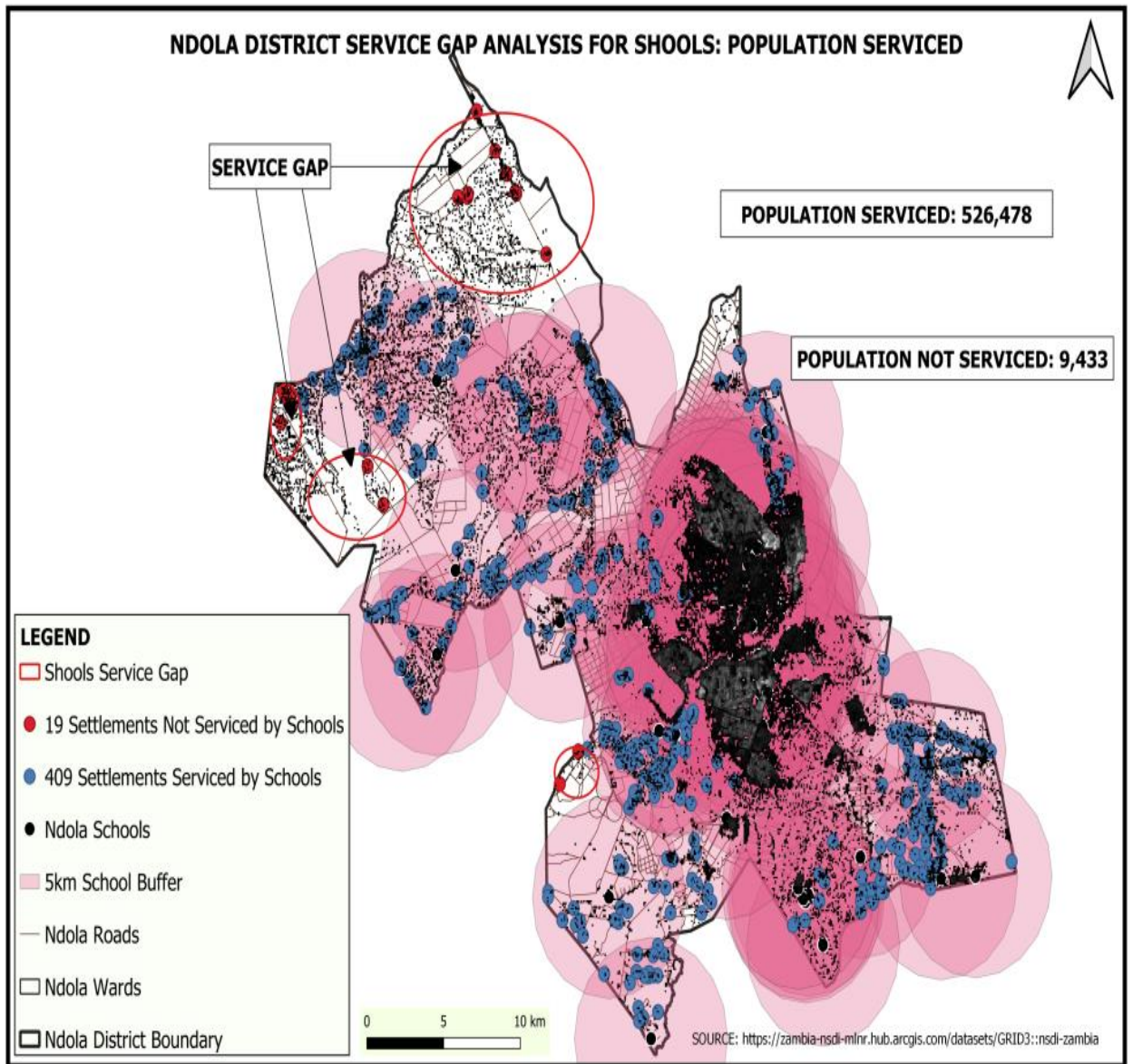
To enable the learners access tertiary education locally, more public colleges and universities should be opened or have branches in Ndola.

3.5.4.2. Impact of Existing Trends on Land use and Population Distribution Patterns

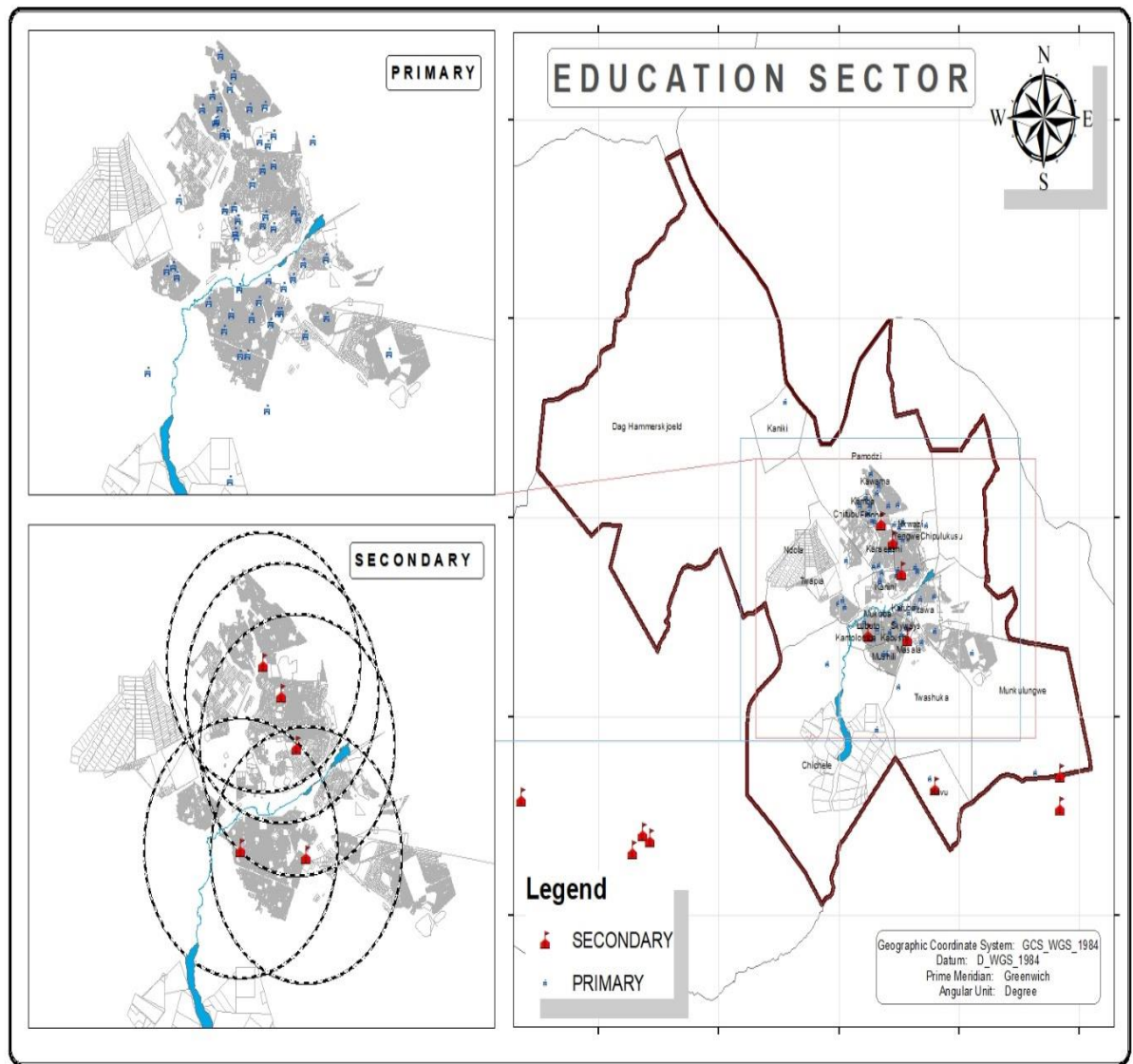
Figure 16: Schools Catchment



Source: Ndola City Council GIS Unit (2021)



Source: Ndola City Council GIS Unit (2021)



Source: Ndola City Council GIS Unit (2021)

The map above shows the distribution of public primary and secondary schools in Ndola City. The secondary schools were subjected to a 5 km radius and it indicates that currently, the secondary schools are not adequate or evenly distributed in the district. With the growing trend of land use, the existing schools will be very distant from the population expansion areas thereby making children travel long distances of more than 5 km to access education facilities. Therefore, the trend of land use and population growth patterns, clearly indicate that more schools or additional classrooms on existing schools are required.

3.5.4.3. Environment and Climate Change Analysis

3.5.4.3.1 The Impact of environmental and Climate Change Issues on the Sector

Climate change has a direct impact on education. The primary impacts of climate change on education arise from the effects of extreme weather events, such as heavy rains accompanied

by flash floods, strong winds and hail storms with short and long-term consequences. Floods cut children from attending classes because they cannot cross streams and flooded areas. Heavy rains prevent children from walking distances to school. Strong winds and hail storms destroy infrastructure as roofs get blown off and trees fall on school infrastructure. The disruptive weather patterns due to climate change set the Government back through rehabilitating destroyed infrastructure.

On the contrary, drought and increasing temperatures lead to poor harvests and food scarcity which impacts negative upon educational attainment. Extreme weather events reduce the availability of safe drinking water, compromise sanitation and increase the incidence of weather-related diseases such as malaria and diarrhoeal diseases, leading to absenteeism and possible withdrawal of children from school. This causes malnutrition among school going children, which impairs their retention and learning performance. Lack of food also increases absenteeism from school. Furthermore, children may be removed from school to support the household income generating activities. The overall result is disruption of the learning process which leads to poor educational outcomes, including higher incidences of repeats and school dropouts.

To avert heavy rains from blowing school roofs, there is need for the sector engineers to improve the architectural plans for classroom blocks so that they are resilient to heavy rains and wind storms. This will greatly lessen Government expenditure on blown-off roofs and missing of classes on the part of the children as classes are being maintained.

3.5.5. Issues Arising Rating to Gender Groups and Vulnerable Groups

Where gender issues are concerned, the education sector aims at having equal number of boys and girls accessing education. This aspect was achieved as the gender parity ratio between boys and girls is 1.04 at primary school and 1.07 at secondary school.

The issue of vulnerability is the most problematic as there are more boys and girls that are becoming vulnerable with time. Currently there are 3,240 orphans and vulnerable children (OVCs). Out of these only 1,710 are being supported by Government and donors while 1,530 are not supported.

3.5.6 Core Issues

- Inadequate classroom space for learners
- Inadequate school furniture
- Inadequate teachers in primary and secondary schools
- Long distance children travel to access education especially in peril-urban areas

- Inadequate public Universities and Colleges

3.6. Community development and social welfare

3.6.1. Review of community development and social welfare

Poverty and vulnerability in Zambia has persisted over the past years recording high rates of poverty standing at 60.5% overall, rural poverty at 80%, extreme poor at 42% (LCMS, 2010). The Zambian government whose mandate is to ensure the well-being of all Zambians has through the ministry of community development, mother and child developing and implementing social protection programs across the country.

The Ministry whose vision is to see ***a middle income nation free from constant or periodic critical levels of poverty, deprivation and extreme vulnerability by 2030*** (National social protection policy 2014) has the mandate to reduce poverty, inequality and vulnerability among all Zambian.

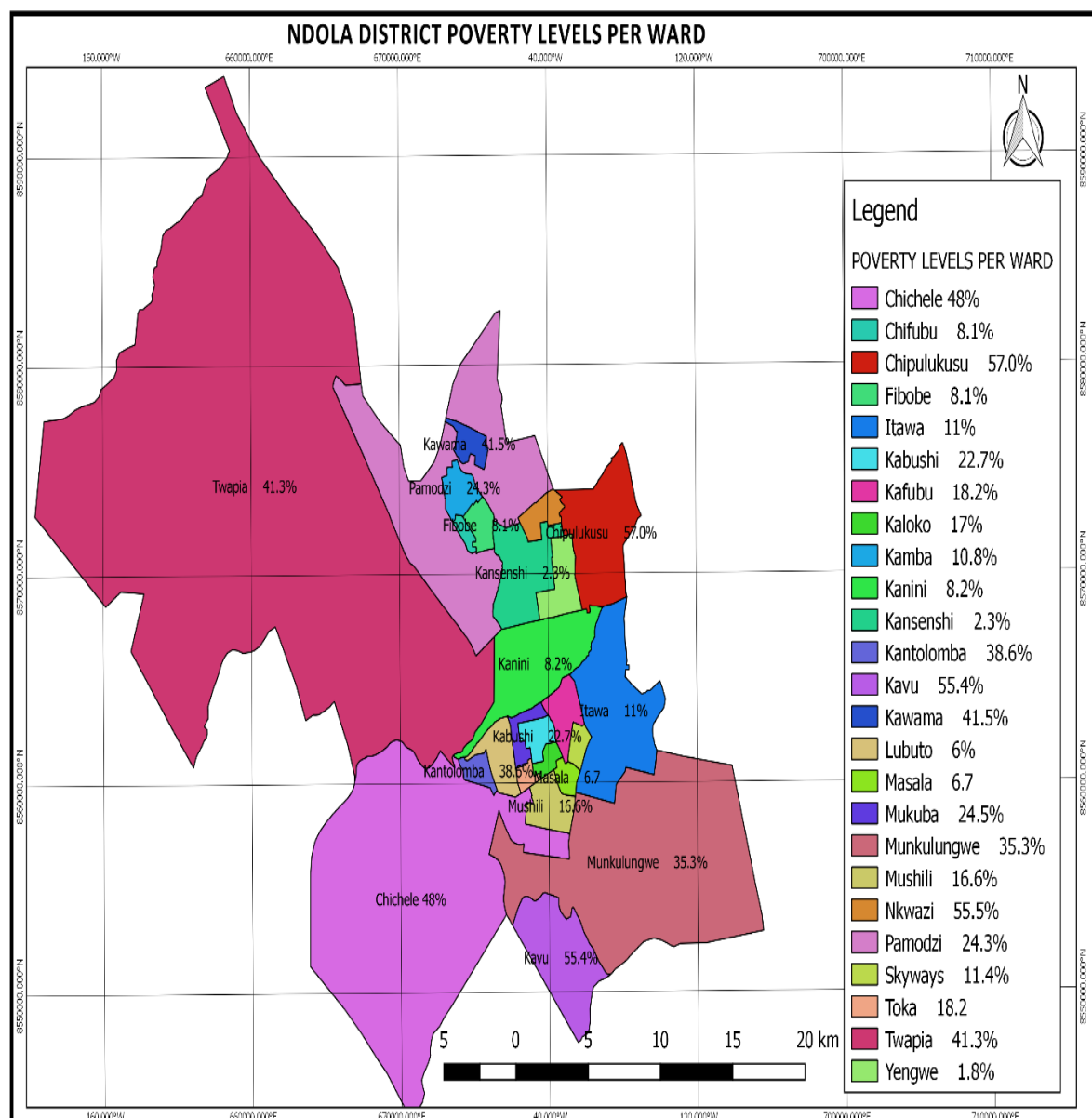
The Ndola district social welfare and community development department has been working hand in hand with partners to ensure that the national vision is realised by 2030, the department has a number of social protection programs offered to various groups of people living in poverty and who are vulnerable that will be discussed in details in subsequent chapters.

3.6.2. Existing State of Development

3.6.2.1. Availability and quality of service

The incidence of poverty levels in the district stands at about 28% of the total population. However, the areas mostly affected are the Peri-urban areas such as Dag Hammarskjold Ward whose poverty level stand at 75% and least being Yengwe situated in the urban area with 1.8% incidence of poverty levels. Most of these areas are characterized by low income levels and lack of basic services such as clean water and poor sanitation. The main economic activities in these areas are subsistence farming. Women and children are the most vulnerable. Below is a map showing the levels of poverty in Ndola District

Figure 16: Poverty per Ward



Source: Ndola City Council GIS Unit (2021)

Ndola district implements a number of social protection interventions targeting the critically poor and most vulnerable people in order to improve their welfare and livelihoods. These have taken a form of non-contributory transfers either in cash or in kind, fee waivers, and subsidies with the view of reducing poverty. These interventions specifically target people who lack the inherent capacity in meeting their needs through short coming in various aspects of their circumstances such as limited access to market, insufficient capital or land.

The current social protection programs are based on empowerment and protection, with the mandate of reducing poverty. The department of social welfare and Community Development have been implementing a number of programmes and these programmes fall under statutory and non-statutory purpose. Statutory mandates protect and promote the rights and welfare of children while non-statutory mandates promote social assistant, social security, and empowerment that include cross cutting livelihood strategies.

Statutory programmes involve the administration of pieces of legislation especially with regard to the protection and welfare of children. These include:

- Probation of Offenders Act, CAP 93,
- Juveniles Act, CAP 53,
- Adoption Act, CAP 54,
- Affiliation and Maintenance Orders Act CAP 64.
- Anti-Gender Based Violence Act No.1 of 2011
- Anti- Human Trafficking Act No. 11 of 2008
- Child Policy (MYSCD)
- National Social Protection Policy

Children's Homes

The institutional care in collaboration with NGOs provide shelter to children in need of care, orphaned, abandoned or whose parents or guardians are unfit to provide them with proper care and guardianship. The Implementation is in accordance with the Juveniles Act CAP 53 of the Laws of Zambia through the provision of Committal Orders in order to maintain the Minimum Standards of Care (MSC) for Child Care Institutions

SEEVCA

The Service Efficiency and Effectiveness for Vulnerable Children and Adolescents (SEEVCA) PROJECT has been introduced in the district. The District has been selected as one of the piloting districts for the project in the district. The project is targeting families under the Social Cash Transfer Programme that have Vulnerable Children and adolescents. The project focuses on enhancing service provision to vulnerable children through the coordination and networking among stakeholders. The Project was introduced to DDCC members, Councillors and other stakeholders in the district. And randomization of wards was conducted with stakeholders and eight piloting wards were selected. Kaniki, Mukuba, Kafubu, Chipulukusu, Kawama Kavu, Lubuto and Darghammaskjot were selected.

Social assistance interventions undertaken by both the department of community development and department of social welfare target specific categories of beneficiaries that meet specific qualifiers. Below are different qualifiers considered during selection of beneficiaries;

- FHH - Female Headed Households
- HKO - Household Keeping Orphans
- DHH - Disabled Headed Households
- UY - Unemployed Youth
- AHH - Aged Headed Household
- TIHH - Terminally Ill Headed Household
- CHH - Child Headed Household
- PWAS - Public Welfare Assistance Scheme

Table 10: Social Protection Program and Beneficiaries

S/NO.	PROGRAM	WARD	F	M	TOTAL
1	FSP	Kavu			50
		Chichele			100
		Chipulukusu			110
		Kaniki			0
		Twashuka			30
		Munkulungwe			0
		Kaloko			20
		Total			310
2	Village Banking	Kawama	25	0	25
		Kavu	30	0	30
		Chifubu	40	0	40
		Chipulukusu			0
		Nkwazi			0
		Kanseshi			0
		Kaniki			0
		Masala			0
		Pamodzi			0
		Kaniki			0
		Total			95
3	Women and Development (64 clubs)	All wards			1224
		Total			1224
4	Adult Literacy	Kawama	28	0	28
		Kaniki	50	0	50
		Chichele	58	2	60
		Kavu	14	7	21
		Chipulukusu	72	04	76
		Nkwazi	17	0	17
5	Service Efficiency and Effectiveness for Vulnerable Children and Adolescents (SEEVCA)	Kavu	279	303	582
		Lubuto	36	30	66
		Kafubu	102	83	185
		Mukuba	178	136	314
		Kaniki	89	52	141
		Chipulukusu	111	128	239
		Kawama	286	275	561
		Dag	111	89	200
		Kaloko	32	20	52
6	Social Cash Transfer	Kawama			535
		Pamodzi			657

	Mushili	569
	Kabushi	422
	Lubuto	423
	Kaloko	232
	Tugargun	278
	Ndeke	572
	Kavu	511
	Twapia	434
	Kafubu	289
	Kansenshi	29
	Chipulukusu	893
	Chifubu	674
	Nkwazi	676
	Kaniki	422
	Masala	581
	Total	7629

Source: District Social Warefare Data (2021)

Social Cash Current Caseload Profile

The current caseload is at 7,629. Below is the breakdown of the current caseload for Ndola district disaggregated by gender and disability.

Table 11: Summary of Social Cash Current Caseload Profile

DISTRICT	TYPE OF MODEL	MALE BODIED	FEMALE BODIED	MALE DISABLED	FEMALE DISABLE	TOTAL
Ndola	2014 Model	0	0	703	744	1447
	2017 Inclusive model	1619	3826	295	442	6182
TOTAL		1619	3826	998	1186	7629

Source: District Social Warefare Data (2021)

3.6.3. Issues arising from public participation

3.6.3.1. Available service

Following the public consultation that was undertaken by the local authority and partners below are the services available according to the public brought up;

- Social cash transfer
- School support for vulnerable children
- Women Empowerment
- Food security pack
- Social safety nets for the vulnerable and disabled

3.6.3.2. Quality of service

Table 12: Showing quality of social services according to the public

S/N	Service	Expected (%)	Achieved (%)	Variance (%)
1	Social cash transfer	100	89	11
2	School support for vulnerable children	100	89	11
3	Food security pack	100	52	48
4	Social safety nets for the vulnerable and disabled	100	54	46
5	Women Empowerment	100	50	50

Source: District Social Welfare Data (2021)

3.6.4. Impact of Changes Anticipated over the Next Ten Years

Ndola district is experiencing a large demographic shift; the estimated population as of 2019 was standing at 574,437 with the growth rate of 1.9%. The projected population 760,000, for the social services this means more demand on the existing social protection programs being offered. Currently the country has been hit by the Covid-19 pandemic which has claimed many lives; the case of Ndola is not very different from other parts of the country. The high death rate that is being experienced has left a number of vulnerable children and families as most of the breadwinners are no more, with the high number of vulnerability that the pandemic has caused the ministry of community development mother and child health care will need more funding to increase the number of beneficiaries on the existing social protection programs and also come up with other social protection programs that will meet the basic needs of the poor of the district.

3.6.5. Summary

The social service is one sector that is mandated to provide social protection for the poor and vulnerable of the district. The government provides these programs through the ministry of community development, mother and child health care. The programs at district level offers a number protection programs that include social cash transfer for the old and other vulnerable groups, SEEVCA for the vulnerable children and adolescents and many other programs. However, the public is of the view that the government through the responsible ministry can do more to help the poor of the district.

3.6.6. Underlying factors contributing to the issues identified

- Lack of adequate funding

Core problems

- Inadequate Women Empowerment
- Inadequate School support programs for vulnerable children
- Inadequate Food security pack program
- Limited Social safety nets for the vulnerable and disabled
- Limited Women Empowerment programs

3.7. Housing

International human rights law recognizes everyone's right to an adequate standard of living, including adequate housing (UN Habitat, right to adequate housing, fact sheet 21). All Ndola residents are entitled to decent housing as this has a direct effect on the community and its local economy. Over the years Ndola district has recorded an increase in its population just like any other city in Zambia. People in the rural districts have rapidly been shifting to urban areas in search of jobs and better living standards. This increase in population growth in urban areas like Ndola has given a rise in demand for decent housing which the Local authority is stressed and lack capacity to meet.

Ndola has two forms of settlements been formal and informal settlements. The increase in population has forced majority of its residents to settle in unplanned settlements translating to 60% and 40% of the population in planned settlements.

Housing as an Economic Contributor

The housing sector has a direct relationship with economic growth. Investment in this sector has a positive impact to the local economy through job creation and increase in individual disposable income as can be evidenced after the liberalizing of the economy in the second regime of government in 1991 which encouraged the private sector and individual owners to build houses to fill in the housing deficit.

This document will look into providing a strategic mechanism in which the projected housing deficit of 3.3 million for the country and the deficit of Ndola district can be managed in the face of a growing population.

3.7.1 Review of housing Policies and Plans

Vision 2030: In the Vision 2030, Zambians aspire for access for all to good quality basic human necessities such as shelter, titled land, health and education facilities and clothing. The vision therefore seeks to have planned resettlement with adequate, affordable and quality housing by 2030. One of its goals is to increase the number of people accessing planned urban and rural settlement to the population of 75 percent by 2030.

The vision acknowledges that Zambia is faced with a critical shortage of housing and an enormous housing backlog dating back to the time before independence, mainly due to lack of an effective system of housing finance, inadequate land titling, rapid population growth and rural-urban migration.

In this document, Ndola district therefore, aligns its plan and strategies to that of the Vision 2030 in the area of providing adequate decent housing for all.

The Eighth National Development Plan: Human and Social Development is one of Strategic Development Areas of the 8NDP. In an effort to address the need to develop human capacities, the 8NDP seeks to increase access to decent and affordable housing for all during the plan period. The government will facilitate the provision of affordable housing finance, provide incentives for the private sector participation and promote investments in research on alternative building materials and technologies. Consequently, during the Plan period, the housing deficit is expected to reduce from 1,539,000 to 1,378,000 housing units. The district aims to provide adequate housing that meets the growing demand and population.

National Housing Policy; The National Housing Policy seeks to provide decent housing to the growing population in Zambia and it recognises collaboration between government, cooperating partners, private sector, traditional authorities and Non-Governmental Organisations to find solutions of the current housing deficit estimated at 1,539,000.

The policy has the following goals:

- Promote sustainable human settlement
- Manage urbanisation
- Facilitate development of decent and affordable housing
- Promote equitable access to land for housing development
- Facilitate access to housing finance
- Increase the capacities of the Local Authorities
- Mainstream crosscutting issues in housing development

Projects

The district has ten unplanned settlements namely Chipulukusu, Twapia, George, Mackenzi, Kaloko, Kantolomba, Kavukang'onga, Sakanya, Nkwanzu, Mwenye which are recognised by the local authority. Service provision in these areas is a challenge as most of these areas are not serviced making it difficult for service providers to deliver their services. In addition, settlers lack legal occupancy documents which can provide security to protect their properties.

Table 13: List of Projects

SN	Projects	Activities	Area	Partners	Status
1	Social tenure domain model meant to support acquisition of occupancy licenses for informal settlements	• sensitization-community meetings • registration • application of occupancy licences • coordination of stakeholders	All 10 unplanned settlements	• UN Habitants • Habitants for Humanity	To be implemented
2	support acquisition of occupancy licenses for	• sensitization-community meetings • registration	All 10 unplanned settlements	Medeem	To be implemented

	informal settlements	• application of occupancy licences • coordination of stakeholders			
3	Newly planned formal settlements	• Planning for high cost, medium and low cost • Provision of service roads • Provision of social amenities such as hospitals, multi-purpose facility zones	Dola Hill Chichele Pine grove Kafubu East Dam	Community	Been implemented
	Heinan Gouge Housing project	• Construct 500 housing units	Dola Hill	Heinan Gouge developers ZAFFICO	Been implemented

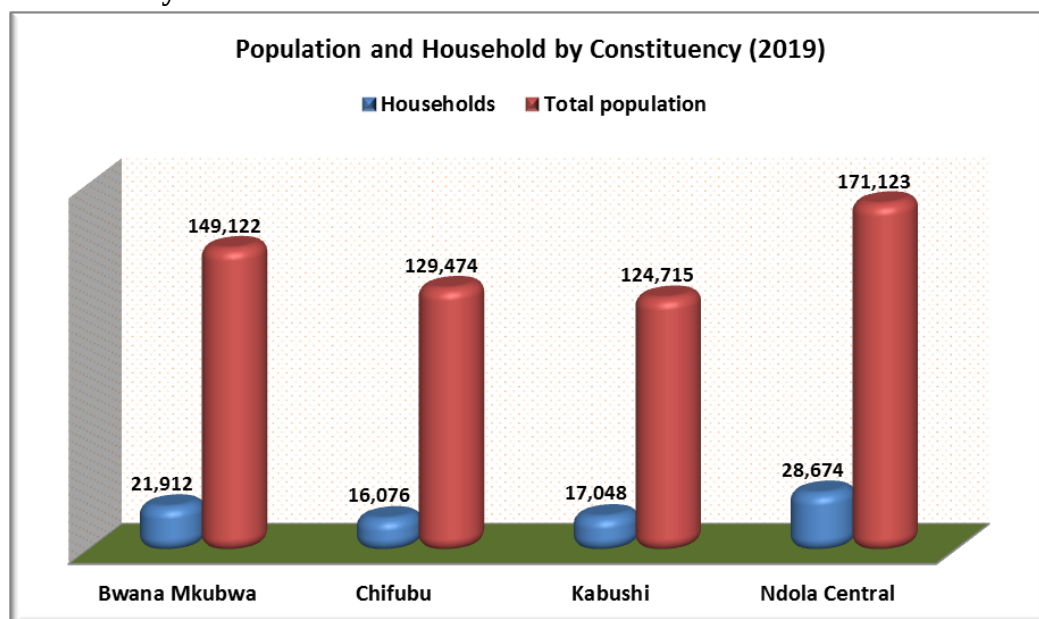
Source: Ndola City Council Department of Housing and Social Services / City Planning (2021)

3.7.2. Existing State of Development

3.7.2.1. Availability of Service

Ndola district is on a rapid rise in development of housing units in an attempt to meet the growing population and demand. The district has a housing stock of 85,707 comprising of both formal and informal settlements. Ndola central constituency stands out to have the majority share of housing units standing at 28,674 with Chifubu constituency having the least number of households at 16,073.

Figure 17: Below is a breakdown of houses against the population in each constituency.



Source: CSO Data (2019)

New formal settlements in Hillcrest extension, Mitengo, Dola Hill are developing improved and modern housing units. The high cost of construction materials has influenced the price of rentals in these areas to be on the higher side making it difficult for low and middle income earners to occupy these houses.

Housing units in informal settlements are in a poor state as these units are in unplanned areas and the materials used in construction of these houses are not durable and not resilient to withstand climatic challenges. Majority of the settlers in the informal settlements do not have legal occupancy documents that recognise them as formal owners of their property and disadvantages them from accessing financial services such as loans from lending institutions, access other services such as connection of water and electricity.

Figure18: Formal Settlements

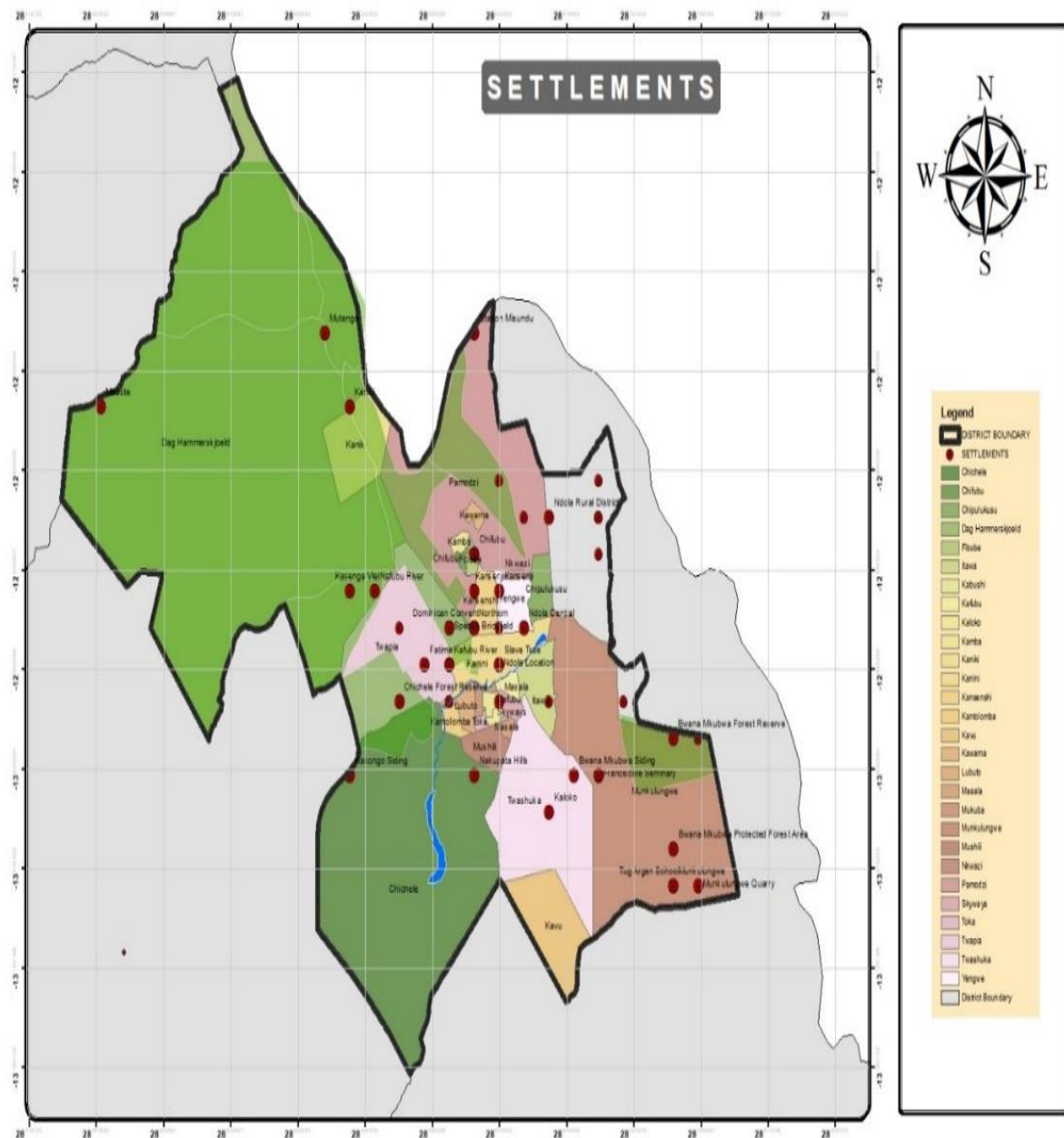
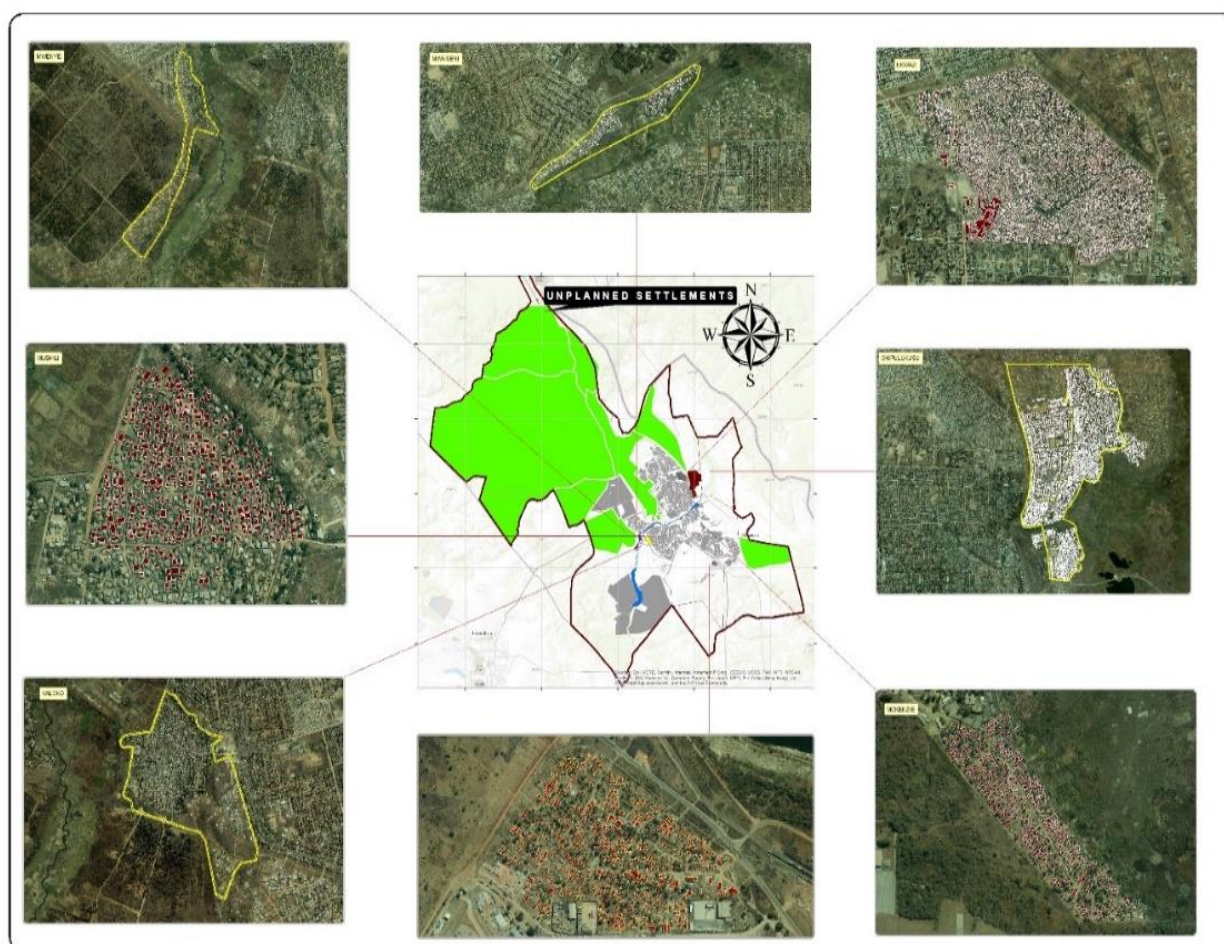


Figure 19: Unplanned Settlements



Source: Ndola City Council GIS Unit (2020)

Table: Households in Informal Settlements

N0.	Name	Area Size MT Sq.	Perimeter	No. of Household
1	Chipulukusu	2023335.72	9726	8434
2	Kwanzi -	1050852.65	4240.874	4390
4	Kantolomba	522345.14	3730.19	995
5	George -	56961	3000	405
6	Mwaiseni Kayala	279825.983	3466.1	321
7	Mwenye	349581.63	4640.33	668
8	Mackenzie	150362.845	2133.8	451
9	Old Regiment	332007	2388.865	598

Total	5308001.968	33326.159	16262
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Source: Ndola City Council Department of Housing and Social Services / City Planning (2021)

3.7.2.2. Quality of Service

Ndola district is faced with the rise in population growth and expansion of illegal settlements like any other urban district. Therefore to arrest the current trend there is need to formally upgrade the already existing settlements. The standard of services such as water, roads, electricity, streetlights, health and security is poor in these areas leading to hardships and a high cost of living.

The district needs to implement programs that upgrades illegal settlements to attain a status where service provision is easily delivered to the communities. The number of applicants for occupancy licenses and number of occupancy licences issued are some of the indicators used to analyse the development of illegal settlements.

3.7.3. Issues Arising from the Public Participation Process

3.7.3.1. Availability of Service

The weakening of the currency and the high inflation have led to majority Zambians who are low income earners endure economic hardships as the cost of various goods and services have gone up. Most community members do not have capacity to develop their own houses and as well cannot afford to pay rentals for the houses available on the market.

This shortage for decent accommodation for low cost houses contributes to the expansion and creation of informal settlements.

Most financial and lending institutions do not have provisions for low income earners to access loans and other financial services for the poor to construct houses. This challenge disadvantages the majority Zambians to have an opportunity to develop their own houses and invest in other businesses.

The local authority is mandated to provide services for issuance of occupancy licences and change of ownership to the residents of its locality. It was noted however, that duration taken to process these documents after application takes too long which causes a lot of problems especially when it comes to change of ownership documents.

3.7.4. Impact of Changes Anticipated Over the Next Ten Years

Population Change in Ndola has steadily grown from 451,246 in 2010 to 574, 437 in 2019 at a growth rate of 1.9% and it is projected that the population will hit a figure of 760,000 by the year 2031. The district faces inadequate capacity to respond to population growth which reflects the entire country's situation. Zambia has an estimated current national housing deficit of 1,539,000 units and projects a shortage of 3.3 million by 2030.

The district needs to bridge the gap in housing units through engagement of conventional construction, private developers and the local authority to collectively work together.

Below is a table depicting a projection of the housing needed in Ndola for the next ten years.

	2010	2019	2025	2031
Population	451,246	574,437	620,000	760,000
Expected No. of households	90,249	114,887	124,000	152,000
Existing No. of households	85,707	109,105		
Deficit				

Source: ZamStat (2020)

The table above shows a rise in population growth and the same can be said on development of housing units which has seen a rise from 85,707 as of 2010 to an estimated 109,105 in 2019. The current trend in housing market indicates deficits and a rise in demand in the future for services such as water connection, electricity and other required services in development of houses.

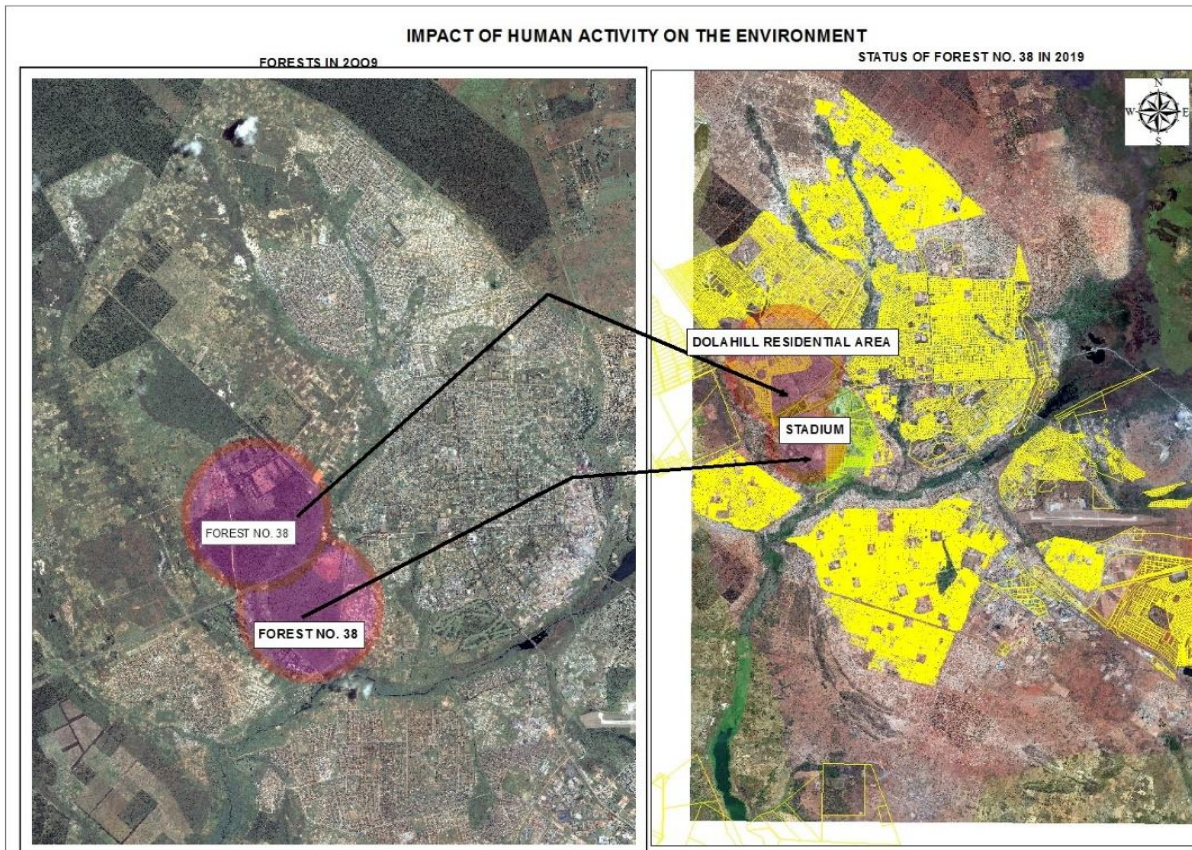
3.7.4.1. Existing and Proposed Investment and Development Programmes

The district has embarked on planning of new settlements that which will include land uses such as residential plots to cater for high cost, medium cost and low cost developers to build decent housing units. This project will enable for organised development of the planned settlement areas and effectively see a rise in the number of decent housing units.

Support to acquisition of occupancy licenses for informal settlements programmes will enable upgrading of illegal settlements and will enable delivery of adequate water points such as kiosks and sanitation, upgrading of roads, street lights and drainages, provide waste management programs such as garbage collection.

Conventional housing projects like Henan Gorge Housing project will have a positive impact on the provision of decent houses in Ndola. It is therefore the plan for the district to identify financing partners for similar projects to come on board and beef up the already existing stocks to meet the unsaturated market.

Impact of Existing Trends on Land Use and Population Distribution Patterns



Source: Ndola City Council GIS Unit (2020)

The anticipated demand for housing will however result into negative practices and outcomes caused by human behaviour such as excessive cutting of forests, encroachment in public and private lands such as ecological areas. These practises will have adverse effects on the environment such as land degradation, air pollution and contamination of both surface and underground water caused by improper disposal of sewage and pit latrines. In addition, there is a likelihood of an increase and expansion of informal settlements across the district.

3.7.5. Environment and Climate Change Analysis

The nature of provision of decent accommodation demands to sit on land and with the rapid construction the country has witnessed in the recent years has come at an expense of converting more land for settlement. Forest reserves total land coverage in hectares amounts to 39.3% in Ndola district. The significance of trees in the eco-system cannot be over emphasized as they play a critical role in the regulation of carbon in the atmosphere and rain precipitation. The existing trends projects that more land will be changed in use for the purpose of settlement by 2031 leading to deforestation.

However, with the rise in population the district is experiencing land invasions in some of these reserves. This trend of cutting down trees will lead to change in climatic conditions resulting to excessive heat, delayed or poor rain patterns, long dry seasons.

Environmental degradation will be on the rise as more settlements are been developed. This usually is expected to occur more in informal settlements as a result of overcrowding and improper use of land such as construction of pit latrines near water bodies and poor solid waste management will lead to contamination of water. Additionally, scramble for land for

agricultural purposes will lead to over cultivation on the same pieces land making them loose soil fertility. This will lead to low levels of production of agricultural produce and intern lead to shortage of food and low household income.

The unsatisfied market for houses is attracting a lot of investors to invest in construction of housing units. This has a multiplier effect in intriguing companies that supply construction material such as cement, steel bars, and roofing sheets to produce more. Majority of these producers are high emitters of carbon dioxide and that has negative effects on the environment.

3.7.5.1. The Impact of Environmental and Climate Change Issues on the Sector

The human actions over the years have led to global climatic change conditions that are having adverse effects on humanity. Zambia and Ndola district to be precise is not left out of this global challenge. The average temperature recorded for the Copperbelt province has risen over the years and that means long days and nights and months of high temperature and heat. Majority of the houses in Ndola are not built to withstand such intense heat and therefore end up expanding causing cracks on walls making those houses unsafe to stay in. This then means that more resilient construction material has to be produced but it comes at high purchasing costs and pollution.

Unplanned settlements in most cases are affected by floods because settlers settle in areas that are prone to floods. Property damage as a result of poor infrastructure that cannot withstand floods collapse. Outbreaks of diseases or epidemics such as cholera emanate from floods and additionally, households are affected by inadequate water supply, food shortage during such disasters caused by climate change.

3.7.6. Issues Arising Relating to Gender Groups and Vulnerable Groups

Our societies have for a long time considered men to be natural leaders and have assigned women with tasks of taking care of homes and children. This has disadvantaged women when it comes to ownership of property such as land and houses. Male dominance to ownership of houses is a social issue that needs to be addressed and create an environment that enables and promotes women to equally own land and houses.

More policies surrounding women empowerment of land and property need to be provided to bridge this inequality gap between women and their male counterparts. In addition, the youth and the disabled need supporting policies that equip them with equal opportunities to own land and houses.

3.7.7. Summary

Every Zambian is entitled to decent housing as housing is listed amongst the global basic human rights. The current standings of housing stock indicate that the country has an estimated deficit of 1,539,000 of which 40% is urban and 60% rural. Ndola district is not spared to these crises and therefore at local level there is need to ensure that adequate housing is provided for its residents.

The expected increase in population from 547,437 at a growth rate of 1.9% to 760,000 in the year 2031 has a direct relationship on infrastructure development and housing to be particular. The district will need to match the growing population by planning and constructing more

housing units. Government, the local authority and other stakeholder need to engage into construction of housing through conventional agreements.

Human behaviour has over the years led to change in climate caused by pollution and cutting down of trees amongst other environmental degrading factors. This change in climatic conditions has negative effects on housing as most houses are constructed with material that is not climate resilient.

Policies that support gender inclusion and equality to ownership of land and houses is needed to be formulated both at local and central levels of government to bridge the social inequality gap to ownership of property between men and women. Other minority groups such as the youth and disabled need similar support to allow them participate in provision of this service.

The IDP will enhance the implementation, monitoring and evaluation of proper construction of decent housing units for the Ndola district because it informs and prepares the district to focus on strategies that will beef up the current housing stakes for the growing population and as a result attain the vision 2030 that also requires that the poor and vulnerable are safeguarded through reducing poverty, inequality and vulnerability.

3.7.8. Factors Contributing To the Issues Identified

- Less control as pieces of land not allocated by the authorities
- Rapid population growth leading to expansion and development of new illegal settlements
- Land encroachments in reserved areas causing deforestation
- Inadequate policies that empower minority groups to own property
- Unavailability of integrated development plan
- High cost of construction material

3.7.9 Core Issues

- Inadequate housing stock
- Increase in population
- construction of more decent housing units
- underperforming economy and high cost of construction
- deforestation and environmental degradation

3.8. Forestry sector

Forests are among the nation's most extensive natural heritage resources which require sustainable management. They are renewable assets and continuous availability depends on anthropogenic actions.

Forestry Department is mandated to ensure continue flow of wood and non-wood forest products and sometime ensures environmental protection and sustainable utilisation of forest resources for the benefit of the present and future generations through active participation of stakeholders.

Ndola District continued to have a total 49,088ha of forest reserves. Over 19,000ha is covered by ZAFFICO pine and eucalyptus plantations. The remaining indigenous tree cover is mainly characterized by encroachments and squatting.



Name of forest reserve	National forest ha	Local forest ha	Status
Misaka National Forest no. 5	24,150		Partly covered by ZAFFICO plantation and the remaining area is characterized by encroachment and squatting. 1500 near Kanfinsa and 700 near Kanike area are earmarked for excision.
Ndola-National Forest no. 38	2,410		Partly covered with ZAFFICO plantation and the remaining area is characterized by illegal residential plot demarcation, cultivation and squatting. The portion opposite Mitengo cemetery is under rehabilitation by allowing natural regeneration. 500ha is excised for the stadium and to get rid of Dola hill forest camp of which the housing units were sold to the sitting tenant.

Ndola-West National Forest No. 54	2,140		Partly covered with ZAFFICO plantation and the remaining areas are characterized with encroachment and squatting.
Ndola-East National Forest No. 192	2,367		This is mainly covered by ZAFFICO plantation.
Chichele national forest no. 89	2,560		Partly covered by ZAFFICO plantation and the remaining areas are characterized with illegal cultivation and squatting. 810ha has been here marked for excision.
Mwekera-National Forest no. F84	2,072		Partly covered with ZAFFICO plantation and the remaining areas are characterized by illegal cultivation and squatting. Mwekera stream source area is under rehabilitation by allowing natural regeneration and through enrichment planting.
Mwekera-National Forest no. 6	6,884		Partly covered by ZAFFICO plantation and the remaining areas are characterized by illegal cultivation and squatting.
Bwana Mkubwa Local Forest no. 53		1,704	Depleted and it's earmarked for degazation.

Source: Ndola District Forestry Department (2020)

3.8.1. Reviewed Forestry Policies and Plans

National Forestry Policy; The aim of the 2014 National Forestry Policy is to attain sustainable forest management of types of forests to enhance forest products and services that will contribute to mitigation of climate change, income generation, poverty reduction, job creation, protection and maintenance of biodiversity.

The policy is part of the government reform programme aimed at changing the centralist approach to forest management through broad participation of stakeholders in order to deal with the drivers of deforestation and forest degradation.

The Forests Act, 2015; The Forests ACT No. 4 of 2015 provide for the

- Establishment and declaration of National Forests, Local Forest, Joint Forest management areas botanical reserves, private forests and community forests through participation of the local communities, local authorities, traditional institutions, non-governmental organisations and other stakeholders in sustainable forest management
- Conservation and land use of forest and trees for sustainable ecosystem management and biodiversity; establish the Forest Development Fund
- Implementations of the united Nations Framework conservation on climate change, convention on international Trade in Endangered species of wild flora and fauna, the convention on wetlands of international importance, especially as water fowls habitat, the convention on biological diversity, convention to combat desertification in those

countries serious drought or desertification, particularly in Africa and any other relevant international agreement to which Zambia is a party.

National Strategy to Reduce Deforestation and Forest Degradation; Zambia has an estimated deforestation rate ranging from 250,000 to 300,000 hectares per annum. The identified drivers of deforestation are wood fuel, agricultural expansion, mining, timber extraction, bush fires and infrastructure development.

Zambia has developed Reducing Emission from Deforestation and Forest Degradation (REDD+) strategy focusing on dealing with drivers of deforestation in both forestry and identified key sectors such as agriculture, energy, mining and infrastructure. Thus, the vision of this strategy is to contribute to a prosperous climate change resilient economy by 2030 depended upon sustainable management and utilisation of nation's natural resources aiming at improved livelihood. The strategy is guided by seven core principles: effectiveness, efficiency, fairness, transparency, accountability, inclusiveness and sustainability.

The ten strategic objectives of the National Strategy to reduce Emissions from Deforestation and forest Degradation are:

- i. By 2030 threatened and unsustainably managed protected areas are effectively managed and protected to reduce emissions from deforestation and forest degradation and contribute to ecosystem services across selected landscape
- ii. By 2030 forests in open areas are effectively managed and monitored
- iii. By 2030 all timber concessions areas have management plans that are enforced and monitored with full participation of local communities
- iv. By 2030 good agricultural practice that mitigate carbon emission are adopted
- v. By 2030 regulated production of wood fuel (charcoal and fuel) and its utilisation is in place
- vi. By 2020 appropriate alternative energy sources are widely adopted
- vii. By 2020 threatened and ecologically sensitive areas are legislated against extractive and infrastructure development
- viii. By 2025 mining industry contribute to management of the surrounding indigenous forest and establishment of forest plantations for own timber needs
- ix. By 2025 integrated land-use planning is in place
- x. By 2030 relevant institutions are capacitated to enable them plan, manage, implement and monitor REDD+ initiatives

National Investment Plan to Reduce Deforestation and Forest Degradation (2018-2022); Zambia's investment plan reduce deforestation and forest degradation is aimed at reducing rural poverty and improving livelihoods. Therefore, implementation of the investment plan focuses on tackling different drivers of deforestation in forestry, mining, agriculture and land use through a landscape approach at watershed level and through policy reforms at national level. This approach will ensure that best possible balance is achieved among a range of different development objectives including climate change mitigation adoption, environmental and biodiversity conservation, improved economic productivity and livelihoods.

To actualise this Investment plan will require mobilisation of funds (public and private, multilateral and bilateral).

Keep Zambia Green, Clean and Health Campaigns; Following the pronouncement made on keep Zambia Green, Clean and Healthy Campaigns and plant 10 000 000 plants project which was launched in March 2018 by His Excellence the President of the Republic of Zambia, Forestry Department intends to raise and manage 300 000 assorted seedlings at Chichele through stakeholder participation for 2021/22 planting season. The seedlings will be sold at minimal fee and others will be donated in order to actualise the two projects mentioned above. Production of these seedlings will create jobs to the local communities.

3.8.2. Existing State of Development

3.8.2.1. Availability of service

Forestry Department offers service to the general public in line with Departmental Key Result Areas ranging from Forest Protection and Management, Forestry Extension, Rehabilitation of Degraded Areas, Plantation Establishment and Management to Capacity Building. The consumption of tangible forest products/services such as industrial round wood, fuel wood (firewood and charcoal) and non-forest products is ever increasing posing a threat to forests.

3.8.2.1. Quality of Service

The department is mandated to offer quality services in the district in order to promote sustainable forest management and utilisation by carrying out environmental awareness campaigns in schools and communities, establishing and managing exotic timber plantations on private land to satisfy the demand for timber and training communities in modern nursery management and beekeeping as outlined in Table 1 below.

Table: description of quality services offered, key activities and indicators

Services offered	COMPONENTS	Key activities	Indicators
3.2.2.1 Forest Protection And Management -	Sustainable management of forest in protected areas(forest reserves, national parks etc), open areas(customary lands) and watersheds	Promote participatory approaches to local forest management in protected and open areas through community forest management(JFM) and private forest management (PFM)	• Number of protected and open area under management • Number of forest management plans developed • Number of communities involved in participatory forest management

		Promote sustainable management of forests adjacent to mining areas by mining companies including support to community forest management initiatives	• Area (ha) managed for natural regeneration • Area (ha) conserved • Size (ha) of forests managed by mining firms
3.2.2.2 FORESTRY EXTENSION	Offer school and farm forestry extension during commemoration of International Day of Forests in march every year	Promote planting of trees in school and on farm land and sustainable management of woodlots through awareness and sensitisation meetings	• No of trees planted • Area (ha) planted • No of schools or individual farms/ institution participated • No of sensitisation meetings held
	Establish nurseries for raising assorted seedlings	Promote raising of assorted seedlings in schools, community areas in order to encourage planting of trees	• No. of seedlings raised in the nursery • No of community nurseries established
3.2.2.3 Plantation Establishment and Management	Establish and manage exotic plantations on state and customary lands	Promote establishing and managing plantations on state land and private land in to encourage Keep Zambia Green, clean and Healthy campaigns	• No of plantations established and managed • No of trees planted • Total number of hectares covered during tree planting
	Raise and manage exotic seedlings in the nursery	Promote establishing and managing exotic timber plantations of pine and Eucalyptus	• No of exotic seedling raised in the nursery • No of seedlings planted out in the field • No of hectares planted
3.2.2.4 Capacity Building	Train communities in nursery management	Promote training in modern nursery management	• No of local communities trained • No of participants trained • No of assorted seedlings raised
	Train farmers in modern beekeeping	Promote training in modern Bee Keeping in order to conserve forests	• No of farmers/communities trained • No of frame hives occupied by bees • Quantity and quality of honey (Kg) harvested

Source: Ndola District Forestry Department (2020)

3.8.3. Issues Arising from the Public Participation Process

The purpose of allowing public participation was to encourage the public to have a strong voice in decision making and processes so that their input is taken on board in trying to find solutions

on the impact of climate change and forest degradation so that there is continuous flow of the available services.

3.8.3.1. Availability of service

It was evident from consultative meeting that 90% of population in Ndola District depend on wood fuel (Charcoal and firewood) as source of energy for cooking and timber for construction and carpentry industries which is harvested from pine/eucalyptus plantations and natural forests. In most cases these services or products are accessed using unsustainable methods of harvesting forest resources.

3.8.3.2. Quality of service

A lot of fuel wood especially charcoal is coming from the neighbouring districts hence increasing the consumption as a result of electricity load shading. Nearly all households in the district can afford to buy charcoal because of its wide range of packaging standards 2kg, 5kg, 10kg, 25kg and 50kg just to mention a few. Firewood is easily accessed from the bush by local community after walking long distance due to deforestation. Fire wood is measured in cords (**1mx1mx3m stack of wood fuel**) or Head loads. The demand for Softwood and hard wood timber is high in construction, carpentry and tourism (curios) industries. The consumption non-wood forest product such mushrooms, caterpillars and other edible fruits is on the increase.

3.8.4. Impact of Changes Anticipated Over the Next Ten Year

3.8.4.1. Future demand for services and facilities

The demand for forest services and facilities is expected to increase by over 100% due to population explosion in the next ten (10) years. It is therefore estimated by Zambia Central Statistical Office that the population in the district shall increase from 574 437 in 2019 to 780, 000 in the year 2031 which will have negative impact on the vegetation as the consumption of wood, non-wood forest products /services and facilities will be increased.

3.8.4.2. Existing and proposed Investment and Development

Plantation Establishment and Management

Afforestation and reforestation projects are carried out in all parts of the district by Government (Forest Department and ZAFFICO Plc) and the private sector in order to satisfy the demand for timber. Forestry Department is managing 95ha of pine plantations while ZAFFICO is managing over 1000ha of pine and Eucalyptus plantations in Ndola District. These projects are improving living standards of the local communities through job creation targeting 1600 people.

Production of seedlings at Chichele Forestry Nursery

Forestry Department with support from cooperating partners brought Chichele Forestry Nursery Station back to life aiming at raising over 300,000 assorted seedling in to order to mitigate the negative impacts of climate change and global warming through planting of trees in the district in line with the Keep Zambia Green, Clean and Healthy Campaigns.



Management of seedlings at Chichele

Therefore, by 2030 the department is expected to raise 3 000 000 assorted seedlings for environmental restoration programmes.

Local ownership

Forestry Department has started implementing the investment plan to reduce deforestation and Forest degradation by engaging the intended beneficiaries and most vulnerable to the impacts of climate change and vulnerability. The local communities shall influence the conception, design, implementation and review of development intervention. International cooperating partners shall play a vital a supportive role to ensure that there is transparency and accountability among various stakeholders.

Improvement local livelihood and environmental security

The implementation of interventions outlined in the investment plan has resulted in improved local livelihoods and promote inclusive economic growth while at the same time sustaining key ecological systems especially under variable climate regime.

Gender equity

Women have been integrated into all forestry activities and empowered through access to technologies and assured of equal representation in economic participation and decision making.

3.8.5. The Impact of Existing Trends on Land Use and Population Distribution Patterns

The consumption of wood and non-wood forest products/services will increase enormously causing a multiplier effect as a result of the continuation of existing trends on land use and population distribution patterns in the next 10years. Population increase distribution patterns will impact negatively on land use as big tract forest estates will be exploited in order to provide goods and service to the majority Zambians in the district.

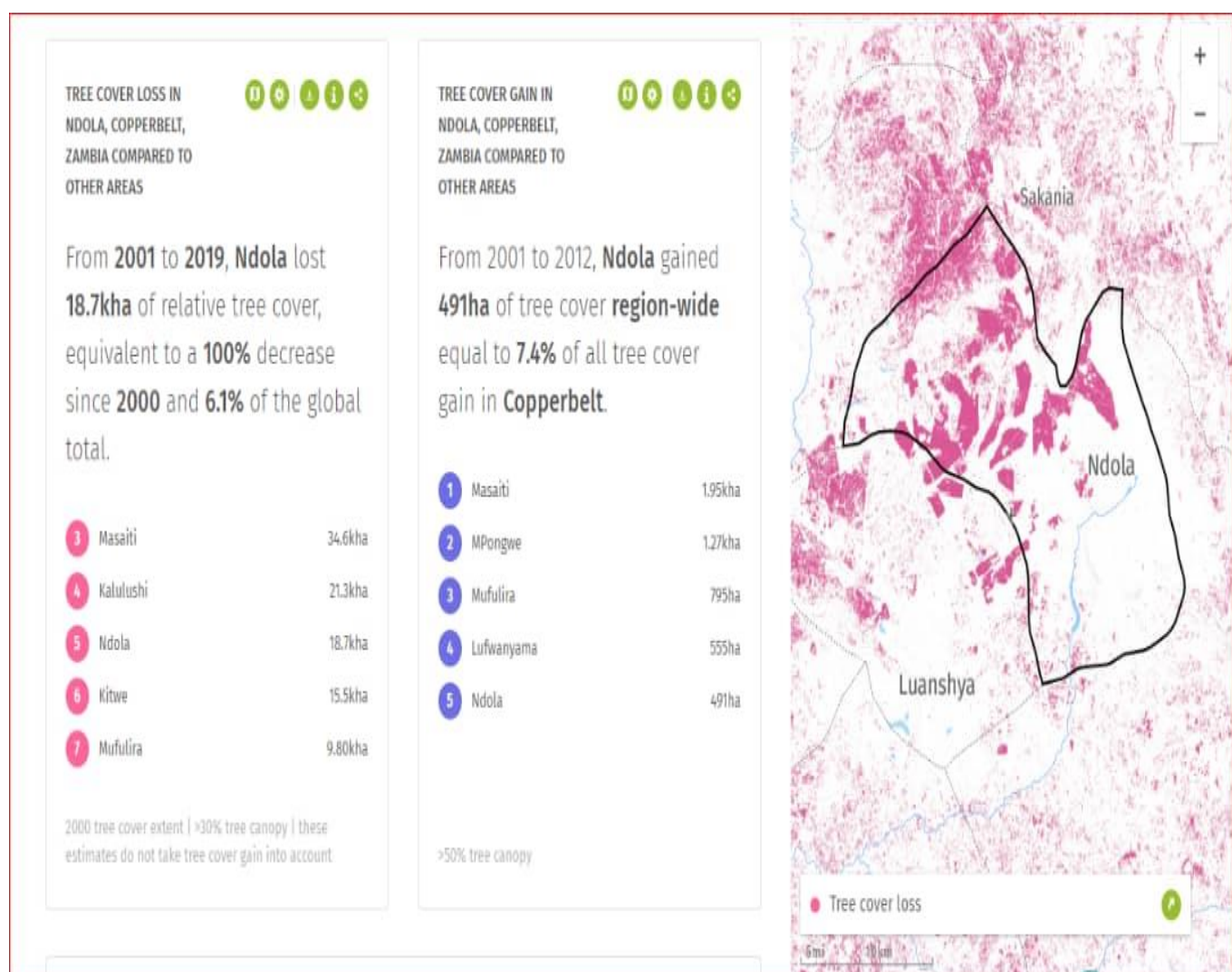
3.8.6. Environment and Climate Change Analysis

A forest resource assessment undertaken through the integrated Land Use Assessment (ILUA) from 2005 to 2008 revealed that 63% of forest – land is relatively undisturbed, 26% moderately disturbed and 5.6% disturbed. The rate of deforestation is 250 000 – 300 000 hectares per annum. Permanent land changes are caused by agricultural practices, settlements, immigration, forest production, infrastructure development, mining and mineral exploration have therefore been the main causes of deforestation and forest degradation as drivers of climate change.

3.8.6.1. The Impact of Existing Trends on the Environment and Climate Change

Loss of vegetation

The vegetation loss in the district will be accelerated by practicing unsustainable harvesting of forest resources, clearing land for agricultural purposes, mining activities and infrastructure development. For instance From 2001 to 2019, Ndola lost 18.7kha of relative tree cover, equivalent to a 100% decrease since 2000 and 6.1% of the global total. This trend is expected to continue in future if effective control strategies are not put in place as shown below.



Source: <https://www.globalforestwatch.org>

Vegetation Gain

The district recorded encouraging results as regard to vegetation cover from 2001 to 2012, Ndola gained 491ha of tree cover region-wide equal to 7.4% of tree cover gain in Copperbelt as shown above. The vegetation gain came about as result of plantation establishment, encouraging enrichment planting in heavily deforested areas and allowing natural regeneration in water catchment areas. The department is working hard in promoting climate change restoration programmes by planting trees with the private sector. Companies, cooperatives and individual farmers are encouraged to establish and manage woodlots in all parts of the district.

3.8. 6.2.The Impact of Environment and Climate Change Issues on the Sector

Forestry Department has noted that unsustainable harvesting system relating to charcoal and fuel-wood production, excessive and increasing forest clearance for farmland, mining and infrastructure development, forest degradation, uncontrolled annual burning have led to the loss of biodiversity and contributing to climate change. Inadequate integrated a approach to forest resource management and coordinated land use planning and management that has contributed to deforestation and forest degradation.

The department is experiencing reduced local level monitoring and law enforcement due to inadequate local level structures and participatory approach which has resulted into excess productivity, erosion, siltation, reduction in stream flow and other negative impacts in other places causing desertification.

There is need to motivate local communities and other stakeholders to participate effectively in sustainable forest management through incentive based mechanism.

3.8.7. Issues Arising From Relating To Gender Groups and Vulnerable Groups

The department is pursuing the issue of gender equity and vulnerability to ensure that both men and women have equal range of opportunities and benefits arising from the implementation of forestry activities or projects. Women are viewed as agents of change and not simply beneficiaries. Gender consideration has been integrated in the design of forestry projects and activities in line with the **National Gender Policy** and **Climate Change Action Plan**.



Gender and vulnerability consideration in implementing projects

3.9. Tourism sector

3.9.1. Reviewed policies and plans

The major policy documents on tourism include the Tourism and Hospitality Act of 2015 and the Zambia Tourism Master Plan (ZTMP).

Tourism and Hospitality Act, 2015; This act provides for sustainable development of the tourism sector through effective planning, management, promotion, and coordination. In line with this act, there is need to exploit the tourism industry in Ndola through careful planning, promotion of tourism products and services, and also monitoring and evaluation of tourism related programs. Notably, this can be achieved by strengthening the linkages and coordination between tourism development in the district and key subsectors such as arts and culture, heritage, transport, education, energy, forestry, fisheries, wildlife, and water resource management. Likewise, in accordance with the Tourism and Hospitality Act, there is need to fully integrate the tourism industry into national development planning, decision making, and budgeting process related to infrastructure development, promotion of tourism related products, and market segmentation.

Zambia Tourism Master Plan (ZTMP); Ideally, ZTMP was developed to exploit the country's vast tourism assets. Developing suitable tourism products in line with clearly defined spatial structure plan. It is a major plan for tourism planning and development. In accordance with

ZTMP, Ndola district intends to provide easy and affordable tourist access, create a favorable business and investment climate, and also raise demand for destination of tourists.

National Gender Policy, 2014; In any sector of the economy, gender issues are inevitable. In this regard, the National Gender Policy (NGP) was developed to address such issues. In accordance with this important policy document, development and exploitation of tourism sector in Ndola district will be facilitated by aiming for gender equality and readdressing gender imbalances. For the tourism sector to fully develop in the district there is need for participation by both men and women. Ultimately, tourism in the district will be based on equal opportunities for men and women to actively participate and contribute not only to the development of the district but also for Nation Development.

Eighth National Development Plan (2022-2026); The Eighth National Development Plan (8NDP) has seen the need to revive the tourism sector by encouraging domestic tourism as well as addressing the challenges of low product diversification, low investment, poor infrastructure and the high cost of doing business. In line with this plan, tourism industry in Ndola will be exploited by addressing major challenges such as lack of tourism related infrastructure; limited investment in the tourism sector by both local and foreign investors; limited tourism product offering range and scope; inadequate tourism promotion and marketing, and low participation by locals in direct and indirect tourism development.

3.9.2. Existing State of Development

The existing state of development entails the availability and quality of services currently existing. In Ndola, however, tourism still remains as one of the most significant sector that has not yet been fully exploited. In this respect, it is important to mention that the sector has the potential and opportunities to attract local and foreign direct investment which is a key ingredient for social and economic development. In this section, the available tourism

3.9.2.1. Availability of Services

Nonetheless, tourism sector in Ndola district has not completely been left out. In this accord, it is important to note that there are a number of tourist attraction sites. Notably, the major ones include Nsoke Game Camp, Levy Mwanawasa Stadium, Dag Hammarskjöld Memorial Site, Copperbelt National Museum, Polish World War I and II Memorial Site, World War Memorial Site, Chichele Mofue Tree National Monument, Masala Cultural Village, Ndola Slave Tree, Arts and Crafts Markets, and Culture and Heritage Sites.

Nsoke Game Camp

This is a small game reserve situated approximately 60 km south of the city. The game has different types of animals such as giraffes, zebras, and waterbucks among others. Likewise, the

game also offers activities such as fishing, cycling, bush spa, and conference facilities to mention but a few. In particular, the camp has tents and chalets for those who might want to spend their night with 100% security guaranteed by game guards. In addition, the camp also has a school called Nsobe Trust School. The school offers Christ-Centered education, preparing pupils not only for jobs but also life skills.

Sports Facilities

The stadium was named after Levy Mwanawasa who served as the third president of Zambia from 2002 until his death in 2008. This is an ultra-modern recently constructed stadium along Kitwe road. It has a capacity of 49, 800 and attracts a lot of attention from local tourists of all ages. Ideally, the stadium is well designed with a ramp at each gate that allows easy access. However, to pass through those gates and see the inside, there is need to wait until there is an activity. In this accord, the entry charges depend on the type of event taking place but the periphery region can be viewed anytime without charge.

Figure showing Levy Mwanawasa Stadium



Dag Hammarskjöld Memorial Site

Dag Hammarskjöld memorial site is one of the tourist attraction sites in Ndola District. Historically, Dag Hammarskjöld was a former United Nations (UN) Secretary General who died in 1961 together with other 18 colleagues in a plane crash. In this regard, the living memorial initiative was conceptualized in 1999 by the UN Country team. Ultimately, in 1970, the Zambian government declared the crash site as a historic landmark and national monument. This is one of the rich history in Ndola and can attract a number of tourists from different parts

of the world if properly managed, developed, and maintained. A glimpse of Dag Hammarskjöld Memorial Site is depicted in Figure 2 as shown below.

Figure showing Dag Hammarskjöld Memorial Site



The Copperbelt Museum

Another important tourist attraction site in Ndola district is the Copperbelt Museum. This is located along Buteko Avenue. Historically, it was established in 1962. The mission of the Copperbelt Museum is to protect and conserve Zambia's heritage for posterity emphasis on natural, resources, geology, mining, and local history. Figure 3 below shows the front part of the Copperbelt Museum.

Figure showing The Copperbelt Museum



Polish World War II Memorial Site

The Polish World War II memorial site depicts the flag of Poland. This has been erected at the place that used to be a refugee camp for the Polish people during World War II. Historically, approximately 18,000 Polish asylum seekers found their way in Northern Rhodesia (i.e. Zambia) after Germany and Russian soldiers under the leaderships of Hitler and Joseph Stalin respectively invaded Poland in 1939 during World War II. Geographically, the Polish World War II memorial site is situated on the outskirts of Ndola city near Bwana Mkubwa Mine area. Apparently, near the site, there is a graveyard where the first Europeans, including the British, who established Bwana Mkubwa Mine and some Polish refugees are buried. However, some of the Polish people who lived at the refugee camp left the country after World War II while others remained in Zambia, where they worked in the mines and industries on the Copperbelt. A glimpse of how the Polish World War II looks like is shown in Figure 4 below.

Figure showing Polish World War II Memorial Site



World War Memorial Cenotaph

This is located across independence way near Ndola City Council (NCC). It is a monumental Cenotaph in honor of World War I and II heroes. Usually, it commemorates all who fought and died during the two world wars for Northern Rhodesia and her majesty's Government. Figure 5 below shows a picture of the World War Cenotaph.

Figure showing World War Memorial Site



Chichele Mofu Tree National Monument

This tourist site is located along the Ndola to Kitwe dual carriage way approximately 13km from Ndola Central Business District. The site was declared a Natural and Historical Monument on 21st March, 1976 to commemorate Zambia's first year of participation in the World Forestry Day. It symbolized the need to conserve trees and use them wisely. The original Mofu tree (said to be about 200 years old) collapsed in 2008 due to strong winds, and the offspring of the fallen tree have since been replanted to take up the position of the fallen tree. The current status of the tree can be seen in Figure 6 as shown below.

Figure showing Chichele Mofu Tree National Monument



Masala Cultural Village

Masala cultural village as shown in Figure 6, was launched in May 23, 2014 in Masala. The main purpose of the cultural village is to promote culture tourism product in the city of Ndola. The main structure of the village at the moment include a theatre which comprises of 340 sitter auditorium that is further enhanced by its large open surrounding meant for outdoor activities and functions (e.g. parties, weddings, meetings, seminars, and religious gatherings). Furthermore, the village structure also include changing and wash rooms for males and females, space for six (6) curio shops, administrative office, VIP holding room, and a bar.

On the contrary, it is important to mention that Masala cultural village usually hosts a number of indoor and outdoor activities. In this regard, some of the activities usually hosted include dances, music, drama, and poetry to mention but a few. On the other hand, it is also important to note that the cultural village is also a venue for visual arts exhibitions and sales. As mentioned earlier on, all forms of artistic events and creative cultural practices can either be held indoor or outdoor. Here, it is important to note that these two forms of hosting events have their unique features. For instance, an excellent feature for outdoor events is the braai type and open-air cultural dancing functions. In essence, Masala Cultural Village is one of the most important tourist attraction venture which needs to be managed and developed fully. One way

of developing it fully is to introduce some other business near the vicinity to generate extra revenue. In this respect, the Ministry of Tourism and Arts should advertise for lease of the curio shops and bar accordingly. Figures 7 and 8 show the Masala cultural village building and ladies performing a dance activity respectively.

Figure showing Masala Cultural Village Building



Figure showing Ladies Performing a Dance at Masala Cultural Village



Arts and Crafts Markets

This is one market that attracts tourists. For example, when tourists visit the Victoria Falls in Livingstone, they are usually welcomed by different arts and crafts that people sell. Similarly, there are also various arts and crafts in Ndola. Good examples include the Twapia arts market which is along Ndola to Kitwe dual carriage way; Kafubu Mall crafts shops, and Wiphan market situated along Misundu road and Nkwazi compound. Furthermore, there are some people who sale arts and crafts works outside the shops near Michelangelo Guest House and also near Spar along Broadway road.

The Ndola Slave Tree National Monument

The Slave Tree National Monument is located along Makoli Avenue that is, former Moffat road in Ndola. It is a place marking the spot where Arab Slave traders held slave markets between 1440s and 1880s. Some of the Africans who were taken as slaves to work in plantations in Europe and Americans passed through Ndola. It is at this place where the Swahili traders including Mwalabu, Chiwala and his warrior general Chipembere held councils of war and a number of slaves were occasionally sold to the Mambundu from Angola for onward shipment to the Americas, others sold to the Arabs were shipped to Europe through Zanzibar, whilst the remainder of the captives were shared among themselves and usually used for warfare. An estimated 10 million Africans were taken as slaves during the Atlantic Slave Trade,

making it the largest transcontinental migration of people in human history. The British Colonial Government declared the offshoot of the Mupapa Tree, a National monument in 1954 as a reminder of the evils of the foregone era of slave trading. Figure 9 below shows a picture of the Ndola Slave Tree Monument.

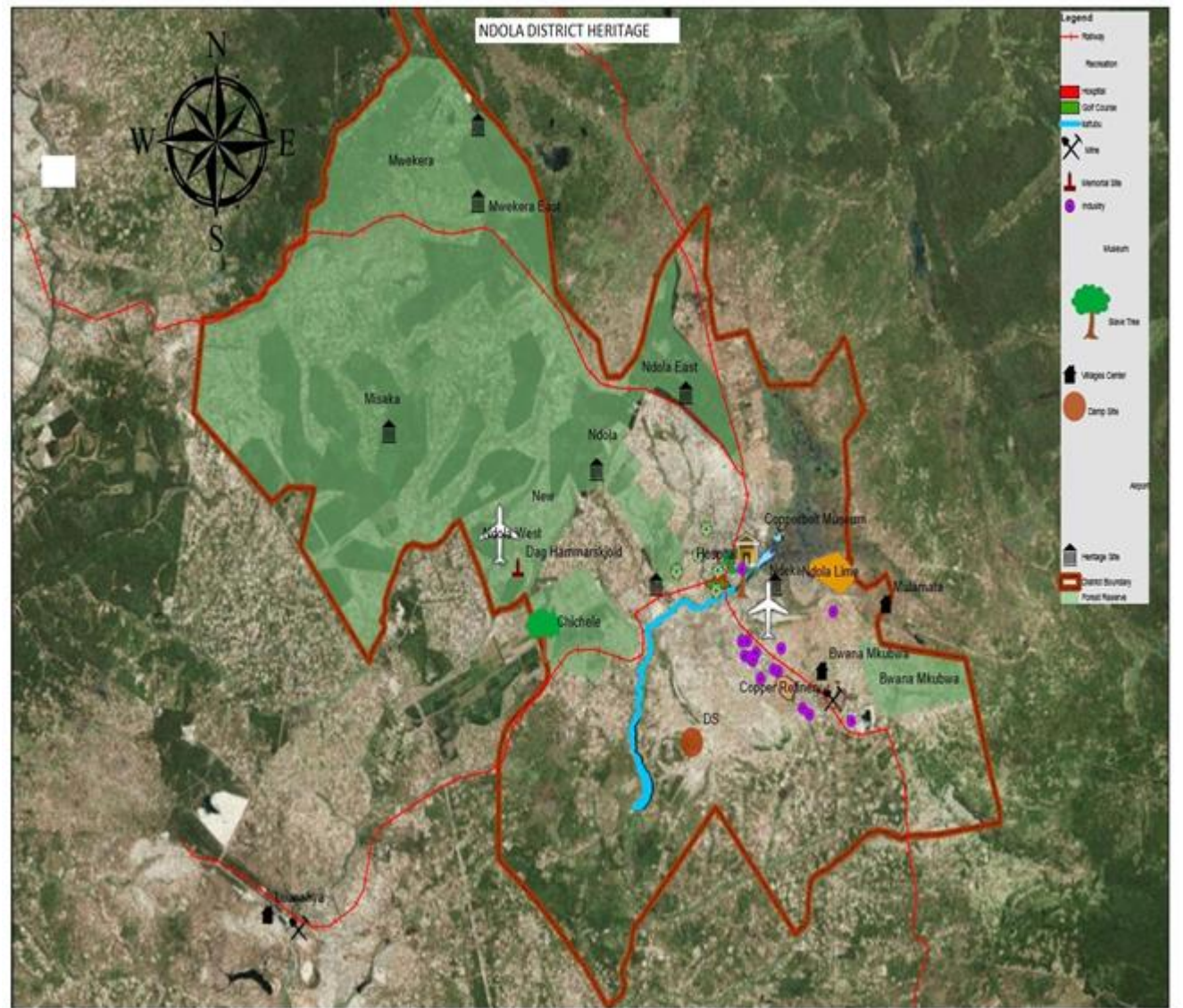
Figure showing Slave Tree National Monument



Culture and Heritage

In addition to various tourism attraction sites, Ndola city has a fair share of the country's cultural and heritage sites with rich history. Notably, the city is a host to historical sites some of which are connected to the Atlantic Slave Trade and World War II. There are also cultural events that take place annually at the cultural village which are intended to promote arts and cultural tourism in the city. The Map below depicts the culture and heritage sites in Ndola district.

Map showing Culture and Heritage Sites in Ndola District



3.9.3. Issues Arising from the Public Participation Process

There are various issues which have been raised from the public participation process. The first issue is on Dag Hammarskjöld memorial site. Notably, this site has approximately seven hectare piece of land available which can be used for investment in hotels, lodges, and shopping mall. This is very important for the purpose of attracting more tourists. Even though there are some hotels nearby Dag Hammarskjöld Memorial Site such as urban hotel, Protea hotel, and Mukuba hotel among others, there is need for more hotels and lodges to increase the number of stay of tourists.

The second issue arising from the public participation process is with regards to the Copperbelt museum. In this respect, the current size of the Museum has been a challenge to many public programs that cannot be held at the museum. In other words, the museum is small to carter for various programmes. Therefore, there is need to build a vibrant, innovative, and popular

museum. Apparently, it was reported that land was acquired in a strategic location near the Mufulira round about closer to the new football stadium and route to the new shopping complex in Mitengo area. Despite this fact, there is no progress in terms of constructions.

The other issue raised from public participation process is inadequate tourism promotion and marketing in Ndola. This is a very big challenge which needs to be addressed. As mentioned earlier on, tourism in Ndola is not fully exploited, yet, it has potential for improvement. In order to fully exploit the tourism industry in Ndola, proper promotion and marketing is key.

On the contrary, limited investment in the tourism in the city by both local and foreign investors is another issue which needs to be addresses. Venturing into the tourism business in Ndola is quite expensive. For instance, some tourism businesses require about fifty eight (58) licenses to operate in the sector. In fact, there are instances were a license in needed from both the local council and central government agency.

3.9.4. Impact of Changes Anticipated over the Next Ten Years

3.9.4.1. Future Demand for Tourism Services and Facilities

Currently, there is construction of a New Ndola Airport also known as Copper-belt International Airport. This Airport will be capable of handling two million passengers per annum. From this we can already conclude that the demand for tourism services and facilities will significantly increase over the next 10 years as more tourists will now be visiting the city of Ndola. Adding to this, the population of Ndola has been projected by CSO to be growing at an annual rate of 1.9 percent. Therefore, in the next 10 years, the population is expected to grow to approximately 926,000 people. Ultimately, with such an increase in population and the construction of New Ndola Airport which will attract more tourists in the city, it is anticipated that there will be increased demand for tourism products, services, and facilities such as accommodation (e.g. lodges, hotels, and guest houses); retail travel agents; tour operators; shopping malls; cinema theatres; restaurants for food and beverages, and tourism information centers among others.

3.9.4.2. Existing and Proposed Investment and Development Programs

The existing investment and development program in Ndola is to build a modern Museum at a land that was acquired along Mufulira road. As noted from the issues raised by the public, the current museum in Ndola is small. Therefore, there is need to build a big museum which will have enough storage space to keep safe the rich history of Ndola. Also, there is an investment opportunity at Dag Hammarskjöld Memorial Site where there is seven hectare of land which

can be used to construct lodges, restaurants or even a shopping mall. Furthermore, taking into account that tourism in the district is not fully exploited; the following are the proposed developmental programs that need to be achieved in the next 10 years;

- Promoting tourism related infrastructure. It was noted that most tourism related infrastructure in Ndola is not in good condition. Therefore, there is need to improve infrastructure. In this regard, some of the major programs include arts and cultural infrastructure development and, roads and local air travel infrastructure development.
- Promote innovative and diversification of tourism products and attractions. This is very important and is among the key priorities. As noted in the existing state of development, there are quite a number of tourism services. However, many tourists do not have such information. In this regard, there is need to promote diversification of tourism products through programs such as tourism products promotion, music and arts festivals, theme parks, accommodation and conference facilities development promotion and tourism investment promotions among others
- Promote tourism source market segmentation. This implies dividing the tourists into market groups for the purpose of communicating with them and providing specific needs. The tourism market segmentation can be divided in term of geographic, demographic, psychographic, and classes of tourists. In this regard, the major proposed development program is tourism promotion and marketing.

3.9.5. Impact of Existing Trends on Land Use and Population Distribution Patterns

Since tourism is a sector which has not yet been fully exploited in Ndola, we can expect that in the next 10 years, there will be more demand for land to build tourism related infrastructure such as hotels, lodges, conferences facilities, shopping malls, tourism information centers among others. Therefore, more land will be used in the district. Furthermore, with improved infrastructure coupled with adequate tourism promotion and marketing, it is anticipated that the number of tourists visiting Ndola district will increase. In future, we would expect an evenly spaced population in terms of tourism in Ndola.

3.9.6. Environment and Climate Change Analysis

In general, climate change is a significant variation of average weather conditions such as conditions becoming warmer, wetter, or drier over several decades or more. In Ndola district, the major climate variables include rainfall variations and temperature variations. Ndola district enjoys a typical savannah climate with three distinct seasons. These include warm rainy season (November to April), cool dry season (May to August), and hot season (September to October). Usually, the rainfall distribution is mainly determined by the topography and prevailing winds

from the Congo. Ultimately, the district as a whole receives an average annual rainfall of approximately 400 millimeters.

On the other hand, the mean annual temperature is between 20°C and 25°C with the absolute maximum temperatures of 32°C in January and the absolute minimum temperature of 12 degrees in June and July. The mean humidity rate is at 81% with the highest in January averaging 96% in the early morning and 66% in the early afternoons.

In Ndola district, the tourism industry is also affected by climate change. In this regard, the extreme climate events that hamper tourism in Ndola include droughts and floods. For instance, arts market is Twapia is usually affected by heavy rains which causes floods. Adding to this, most of the drainages at the market area are blocked. As a result, residents become vulnerable to Malaria and Cholera. The other tourist attraction site which is affected by floods is the Polish World War II memorial site. This is situated in Bwana Mukubwa. Ideally, Bwana Mukubwa is usually affected by floods during the rainy season. As a result of this, houses with weak foundations collapse causing displacement of the people in the area. In essence, not so many tourists visit the site during the rainy season.

Furthermore, the other tourist attraction site which has not been spared to the negative impacts of climate change in Ndola district is Chichele Mofu Tree National Monument. The current status of the tree is that it collapsed due to the fact that it was attacked by termites. In Chichele, the major activity is farming. However, the major hazard is that the area is infested with pests, termites, and army worm among others.

In a nutshell, it is important to mention that the tourism industry in Ndola is not spared from the negative impacts of climate change. Even though the district is blessed with good rainfalls, floods usually affect some tourist sites. Therefore, there is need to improve on the drainage system especially in those areas which are affected by floods.

3.9.7. Issues Relating to Gender Groups and Vulnerable Groups

There are various issues related to gender groups and vulnerable groups in relation to the tourism industry in Ndola district. However, the major ones include the following;

- **Lack of Policy to Encourage More Women to Invest In the Industry.** There is no deliberate policy put in place to ensure that more women access financial assistance which would encourage them to invest in the hospitality industry, micro enterprises and small businesses. This has resulted in untapped potential for tourism products and

services. Notably, under the Hospitality industry, most Hotels and Lodges are owned by men as compared to women.

- **Unequal Access to Higher Positions.** Majority of the higher positions in tourism industry in Ndola are dominated by males than females. In other words, women have unequal access to opportunities compared to their male counterparts. Ultimately, in terms of access to opportunities in the tourism sector, majority of the women are in low paying jobs.
- **Sexual Exploitation.** Tourism in Ndola district is also hampered by sexual exploitations. In fact, experiences worldwide show that women and children have a high potential of being exploited sexually and otherwise in the tourism industry. Therefore, there is need to safeguard the lives of women and children among others.
- **Lack of Gender Mainstreaming.** In general, gender mainstreaming is an approach to policy making that takes into account both women's and men's interests and concerns. In Ndola, tourism is also affected by lack of gender mainstreaming. Most women are not involved in the preparation, design, implementation, monitoring and evaluation of policies aimed at bettering the tourism industry. Therefore, there is lack of gender equality in decision making.
- **Lack of Employment Opportunities for the Disabled in the Sector.** In Ndola district, there are few or no job opportunities for the disables in the tourism industry. In this regard, most disabled people are denied opportunities for employment in places such as hotels and lodges.

3.9.8. Summary

The tourism industry in Ndola district has not yet been fully exploited. However, it has potential for improvement. Notably, there are various tourist attraction places such as Dag Hammarskjold memorial site, Levy Mwanawasa Stadium, Copperbelt Museum, Masala cultural village, Chichele Mofu Tree National Monument, Nsoke Game Camp, and World War Memorial Site among others. However, tourism industry in Ndola lack adequate infrastructure, limited opportunities for investment, inadequate promotion of tourism source market segmentation, and inadequate diversification of tourism products. Therefore, there is need work on these challenges because demand for tourism in the district in the next 10 years will increase. This is supported by the fact that the population of the district is currently growing at 1.9 percent. Therefore, in the next 10 years the population is projected to grow to approximately 926000. In addition, there is a construction of a new Ndola International Airport which will be accommodating approximately two million tourists. Indeed the demand for tourism services

and facilities such as hotels and lodges, shopping malls, cinema theatres, restaurants for food and beverages, and tourism information to mention but a few.

3.9.9. Consideration of Underlying Factors Contributing to the Issues Identified

The major issues identified with regards to tourism sector include inadequate diversification of tourism products, inadequate tourism related infrastructure development, inadequate promotion of tourism source market segmentation, limited investment in tourism, lack of gender mainstreaming, sexual exploitation, and limited employment opportunities for the disabled to mention but a few. However, the following are the factors that are contributing to the issues identified.

- **Lack of resources.** For instance, most tourism related infrastructure in the district is not fully developed due to limited funds.
- **Lack of a deliberate policy to encourage more women to invest in the tourism sector.** This is one of the factors which is contributing to limited investment in tourism in the district. If the government can come up with a deliberate policy to motivate more women to invest, the tourism industry in the district can greatly improve
- **Culture.** Some of the issues identifies such as limited opportunities for women to be put in higher positions in the sector is as a result of cultural believes that men are better than women in those area.

3.10.0 Core Problems

In Ndola district, the major problem with regards to tourism is that it is not fully exploited. In this regard, the core problems with regards to tourism are as follows:

- Limited innovative tourism products and attractions
- Inadequate diversification of tourism products
- Tourism related infrastructure is not fully developed
- Inadequate promotion of tourism source market segmentation
- Limited investment in tourism

3.10.1 Agriculture sector

The impact of agriculture on economic growth is significant in both the short-run and long run in Ndola just as it is countrywide. In Ndola, the effects are strong because more than two-thirds of the population rely on farming, and agriculture has stood as a catalyst for food security and household income. From the last ten years, in order that the effect of agriculture is much more profound, farmers have and must be supported with adequate infrastructure, accessibility to markets, farming inputs, better irrigation techniques, which address the problem of reliance on rain, all of which have been inconsistent in the last decade. Additionally, government is

ensuring the institutionalization of food processing industries, which add more value to the district income through the promotion of the diversification program.

The main objective is to attain sustainable **agricultural production**, which will enhance competitiveness, profitability, and food and nutrition security, contribute to employment and income generation, national economic development, and contribute to poverty reduction. Ndola district draws all its plans from this main objective.

Policy and Legal framework

The agricultural sector is guided by the National Agricultural Policy, which undergoes periodic reviews to ensure its relevance to prevailing climatic, social and economic conditions of the country. In addition, the sector has a number of pieces of legislation some of which are outdated. A process has been initiated in the recent past to repeal, review, amend and enact new legislation aimed at providing a legal framework that will maximize sector development and growth.

The following policy statements were arrived at by consensus of all the key stakeholder categories in the sector and are contained in both the succeeding NDPs, as well as the Zambia CAADP Compact of January 2011. Consequently, these policies are the basis for the implementation of NAIP.

3.10.2 Reviewed agriculture policies and plans

The government is promoting improved access to agricultural inputs and services as a way of increasing production and productivity among smallholder farmers in NDOLA district. Through a participatory extension approach (PEA) solution to poverty Government intends to develop the capacity of district based agriculture staff, entrepreneurs, and businesspersons to enhance their capacity to provide adequate support for agriculture activities. The table below gives the summary of the review.

Key Policies

NAIP Structure	Programs	Policy Statements
Natural Resources Management	Sustainable Land Management Program	Government will facilitate equitable access to land for agricultural purposes, in Ndola, Misaka is one such program being done to avail land for agriculture. A total of 2,340 Ha has been excised for this purpose and will cover about 792 smallholdings'
Agricultural Production and Productivity Improvement	Agricultural Productivity Improvement Program	Government will develop and implement policies and programs that support crop diversification, livestock and fisheries production, increased productivity in crops and livestock, sustainable land and water management, including forestry, agroforestry, climate change adaptation and mitigation and other environmentally friendly agricultural systems;
Market Access and Services Development	- Agricultural Marketing Development Program - Agricultural Investment Promotion Program	- Government and the private sector will implement and adhere to predictable, rule-based market and trade policies and strengthen public-private coordination and dialogue - Government in consultation with stakeholders will identify investment priorities in infrastructure development that support the sector - Government will facilitate private sector to scale-up investments in production, input and output markets, processing and value addition in crops, livestock and fisheries
Food and Nutrition Security and Disaster Risk Management	Food and Nutrition Security Program	Government will explore social protection instruments in partnership with private sector and civil society

NAIP Structure	Programs	Policy Statements
Key Support Services (Knowledge Support Systems)	Research and Extension Enhancement Program	- Government in conjunction with private sector will promote diversified extension messages for all categories of farmers (crops, livestock and fisheries) with emphasis on the small-scale farmers - Government in collaboration with private sector and Cooperating Partners will mobilize resources in order to develop cost-effective, demand-driven research and extension linkages focusing on Public-Private Partnerships
Key Support Services (Institutional Strengthening)	Cross-cutting	- The sector will also collaborate with relevant stakeholders to speed up the implementation of the National Decentralization Policy in order to facilitate improved service delivery - Government in collaboration with private sector and Cooperating Partners will promote and strengthen cooperatives and other farmer organizations as a vehicle for agricultural development

3.10.3. Existing State Of Development

3.10.3.1 Availability of Service

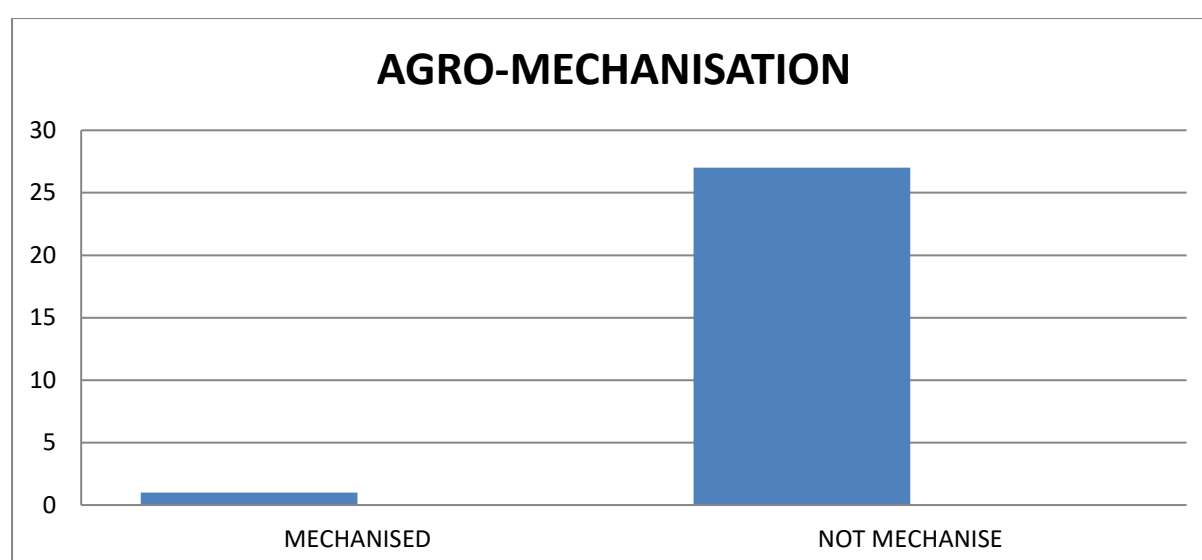
ITEM/ Issue/Type of Service	NUMBER	STATUS	KEY PERFORMANCE INDICATOR	AVAILABILITY OF SERVICES
Irrigation development	5	Active (temporal weir)	Number of farmers directly benefiting	Available in all the 3 agricultural blocks
Agricultural camp housing	17	Only 1 dilapidated house. The rest there is none	Number of camp houses constructed	Available in 1 camp out of 17
Market research and linkages		Active	Number of farmers accessing market linkages	Available through the marketing department
Land husbandry		Active	Number of farms demarcated	Available through Technical services
Crop diversification		Active	Number of crops being adopted	4 different main crops being supported
Conservation agriculture		Active	Number of farmers practicing conservation agriculture	Available through extension services in all camps
Farm power and mechanization		Active	Number of farmers adopting mechanized farming	Available through extension services in all camps
Food processing and value addition		Active	Number of farmers adopting the technologies	Available through extension services in all camps
Fish clusters formation and capacity building	3		Number of fish cluster farmers formed	
Livestock production			Number of livestock farmers	
Veterinary camp houses	2	1 under construction	Number of houses occupied by VA	1 occupied; 1 under construction (70% complete)
dip tanks	3	3 complete	number of dip tanks in use	not in use
spray race	1		number of spray races	not in use

3.10. 4. Issues Arising From the Public Participation Process

Public Consultation was a fundamental part of the planning process. It ensures the involvement of all stakeholders in order to enhance issues survey, awareness, raise the quality of the plan, and increase the likelihood of the plan's acceptability and success of implementation, where stakeholders including (government, parastatals and private sector), and the active involvement of the community in identifying issues, evaluating options, and formulating strategies

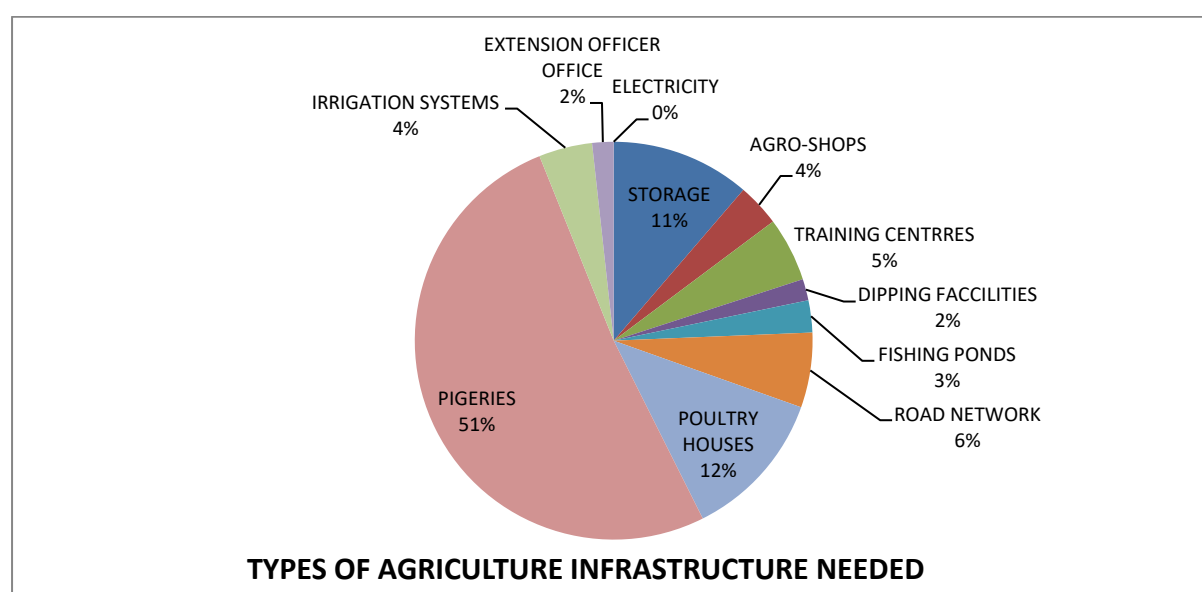
Communities were engaged through established grass root structure (Ward Development Committees) with the help of Local leadership by informing on the plan in line with physical, environmental, socio economic profile of a particular space. Below is the analysis of the findings in the community

Agro - Mechanisation



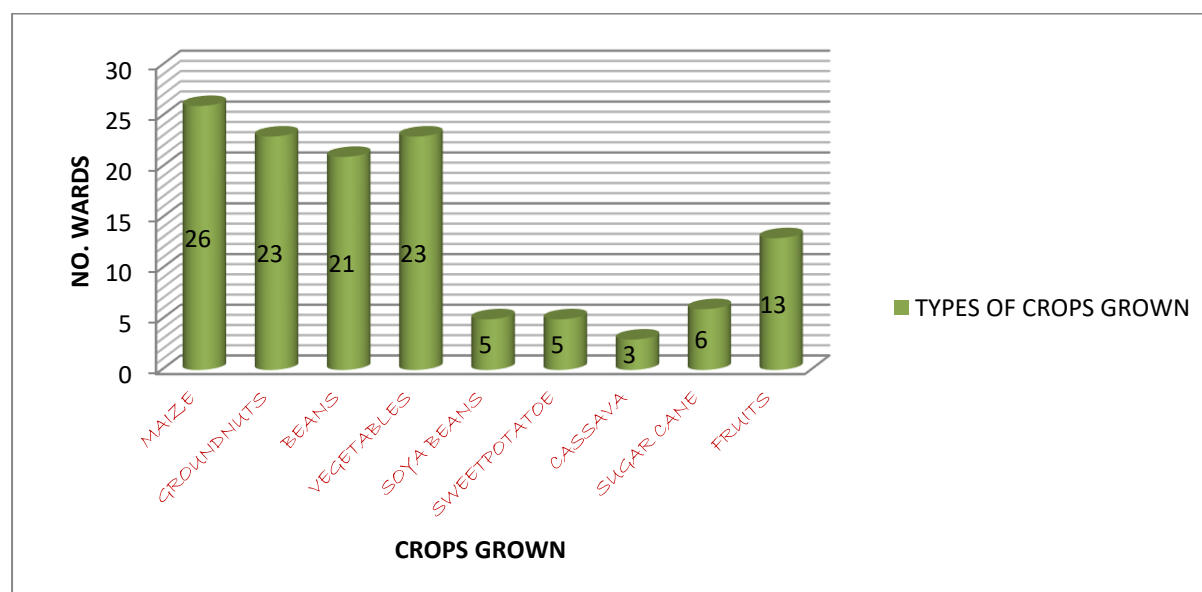
From the figure above, it can be clearly seen that agriculture in the district is not highly mechanised as much of the famers are still into the traditional way of cultivation were they use tools like hoes, ploughs and simple irrigation systems.

Agricultural Infrastructure Needed



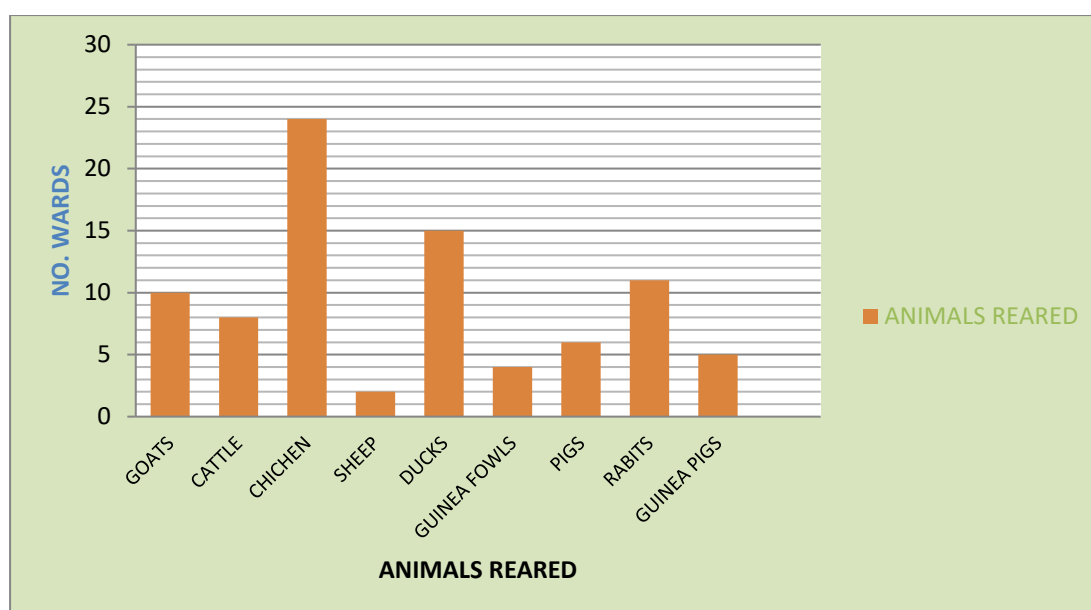
The figure above shows that the district is so much in need of infrastructure in piggeries according to the communities. Poultry is another area the district wants to dwell so much in as there is already available market. Dipping facilities and extension officers seem not so much a priority for most communities within the district.

Type of Crops Grown



Ndola like most of the districts in Copper-belt province were 26 wards has ventured so much in the cultivation of maize as depicted above. However, there other crops like groundnuts and vegetables which farmers also ventures into for a good yield. Cassava is not so much produced at a large scale due to unfavourable conditions for the crop in the district.

Type of Animals Reared



The most reared animals in Ndola district are chickens, ducks and rabbits because they do not need a lot of space to be reared on, less maintenance cost as compared to cattle, pigs, sheep or goats and most of them are reared in residential areas. Due to limited space expansion is impossible and the activity itself in the residential area is illegal according to the Public Health Act.

The need to boost the poultry industry can be observed from the number of wards that have ventured into this business. The district is also trying to meet the market demand of rabbits and goats which including guinea pigs. However, the district has not done so much in the rearing of sheep.

3.10.5. Impact of Changes Anticipated Over the Next Ten Years

3.10.5.1. Future Demand for Services and Facilities

Crop Production and advisory services

- I. shortage of agriculture land, e.g. Kaloko gone residential
- II. low food production
- III. increase in poverty
- IV. change in biodiversity due to encroachment of natural habitats
- V. Rivers will dry up due to increased human activity ie. Houses, gardens
- VI. shortage of unemployment
- VII. migration from one place to another, eg farmers find new land in far places
- VIII. increase in disease incidences e.g. COVID-19 spreads fast where there is high population
- IX. slow development due to pandemics that may arise

Fisheries and Livestock

- X. Impact of changes anticipated over the next 10years.
 - a. Increased in the number of fish farmers.
 - b. Increased in the tonnage of fish harvested.
 - c. Increased number of livestock.
 - d. Increased number of livestock products.
 - e. Increased number of village chicken due to the 1st 1000 days program that has incorporated village chicken production for the next 5 years.
 - f. Improved nutrition due to increase in production of protein source foods (fish, village chicken, shoats, pork and beef)

Population change	Future demand for services and facilities
1. Population increase	1. Increase in demand for extension services i.e. number of trainings will increase, number of monitoring services on agriculture activities will increase 2. Consumption of food will increase 3. Production and productivity will increase 4. Productivity might decrease due to diseases that may come up due to population increase, which increases chances of disease pandemics. E.g. corona virus and HIV?AIDS 5. Increase in human activities on natural resources leading to climate change 6. Increase in demand for social amenities such as housing and the need for new infrastructure, will cause damage to the natural resources such as land and will lead to climate change, which may lead to low production and productivity. 7. Might bring about policy change to suit the existing occurrence in agriculture trends that could push down the positive gains achieved over a length of time.

3.10.5.2. Existing and Proposed Investment and Development Programmes

Market Information and Research	Field Visits	Farm power and Mechanisation
Creating market linkages	Crop diversification	Irrigation development
Training Programs	Conservation agriculture	Livestock production
Aquaculture and capture fisheries Management and Development	Food processing and value addition	Value addition on locally produced materials

3.10.5. The Impact of Existing Trends on Land Use and Population Distribution Patterns

- Degradation of land and the environment due to increased human activity on land
- Rapid expansion of agriculture related activities
- Alteration of the natural ecosystems, for example dambos and springs may dry up
- Increase in ecological problems
- Increase in climate related hazards including droughts and dry spells because of increase in human activities
- Land degradation due to pond abandonment and overgrazing.
- Increased numbers of fishponds in the district.

3.10.6. Environment and Climate Change Analysis

3.10.6.1 The Impact of Existing Trends on the Environment and Climate Change

- Negative change in biodiversity of the district
- Increase on the demand for environmental based products such herbs causing damage to the environment
- Change in weather elements and patterns will cause change in climate
- Change in soil structures and textures will lead to change in productive potential of the land causing crop yield reductions
- Continued unsustainable human activity will lead to reduction in arable land.
- Increase in excision of forest reverse to human resettlement

3.10.6.2. The Impact of Environmental and Climate Change Issues on the Sector

- Water shortages for agriculture activities
- Shortage of arable land due to land degradation
- Reduction in agriculture production and productivity
- Declines in precipitation
- Reduction in the flow of surface water and ground water for drinking and agriculture purposes
- Rising temperatures causing high levels of evaporation and evapotranspiration
- Shortage of food due to increased crop failure increasing poverty levels

3.11. Water and Sanitation

3.11.1. Review of water and sanitation policies and plans

Government of the Republic of Zambia had characterized sanitation as a subsector of water supply, rather than developing and implementing strategies that were specific to sanitation.

Vision 2030; The Zambia Vision 2030 outlines the overall development goals for Zambia, including a Socio-Economic Development Objective to provide “improved water and sanitation facilities to 100% of the population,” Specifically, the sanitation targets in Vision 2030 aim to achieve access to “appropriate, environmentally friendly” sanitation for 68% of the total population by 2015 and 90% by 2030 (ibid., p.33) in view of this vision Ndola district through Kafubu Water and Sanitation main objective is to improve staff productivity in order

to improve quality overall performance to increase to 95% and improve water supply hours from 18 hours to 20 hours per day.

The Eighth National Development Plan (8NDP) 2022 – 2026; The 8th National Development supports the implementation of Vision 2030 through objectives under a five-year period. As such, under the Strategic Development Area ‘Human and Social Development’, the Development outcome No. 3 calls for “improved water supply and sanitation”. In the plan period, the Government will prioritise the provision of safe water and improved sanitation across the country, especially in districts, constituencies and wards where there have been water deficiencies. Key programmes to be implemented will include climate resilient water supply infrastructure development and maintenance, water quality monitoring and protection of aquifers and protected water sources.

The implementation of these measures will lead to an increase in access to safe water to 67 percent by 2026 from 58 percent in 2018 for rural areas and to 98 percent by 2026 from 91.8 percent in 2018 for urban areas. Further, access to sanitation will increase to 55 percent by 2026 from 37.2 percent in 2018 and to 90 percent by 2026 from 77.7 percent in 2018 for rural and urban areas, respectively.

Additionally, the Government will strengthen sewerage and solid waste management by enhancing financing to address the constraint of inadequate infrastructure in the sanitation sub-sector. Promotion of sustainable sanitation practices will also be scaled-up. With these interventions, the proportion of households with access to improved sanitation is expected to improve from 37.2 percent and 77.7 percent in 2018 to 55 percent and 90 percent in 2026 in rural and urban areas, respectively. Constituencies and wards will identify areas of need for water and sanitation services and plan for provision of such services to reduce developmental inequalities related to access to these services

To achieve this policy, the district main objective is ensure that all the customers in the catchment areas receive good quality water and sanitation services.

The National Water Policy of 2010; The National Water Policy of 2010 guides optimal harnessing of water resources in all climate situations particularly during dry spells (MEWD, 2010). The Water Resources Management sub sector’s policy statement takes into account catchment management of water resources, stakeholder consultation and involvement, assessment, monitoring, water conservation and preservation of its acceptable quality and quantity, efficient and equitable water allocation to all users and disaster preparedness (MEWD, 2010). To achieve the following objectives the policy outlines a number of objectives and this include;

- Promoting water resources development through an integrated management approach
- Recognising water as an economic good

- Recognising the important role of the water sector in the overall socio-economic development of the country

To ensure better water management and water supply and sanitation services as the main objective of the policy, the district through Kafubu Water and Sanitation's main objective is to improve infrastructure development and sustainability in order to give attention to key operational factors such as replacement of leaking network pipes, meter installation reservoir repair, mechanical equipment repair and general work expansion to meet the current demand level. To this effect, 99% equipment in the water treatment plants has been installed and is operational.

3.11.2. Existing state of development

3.11.2.1. Availability of service

In view of the above policies, Ndola District has undertaken a number of priority projects in relation to water and sanitation as tabulated in table below.

S/N	PROJECTS	Activities being implemented
1.	Kafulafuta Dam Water Project (Masaiti) Under Kafubu Water and Sanitation. A combined project to benefit; Mpongwe, Luanshya, Ndola Masaiti	• Dam construction • Construction of water supply • Network extension in un-serviced areas • Network repair and replacements • Training • Advocacy and sensitization • Increase service coverage to 100%

2.	Emergency Water and Sanitation Intervention (ESWI)	This project commenced in the year 2011 and was completed in 2013 funded by the Zambian government. • Installation of 5,000 water meters in the main masala, Kabushi, And Twapia Townships of Ndola. Installation of new water distribution pipes in selected areas. • Replacement of 1,392 metres of sewer pipes in Masala and Kabushi Township
3.	Improvement of Water Supply in Ndola	The project is funded in partnership with JICA • Rehabilitation of Kafubu Dam Water Treatment Plant • Replacement of Nakaputa, Northrise and Chifubu Falling Main • Construction of Kiosk in Kaloko Township • Laying of supply lines to Mushili Township

Surface water sources

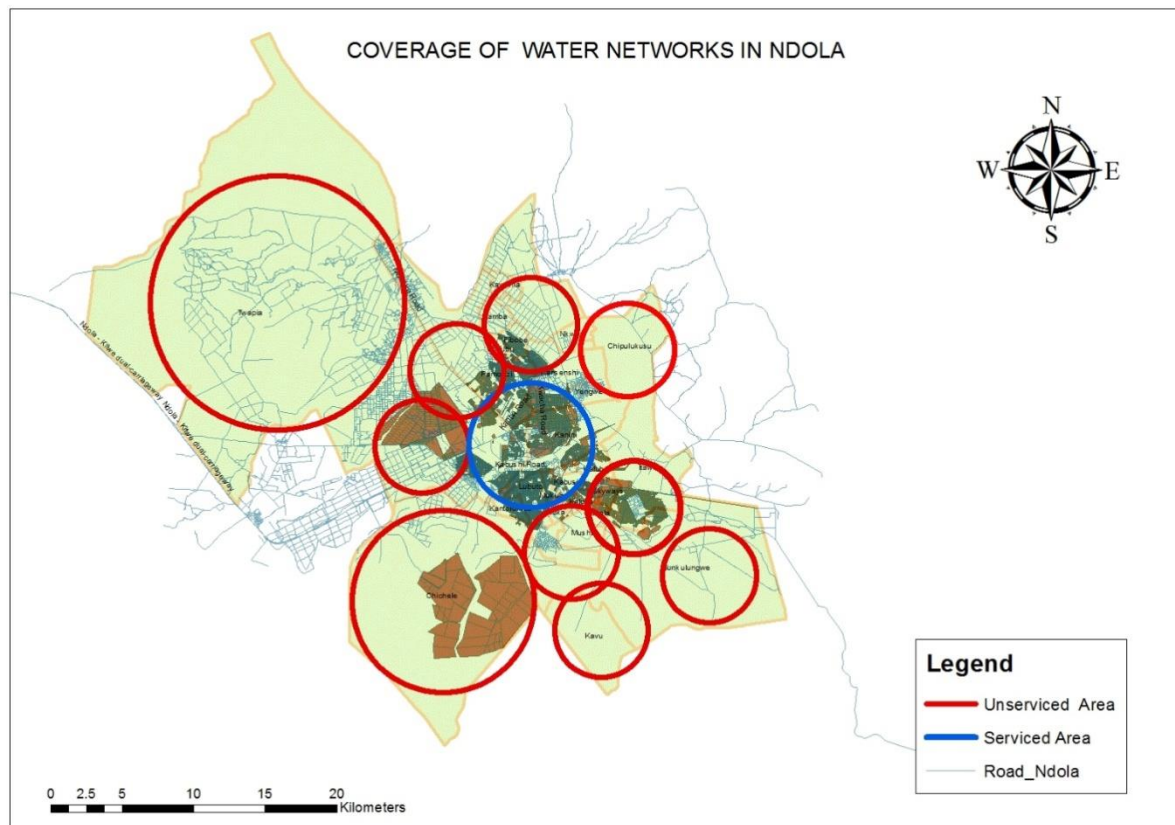
Surface supplies are derived from the Itawa and Kafubu impoundments both of which are situated at the Kafubu River while underground supplies are derived from two well fields in the Minsundu area developed in the 1980's to augment existing surface supplies. Kafubu dam has a designed storage capacity of 15 million m³. The actual useful capacity of the dam has diminished due to siltation resulting in the company being compelled to reduce the water amounts discharged downstream. Designed storage capacity of Itawa dam has never been quantified. As a result of low recharge and due to siltation in this dam and in an effort to improve raw water supply, a channel was excavated recently to direct excess water from two mining companies namely; Chilanga Cement, and Ndola Lime and also a nearby disused quarry into the dam.

Underground water sources

There are twenty-one boreholes used to abstract water from the two well fields for the underground supply. The borehole yields are yet to be assessed. The combined rated capacities for all the four water treatment plants is 202Mld, whereas the total combined production is

estimated to be 150Mld representing 74% of the installed capacity. Below is a figure showing the coverage of water network in Ndola District.

Water Service Coverage



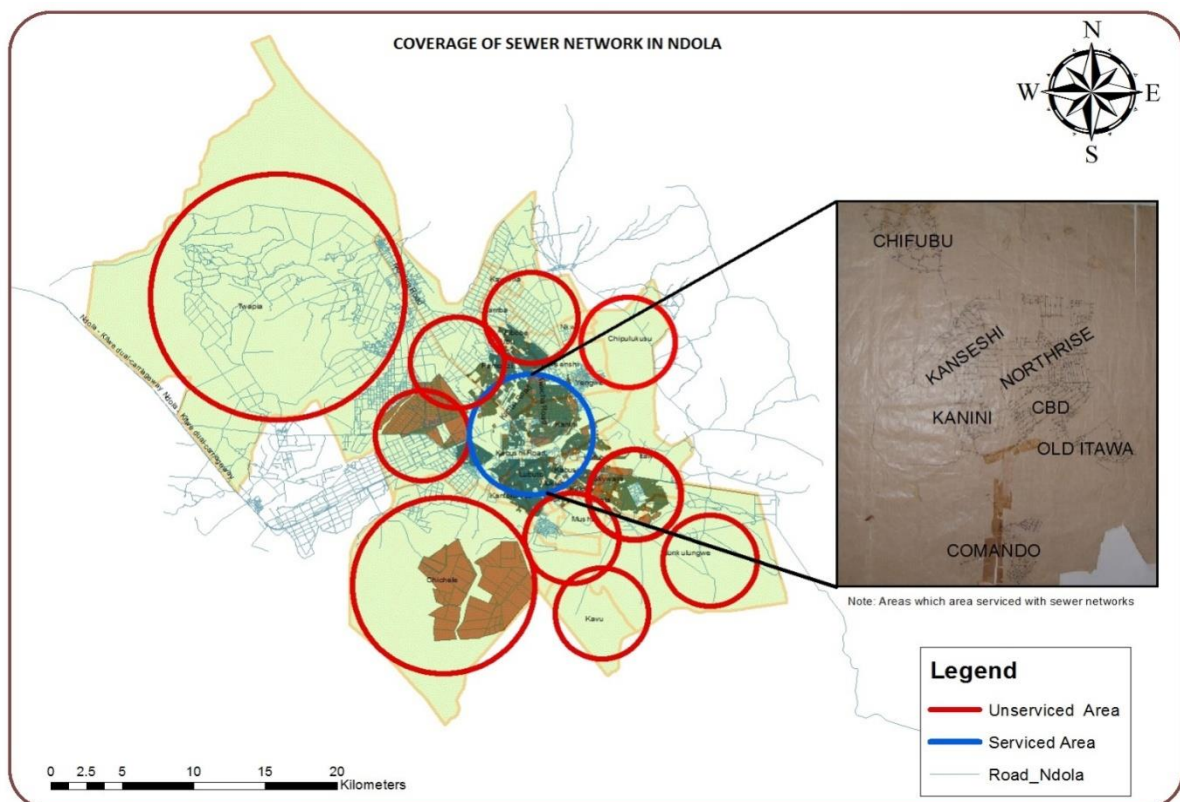
Source: Ndola City Council GIS Unit (2020)

Sanitation (Sewerage System)

Ndola is serviced by three sewage collection systems within two primary sewage catchment areas. The first system serves Pamodzi, Chifubu, Kawama, Kansenshi and parts of Northrise in the northern catchment area and conveys sewage to New Kanini Sewage Treatment Plant, whilst the second system also within the northern catchment caters for the Central Town area, Kanini, Hillcrest and remainder of Northrise, Itawa and Kansenshi, with sewage being pumped to Old Kanini Sewage Treatment Plant through Itawa and Dambo pumping stations. The third system caters for the southern supply district, and conveys sewage by gravity to Lubuto Sewage Treatment Plant.

There are three pumping stations in the northern catchment for conveyance of sewage, but only two, Itawa and Dambo are in an operable state. Operations at the third station 604 ceased some years back because of a crack in the sump and eventual vandalism of the equipment. The other four sewage treatment plants namely Old and New Kanini in the northern zone and old and new Lubuto in the southern zone. All these plants are of the conventional trickling filter type. Old Kanini due to equipment breakdown, operations ceased several years ago and the plant lies derelict as most of the equipment was stolen or vandalized due lack of security. As a consequence the, New Kanini is the only treatment plant carrying out grit removal and biological treatment as shown in figure below.

Sanitation Services



Source: Ndola City Council GIS Unit (2021)

KWSC has established the average water demand for households in 2010 and 2030, water demand for institutional use has been added as 20% of household demand and water for industrial and commercial use has been added as 45 % of household demand in Kafubu WTP service area and 20% in all other areas. Water loss or NRW has been added as 30% of the total demand, being the assumed NRW level after the Kafulafuta Water project. While the daily

maximum demand is based on peak factors 1.2 for Ndola, Dimensioning of intake, treatment plants, transmission system and reservoirs shall be based on daily maximum demand.

Highest Average Daily Water Consumption in Ndola

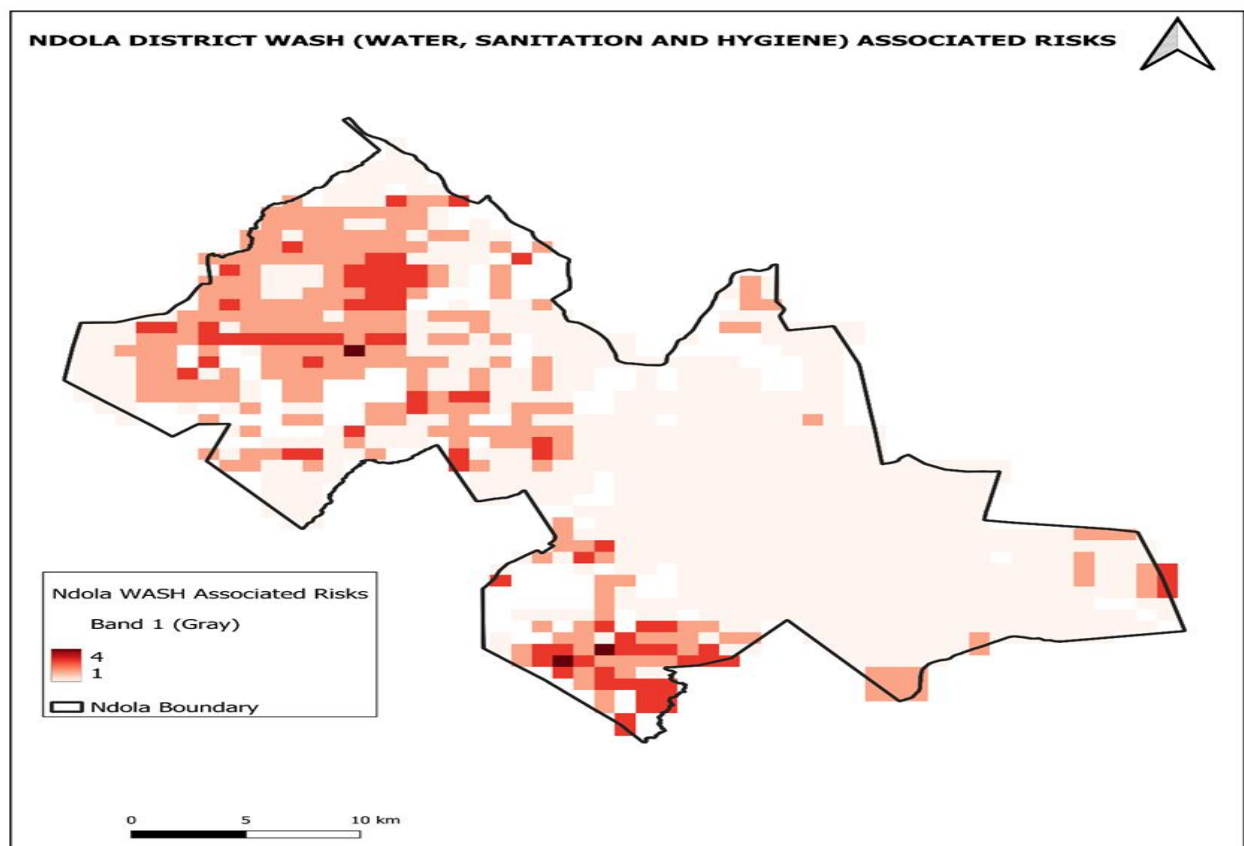
No.	Item	Short term	Long Term
		2025	2030
1	Population	1,001,264	1,426,918
2	Average water consumption per day (m3/d)	165,859	236,369
3	Proportion of industrial water consumption (%)	20	20
	Institutional water consumption (m3/d)	33,172	47,274
4	Proportion of industrial and commercial water Consumption (%)	30	30
5	Industrial and commercial water consumption (m3/d)	49,758	70,911
6	Proportion of water loss or NRW consumption (%)	30	20
7	water loss or NRW consumption (m3/d)	74,637	106,366
8	Total water demand (m3/d)	323,426	460,920

Source: Kafubu water and sewerage co.2019

Ndola District WASH (Water, Sanitation and Hygiene) Associated risk

WASH subcomponents include proportion of population with no access to soap, water for hand washing, piped-in drinking water, live in a household that takes more than 30 minutes to fetch and retrieve water, and households that shares toilets with other households or do not have a toilet. This map above demonstrates an average WASH risk score from a 1 to 5 scale, with 5 indicating the highest level of risk, at a 1km squared resolution. In Ndola district, the further away from the Central business district, the higher the WASH associated risks, and the closer to the Central Business District, the lower the WASH associated risks as indicated on the map

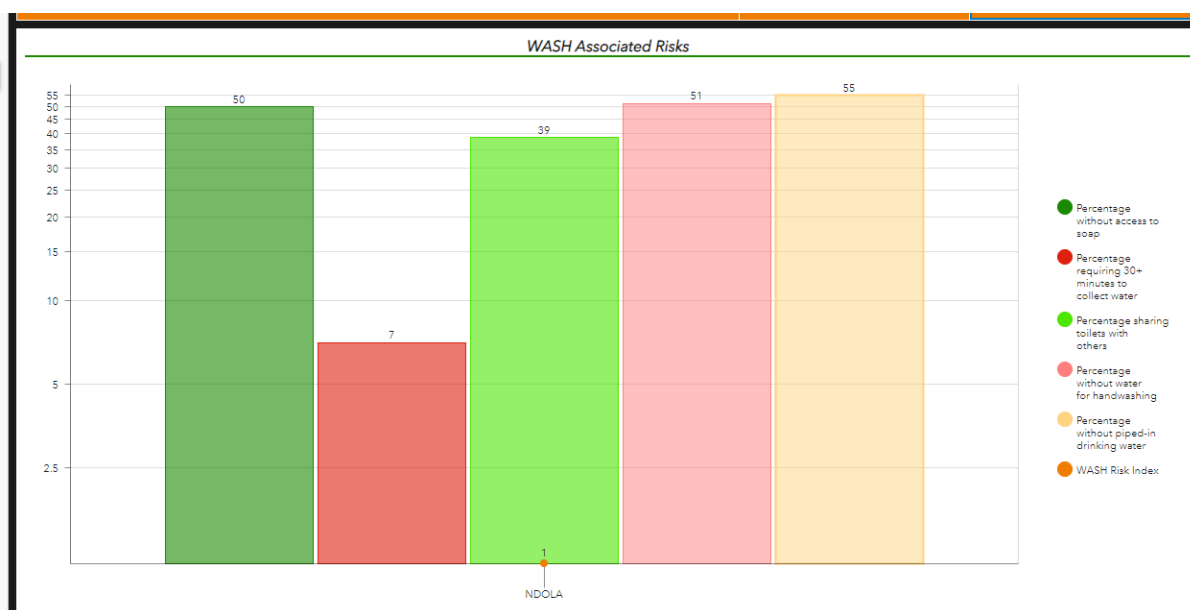
below;



Source: <https://data.grid3.org/maps/GRID3::grid3-zambia-z-scores-wash-risk>

The figure below shows population percentages with no access to soap, water for hand washing, piped-in drinking water, live in a household that takes more than 30 minutes to fetch and retrieve water, and households that shares toilets with other households or do not have a toilet in Ndola district;

- ❖ Percentage without access to soap: **50**
- ❖ Percentage requiring 30+ minutes to collect water: **7**
- ❖ Percentage sharing toilets with others: **39**
- ❖ Percentage without water for hand washing: **51**
- ❖ Percentage without piped-in drinking water: **55**
- ❖ Overall WASH risk index for Ndola District: **1**



Source: <https://data.grid3.org/maps/GRID3::grid3-zambia-z-scores-wash-risk>

3.11.2.2. Quality of service

Hours of Supply

New Un-serviced Areas

Over the years a number of residential areas have developed which are not services with water. This will require huge capital injection to extend the service to these areas. Nonetheless Danida project provided funds which have embarked water infrastructure development to extend water services for new areas and rehabilitate infrastructure within old residential localities. The table below give an insight on these areas.

Table: Water and Sanitation Un-serviced Areas

No	Area	Estimated Households	Estimated Cost Water (ZMW)	Estimated Cost Sewerage
1	Mitengo Extension	521	1,800,000	1,200,000
2	Hillcrest	337	800,000	900,000
3	Kalewa Extension	249	200,000	200,000
4	Pamodzi Extension	200	1,500,000	300,000
5	Kansenshi Extension & infill& Extension	430	1,600,000	300,000
6	Nkwazi Extension	220	710,000	400,000
7	Kabushi Extension	200	600,000	300,000
8	Minsundu Bwana-Mkubwa	100	200,000	500,000
9	Twapia PHI	100	1,800,000	500,000

10	Lubuto/Kabushi Extension	95	1,900,000	300,000
11	Mushili Extension	341	1000,000	800,000
12	Itawa Extension	85	1,200,000	700,000
13	Jacaranda Villa	211	800,000	800,000

Source: Kafubu water and sewerage co.2019

It has been assumed that the residents in high-cost, medium-cost and low cost houses will decrease their unit water consumption gradually to 200 lpcd, 125 lpcd and 75 lpcd in 2025 and remains almost constant with very little variations when the projects are fully implemented, while the unit consumption in the peri-urban is kept at 50 lpcd.

Table showing: Development in unit water consumption

Category	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
High-cost houses	250	250	240	230	220	210	200	200	200	200	200	200	200
Medium-cost houses	150	150	145	140	135	130	125	125	125	125	125	125	125
Low-cost houses	90	90	90	90	85	80	75	75	75	75	75	75	75
People living in peri-urban areas	50	50	50	50	50	50	50	50	50	50	50	50	50

Source: Kafubu water and sewerage company.2019

The water demands are projected that the total water demand will increase gradually from the present level of 33,000m³/day to 40,000m³/day in 2020 and further approximately 44,000 m³/day in 2031. Domestic or household demand constitutes 54%, whereas commercial/industrial demand constitutes 37% of total demand. The effect of the adjustment in unit water consumption is seen clearly until the project is fully implemented, where only a minor increase in domestic demand is experienced even though population is expected to grow. After the adjustment in unit water consumption (the metering effect) water demand starts increasing again with the population growth assumption at lower unit consumption levels.

NRW has been assumed to gradually decrease to 20% from the present level of 60%. This is due to the rehabilitation of the distribution network, transmission mains, and implementation of effective NRW management (water losses and wastage reduced strategies), including leak detection and repair measures, etc. This eventually reduce the water production from the

present 158,000 m³/day to 90,000m³/day in the year 2030. The reduction in water production will subsequently reduce the operation and maintenance (O&M) costs. Capacity utilization will decrease following rehabilitation of the WTPs and the expansion of the capacity at both Kafubu WTP and the Misundu-I, Misundu II and Itawa WTPs under the completion of the Projects. The capacity utilization will on average decrease to around 56%, from where it continues to increase and it will attain a level of 63% in 2025 and 71% in 2030. The above development in capacity utilization depends on the assumptions made. If KWSC does not manage to reduce NRW as expected the required water production will increase to meet the demand, and capacity utilization will increase and thus reduce spare water production capacity. If KWSC manage to connect more households than envisaged (following population growth in the service area), or the number of people in each household has been underestimated, total water demand will increase and similarly increase the required water production and thus increasing the capacity utilization rate - and vice versa.

In Old Lubuto, most units are also out of operation due to lack of maintenance and theft of some equipment. This has compelled the company to divert the sewage inflow to the New Lubuto plant, which although operational has also been affected by poor maintenance practices, and also has become overloaded beyond design capacity.

The sewers lines in the high cost and central business district are generally in good condition with only a few blockages being reported. However most of the sewers in the low cost housing areas no longer have the capacity to effectively convey the high sewage flows as they now cater for populations far in excess of the design, resulting in frequent flooding with most affected areas being Chifubu, Main and Mine Masala and Kabushi. The District Health Management Team (DHMT), Public Health Department and some Non-Governmental Organizations like CARE International and PUSH have been very instrumental in educating the public both in Urban and Peri Urban areas on the importance of safe keeping of drinking water and maintaining the surroundings clean to avoid outbreaks of unsanitary diseases.

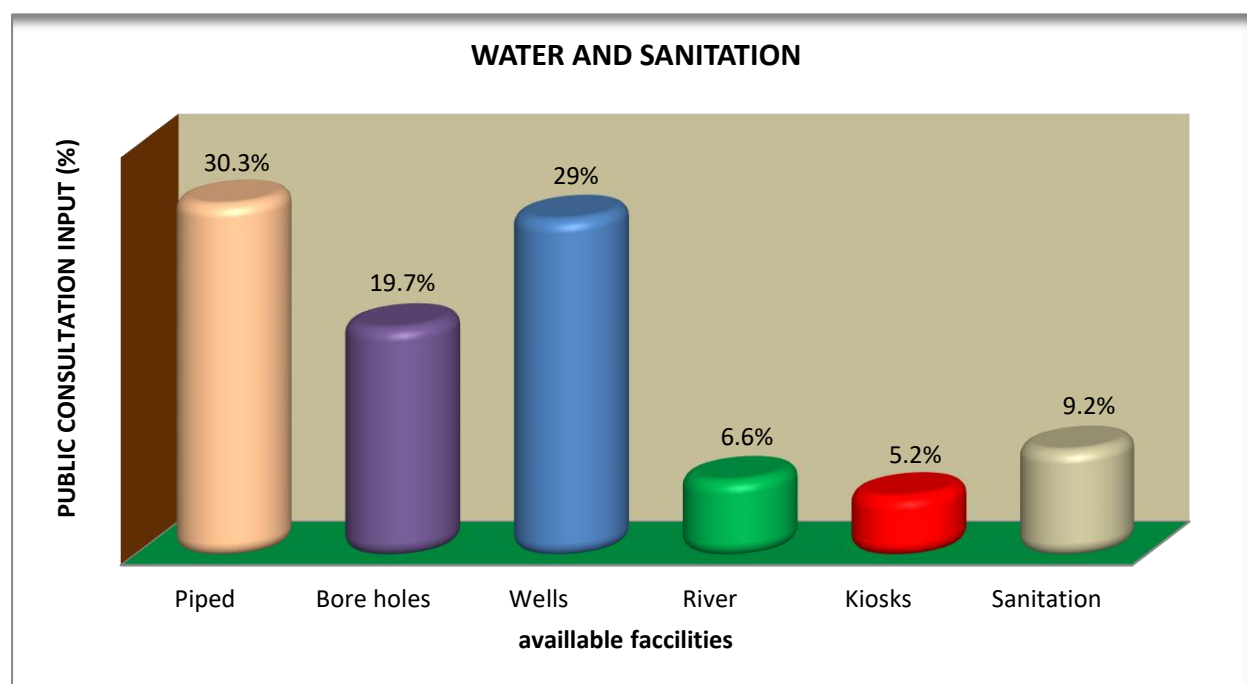
3.11.3. Issues arising from the public participation process

The main aim of conducting a public participation was to encourage the public to have a meaningful input into the decision-making process. Public participation thus provided the platform for community members who were engaged through established grass root structure (Ward Development Committees) with the help of Local leadership. Consultations with stakeholders' involved government institutions, non-governmental organisations, Parastatals institutions and the private sector.

3.11.3.1. Availability of services

From the consultative meetings held, it was evident that the district's largest source of water has been piped water. Wells have also dominated in peri-urban areas with over 29%. The use of kiosks is not so much as this facility is only found in fewer communities making community members to sink bore hole as an alternative source of the commodity. Sanitation was another major issue which was raised from across the district where it was reported that only about

9.2% of the district has been serviced and the rest use pit-latrines and Soak-ways. This information is depicted below.



Source: Kafubu water and sewerage company (2019)

3.11.3.2. Quality of service

In Ndola District, limited access to adequate water and sanitation is a key developmental challenge particularly for rapidly expanding peri-urban areas. From the district public consultation submissions on the quality of services offered on water and sanitation, it is evident enough that communities feel that water is supplied for less hours and in some areas which are already serviced, there is literary no water that runs through the pipes hence it being pointed out as the major contributor to disease outbreaks. The public also observed that lack of improved proper sanitation has negatively affected individuals, environment and communities at large. It was pointed out that sewer influence runs on surface grounds from broken pipes posing a health hazard especially to children.

3.11.4. Impact of changes anticipated over the next ten years

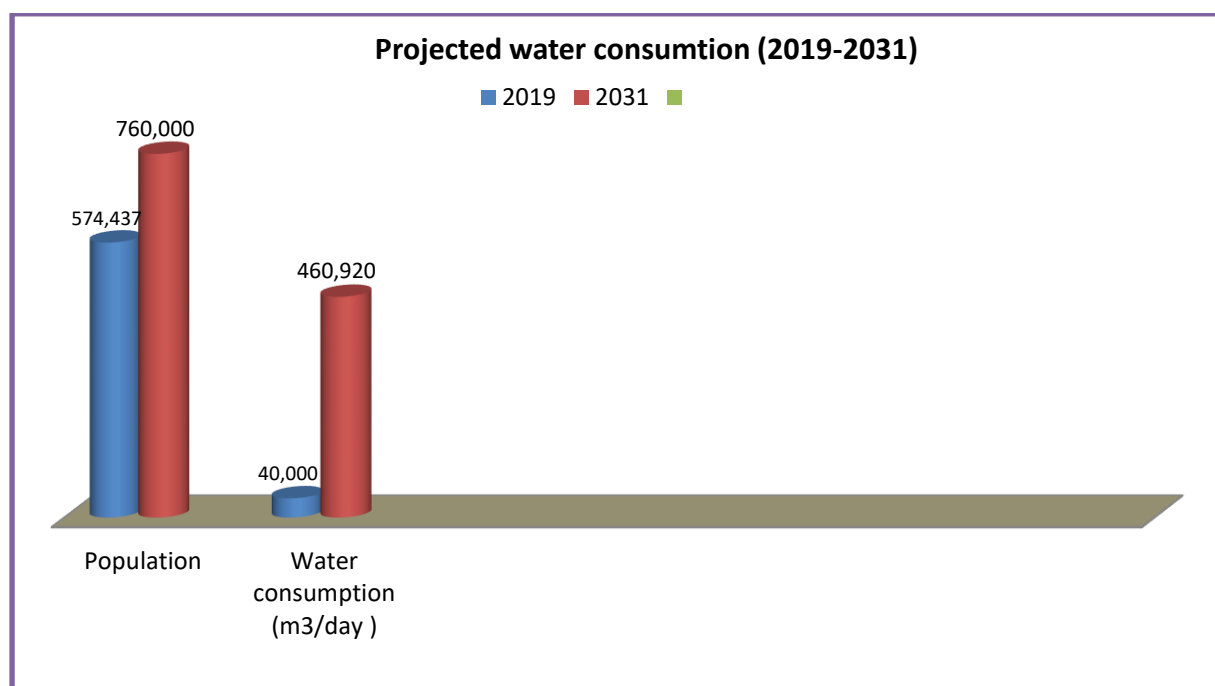
Rapid and often unplanned population growth is often associated with population demands that outstrip infrastructure and service delivery. As the population of the district grows, the demand for water and sanitation facilities intensifies. In rapid growing districts like Ndola were it has been observed to have few facilities to offer quality services, the number of people living without accesses to basic clean water and sanitation is still increasing. The accessibility to

improve water and sanitation in the district has been understood as a crucial mechanism to save communities from the adverse health outcomes associated to disease outbreaks

3.11.4.1. Future demand for services and facilities

Water demand

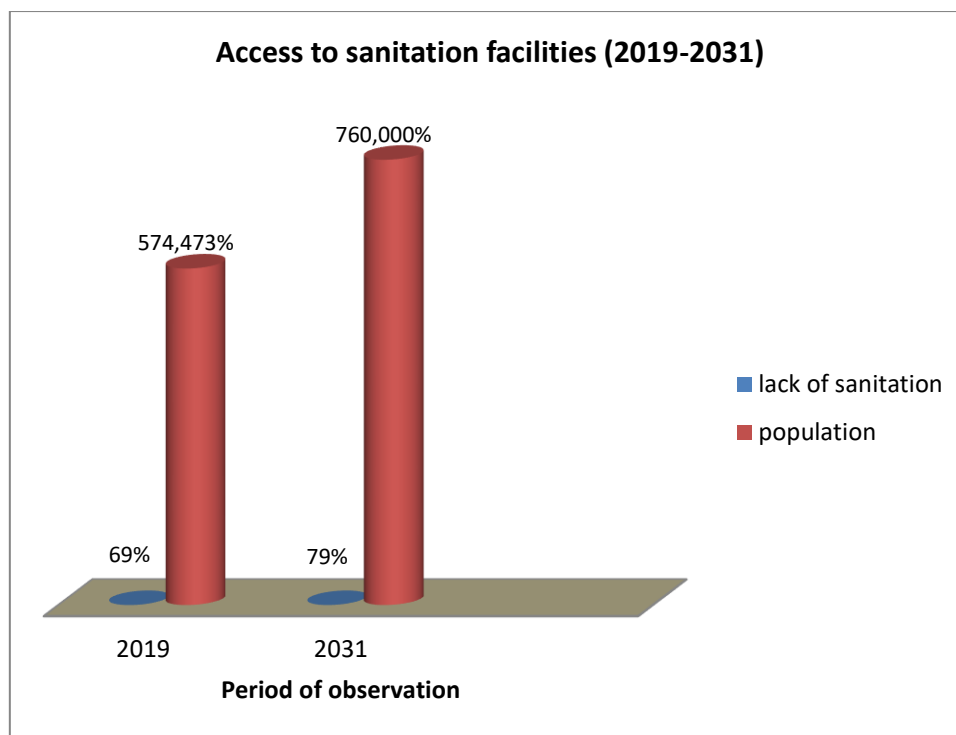
With the projected population increase of the district from 574,437 in 2019 to 780,000 in the year 2031 representing a 36% increase in population. The quantity demand for water is equally likely to increase from 40, 000 m³/day to 460, 000 m³/day by 2031 as shown below.



Source: Kafubu water and sewerage company (2019)

Sanitation demand

Proper sanitation facilities promote good health because it allows people to dispose their waste appropriately. Without adequate sanitation facilities, waste from infected individuals can contaminate a community's land and water increasing the risk of infection for other individuals. With the projected population increase to 760,000 by the year 2031 from 574,437 in 2019, it is can be concluded that the current infrastructure cannot meet the growing population hence need for expansion. It is for this reason that the district is planning to lay out its infrastructure in the un-serviced areas and repair the existing ones to mitigate the shortfall.



Source: Kafubu water and sewerage company (2019)

In view of this, more facilities are needed to support the growing demand for the commodity. This deficit will lead to a number of consequences such as;

- Pollution of water sources- due to shortage of sanitation facilities, pit latrines are often constructed close to ground water sources such as wells, hand pumps, or springs. During heavy rains, the latrines tend to get flooded, resulting in water seeping through the soil contaminating ground water.
- Health-most diseases in developing countries are caused by polluted water and insufficient water supply and sanitation. With the shortage of the service Ndola district is likely to experience over the next 10 year period, the district is likely to face challenges in eradicating diseases such as diarrhea, cholera, malaria and dysentery.

3.11.4.2. Existing and proposed investment and development programs

With the district growing population from 2019-2031 of about 36%, the existing and ongoing infrastructure in the water and sanitation sector such as Kafulafuta Dam Water Project to have capacity of 430,000m³/day and another auxiliary facilities such as pump stations, water distribution pipelines, water intake pump station to be completion in the year 2021. Emergency Water and Sanitation Intervention, Improved Water Supply in Ndola Project, Kamuchanga Water Supply Project and Metering Project which are on-going are likely to help cushion the

projected deficit in the sector. These projects involve upgrade, installation of new equipment, repair of the existing infrastructure and advocacy to the general public on safeguarding the facility.

3.11.5. The impact of existing trends on land use and population distribution patterns

Increase in population does not only depend on the size or density of the population but on the ratio of population to availability of sustainable resources. It is also estimated that increase in population affect the national land use pattern available for agriculture, forest land coverage, nearby area of different types of water bodies

3.11.6. Environment and climate change analysis

It is widely recognized that climate change constitutes a significant and serious threat to sustainable development of any country, including Zambia. Evident shows that Zambia has over the past years experienced a number of climate related hazards including droughts and dry spells, seasonal flash floods, and extreme temperatures. Some of these, especially droughts and floods have increased in frequency and intensity over the last two decades and have adversely impacted on food and water security, energy and livelihoods of communities. Temperatures also indicate a rising trend with potential for increased heat stress, land degradation and desertification. Such impacts are likely to compound the daunting economic and social challenges the country already faces, (MNDP, 2016).

Copper-belt Province and Ndola district in particular has not been spared to climate related hazards mentioned above. Impacts like droughts and dry spells have threatened the food security in the district while floods and temperature rise has given rise to many health issues for the community. Some of the areas of concern related to climate change include the following.

Water Resources

Ndola has a restively abundant supply of surface and ground water. However, surface water is unevenly distributed throughout the city and the Western region often experiences water shortages. During drought periods and following declines in precipitation, they have been reductions in the flow and volume of rivers and streams, which have affected the accessibility and availability of surface water and groundwater for drinking, agriculture, livestock rearing other human activities and fisheries.

3.11.6.1. The impact of existing of trends on the environment and climate change

Increase in Human population is one of the biggest causes of the depletion of natural resources. Ndola district population has been increasing by 1.9% since 2010. As the district population is

increasing, means more requirement and thus more usage of resources. At the current growth rate, it is expected that the district population will reach about 760,000 by 2031. Increase in population is one of the root causes of environmental issues like climate change. Human activities especially the ones leaned towards economic gain are leading to the alteration of the environment at a very high speed. Economic activities are supported by the natural environment which plays a significant role. For instance the natural environment contributes to economic development by availing natural resources and raw materials such as timber, minerals and water. These raw materials are utilized as input for production of services and goods which make up the economy of the district like Ndola.

With the number of projects being implemented by the district to mitigate the deficit of water and sanitation services such as the Dam project in Masaiti District will bring out some of the environmental challenges mentioned like land degradations as excavation works are being done trees and land is disturbed. In so doing, problems of climate change is being escalated.

3.11.6.2. The impact of environmental and climate change issues on the sector

Depletion of natural water resources for urban supply Urbanization goes hand in hand with an increase in population density, which means that more people have to rely on existing water sources. New sources are also difficult to find, and/or expensive in exploitation. Many cities have come to the limits of water availability from natural sources Boreholes and wells are also very important in supplementing available water for consumption. Unfortunately, they are endangered by increasingly being depleted and not renewed .This is particularly rampant in Ndola District where due to presence of roads and buildings; infiltration capacity of the soil is hampered to a minimum which further decreases the capacity for ground water renewal. To this effect, the sector is disadvantaged in that if ground water is disturbed, there will be less water to supply to the ever increasing population of the district.

3.11.7. Issues arising relating to gender groups and vulnerable groups

The 2014 National Gender Policy is closely aligned to the Social Protection Policy and contains specific measures to “enable women at community levels to participate actively in decisions on construction and management of infrastructural facilities for water supply and sanitation,” the district’s one of its objective is on health status of women, men, girls and boys dependence on access to clean water and sanitation.

Thus, rural communities (along with peri-urban) face enormous challenges in securing clean water and are continually exposed to hazards of poor sanitation.

Women and girls are disproportionately affected by lack of access to improved water supplies and total sanitation. They bear the burden of carrying water and also of poor health and security risks that arise when they are forced to walk long distances and/or go out at night.

3.11.8. Summary

With national Vision 2030 outlining the overall development goals for Zambia, including a socio-economic development objective to provide “improved water and sanitation facilities to 100% of the population, “Ndola district through Kafubu Water and Sanitation Co. Ltd. Have been implementing various developmental projects meant to mitigate the deficit caused by ever increasing population which is putting pressure on demand of water and sanitation facilities.

However, the public feels much need to be done to improve sanitation and provision of safe drinking water in the district. To this effect, the district has been implementing a number of projects such as repair works, installation of new facilities and implementation of new projects as a measure to mitigate the public concerns. However, this has not come without challenges like environmental degradation, pollution, and climate change which as greatly affected the quality service delivery.

3.11.9. Factors contributing to the issues identified

- Demographic pressure on water and sanitation management
- Dilapidated water infrastructure
- Environmental degradation
- Rapid increase in population leading to amplified demand for services
- Climate change
- Encroachments in environmentally sensitive areas
- Unplanned settlements hampering service delivery
- Vandalism
- Lack of campaigns for pro-environment activities that ensure sustainability such as responsible use of the water resources and sanitation facilities

3.11.10 Core Issues

- low quality of services provide in the district
- Few households are connected to sewer line and piped water
- Erratic water supply
- Polluted communities due to leaking sewer lines

3.12. Energy Sector

The Energy sector is a vital input in the socio-economic development of the country. All the productive sectors in the economy rely on energy. The Country is endowed with a range of energy resources, predominantly woodlands and forests, water, coal and renewable sources, such as geothermal, wind and solar energy that are the primary source of energy. The energy sector in Zambia is dominated by Electricity produced by hydro energy. Other sources of energy are Petroleum products and alternative sources of energy such as biomass, coal, wind, geo-thermal, solar, uranium, waste (including municipal solid and agricultural waste) and petroleum.

3.12.1. Reviewed energy policies and plans

The Energy sector comprises both public and private actors. The Ministry of Energy supervises the following statutory bodies: Energy Regulation Board (ERB); Zambezi River Authority (ZRA); and Rural Electrification Authority (REA). The Operations and development of the energy sector is guided by National Energy Policy 2019 (NEP 2019) is anchored on the Vision 2030 and consequently the National Development Plan. *The overall objective of NEP 2019 is to achieve an optimal energy resources utilization to meet Zambia's domestic and non-domestic needs at the lowest total economic, financial, social, and environmental and opportunity cost and establish Zambia as a net exporter of energy.*

The Eighth National Development Plan (8NDP) seeks to undertake Energy Sector Reforms: The aim of the reforms is to improve performance of the energy sector. The implementation of the reforms will remove the inefficiencies in the fuel supply chain by among others, supplying of diesel through the pipelines and procurement reforms to standardise supply contract prices. In the electricity sub-sector, reforms will focus on enhancing the operational efficiency of ZESCO Limited including its cost structure.

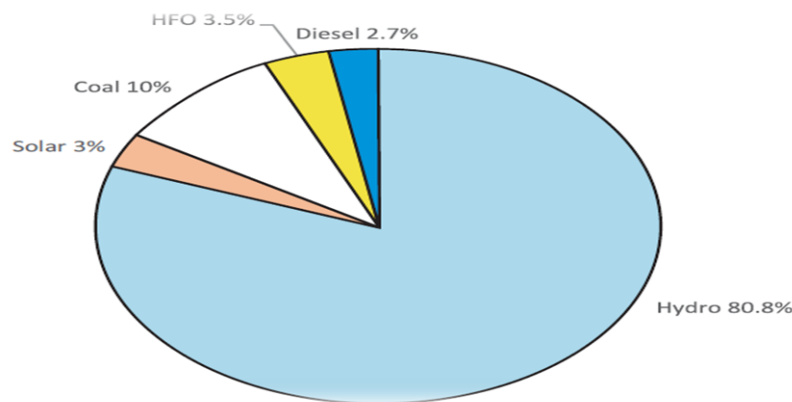
The District is implementing policies in the NEP 2019 to achieve its objective for the benefit of Ndola City.

3.12.2. Existing state of development

Electricity

ZESCO Limited is a vertically integrated power utility and is responsible for generation, transmission, distribution and supply of electricity in the country. The company is the dominant participant in the electricity market in Zambia. According to NEP 2019 “The installed generation capacity in Zambia is 2,976.3 MW. The installed capacity comprises of 80.8 percent

of hydro, 10 percent of coal, 3.5 percent of heavy fuel oil, 2.7 percent of diesel and 3 percent solar PV as of September 2019 as shown in the figure below.

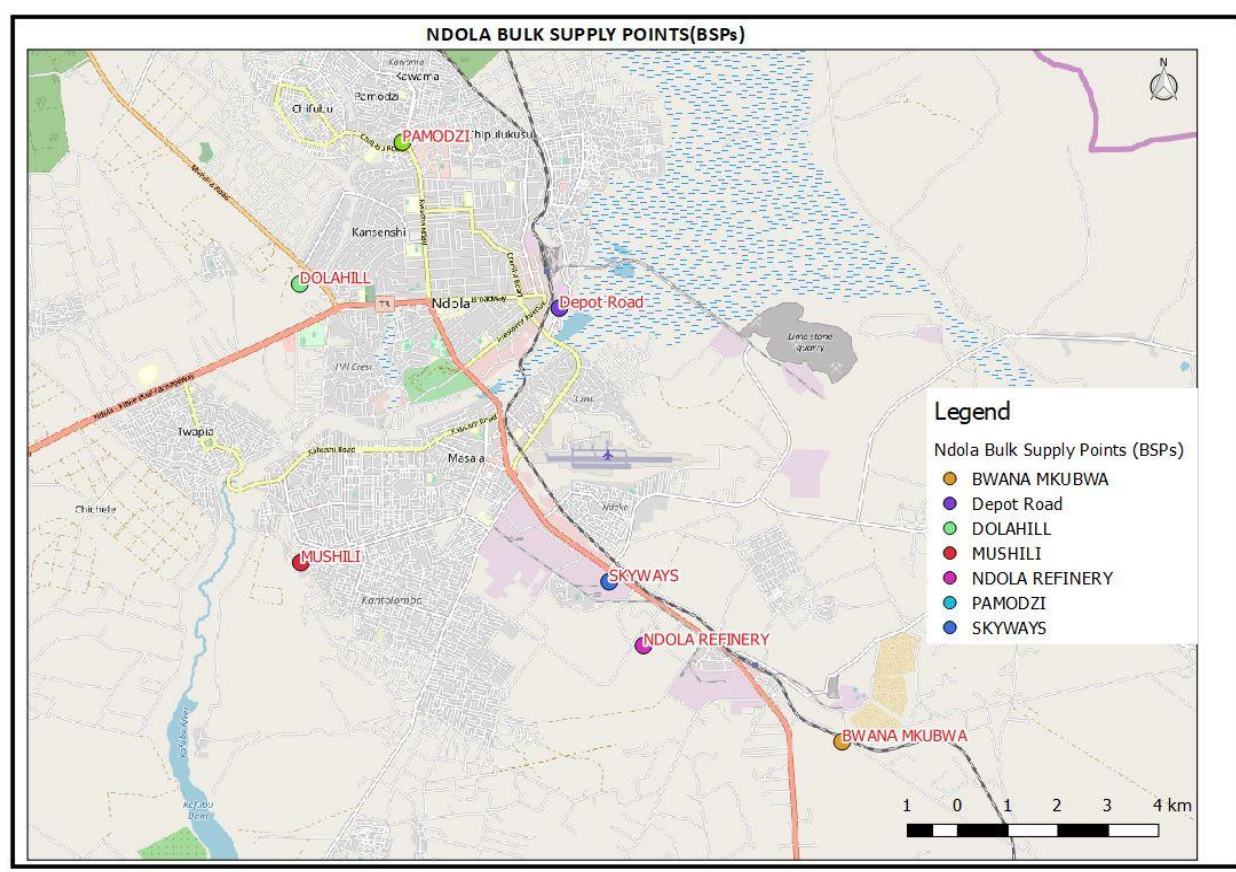


Source: ZESCO Copperbelt Regional Office (2019)

Ndola District, the Provincial Capital of Copper-belt Province is supplied from Maposa CEC substation 220/66KV via 66 KV overhead line. The City is also supplied with power from HFO from Ndola Energy.

The bulk supply points (BSPs) are evenly distributed across the City of Ndola. In the Northern part of the City, there is Depot Road substation, Pamodzi Substation and Dolahill Substation. In the Southern part, there is Skyways Substation, Bwana Mukubwa Substation, Ndola Refinery Substation and Mushili Substation. The current total power capacity for Ndola is 348 MVA (hydro) and 105 MW (HFO).

Distribution of Energy Sources (Map)



Source: Ministry of Energy (2019)

3.12.3. Issues Arising From the Public Participation Process

- Most households are not connected to electricity supply
- Unreliable power supply

The public participation revealed that most households in various locations in the district are not connected to the Electricity grid. This situation exists in both old and new settlements. Unreliable power supplier is a major challenge affecting the Country Ndola is no exception. ZESCO has had to implement Electricity management through load management throughout the country. This has had an adverse impact on the economy. Load Management has affected industry as well as home users.

3.12.4. Impact of Changes Anticipated over the Next 10 years

Ndola's total population is expected to grow from 451,246 in 2010 to 926,000 by 2035 leading to an increase in the demand for energy services. The energy sector consequently needs to put in place measures and interventions to meet the future demand. Zambia aspires to be a prosperous middle-income country by 2030. This will require the supply of energy services to Zambians as an input to their productive activities. Ndola district is generally an industrial

district. This implies that the economy in the district is largely dependent on the manufacturing, service and construction industries. Growth in population and industrialization in the district will lead to increase in Electricity requirements.

3.12.4.1. Existing and Proposed Investments and Development Programmes

There are capital projects under way to firm up Power supply (reliability) in the City. The main aim of the Projects is to support increased industrial, agricultural and commercial activities in the District. The following are the projects;

1. Refurbishment of Depot Road Substation
2. Refurbishment of the Itawa Water Works (installation of latest switch gear)
3. Refurbishment of Boat along year Substation (installation of latest switch gear)
4. Bwana Mukubwa Substation (installation of latest switch gear)
5. Construction of Substation (New) at Copper-belt International Airport (CIA)

In addition, to these Projects, ZESCO is undertaking an ambitious program of replacing old 11KW supply cables from bulk supply points to ensure that customers have reliable supply in the industrial area, agricultural areas and Central Business District (CBD) of Ndola.

3.12.5. Environment and Climate Change Analysis

3.12.5.1. The Impact of Existing Trend on the Environment

Carbon emissions from Coal, heavy fuel oil and Diesel contribute to Climate change. This in turn has contributed to changing weather patterns such as rainfall. Reduced rainfall affects water level that lead to reduced hydro power generation.

3.12.5.2. The Impact of Environment and Climate Change issues on the Sector

Climate change has adversely affected the district, for example rainfall patterns are no longer predictable. The power generation from hydropower has been compromised as a result of this the district has been facing unprecedented power outages through load management. The district is undertaking various initiatives aimed at mitigating the impact of climate change which include but not limited to: efficient management of forests; and promotion of climate smart agriculture

3.12.6. Issues Arising Relating to Gender Groups

At household level poverty is highest among vulnerable groups headed households. These include Women, People living with Disabilities, Children / Youths. In addition to Poverty these groups have limited to no access to electricity. Vulnerable groups are unable to afford the cost of being connected to the grid as well as paying for the service. This situation has negative

consequences on the environment. These groups heavily rely on firewood and charcoal for sources of energy. This leads to uncontrolled tree cutting and deforestation. There by contributing to climate change. Energy is sensitive to gender. Most women, particularly in shanty compounds and rural communities spend their productive time collecting firewood to prepare a meal for families.

3.12.7. Summary

Electricity plays an important role in facilitating activity in all sectors of the economy. Electricity is the major driver for social and Economic growth in Zambia. This has been an important driver for economic development. The generation, transmission and distribution of electric is key to attainment of middle-income status by 2030. To increase supply, there is need to promote investment in hydro, nuclear, geothermal, wind and solar energy generation and all energy sources that will increase electricity generation in the country.

3.12.8. Factors Contributing To the Issues

- Unreliable power supply has been caused by low water level as a of effects of climate change result.
- Rapid growth in demand caused demographic and socioeconomic factors

Renewable Sources of Energy

Ndola District has an opportunity to develop its sources of renewable energy by creating an enabling environment for investment as well as research and development. According to the NEP 2019 the sources of renewable energy include solar, wind, geothermal, small hydro and biomass.

(a) Solar

In Zambia the average solar irradiation of 5.5kWh/m²/day with approximately 3,000 sunshine hours annually creating potential for photovoltaic and solar thermal appliances. The potential for Solar power is able to cover the generation of electricity, solar domestic appliances, solar water pumps. In Zambia Solar power generation accounts for 3 percent of total generation capacity. Investment in solar electricity generation is low because of the high cost of capital requirements and weak regulatory frameworks on renewable energy. NEP 2019

(b) Wind

The average wind speeds in Zambia is sufficient for electricity generation, pumping of water for domestic and irrigation use. Power generation from the wind energy is low. This is mainly due to low investments. The district can create an enabling investment for research and development as well as investment in wind energy generation.

(c) Geothermal

The potential for geothermal energy exists in Zambia. Ndola does not have any of these sites. The District can benefit from growth in this sector when it is joined to the national grid. According to NEP 2019 There are more than eighty (80) hot springs spread out in different parts of the country which indicate potential for geothermal for both power generation

(d) Small hydropower

Small hydropower generation contributes 1.5 percent to the national installed electricity generation capacity. Small hydropower generation plays an important role in rural electrification and the off-grid areas where large hydropower and grid extension may not be economically and financially viable.

(e) Biomass

Biomass is another source of energy that exists in Zambia. This source of energy includes wood fuel (charcoal and firewood), biogas, pellets, briquettes, biofuels and gel fuel that are predominantly used for domestic source of energy for cooking and heating. Wood is commonly used for cooking. Its uncontrolled exploitation has an adverse effect on the environment and its continued use is unsustainable because the rate of depletion exceeds regeneration of biomass thus contributing to climate change. The high dependence on wood fuel is due to low access and the cost of connecting to Hydro and other electricity connections in most of households especially in the unplanned settlements.

(f) Waste to Energy

Waste is another source of energy that has potential to contribute the growth of the energy sector. In its quest to be a *Prosperous, Green diversified and sustainable economy that provides equal opportunities for all by 2032*. Sustainable power supply is a must. Waste management can contribute to the generation of power. The district is struggling with solid waste management. The use of waste will not only contribute to power generation but also waste management. The waste includes Solid waste, Sewerage wastewater, agricultural crop residues, livestock manure or waste, wood chips or biomass and industrial waste. The district is reviewing some proposal for possible Public Private Partners.

(g) Coal

According to the NEP 2019, the proven reserves of coal in Zambia are estimated to be over 30 million tones, Coal is the second highest contributor for energy production in electricity. The generation capacity from coal accounts for 10 percent of the total installed capacity. More exploration work is required to ascertain the quality and extent of the coal deposits for electricity generation. Increased production in electricity benefits the district through the grid

(h) Uranium

Uranium is another potential source of energy that Zambia can benefit from. This source has not been exploited for energy generation purposes.

(i) Petroleum

Petroleum products are a major driver of the economy. They contribute 9.4% to the total national energy demand. Zambia imports petroleum products that include petroleum feedstock and finished petroleum products. This feedstock is imported as a comingled crude oil which comes in to the country through the TAZAMA pipelines and refined by INDENI Petroleum Refinery in Ndola. Refined petroleum products are also imported to supplement what is produced by INDENI. Finished Petroleum products are imported by road, and rail through private suppliers. Petroleum products are distributed from various Government owned depots by Oil Marketing Companies that distribute finished products to the public from their own depots and, service stations and commercial customers. The major players in the distribution of petroleum oil are based in Ndola. This has a significant contribution to the Economy and creation in the district.

3.13. Communication and Transportation Sector

3.13.1. Review of Transport and Communication Policies and Plans

Transport

Social and economic development in Zambia has recorded about 6% growth in the past 10 years as recorded by sustained annual growth Domestic product. The transport sector has particularly played a critical role in facilitating the development process in all economic and social sectors. In Zambia transport infrastructure provision has been steered mainly by the public sector and its development partners while transport services have been largely provided by the private sector.

According to the *2019 National Transport Policy* covering the period 2019 to 2028, the government is aiming at creating an Intermodal transport system that will provide for interlink ages among the four models (Rail, Roads, Air and Water) of transport and ultimately transform

Zambia into a regional transport hub. The sector runs with the vision “***to have an efficient and integrated transport system in Zambia by 2028***”. The rationale for the National transport policy is to ensure facilitation of development process in all social and economic sectors of Zambia.

The world over Information and Communication Technology (ICT) has been identified as an enabler of social and economic development. The most technologically advanced economies are also some of the most developed with advancements in areas of easy access to information and communication technology. Information and Communication is a very fast-growing industry that has gained worldwide recognition in its contribution to global trade and investment. This is noted by the United Nations Millennium Development Goals (MDGs) and the World Summit on the Information Society (WSIS). The push is in the advancement of information and communication as a tool for social and economic development. The development of ICT is key in the enhancement of service delivery such as education, health care, library and information services.

Communication

In recognition of the key role that information and communication plays the Zambian Government developed the **Zambia Information and Communication Policy of 2006**. The policy document provides the Legal and Regulatory framework of Information and Communication development in Zambia. The Vision of this policy is “*A Zambia transformed into an information and knowledge-based society and economy supported by consistent development of, and pervasive access to ICTs by all citizens by 2030.*” Government’s commitment on the development of ICTs is further demonstrated by the 8th National development plan whose aim is to facilitate investment in information and communication technology (ICT) and science and technology to support digital transformation and innovation as key enablers under Economic Transformation and Job Creation.

The integration of digital technologies into business processes will be key to enhancing efficiency and productivity for economic transformation. To advance the digitalisation agenda, the Government will continue with the roll-out of digital communications infrastructure, including the launch of a satellite. This will also ensure that citizens in rural areas are connected to socio-economic services through mobile phones and other electronic devices. Further, the Government will facilitate additional investments in building digital skills, especially among the youth to provide a sound foundation for innovation, including in science and technology. As a result, internet penetration is targeted to increase to 80 percent by 2026 from 52.9 percent in 2020

In line with the national aspirations the district is creating an enabling environment for investment in ICT through the strengthening of the Ndola Chambers of Commerce. The district is prioritizing growth in e-commerce and digitalizing the provision of government services across the district. The district aims at creating a conducive business environment that attracts all sorts of investment including ICT. The district houses the Zambia Telecommunications and ZESCO ICT colleges. Increasing capacity in these institutions will enhance ICT human capital development for not only the district but the Country as whole.

3.13.2. Existing State of Development

3.13.2.1. Availability and Quality of Services

TRANSPORT (Roads)

Legal framework - The Zambia road management is highly centralised in Zambia and it requires more reforms. The public act No 12 of 2002 gives Road Development Agency (RDA) overall all the public roads in Zambia while at the sometime recognising Local Authorities as authorities for rural and urban roads. The Local Government Act Cap 281 empowers local authorities to manage all urban and rural roads. This has lead coordination and financing challenges for the urban and rural roads because of this unbalanced development of various road classifications has been observed. With this said, there is need to harmonise the various pieces of legislation in the Road sector to address the institutional overlaps among key players in the road sector.

State of Roads and quality of service – The socio-economic development and subsequent economic growth of any district is strongly linked to its transport infrastructure. Lack of adequate transport systems and proper road network are some of the factors that may hamper development in an area. Ndola district has an extensive road network that connects and spreads through the settlements. It has an estimated combined total of 552.92 Km of roads broken down into 344.9 km of district roads, 165km urban roads and 43 km feeder roads.

Urban Roads

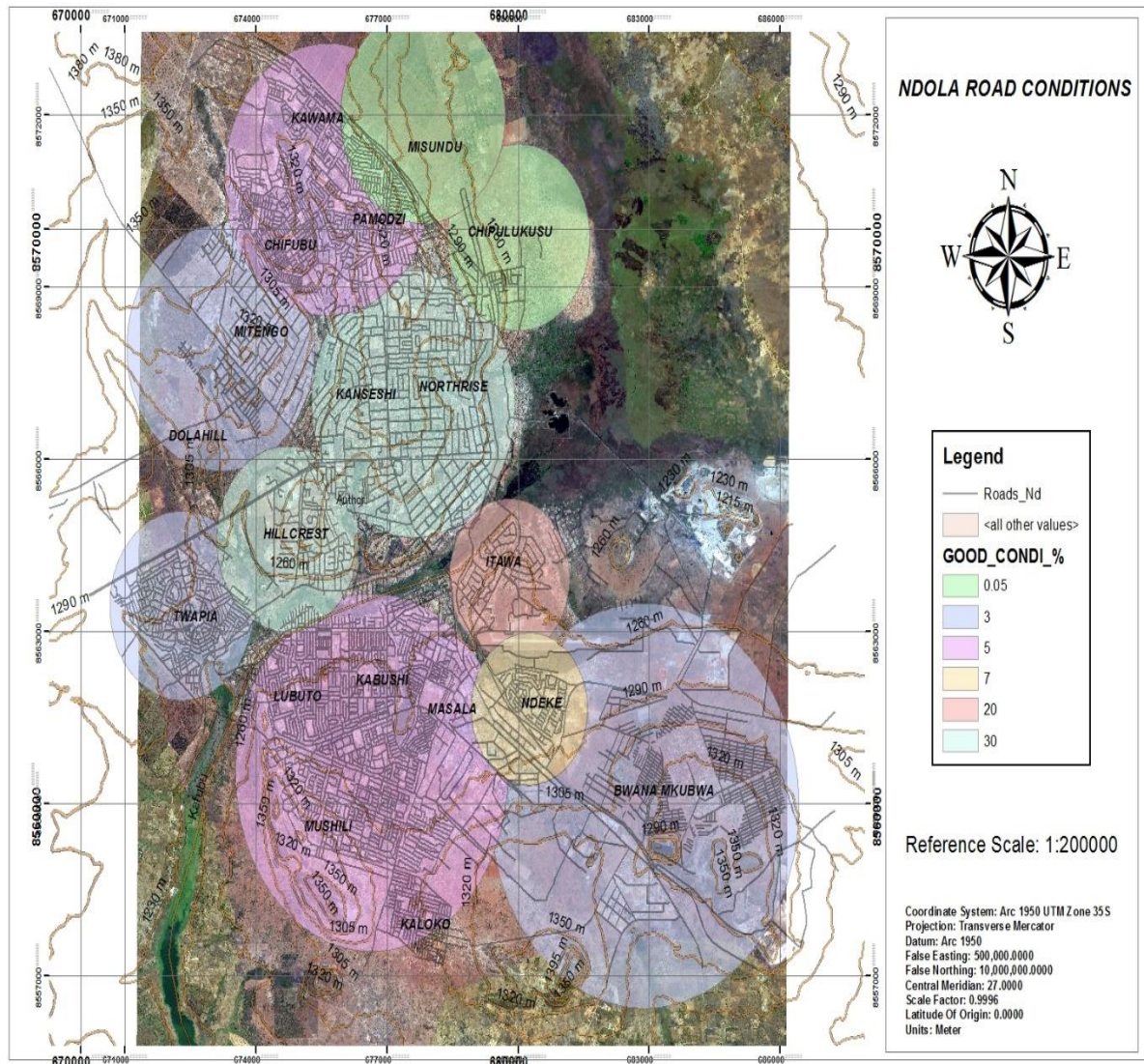
The Local authorities are charged with the responsibility of implementing and maintenance of roads and road related projects. Ndola District has a total of 165km of` urban roads and of which 145 km are paved (bituminous). These roads connect the different settlements with each other thereby providing ease of movement of goods and services. The roads in the Central business district link the business premises such as banks, shops markets and health facilities such as the Ndola Teaching Hospital. The tables below give a list of urban roads in perfect

conditions and those that require rehabilitation and upgrading to bituminous standards. ***(Refer to Annexure 1)***

Feeder Roads

Rural communities have the potential to effectively drive the socio-economic development of districts. Even though urban communities produce food for their own consumption and for commercial sale, this is not sufficient to meet the current demand. It is therefore essential to link the rural producers to the urban market. Rural roads are of a specific significance in achieving this. The Government has been instrumental in spearheading the construction and rehabilitation of feeder roads. Ndola City Currently has a total of 43 km of feeder roads. Rural communities particularly take advantage of this road network to deliver their inputs to the urban communities. A good feeder road network also promotes easy access to facilities such as health posts and schools. The table below shows a consolidated list of priority feeder roads.

S/N	ROAD NAME	DISTANCE (KM)	STATUS
1	Periodic maintenance of Creso – Graeyard Rd	25.00	Contractor mobilized the site/Equipment. Clearing/grubbing (25km). Currently installing Culverts.
2	Contract for the periodic maintenance of Kevu Rd.	18.00	Contractor mobilized the site/Equipment. Currently working on a bridge.



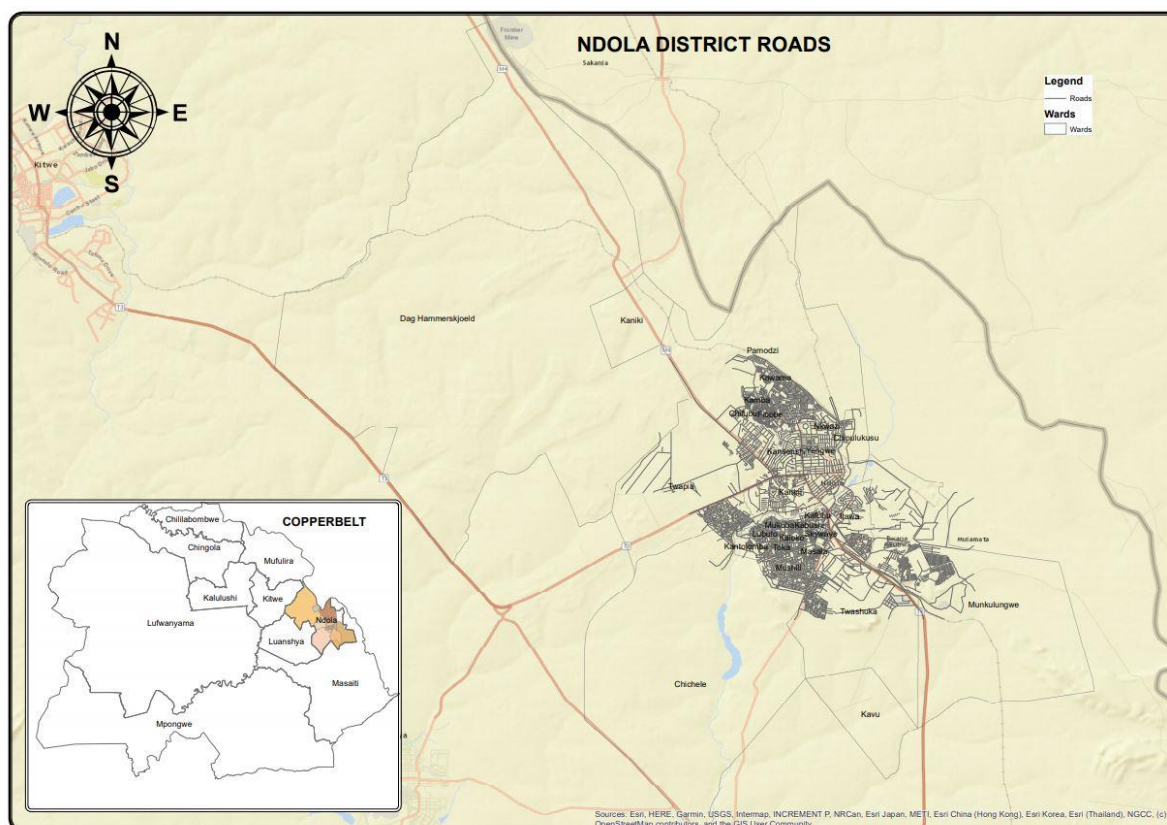
District Roads

The District Roads have the important role of connecting various districts and enhancing transit and movement of goods and services in-between districts. Ndola district inter connects with five (5) other districts in the province namely Kitwe, Masaiti, Mpongwe, Mufulira and Luanshya respectively. The table below shows a consolidated list of District roads.

S/N	ROAD NAME	LINK	DISTANCE (Km)	STATUS
1	T3 SOUTH	NDOLA - MASAITI	51.5	Bituminous road in good condition. Certain sections of the road surface bleeding.
2	T3 NORTH	NDOLA - KITWE	62.3	Bituminous road in good condition. Certain sections of the road surface bleeding.

3	M4 MUFULIRA ROAD	NDOLA - MUFULIRA	69.8	Road in poor state with many potholes on road surface. Road undergoing rehabilitation.
4	D468	NDOLA - MPONGWE	125	
5	M6 VIA T3	NDOLA - LUANSHYA	36.3	Bituminous road in good condition. Certain sections of the road surface bleeding.
	TOTAL		344.9	

The road network of Ndola District springs from 5 main roads which stream out into 46 township roads linking the entire district together with a volume of about 6,325 vehicles passing through daily



Source: Ndola City Council GIS Unit (2021)

Air transport

Legal framework – the Zambian Government enacted the civil aviation Act No 5 of 2016 which provides for the establishment of; the National Civil Aviation Security Committee, the Aircraft Accident Investigation Board, and the Transport Health Board however these boards are yet to be established. The institutional framework is not clear on the administrative and management arrangements of district and provincial aerodromes.

State – the district has the Simon Mwansa Kapwepwe International Airport which caters for both domestic and international flights. The Airport provides 26 domestic and 30 international flights per week.

The local Airline operating at the airport is Proflight Commuters Zambia. The five foreign Airlines operating at Simon Mwansa Kapwepwe International Airport include South African Air link; Kenya Airways; South African Airways; Ethiopian Airlines and Inter air South Africa. However the airport is highly underutilized resulting in low revenue to operation cost ratio and this is mainly because both domestic and international air fares are comparatively high due to lack of competition and utilization rates which are low. **Railways**

The railway sector has over the years had limited public investment programmes. Challenges include the dilapidated permanent way, obsolete locomotives and wagons, inadequate corridor connectivity and development. Due to dilapidation of the permanent way, there are so many speed restriction areas which pose security risks to cargo. As a result, the performance of the railway operators has declined in terms of freight volumes and passenger counts.

Legal framework - the current legislative framework for the railway sector in Zambia principally consists of the railway Act of 1994 and TAZARA Act of 1995. However, the legislative framework is outdated and does not meet the regulatory requirements and railways occupational health and safety standards.

State: The main railway spine through the country is the Zambia railways line from Livingstone, via Lusaka through Ndola to Kitwe. The Zambia railways offer two types of services, passengers' and parcels' services

Parcels services – under this service the intuition conveys parcels or unaccompanied luggage at very affordable rates, goods transported include household items agriculture products, building materials and many other things.

Passengers' services - the transport of passengers, the charge are on average, but the quality of service is lower than that of buses.

The district does not own any railway line in the communities, only benefits from the main railway that spine from Livingstone via Lusaka to Kitwe

Communication

The ICT sector in Ndola District is dominated by four main sub-sectors, namely;

Telecommunications, information technology, electronic media and postal communication

System. In the communication sector, the district has the Zambia Telecommunications Corporation (Zamtel) which is a state owned and sole provider of landlines. The corporation provides telephone, mobile phone and internet services. Zamtel is the biggest provider of communication services. The Mobile Telecommunications sector also has Airtel and MTN. They provide mobile phone communication and mobile phone internet services. Other companies providing internet services in the district include as Iconnect, ZAMNET Africonnect, Liquide Telecom, Copper Net Solutions and many more.

In the postal and courier services we have ZAMPOST, DHL, Mercury couriers, FedEx, Times of Zambia and a lot more other start-ups that include intercity bus transporters that are also offer delivery service.

The District has the following Media houses: Zambia News and Information Services (ZANIS), Times of Zambia, Zambia Daily Mail, Mast News Paper, Mission Press, Muvi TV, Copperbelt Broadcasting Services (CBS), Sun FM, Radio Ichengelo and Radio Phoenix. Komboni Radio and UCZ Radio stations are other radio stations expected to start operating in the district later in the year.

Table: Media Type and Coverage

No.	Media Type	Coverage
1	Zambia News and Information Services (ZANIS)	Entire District
2	Times of Zambia	Entire District
3	Zambia Daily Mail	Entire District
4	Mission Press	Entire District
5	Muvi TV	Entire District
6	Copperbelt Broadcasting Services (CBS)	On Test Transmission
7	Sun FM	Entire District
8	Radio Ichengelo	Entire District
9	Radio Phoenix	Entire District
10	ZNBC (TV LEVY OFFICE)	

Sources: ministry of information & broadcasting services (2019).

The district has six (6) Public Libraries run by Ndola City Council. These are located in Masala, Mushili, Lubuto, Kabushi, Twapia and the main library in the central business district. The Libraries are in dilapidated buildings and have outdated reading material.

3.13.3. Issues from the Public Consultations

3.13.3.1. Availability of service

These are the available services according to the respondents from the 28 wards in the district.

- Feeder roads
- Street lighting
- Drainages
- Bus stations/shelters
- Bridges
- Pathways

3.13.3.2. Quality of service

According to the public about 43% of the respondents from different wards indicated that feeder roads in the wards are in very bad state and some roads are almost impassable during rainy season, while 39% have no bus station or bus shelters as a result public transport like mini buses stop have no designated passenger pickup points along the road leading to unnecessary congestion and accidents.

The public also brought up the issue of drainages, 11 % of the wards presented that they have no proper drainages resulting into floods and numerous port holes and cut away earth in the long ran. 11% of respondents indicated lack of bridges in the wards, while 11 % highlighted the need for direct access routes to their wards. 50% on the other had indicted the need for street lighting in wards across the district and lastly 28% brought up the need for pathways, for further information refer to the annex on issues raised from the community.

Table: summary of quality of services with key indicators according to public consultation

S/N	Service	Expected (%)	Achieved (%)	Variance (%)
1	Feeder Roads	100	57	43
2	Bus station/shelter	100	61	39
3	Drainages	100	89	11
4	Street lighting	100	50	50
5	No direct rout	100	89	11
6	pathways	100	72	28
7	Bridges	100	89	11

Communication

Zambia has continued to record increases in the access and use of information and telecommunication communications. This is evident through the increased numbers of mobile telecommunication and internet access subscribers. In spite of the obvious growth there exist challenges that hinder development in this sector notably ICT infrastructure. There has been slow infrastructure development in the area of Telecommunications such as the major international communication gate ways that are run by Zamtel are not up to modern technological standards, infrastructure both in public and private, is inadequate and fragmented, resulting in poor connectivity, communication and high cost of telecommunication and internet access. Library and information services are in a poor state

3.13.4. Impact of Changes Anticipated Over the Next 10 Years

3.13.4.1. Future demand for services and facilities

Transport

Roads

Currently the population projected population for the district as of 2019 was standing at 5,744,037, with the growth rate of 1.9%. With this growth rate, the population of Ndola by 2031 is projected to be standing at 760,000 this means that they will be more demand in the district on the existing road infrastructure and services. The district will need opening of new roads in certain parts of the district, expanding certain roads from a single lane to double lane, up-grading certain roads to bituminous Standards to avoid have congestion on the existing roads.

Air transport

In the next ten years the population for the district is projected to be standing at 760,000 from the 574,437 projected in 2019. The increase in population will result in increased economic activities and increased demand for air transport as the quickest mode of transport for both cargo and passengers.

The Zambia Airports Corporation limited has for the past few years been engaged in infrastructure developments to upgrade the 4 main international airports in the country. The infrastructure development is meant to accommodate the anticipated increase in traffic, cargo volumes and passengers.

Currently the district has a new airport under construction that upon completion will replace the current Simon Mwansa Kapwepwe international airport. The new airport will have a capacity of one million passengers per annum and the terminal building will include three aero bridges, other facilities will include a 50 roomed luxury hotel, a new fire and rescue service station, a cargo terminal, aircraft maintenance hangar and an airport fuel farm. The overall completion of the project is at 88% as of February 2021.

Communication

The Population of the District is projected to grow at an annual growth rate of 1.9 %. It is estimated that at this growth rate Ndola is projected to have a population of 926,000 by the year 2035. This projected population growth will require significant investment in ICTs infrastructure (infrastructure encompasses telecommunications networks; radio and TV transmission systems; the Internet and other multimedia delivery platforms.) the rehabilitation, construction and modernizing of Library and information services, Human capital development and investments in Research and development.

As espoused in the Zambia Information and Communication Policy, in terms of this agenda Government shall take a leadership role mainly in facilitating the mobilization of investment required for development of a national telecommunications infrastructure backbone as well as the implementation of this policy. Ndola City Council is committed to playing its role in implementing the Zambia Information and Communication Policy at district level collaborating with sector stakeholders.

The District will provide an enabling environment for Public Private Partnerships to support infrastructure development. The District aims at attracting investments in this sector through the provision of land and enhanced service delivery by the local Authority. The District is committed to incorporating ICT in service delivery to enhance efficiency and productivity.

3.13.4.2. Existing and Proposed investment and development programs

The Zambian national railway is very important to the economy of the country as it is bulk carrier with less effect on the environment than any other transport modes. The government is planning on extending the railway line network in the country to develop the surface transport sector. The development of the rail routes linking important exit points is not only vital for facilitating smooth access to the outside but also for the overall boosting of the trade in the sub-region and making Zambia a competitive country for business.

The government intends to rehabilitate the Zambia railways mainline which is 890km and all the other rail lines that are in bad conditions across the country. Installation of an advanced signaling and communication system is another project that the government intends to do to revamp the railway transport system. However the main constraints are the tracks and maintenance as the train speeds are far slower than Road Freight trucks. The main line from Lusaka to Kitwe is yet to be maintained, a distance of about 308kms.

The government also has plans of shifting a minim of 30% from road to railway transport of bulk and heavy cargo, bulk and containers cargo for both local and international transport.

3.13.5. Environment and Climate Change Analysis

Environment and climate change is of greater concern in every developing economy or society especially the construction and expansion of road infrastructure which is directly linked to economic growth, development and social welfare. The constant exposure of environment conditions and climate change to the roads threaten the anticipated economic growth, development and social welfare benefits of the infrastructure benefit.

Opening up of previously accessible areas may have a positive economic benefits but development also comes with a cost e.g. deforestation becomes of the consequence of road networks in previously protected or inaccessible forest areas. Also heavy machines such as bulldozers and hydraulic cause environmental damage by weakening the soil structure when under taking excavation activities.

Currently, there is lack of motorist rest places and solid waste storage and disposal facilities along major transport corridors and roadways leading to pollution of the environment and human distress in general. Further, there is inadequate enforcement of the law that prohibits disposal of waste from moving vehicles

3.13. 5.1. Impact of environment and climate change on the sector

In as much as road construction causes damage to the environment, climate change equally has a negative impact on the roads e.g. heavy rains that lead to massive flooding it damages the road infrastructure and the bridges.

3.13.6. Issues arising relating to gender and vulnerable groups in the sector

Transport

Gender - has not sufficiently been mainstreamed in the transport sector as there are no deliberate programmes in the sector for gender mainstreaming. Most of the jobs ranging from construction to service provision are dominated by males. Most transport is not gender sensitive as it has no provision to cater for the needs of women.

HIV/AIDS - the transport sector has been identified as a key driver of HIV/AIDS. Drivers are mostly made vulnerable due to prolonged period of time spent away from home. International truck drivers are more at risk and are more vulnerable due to distances and exposure to more risk factors. There is also an inherent low health seeking behaviour among men.

Disabled – the transport sector has not come up with deliberate policies that consider persons with disabilities in the transportation infrastructure development. The facilities and construction of both outdoor and indoor facilities prevents them mainly from accessing places using public transportation.

Communication

Generally low literacy levels among vulnerable groups such as Women, youths and people living with disabilities has culminated in Low ICT literacy and skills levels especially among women. This results in basic or lack of use of ICT

Low levels of ICT literacy and skills prevent vulnerable groups from participating in income generating activities. This consequently leads to Low opportunities for access to ICT products and services; and there are entry barriers that include lack of finances and lack of, or limited skills for youths and women especially to venture in business activities as entrepreneurs. Low literacy levels are responsible for poor uptake of Library and information services among vulnerable group. This limits their access to information as well as their participation in social, economic and political. Inadequate number of public libraries

3.13.7. Summary on the Transport and Communication Sector

The transport sector is an important sector in the growth and development of any economy. The local authority together with the help of government should consider investing in the transport sector so that all parts of the district are easily accessible also revamping the railway line which is the most affordable mode of transport for both cargo and passengers. Also the local authority should consider coming with by laws that consider the needs of persons with disability in the infrastructure development

Information and Communication Technology is an integral requirement for national development that provides for social and economic development by enhancing commerce and trade through improved productivity, increased efficiency, easy access to information and sharing, equal participation in social, political, and economic spheres, improved access to education and health services, and breaking barriers of isolation. Successful development of the ICT sector is a must in the development agenda.

3.13.7. Underlying factors contributing to the issues identified

- Erratic flow of funds to the sector
- Lack of maintenance of roads
- Political interference
- Lack of public engagement
- Weak enforcement of laws
- Weak Legal and Regulatory framework
- Inadequate investment in infrastructure development
- Low literacy levels in that impede uptake of ICT
- Very High Capital requirements for ICT business Ventures
- The cost of acquiring ICT technology is high, this makes ICT technology and skills costly

- Brain Drain of the few ICT trained personnel leading to loss of skills
- Limited local ICT industry thus offering inadequate services and few job
- Inadequate funding to Library and information services

3.13.8. Summary-Core Issues

- Lack of maintenance on all the roads in the district
- Limited street lighting on most of the roads
- Very few drainages
- Lack of access roads in most farming areas
- Lack of consideration of persons with disability in transport infrastructure
- Lack of motorist rest places and solid waste storage and disposal facilities along major transport
- Limited designated bus stops in the communities
- The district has no formal intercity bus terminus
- The district has no Dry ports for accommodation of trucks (consider PPP arrangements).
- Unreliable railway transport for both cargo and passengers
- Inadequate infrastructure in Information and Communication
- Fragmented information and communication technology
- High-cost access to of information communication technology
- Inadequate number of Libraries

3.14. Health Sector

3.14.1. Review of Health Sector Policies and Plans

Zambia still experiences a high disease burden despite making tremendous progress in some selected indicators. According to the Zambia Demographic Health Survey (ZDHS) 2018 report, infant mortality decreased from 45 to 42 deaths per 1,000 live births, while under-5 mortality decreased from 75 to 61 deaths per 1,000 live births. While the maternal mortality (MMR) decreased from 398 per 100,000 live births to 252 per 100,000 live births, the health sector has also recorded remarkable progress towards HIV/AIDS free-country 2030 target. For instance, the recent ZDHS (2018) survey shows a reduction in HIV prevalence from 11.6% in 2016 to 11.1% in 2018 although, our province Copper-belt leads with the highest HIV prevalence of 15.4% in the country.

In addressing the management of high disease burden, the Ministry of Health has embarked on programs to construct more health facilities and expand the already existing ones in order to meet healthcare service demands of the growing population, and delivery healthcare service to as close to the people as possible. According to the Zambia National Health Strategic Plan 2017-2021, the target of reducing the maternal mortality ratio from 252 deaths per 100,000 live births in 2018 to 100 deaths per 100,000 live births by 2021 still remains the same (MOH 2016). The plan also aims to reduce adult mortality from 24 to 12 deaths per 1,000 populations in the same period.

The Ministry of Health is still committed to and focused on the provision of a continuum of care with particular emphasis placed on strengthening health systems and services using the primary health care (PHC) approach. To this effect, 8 health posts under the 650 Health posts

project were constructed in 8 Ndola district. The ministry has also embarked on the construction of 4 mini hospitals and two district hospitals. The 2017-2021 National Health Sector Strategic Plan (NHSP) covering five-years provides guidance on all health interventions in the health sector. As part of the realizing the vision of the Ministry of Health under the guidance of the 2017-2021 National Health Sector Strategic Plan, the Ndola Teaching Hospital (NTH) underwent through modernization phase with the installation of computerized tomography (CT) scans, mammography equipment. As of December 2020, all the 8 of 650 health posts were built and are now operational. Furthermore, one (1) district hospital will be under construction with support from the JICA of Japan.

Legal framework - The National Health Policy outlines a statement by the Zambian Government to set clear directions for the development of the Health Sector in Zambia. The policy is anchored in the Vision 2030 and shall be implemented through successive National Development Plans and National Health Strategic Plans. It sets out policy measures that guides strategies and programmes in the health sector. This policy document underscores Government's commitment to provision of equitable access to cost effective and quality health services as close to the family as possible in a caring, competent and clean environment. In this regard, Government prioritizes among other programmes, primary health care services, hospital referral services, human resource development and management, medical supplies and logistics, infrastructure development, legal framework and health care financing.

The National Health Policy has taken into consideration emerging issues at both the national and international levels such as communicable and non-communicable diseases, health systems strengthening, gender equality, globalization, and climate change.

State of Health facilities and quality of service – Ndola District has 36 public health facilities which provide different range of healthcare services depending on the level of care. Ndola district does not have a second level hospital at the moment. Ndola Teaching Hospital currently assumes the role of provision of second level hospital services. However, the available health facilities are at different levels such as health post, health centre, and first level referral hospital up to third level hospital.

However, the health sector still faces challenges in the provision of quality healthcare services. For instance, the Covid-19 pandemic disrupted the normal provision of healthcare by diverting the resources (both financial and human resources) towards the existing programs and also disrupted supply chains in countries where many of the medical supplies and drugs are manufactured.

3.14.2. Existing state of Development

Currently, the District has 4 level three (L3) hospitals, out of which 2 are privately owned and 2 level one hospitals, 28 urban health centres of which 19 are government owned and 9 are private, 1 rural health centre and 8 health posts. In addition to the two hospitals, the District is host to the Zambia Flying Doctor Service, which offers outreach activities throughout the

3.14.2.1 Quality of service

INDICATORS	2020		
	TARGET	ACHIEVED	REMARKS
% under one year children fully immunized (0-11 months)	85%	95%	The target was met
% BCG coverage	100%	96%	The country ran out of BCG vaccine in quarter 1 due to Covid-19 pandemic
% new family planning acceptors	-	37%	This indicator doesn't have any target
% pregnant women who received ANC 1+	60%	25%	This was also affected by the Covid-19 pandemic
Screening for TB in HIV Care	100%	98%	This was also affected by the Covid-19 pandemic
Delivered by skilled provider	65%	86%	The target was met
% of people currently on ART (HIV treatment)	95%	106%	The target was met
% of women attending postnatal care visit within 6 days and 6 weeks	20,197	17,543	This was also affected by the Covid-19 pandemic

The health sector performed well in 2020 in many key performance indicators as can be seen in the above table. Although, the quality of services in some indicators such as the BCG coverage and child immunizations were adversely affected by Covid-19 pandemic. For instance, the Statutory Instrument (SI) on public health guidelines restricting social gatherings and fear of the disease discouraged children care takers from visiting the facilities. There also was national stock out of BCG vaccine in the first quarter of 2020 due to supply chain interruption as result of lockdowns in the countries where these supplies came from. Some health facilities had given up some health personnel who were needed to work at the identified

Covid-19 center while; some staff became victims of the disease. This generally negatively affected the quality service delivery across the communities.

On the other hand, the Covid-19 has had a positive impact in that it promoted hand washing hygiene practices which somehow helped to prevent diarrheal diseases like cholera.

3.14.2.2. Issues immerging from public participation

The during public consultations, members of the general public identified the following as major issues affecting health care delivery in Ndola District;

- Long distances covered by members of general public when accessing health care facilities.
- There were concerns about the gender composition of health care providers, especially antenatal services (mostly couples expressed concern on being attended to by males during antenatal visits)..
- Majority participants noted with concern the negative attitudes of some health care providers, a situation which left many wondering the caliber of some health care workers. This further discouraged many from accessing health services from public health delivery facilities. Further, majority of the health facilities in the district lack specialized personnel.
- None availability of essential Drugs, Medical Supplies, motorized transport and equipment, which culminated into inefficient delivery of health care in the district. Further, general health care infrastructure is inadequate especially maternity wards and mortuaries.

3.14.3. Impact of Changes Anticipated Over the Ten Years.

It is anticipated that there will partial breakdown in the delivery of health care services in the district if measures are not put in place to address the issues immerging during public participation. As economic activities improve in the district, so will the population as the district experiences increased migration of people from different parts of the country. This development will further strain the already stressed health care facilities in the district.

3.14.3.1. Future Demand for Services and Facilities

The table below summarizes long term future demand on the delivery health care services in Ndola district as a result of population increase.

Population change	Demand of services
Increase in the Reproductive age (15-49)	• High Demand for family planning services • High Demand for maternity services.
Increase in the number of children below 12 months	• High Demand for Immunization services. • High demand for Nutritional services.
Increase in Non-Communicable Diseases due to lifestyle changes and increased life expectancy.	• More demand for services such as treatment of diabetes and blood pressure

Increased risk of modifiable diseases including those of international health concerns due to the presence of international airport and increased cross border trading	• High demand for port health screening services
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3.14.3.2. Existing investment and proposed development program

Currently, the District has two (2) third-level (specialized referral) hospitals of which both are Government owned facilities, one (1) first level hospital and 36 health centers and health posts. In addition to the two hospitals, the District has Zambia Flying Doctor Service, which offers outreach activities throughout the country, and Tropical Diseases Research Centre (TDRC), which is a disease research Centre. Arthur Davison Children's Hospital is the third level referral Institution which offers exclusively Children's (below 15 years of age) health services while Ndola Teaching Hospital attends to adults (above 15 years' population) in Ndola. In terms of Policy, Arthur Davison Children's Hospital provides health care services to children from Zero (0) up to fourteen (14) years, while Ndola Teaching Hospital attends to residents above 15 years old.

Present Staffing in the District

1. Table : Staffing Levels by Category of Staff

Category of Staff	Establishment			Existing		
	DHO	Hosp.	H/Cs	DHO	Hosp.	H/Cs
Medical Doctors	4	0	6	3	1	6
Nursing Staff	25	17	14	25	38	393
Clinical Officers	7	3	34	7	2	5
Medical Licentiates	0	0	1	0	2	0
Environmental Staff	4	1	32	4	1	48
Paramedical Staff	12	3	93	6	10	93
Non-medical professional staff	20	0	0	20	0	0
Other	37	18	272	35	6	286
Total	109	42	852	100	60	861

3.14.4. Environment and Climate Change Analysis

3.14.4.1 Impact of existing trend on the environment and climate

The construction of the existing Infrastructures (Hospitals and Clinics) has been necessitated by the clearing, or otherwise removal of trees through deliberate means. This has mostly occurred in areas densely populated by trees and other plant life, but the majority of it has happened in the rural areas. This clearing of trees, also called Deforestation has greatly contributed to climate change, desertification, soil erosion, fewer crops, flooding, increased greenhouse gases in the atmosphere, and a host of problems for indigenous people.

The burning of solid and regulated medical waste generated by health care aggravates to the climate change challenges. Medical waste incinerators emit dioxin air pollutants and toxic ash residues that are the major source of toxins in the environment.

3.14.4.2. Impact of environmental and climate change issues on the sector

The health impacts of climate change include increases in heat-related illnesses and death; extreme weather-related injuries and mortality; aggravated chronic illnesses; spread of infectious diseases: vector, zoonotic, water and food borne diseases; increases in asthma, respiratory allergies, and chronic respiratory disorders; growing malnutrition and child development complications; increases in stress-related and mental health disorders; growing health impacts related to both population displacement and migration; as well as climate-triggered instability and conflict. The healthcare sector is also just beginning to understand that climate change will have major impact on health care costs, services and delivery.

3.14.5. Issues arising relating to Gender groups and vulnerable groups

Two to three times more women than men experience mental health problems like depression and anxiety. Harmful stereotypes about sexuality and body image play a part in this. The gender pay gap and Inequality at work puts women at higher risk of physical and mental illness. Gender-based violence and harassment affect how and when women access and use public services and spaces.

As the above cases rise due to increased population in the district, coupled with a number of factors including but limited to economic, social and cultural dynamics, the demand for specialized health care provision will be high with likely disturbance to accessing such services by the greater population.

3.14.6. Summary

The successful attainment of Zambia's goal of being a prosperous, middle-income country by 2030 as stipulated in its Vision 2030 begins with transforming Ndola into a City of healthy and productive people. Therefore, the Ministry of Health (MOH) through the District Health Office, 2nd and 3rd Level Hospitals has continued to prioritize health service provision as a pathway to achieving this.

Each institution has a specific role to play in the provision of the delivery of service. However, quality of service is being compromised as the Department of Health does not achieve all the indicator targets set forth by M.o.H. The issues arising from the Public Participation clearly indicates Challenges the Department of Health is currently facing towards meeting the Ultimate targets. These challenges are but not limited to the Inadequate Medical Personnel, Support Staff and inadequate Equipment's, Infrastructure and commodities to ensure the provision of effective quality service.

As evidenced by the demographics at the beginning of this document, there has been an increase in Population grown. This would mean an increase in demand for the services such as Immunization, family Planning and Delivery services. To accommodate these services, there would be need for more 1st Level Hospitals, Clinics and Health Posts to cater for the increasing population. Failure to expand the services would result in congested Existing Facilities making it difficult for patients to access health services.

The construction of the existing facilities has caused deforestation as a result of the clearance of land; the incineration of Medical waste has also caused the Air Pollution to the People of Ndola. On the other hand, the health impact of climate change includes among others: increases in heat-related illnesses and death; extreme weather-related injuries and mortality; aggravated chronic illnesses; spread of infectious diseases: vector -, zoonotic-, water- and food- borne diseases; increases in asthma, respiratory allergies, and chronic respiratory disorders.

These effects may affect all individuals and notably, it could affect Gender in a different way, for instance, 2 to 3 times more women than men experience mental health problems like depression and anxiety and men are more vulnerable to chronic illnesses such as Stroke and diabetes.

3.15. Cross Cutting Issues

Gender

The pursuit for gender equity has evolved from advocacy, negotiation and consensus building, awareness rising on the importance of gender equity. Gender considerations are an obligation in development programming and implementation. The National gender policy of 2014 aim at “ensuring the attainment of gender equality in the development process by redressing the existing gender imbalances. For Equitable and Sustainable development to be attained gender equality must be achieved. However, gender inequalities are still very high between males and females in the District and the country at large. According to the 2010 Census of Population and Housing Report, Ndola District had a total population of 451,246 with a distribution of 223,020 being male and 228, 226 being female. The populations of females have been slightly more than males.

Unequal Access to Resources Women have unequal access to opportunities compared to their male counterparts in terms of access to jobs, women are in the in the lowest paying sectors and non-technical jobs, which are linked to their lower education levels. This is as a result of high illiteracy levels among women compared to men. In Ndola school attendance between boys and girls is at 93% and 86% respectively. According to the 2010 Census unemployment was

higher among women compared to men with rates of 24.20% - 20.80%. Employment rates indicate 63.2 % for men compared to 37.7 for women. The district has a disproportionate ratio in terms of property ownership such as land between male and females which favors males.

Poverty Due to the existing gender inequalities the Impact of poverty is more devastating among women compared to men. This is as result of significant differences in opportunities available between the sexes. According to the Living Conditions Monitoring Survey (2010), overall poverty is marginally higher among female populations at 80 per cent compared to the male population at 78 per cent.

Decision Making In Ndola decision making have generally been dominated by men. Women have been under-represented at all levels of national decision-making especially in government, parliament, the private sector, special committees, religious groupings, boards and other institutions in the community.. Gender equality can only be archived through women's participation and in decision making at all levels, starting in the home and extending to the highest levels of government. The fact that gender inequalities continue to exist in all sectors at all levels, has negative implications on national development.

Traditional & Culture Beliefs In Ndola, gender inequalities are perpetuated through cultural roles across ethnic groups that are determined and by one's sex, socialization and culture. Cultural norms, values and taboos that lead to the perpetuation of gender imbalances are transmitted through the socialization process. For instance, girls are socialized to be wives and mothers who are generally submissive while boys are socialized to be husbands, fathers and leaders. Girls and women are overburdened than boys and men in the sharing of roles particularly at the household and community levels.

Gender based Violence (GBV) Gender Based Violence (GBV) continues to be a huge problem in the district. This is despite various legislature and policies such as the Anti-Gender Based Violence (GBV) Act No. 1 of 2011. . The 2007 Zambia Demographic Health Survey (ZDHS) indicates that one in five women has reported having experienced sexual violence at some point in her life. The majority of GBV cases are violence against women. These are mainly in the form of physical violence, Sexual assault.

Force, early and child marriages According to the 2014 National gender policy "Zambia has one of the highest child marriage prevalence in the world. On average 2 out of 5 girls are married before their 18th birthday. It is worth noting that of these married girls, 65 per cent have no education while 58 per cent have only primary education as compared to 17 per cent of girls with secondary education".

Legal reforms and Gender inequalities in Ndola not only imitate from traditional, cultural and social norms but also from the dual structure of statutory law and customary law. Rights, which are supposed to be protected under statutory law, are not necessarily observed and women endure unfair treatment in terms of child marriage, unequal distribution of property, etc. Meanwhile, there have been some positive developments at the policy level, including the establishment of an independent Ministry of Gender, introduction of specific national and local

authority level gender policies and revision of certain provisions of the Constitution, which epitomize gender inequality (currently being deliberated).

HIV/AIDS prevalence rate in Zambia is higher among women (16 percent) compared to men at 12 per cent (ZDHS, 2007). The incidence rates are at 2 percent for women and 1.6 per cent for men.

The National HIV and AIDS Strategic Framework (NASF 2018) reveals that i) There is a lack of gender sensitive HIV and AIDS workplace policies including Gender responsive HIV and AIDS national action plans. ii) There is a lack of prioritization of gender in most planning, implementation and evaluation of HIV and AIDS initiatives. iii) There is a lack of representation of gender experts in most of the key national institutions dealing with HIV and AIDS, leading to inadequate capacity to analyze and incorporate gender.

Gender, Environment and Climate Change: Climate change as a development subject matter is critical for the District where the majority of the populations are female small holder farmers and they rely on agriculture for livelihoods. The gender differences between men and women imply that their vulnerabilities differ. Since women are already in a disadvantaged position, effects of climate change threaten to further increase the inequality.

The Government of the Republic of Zambia has ratified 22 regional and international instruments/protocols including SADC Protocol on Gender and Development and the African Charter on Human and People's Rights; and the UN Conventions on the Elimination of all forms of Discrimination Against Women (CEDAW) and the Rights of the Child (CRC) which provide standards and policy guidance on the socio-economic empowerment of women and children. (National Gender policy, 2014). The District is committed to mainstream gender in all aspects of Development in line with these international commitments. Gender mainstreaming is a matter of protection of basic human rights. The District can only attain meaningful and sustainable development by addressing the existing gender imbalances. This will be achieved through the provision of equal opportunities for women and men to actively participate and contribute to their fullest ability and equitably benefit from district and national development.

Gender mainstreaming involves awareness rising, Capacity building, expanding choices and increasing access to, and control over resources. The important instruments of empowerment include information, women acquiring knowledge, skills and a willingness to critically analyze the situation and take appropriate action to change the status quo in society. Empowerment involves practical measures to enhance women's participation in decision-making and in governance processes, and generally to uplift their status through literacy, education, training and raising awareness. Other actions include poverty reduction programmes, that themselves involve income-generating activities and enhancing access to job opportunities. Underlying issues are taken into account such as the protection of a woman's human rights, her reproductive health as well as property ownership.

The district with support from various stakeholders has developed a district Gender policy. The District Gender policy applied with all relevant national laws and policies aims at providing a framework from which Gender equality can be archived. The Policy is premised on the human rights approach that seeks to improve the rights of women, men and children. This Policy rests on the principle that women and men are equal in all respects.

To achieve gender equality the District focuses on six thematic areas:

- 1 **Legal Frame work** ensuring that the constitutional and legislative provisions on gender justice are implemented and gender equality targets set in the national and international and regional protocols, to which Zambia is part, are achieved. This is through the provision of an enabling legal & policy environment where stakeholders operate based on their varying mandates. The district provides leadership through a coordinated gender mainstream committed in conjunction with Government line ministries, NGOs, CSOs Zambia police and Civil Society Organizations.
- 2 **Gender and Economic Empowerment:** promoting equality and equity in access to economic opportunities for men and women. With support from all stakeholders the provision of women Economic empowerment programs. The provision of skills and knowledge acquisitions programs targeted at women. The promotion of equal access to education for boys and girls and their retention at all levels of education and access to training opportunities for men and women, to make possible their equal participation in the workplace, marketplace and in governance structures. The district aims at providing an environment that fosters equal access to employment and resources such as land and basic services. As such, the economic empowerment of women has become a key priority for the district to achieve sustainable economic growth. Programs such as Women's Savings and Lending Schemes by various development agencies, Business platforms and Women's Associations are key. The support of such platforms is a requirement in the elimination of poverty among women thus reducing vulnerability.
- 3 **Gender, Politics and Decision Making:** the district aims to create a supportive environment for gender parity in politics and decision-making positions. The district seeks to provide an enabling environment for women participation in governance and decision making. Participation in decision-making includes the active involvement of women's organizations and the broader civil society, in processes such as participatory planning mechanisms and in budgeting.
- 4 **Gender Based Violence:** The District aims to eliminate all forms of Gender Based Violence. A coordinated GBV response that supports stakeholders such as the Zambia police VSU and the broad spectra of Civil society organizations that provide serves such as Sensitization programs, care and support of Victims, empowerment programs as well as prosecution of perpetrators of the GBV are in place. It is acknowledged that gender mainstreaming is a multi-sectorial issue and as such a coordinated approach is a necessity.

- 5 **Gender, Environment and Climate Change:** the District encourages gender responsiveness of the environment and natural resources management strategies and in climate change adaptation and mitigation initiatives. By undertaking adaptation and mitigation programmes for building resilience that target the most vulnerable who are women and children. Developing strategies for mainstreaming gender into all climate change programmes.
- 6 **Tradition and cultural** practices tend to support the sterol type that men are superior to women. The district aims at targeting traditional and cultural practices which inhibit the advancement of women, men and children in order to curb them. The district through stakeholders is providing awareness and education as a strategy to combat negative traditional and cultural vices.

Disability

There is broad evidence that persons with disabilities are more likely to live in poverty than those without (Banks and Polack, 2014; WHO and World Bank, 2011). A disability affects not only the person who suffers from it but has an impact on all members of a household. According to Mitra, Posarac and Vick (2013), households where a member has a disability were significantly more likely to be classified as living in poverty. Poverty and disability reinforce one another. Poor health and nutrition, poor living conditions, poor access to health services, environmental risks and injuries among persons living in poverty can lead to disability. Equally, the onset of disability can have an adverse effect on education, employment and earnings, increase living costs and result in higher rates of poverty (Groce and others, 2011; Mitra, Posarac and Vick, 2013; WHO and World Bank, 2011; Yeo and Moore, 2003).

Persons with disabilities face physical and social barriers that hinder their access to services or employment and prevent them from enjoying their rights. The design and construction of indoor and outdoor facilities can prevent them from going to school and hospitals, shopping, gaining access to police services and finding or keeping a job. Footpaths, parks and public transportation may also be inaccessible, preventing some persons with disabilities from enjoying the most basic elements of participation in social life. The forms of disability include Blindness, Partial Sightedness, Deaf and Dumbness, Hard of Hearing, Dumbness, Mental Illness, Intellectual disability, Speech Impairment, Physically Disabled, epileptics and leper and Mental Retarded:

The proportion of the population with disability in Ndola district was at 1.4 per cent of the population. Distribution in terms of sex was 1.7 per cent in males and 1.4 per cent in females. Physical disability was the most common type of disability at 32.4 per cent on the Copperbelt province. The major cause of disability was disease at 31.3 per cent. The literacy rate for the population with disability in Ndola District was 74.4 per cent. Generally, the highest level of education attained by the majority of the population with disabilities regardless of sex was primary education. The proportion of the population with disabilities who were employed was

83.7 per cent. Most of the population with disabilities had agricultural related occupations at 59.0 per cent.

The Convention on the Rights of Persons with Disabilities the United Nations Convention on the Rights of Persons with Disabilities, which entered into force in May 2008, was the first international treaty to detail the rights of persons with disabilities and set out a code of implementation. The purpose of the Convention is to promote, protect and ensure the full and equal enjoyment of all human rights and fundamental freedoms by all persons with disabilities, and to promote respect for their inherent dignity

Ensuring mainstreaming of all disability perspective in all aspects of Development, policy and legislation, effective implementation and enforcement of existing disability laws and policies by providing for equal employment and Education opportunities are among the factors that can contribute to the reduction of poverty and to the social and economic inclusion of people with disabilities in Ndola. Ndola District is committed to:

- Equip disabled people with skills to do wood and needle work.
- Provide disabled people with the ability to work and earn and income. Productive and decent work enables people with disabilities to realize their aspirations, improve their living conditions and participate more actively in society.
- Raising self-esteem of disabled people through the Development of life skills and the elimination of stigma and all forms of discrimination.
- Allow disabled people to use and advance their own working skills.
- Opportunity for disabled people to start their own business to provide them with the opportunity to provide for their families.

Youth

According to the 2010 Census of population and Housing Ndola District, has a youthful population accounting for 40.1 per cent of the total population. Ndola has high rates of youth unemployment. According to the 2010 Census of population and Housing Ndola District youth unemployment was at 32.3 per cent. High unemployment rates have increased poverty and vulnerability among youths. Youth unemployment is exacerbated by lack in the kind of education, training and effective vocational guidance that would match industry needs. Many young people drop out of school. As a result, youths survive on informal work such as waiters, taxi drivers or barbers. Crime and drug abuse are a common result of youth unemployment. High rates of HIV/AIDS infections among youth attributed to unemployment that lead to risk behaviors such as prostitution, Substance and drug abuse. Early marriages and early childbearing among the leading challenges among youths.

In line with the 2030 Agenda for Sustainable Development, SDGs and the National youth policy of 2015 the District is committed to the key policy priority areas that include employment and entrepreneurship development; education and skills development; health and cultural, creative industries and sport. Cementing these policy areas are crosscutting issues such

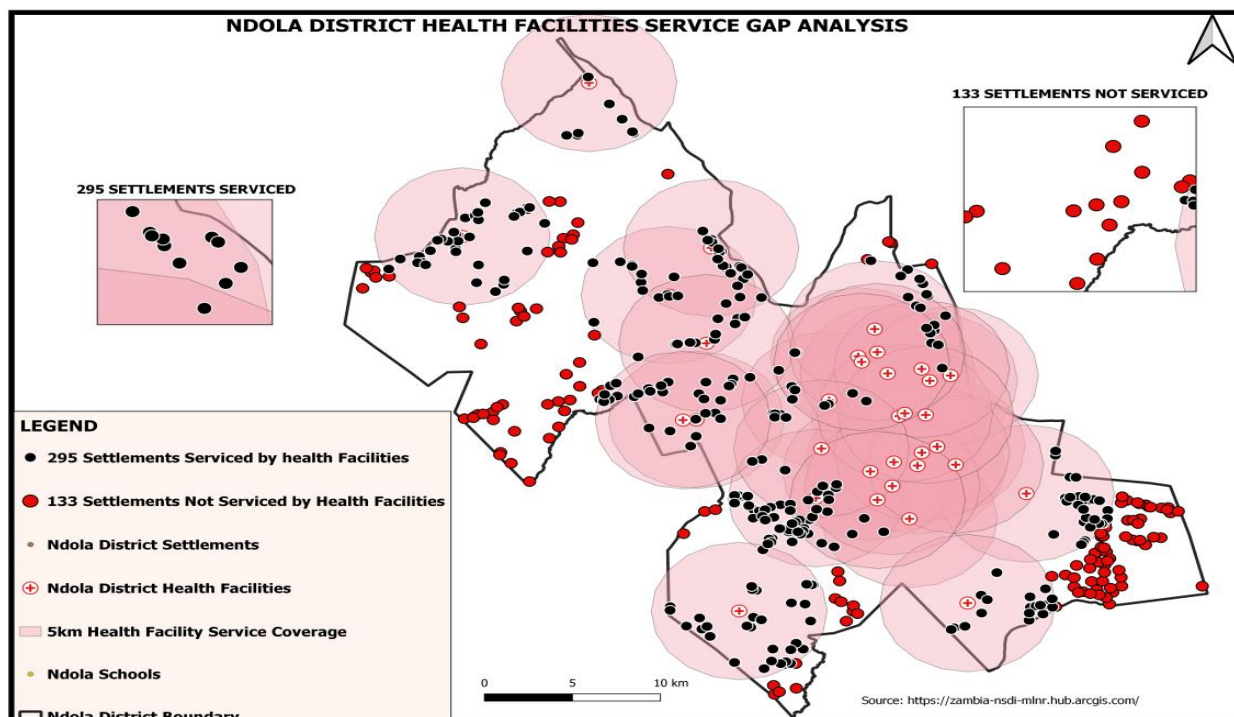
as gender, disability, environment, HIV and AIDS and participation. It is envisaged that the latter can be harnessed by inclusion and active participation of youth in decision-making processes, policy development, programming and project implementation (National Youth Policy 2015). This will be achieved through mainstreaming youth development in all District activities.

- i. Through youth targeted policies with support from both the public and private sector support the creation of employment opportunities through the growth of youth-led initiatives;
- ii. Mainstream youth development through participation of youths in the socio-economic development of the country;
- iii. The provision of skills and vocational training to out-of-school and unemployed youths to venture into sustainable and viable income generating projects;
- iv. Promote the development of competitive sustainable and growth-oriented citizen owned youth-led enterprises;
- v. Provide youth empowerment funds to youth led businesses and continued support services for the development of sustainable youth-led initiatives development
- vi. With support from various stakeholders the provision of Mentorship and Leadership programs.
- vii. Infrastructure development that supports youth empowerment such as, markets, sports and smallholder farming schemes.

District HIV/AIDS Profile

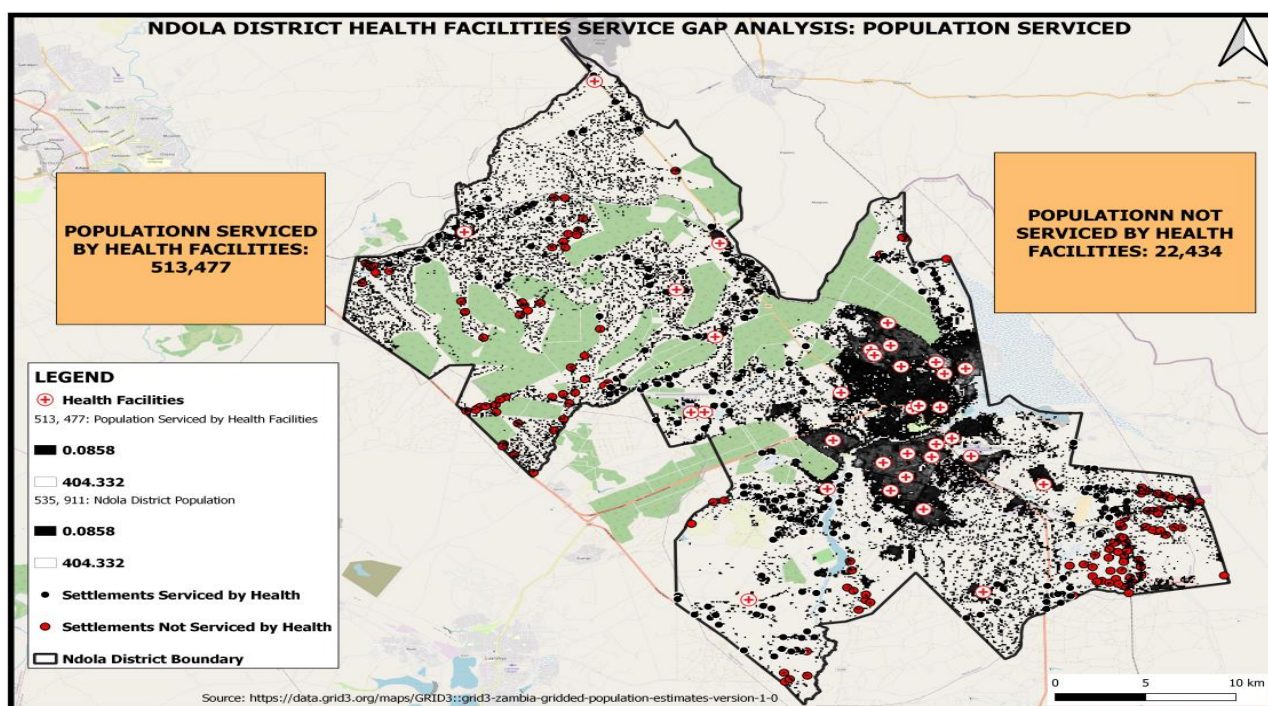
The district has one of the highest HIV prevalence rates in Zambia, with a 2007 adult HIV prevalence rate of 23.5% (NACMIS). This makes it one of the five (5) worst affected districts in the country. The projections of the HIV prevalence rate in the district have been decreasing, from over 29% in 1995 to an estimated 23.5% in 2010 (CSO, 2005). Since 2004 there has been a reduction in the projected number of new HIV cases annually, from 7,808 in 2004 to 7,537 in 2010. Despite these reductions, the HIV figures in the district are still high and as a response to this situation, the District HIV/AIDS Committee continues to mobilise all relevant stakeholders to prevent new infections, provide treatment and mitigate the impact of HIV/AIDS as well as coordinating the response in the district.

General Health Services



Source: Ndola City Council GIS Unit (2021)

The Ministry of Community Development Mother and Child Health (MCDMCH) provides health services through the District Community Health Office (DCHO) in twenty-two (22) health facilities dotted around the townships and peri-urban communities. Nineteen (19) of these are concentrated in the urban communities while the remaining three (3) are in the sparsely populated peri-urban areas. The services include among others, laboratory, radiology; and Voluntary Counselling and Testing (VCT) services including prevention of mother-to-child transmission (PMTCT). The Ministry of Health (MoH) has two hospitals (Ndola central and Arthur Davison Hospital) which cater for the rest of the Copperbelt Province for all the referral services. Ndola Central Hospital offers services to adults whereas Arthur Davison Hospital offers services to children below the age of fourteen (14). The above Government facilities are supplemented by an additional twenty (20) private facilities which provide data and feedback to the DCHO.



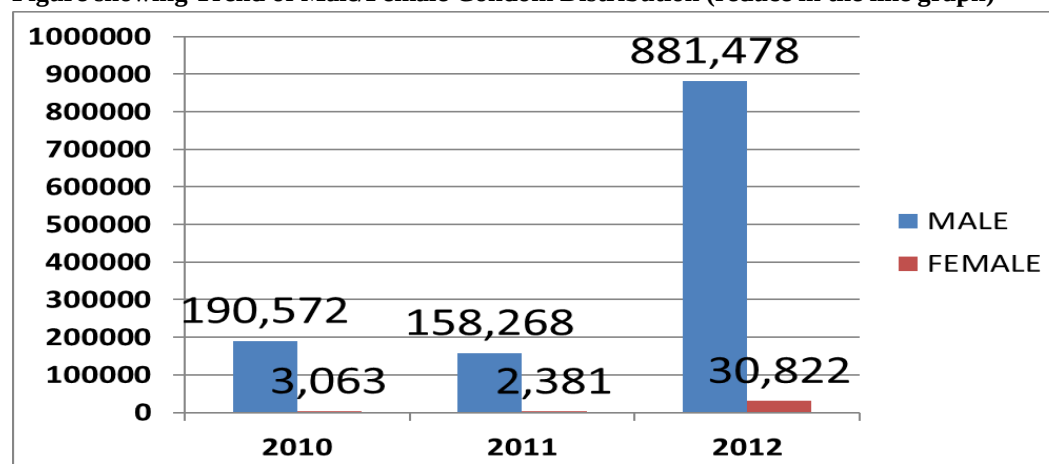
Source: Ndola City Council GIS Unit (2021)

HIV Prevention

The services that are being offered to reduce new infections in the city include condom distribution, Counselling and Testing (CT) Services, Voluntary Medical Male Circumcision (VMMC) and PMTC.

Condom Distribution: Apart from the 22 Government health centres, the city has non-health facilities that distribute condoms to end users. These include CBOs, support groups and the private and public sectors mainly through the workplace programs.

Figure showing Trend of Male/Female Condom Distribution (reduce in the line graph)

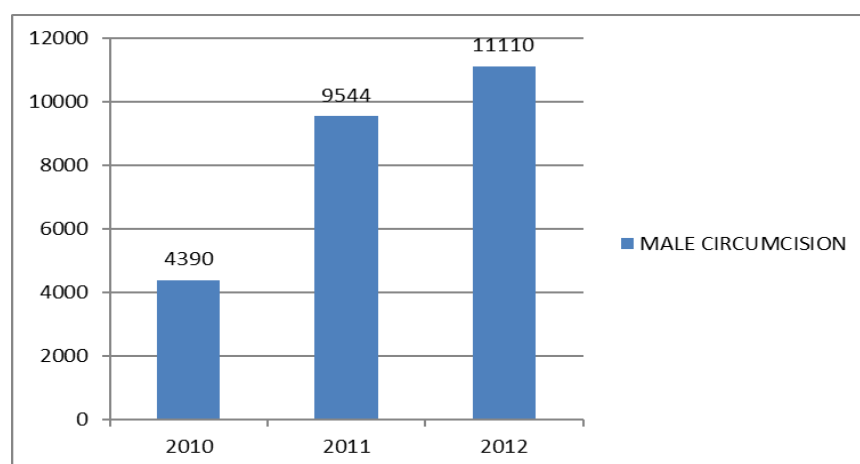


Source: Ndola HMIS 2010-2012

The number of condoms distributed for both male and female has increased by more than 100% and this shows that the district has improved in condom distribution compared to the last three years. According to the CSO 2010 Census Report, there are 116,364 males and 118,557 females in the district between the age of 15 and 49. Based on these statistics, the district should distribute at least 150,000 to 200,000 male condoms per month to meet the target. Research need to be taken to ascertain correct use. There is also need to promote female condoms, as the number of female condoms being distributed in the district is very low despite proof that the female condoms are effective in HIV prevention.

Male Circumcision: Voluntary Medical Male Circumcision (VMMC) is one of the key interventions in the prevention of new HIV infections as it has been scientifically proven to provide 60% protection against the transmission. In Zambia low MC uptake has been identified as a key driver of HIV. In Ndola, there are fourteen (14) facilities offering this service although the surgical operations are done by using equipment that is normally taken away at the end of time by the partners thereby posing a challenge for sustainability of the activity. Ten of these are offered by the government facilities and four (4) others by civil society organizations such Jhpiego, Society for Family Health, Zambia Prevention, Care and Treatment (ZPCT) and Marie Stopes. The Figure below shows trends in accessing of Male Circumcision between 2010-2012 as provided by Society for Family Health and (ZPCT).

Figure showing Male Circumcision



People Living With HIV/AIDS: Some people living with HIV and AIDS work in the informal sector, where services such as health insurance schemes and support services are not always available. Consequently, given the loss of income due to illness, they are often unable to meet their basic needs, and transport and nutrition remain a daunting challenge that contributes to many PLHIV not adhering to ART. However, in Ndola city more people living with HIV/AIDS are accessing community care, share information on ART and other opportunistic infections, adherence and general information on HIV/AIDS, and psychosocial support through support groups.

Coordination and Response Management: The District HIV/AIDS Committee (DHAC)

The DHAC coordinated by the District AIDS Coordinating Advisor (DACA), is responsible for coordinating all HIV related initiatives being undertaken by other institutions and organisations in the district and for overseeing M&E using the National AIDS Stakeholder Reporting Form (SARF).

Ndola DHAC has adopted a multi-sectorial and decentralised coordination and management framework aligned to the Three - Ones principle. The DHAC's composition is multi-sectorial in nature and is drawn from the public and private sector, civil society, and FBOs. This committee is expected to provide technical expertise on the various aspects of the HIV/AIDS response.

Besides coordinating local HIV-related initiatives, DHAC is responsible for facilitating the mainstreaming of HIV and gender in all development plans, mobilise resources required for coordination as well as Monitoring and Evaluation through the collection and submitting the SARF from all implementing organisations in the district on a quarterly basis.

The DHAC, as a subcommittee of the DDCC, is mandated to oversee and coordinate the HIV and AIDS response activities in the district by performing the following functions:

- To plan, promote, administer services to all categories of issues pertaining HIV and AIDS
- To provide guidance on rehabilitation, training and welfare services
- Cooperate and network with all Government ministries and all stakeholders in the provision of prevention, training and education.
- To advice stakeholders on all matters relating to social economic development and the general well-being of infected and affected persons
- To monitor and supervise the provision of services by stakeholders
- To have a data base for all HIV and AIDS related activities
- To encourage stakeholders on the promotion of public awareness relating to the prevention of HIV and AIDS activities

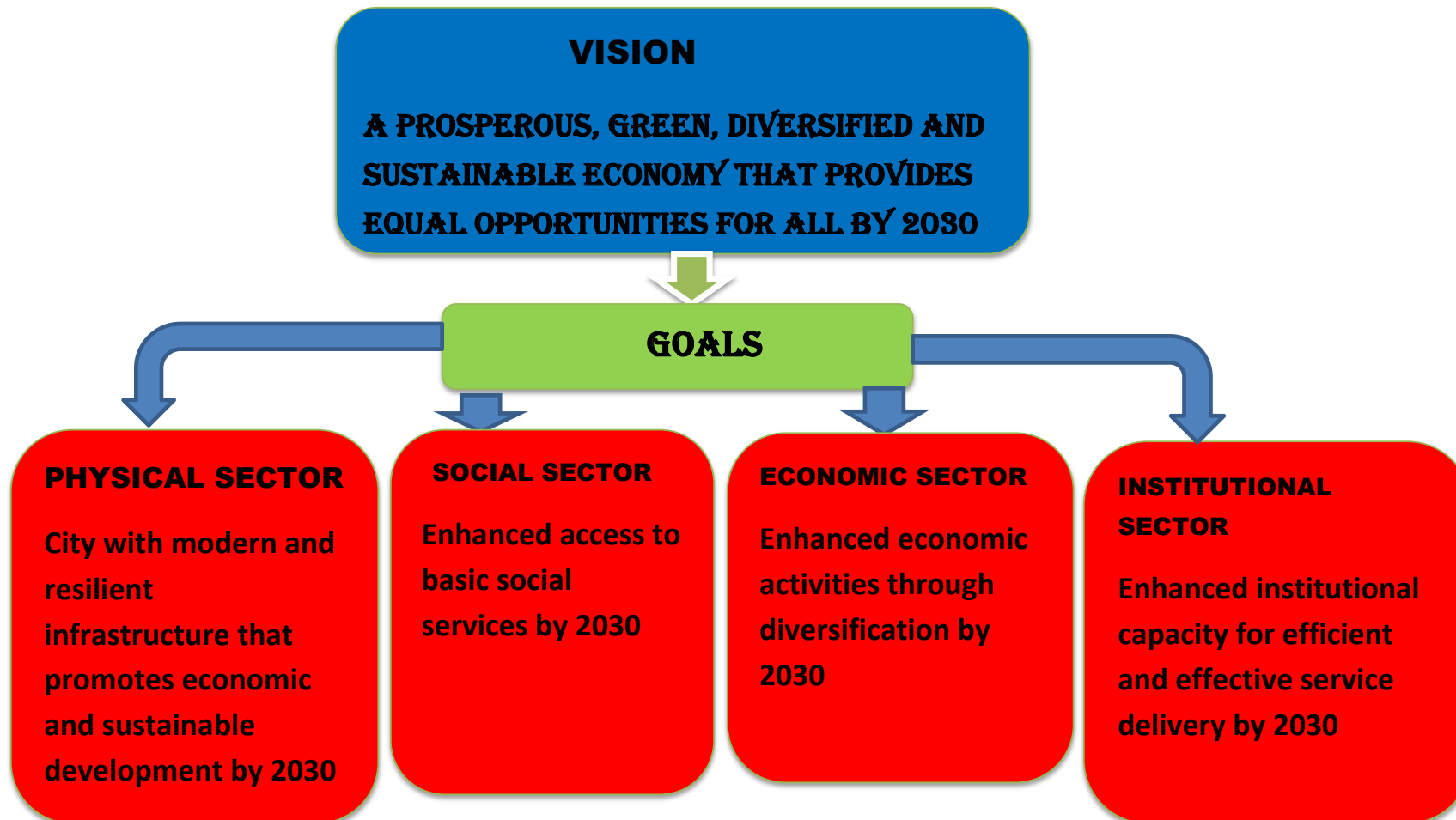
PART THREE DEVELOPMENT FRAMEWORK

3.0 Vision for the Long Term Development of the District

The Development Thrust laid out what the residents of Ndola District would like to see Ndola in terms of its physical, institutional, social and economic development. Agreed Development Thrust exercise binds the city together as a community which recognizes their shared values and purposes, and helps them articulate a shared vision of their locality. It seems to have created a sense of ownership among the stakeholders to the extent that they want to achieve their vision as the plan gets implemented over the next Ten years. The Development Thrust will stand to serve as the driving force that moves the entire city towards the achievement of a common development direction and provides the overall guidance and focus in the succeeding stages of the planning process.

The Vision of Ndola District was formulated based on and in relation to the National Vision for conformity, easy implementation or realisation. In view of this, there has been formulation of Sectorial Goals and Objectives to help in achieving the desired vision of the district as indicated in the figure below:

Figure 3.0: Development Framework: Vision, Goals and Objectives



3. .1 Defining the Long-Term Vision

The Vision Statement

“A Prosperous, Green, Diversified and Sustainable Economy that Provides Equal Opportunity for all by 2032”

3.1.1. Explanation of the Vision

- **Prosperous** – A city that is Prosperous is one that is advancing in the pursuit of anything desirable; making gain, or increase; thriving; successful. Ndola district aims at succeeding in all areas of development to meet the needs and aspirations of community members.
- **Green** - A green city is a community of residents, workers, and visitors who strive together to balance ecological, economic, and social needs to ensure a clean, healthy and safe environment for all members of society and for generations to come. The green city of Ndola means a way to increase the sustainability of urbanized areas. It is a concept of urban planning relying on the ecosystem services that green infrastructure can supply.
- **Diversified** – Means having variety of character or form or components; or having increased variety. A diversified economy is an economy based on a broad spectrum of economic sectors. The city of Ndola has been implementing programs aiming at looking on a variety of sectors in order to sustain the economy of the district.
- **Sustainable** - sustainable means being “**able to be maintained at a certain rate or level.** A city designed with consideration for social, economic, environmental impact and resilient habitat for existing populations, without compromising the ability of future generations to experience the same.

3.1.2. Development of objectives and strategies

The Goals and objectives for development of Ndola District are framed under four sectors namely Physical, Economic, Social and Institution sector.

Physical Sector: *City with Modern and Resilient Infrastructure that Promotes Economic and Sustainable Development*

Objectives

1. To improve and maintain road networks in the district by 2032
2. To construct and upgrade market infrastructure in the district by 2032
3. To improve waste management infrastructure in the district by 2032
4. To establish more health centers in the district by 2032
5. To establish more educational facilities in the district by 2032

6. To expand water and sanitation infrastructure in the district by 2032
7. To promote the development of high rise housing units to maximize land use in the district by 2032
8. To mitigate the impact of climate change in the district by 2032

Social Sector: *Enhanced Access to Basic Social Services*

Objectives

1. To increase access to quality education in the district by 2032.
2. To increase access to quality primary and secondary health care in all wards by 2032.
3. To increase access to improved sanitation and safe drinking water for all residents by 2032
4. To promote gender mainstreaming in all sectors by 2032.
5. To increase social security programs for vulnerable groups in the district by 2032
6. To mitigate the impact of climate change of climate change in the district by 2032

Economic Sector: *- Enhanced Economic Development through Diversification*

Objectives

1. To enhance the development of sustainable domestic and foreign market linkages by 2032.
2. To promote mechanized farming methods in the district by 2032
3. To promote value addition to locally produced raw materials by 2032
4. To expand and strengthen the revenue base in the district by 2032.
5. To promote tourism in the district by 2032
6. To facilitate the establishment of a Multi- Facility Economic Zone in the district by 2032.

Institutional Sector - *Enhanced institutional Capacity for efficient and Effective Service Delivery*

Objectives

1. To increase qualified human resource in public institutions by 2032.
2. To develop a smart city by linking all sectors to the district data bank by 2032.
3. To strengthen the legal and policy framework for various sectors to enhance service delivery in public sectors by 2032.
4. To promote inclusiveness in the formulation of By-laws in the city by 2032.

Strategies

This part outlines the strategies that will be used to achieve the stated objectives as outlined above

Objective1. Physical sector: To improve and maintain transport infrastructure in the district by 2032

Strategies

- Formulate a deliberate policy to allocate a percentage of revenue towards road construction, rehabilitation and maintenance

Objective2. Physical sector: To construct and upgrade market infrastructure in the district by 2032

Strategies

- Engaging in PPP
- Formulate a deliberate policy to allocate 5% of local revenue towards market infrastructure

Objective3. Physical sector: To improve waste management infrastructure in the district by 2032

Strategies

- Create platforms of key stake holders to partner with in Public Private Partnership

Objective4. Physical sector: To establish more health centers in the district by 2032

Strategies

- Lobby for increased budgetary allocation

Objective5 physical sector: To establish more educational facilities in the district by 2032

Strategies

- Lobby for increased budgetary allocation
- Engage in Public Private Partnership

Objective6 physical: To expand water and sanitation infrastructure in the district by 2032

Strategies

- Provision of serviced plots

Objective7 physical sector: To promote the development of high rise housing units to maximize land use in the district by 2032

Strategies

- Develop standardized high rise building plan
- entice potential investors in Public Private Partnership

Objective8. Physical sector: To mitigate the impact of climate change in the district by 2032

Strategies

- Transformation of people's attitudes on issues of environmental protection and restoration

Objective1. Social sector: To increase access to quality education in the district by 2032.

Strategies

- Lobbying for colleges and universities to have branches in the district
- Provision of human resource & materials

Objective2. Social sector: To increase access to quality primary and secondary health care in all wards by 2032.

Strategies

- Lobby for more Human Resource in the Health Sector

Objective3. Social sector: To increase access to improved sanitation and safe drinking water for all residents by 2032

Strategies

- Increase safe, clean, healthy water points

Objective 4. Social sector: To promote gender mainstreaming in all sectors by 2032

Strategies

- Gender mainstreaming

Objective 5. Social sector: To increase social protection programs and access for vulnerable groups in the district by 2032

Strategies

- Increase the coverage of social protection programs
- Lobby for Increased budgetary allocation
- Engage in Public Private Partnership

Objective 6. Social sector: To mitigate the impact of climate change of climate change in the district by 2032

Strategies

- Transformation of people's attitudes on issues of environmental protection and restoration

Objective 1. Economic sector: To enhance the development of sustainable domestic and foreign market linkages by 2032.

Strategies

- Strengthening the chamber of commerce
- Cluster group support and capacity building
- Local product inspection and market information management system

Objective 2. Economic sector: To promote mechanized farming methods in the district by 2032

Strategies

- Promotion of technology adoption; conservation agriculture, crop diversification and mechanization
- Promotion of dairy and small livestock production
- Fish clusters formation and capacity building

Objective 3. Economic sector: To promote value addition to locally produced raw materials by 2032

Strategies

- Promotion of value addition on local products in order to increase shelf life

Objective 4. Economic sector: To expand and strengthen the revenue base in the district by 2032.

Strategies

- Increase investment in ICT in public service delivery

Objective 5. Economic sector: To promote tourism in the district by 2032

Strategies

- Promotion of innovative tourism products and attractions
- Renovation and maintenance of tourism sites

Objective 6. Economic sector: To facilitate the establishment of a Multi- Facility Economic Zone in the district by 2032.

Strategies

- Facilitate the creation of the multi-facility economic zone

Objective1. Institutional sector: To increase qualified human resource in public institutions by 2032.

Strategies

- Institution assessment of human resource needs
-

Objective2. Institutional sector: *To develop a smart city by linking all sectors to the district data bank by 2032.*

Strategies

- Development of a district data bank

Objective3. Institutional sector: *To strengthen the legal and policy framework for various sectors to enhance service delivery in public sectors by 2032.*

Strategies

- Formulation and enforcing of by-laws

3. 2. Spatial Development Framework

The Spatial Development Framework seeks to address the major issues in Ndola's Spatial and Social Landscape:

- ❖ Increased pressure on land
- ❖ Increasing pressure on the natural environment and green infrastructure
- ❖ Urban Sprawl and fragmentation
- ❖ Spatial Inequalities

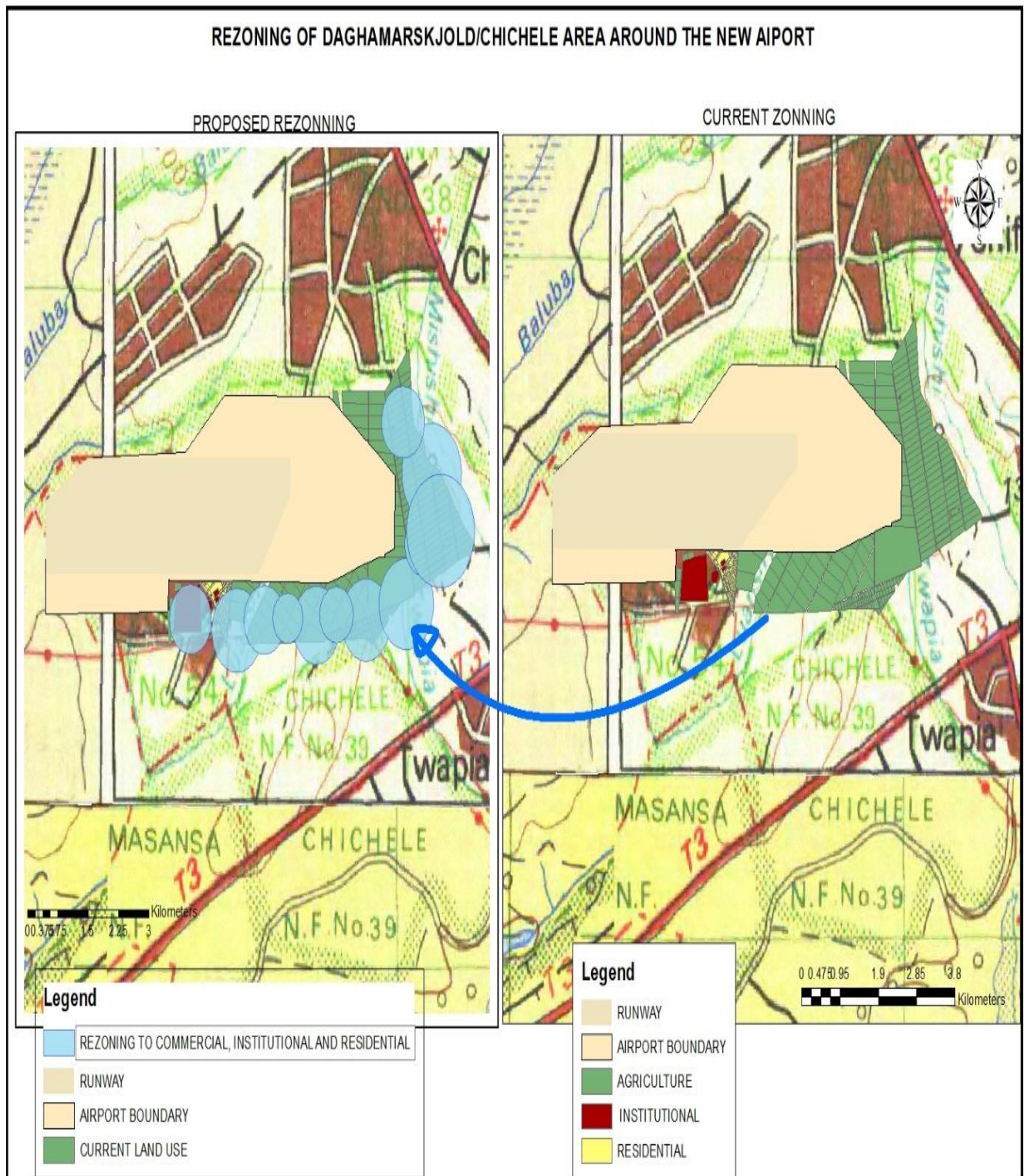
Models

1. The first model describes a '**Business- As- usual**' Scenario with dispersed, Sprawled Growth.
2. The second model describes a '**Linear Development**' scenario where future Development occurs along an expansive public transport network (Corridor Development) linking all marginalized areas of the city. Transport infrastructure in particular is a determinant of where people choose to live, with people generally preferring to live near to transport hubs.
3. The third model describes a '**Compact Polycentric Model**' which concentrates growth in a compact urban core and around priority transformation areas and key urban and transit oriented development nodes.

3.2.1 Alternative Scenarios

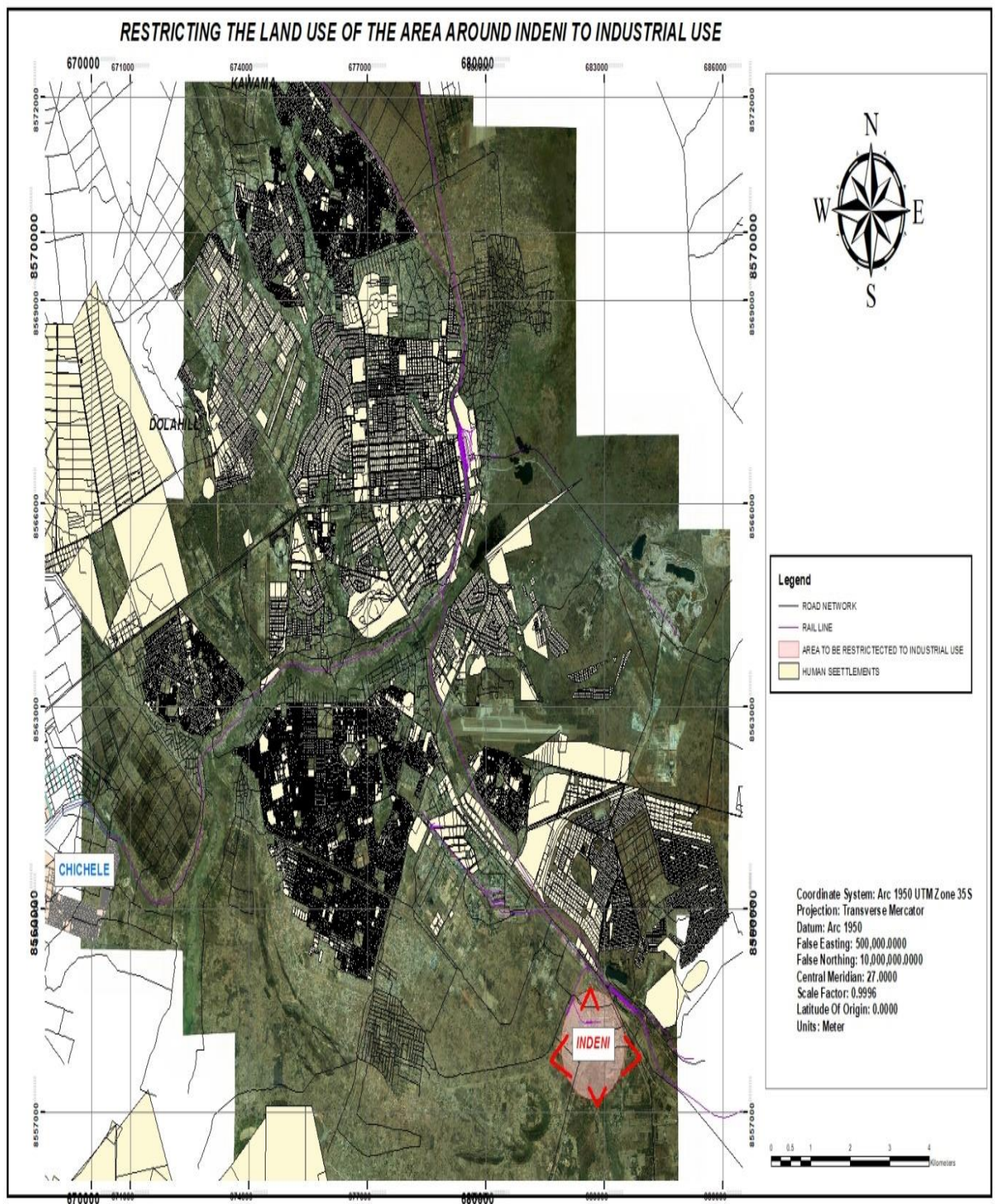
Development Plan: Rezoning Of Agriculture Land to Urban Development

i. Around the new airport- Chichele and Daghamarskjold to mixed use development



Source: Ndola City Council GIS Unit (2021)

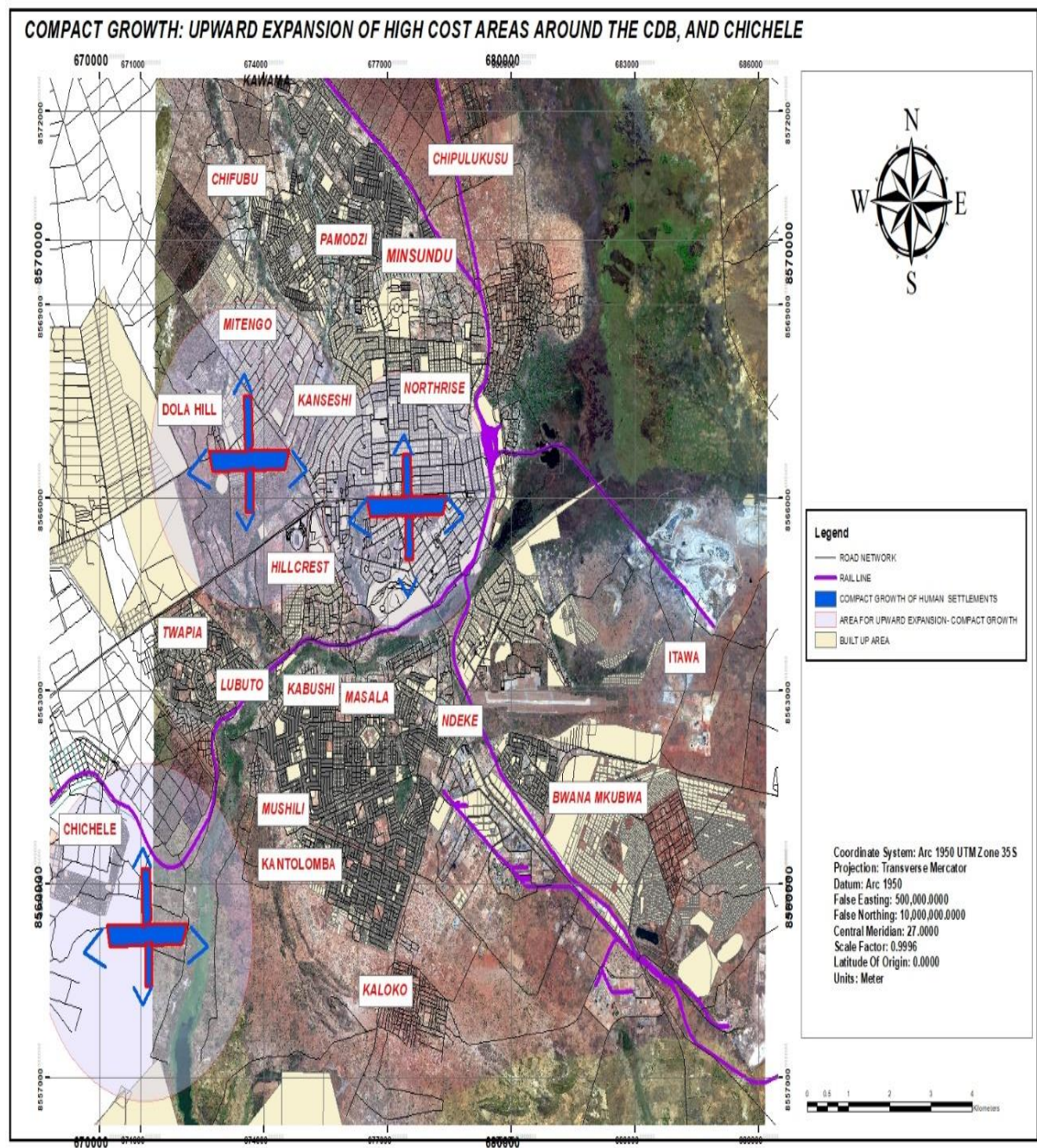
ii. Restricting the land use for the area around Indeni to ‘industrial land use’



Housing development

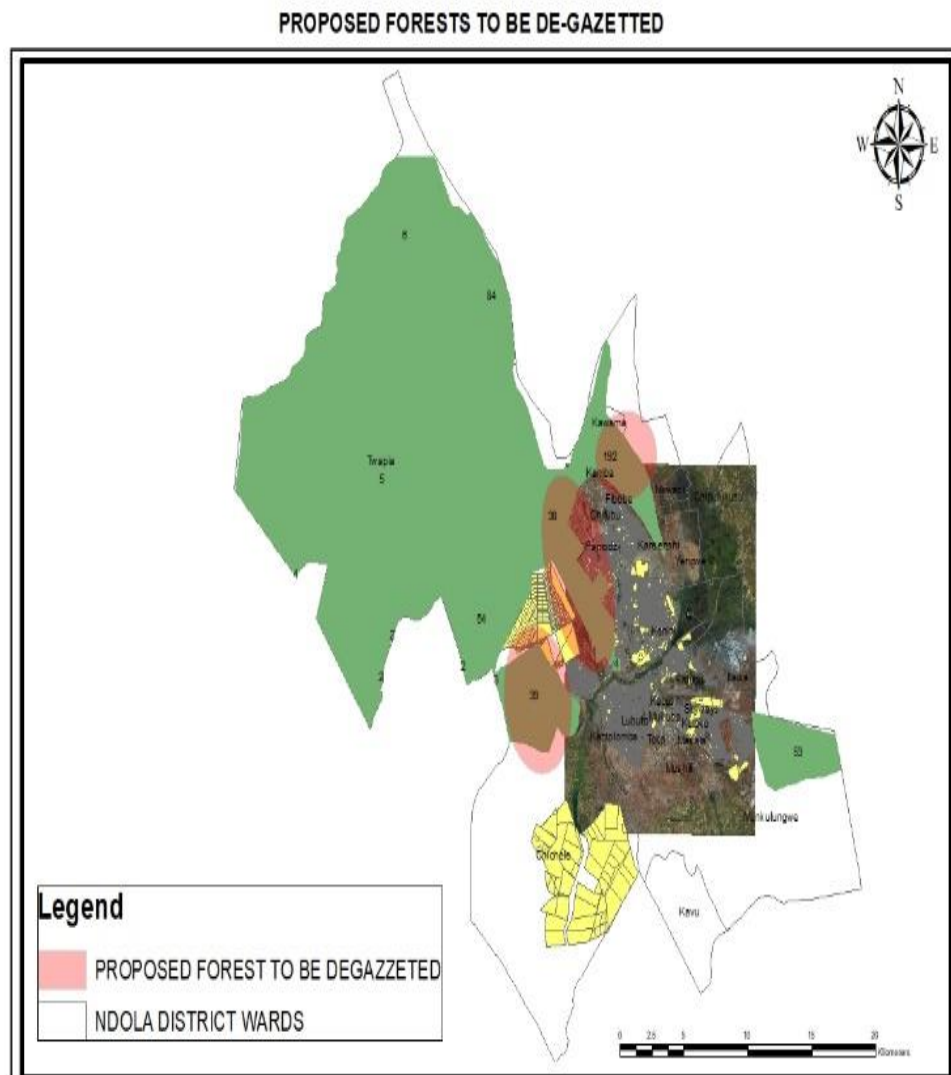
The need to have at least 2000 hectares of land in the next 10 years in order to meet the increasing demand for land for development.

i. Upward expansion in infrastructure development.



Source: Ndola City Council GIS Unit (2021)

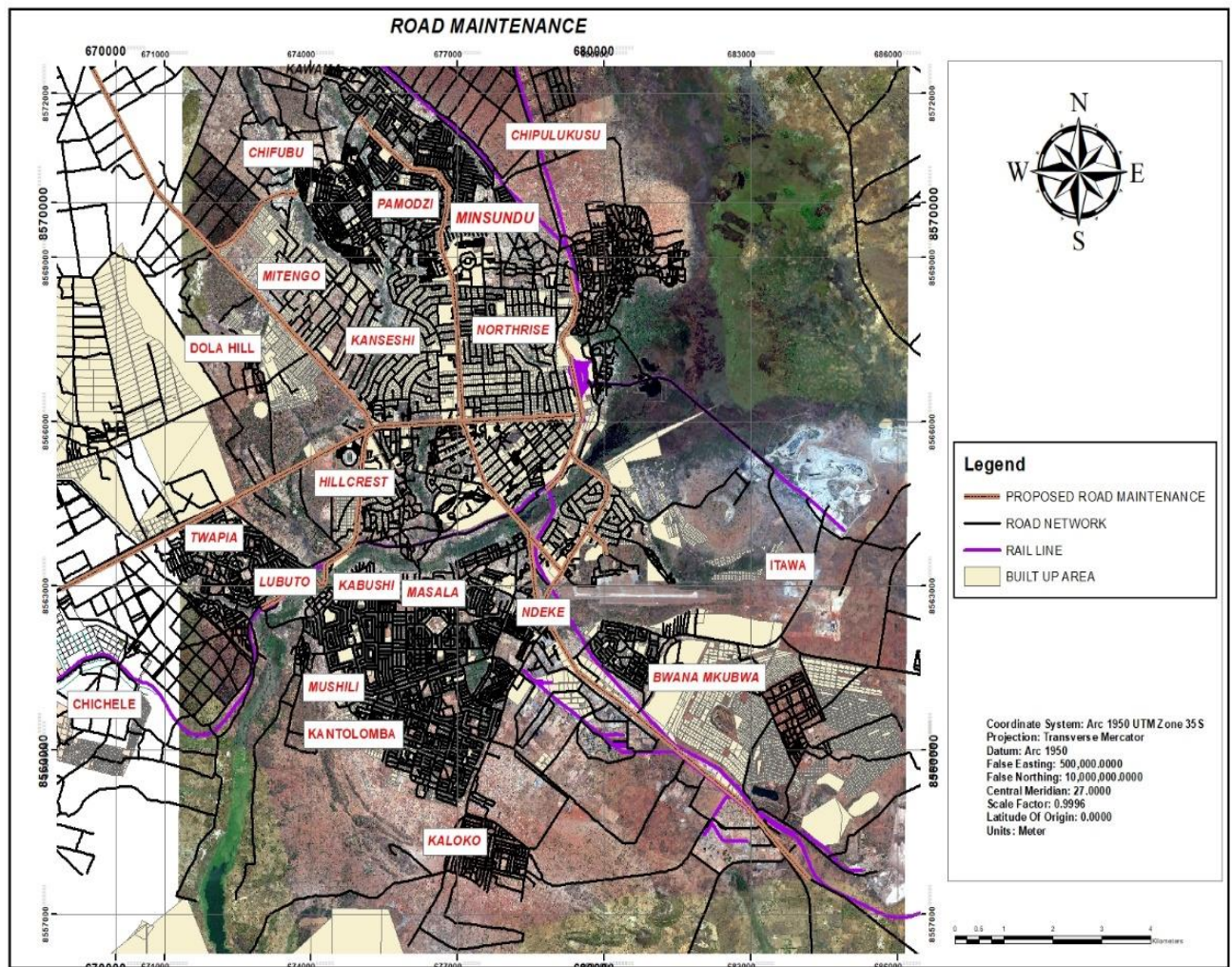
- ii. De-gazetting forest no. 38 and Excision of Chichele forest no.39



Transportation

With the existing scenario of road network in Ndola District, the following are inevitable;

- i. Upgrading of all the five(5)major Roads as well as the other distributor roads in the district



Source: Ndola City Council GIS Unit (2021)

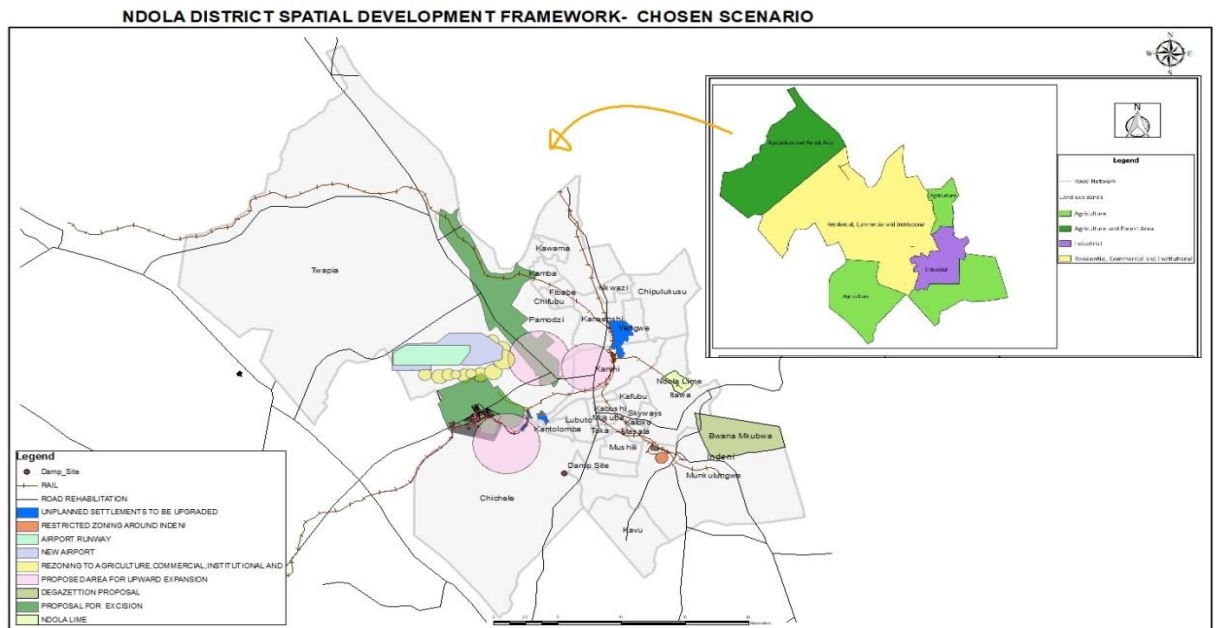
1. Buying off some privately owned farms near the CBD

Create a local area plan for the area to meet the demanding services.

3.2.2. Chosen Scenario

The Compact Polycentric Model performed significantly better than the other two in terms of economic, environmental and social indicators.

Therefore, the Spatial vision envisaged by the Spatial Development Framework for Ndola District is a Compact Polycentric district with a dense urban core linked by efficient public transport networks to dense, mixed use, complimentary sub-centers, situated within a protected and integrated natural environment. To achieve this, all the above scenarios will have to be incorporated.



Source: Ndola City Council GIS Unit (2021)

3.2.3. Policies to protect the environmental, social and economic Development

The following are land use policies used to improve, maintain or protect the environment, social and economic aspects of development:

The Urban and Regional Planning Act No.3 of 2015 ; The Urban and Regional Planning Act No.3 of 2015 is a policy document which provides for a planning framework, guidelines, systems and processes for urban and regional planning for the Republic. This act also ensures sustainable urban and rural development by promoting environmental, social and economic sustainability in development initiatives and controls at all levels of urban and regional planning.

Land Act of 1995 and Land Acquisition Act of 1995; Under the Land Act of 1995, land has been divided into three categories namely; State, Local Authority and Traditional land. The Act provides for efficient and effective land administration system that promotes security of tenure, equitable access and control of land for sustainable socio-economic development of the people of Zambia.

The Local Government Act (Chapter 281 of The Laws of Zambia) ; This Act provides for the establishment of city councils, municipal councils and district councils, township councils or management boards and provincial local government appeals boards and defines their functions and powers including the acquisitions of land. The act sets out rules regarding the organization,

proceedings and administration of local bodies and provides for their supervision and control, some of these functions have to do with pollution prevention, fire and safety, public health, environmental protection in general, inspections of building plans and factories.

Public Health of (1930) 1996; The Act was passed in 1930 and has been amended many times including in 1996. The Act empowers a Council to prevent unhealthy activities. It provides for the prevention of diseases, drainage, latrine and disposal of sewerage and treatment of systems.

The Public Health Act provides for the subdivision and general layout of land intended to be used as building sites, the level construction, number, direction and the width of streets and Thoroughfares, the limitation of the number of dwellings or other buildings to be erected on such land, the proportion of any building site which may be built upon and the establishment of zones within which different limitations shall apply and zones within which may prohibit the establishment or conduct of occupations or trades likely to cause nuisance or annoyance to persons residing in the neighborhood.

In addition, the Act provides for the inspection of the district of any Local Authority by that Local Authority with a view to ascertaining whether the lands and buildings thereon are in a state to be injurious or dangerous to health and the preparation, keeping, and publication of such records as may be required.

National Land Policy; The Council also awaits the approval of the National Land Policy which is currently in draft form. The National Land Policy will provide policy direction on all matters relating to Land Use Planning Policies and Proposals to Improve, Maintain or Protect the Environment, Social and Economic aspects of Development.

3.2.4. Land use Planning Policies for Environmental Sensitivity and Historical

The Environmental Management Act No.12 Of 2011: The Environmental Management Act No of 2011 replaced the repealed Environmental pollution and Protection Control Act (EPPCA) of 1990. This is the main environmental law in Zambia with superiority over other legislation on environmental matters. The Act continues and renames the Zambia Environmental Management Agency (ZEMA) which was established under the EPPCA.

The Act requires that a developer whose project will have a significant impact on the environment must carry out an environmental impact assessment which must be approved by ZEMA prior to the commencement of the development. According to the guidelines provided in environmental impact assessment (SI No. 28, 1997), this project qualifies for an environmental project brief. (EPB)

The Environmental management (Licensing) Regulations, 2013(Statutory instrument No. 122 of 2013), promulgated under this Act provides for control of pollution through licensing of waste generation, storage and disposal and discharge of effluent among others. Accordingly, this project will need to comply with regulations on discharge of emissions to the air and transportation of waste.

Hazardous Waste Management Regulations (S.I. No 125 of 2001); Legislation | Zambia | 2001 | Repealed

This section of the regulations manages: Waste disposal, Waste management, International trade, Authorization/permit, Trans boundary movement of waste

Waste Management (Licensing of Transporters of Wastes and Waste Disposal Sites) Regulations (Cap. 204); Legislation | Zambia | 1993 (2006) | Repealed

This section of the regulations manages: Waste disposal, Waste management, Transport/storage, Registration, Authorization/permit, Enforcement/compliance, and Offences/penalties

Pesticides and Toxic Substances Regulations (Cap. 204); Legislation | Zambia | 1994 (2006) | Repealed

This section of the regulations manages: Pollution control, Hazardous substances, Pesticides, Packaging/labelling, Transport/storage, Inspection, Enforcement/compliance, Offences/penalties

Water Resources Management Act No. 21 Of 2011; An Act to establish the Water Resources Management Authority and define its functions and powers; provide for the management, development, conservation, protection and preservation of the water resource and its ecosystems; provide for the equitable, reasonable and sustainable utilization of the water resource; ensure the right to draw or take water for domestic and noncommercial purposes, and that the poor and vulnerable members of the society have an adequate and sustainable source of water free from any charges; create an enabling environment for adaptation to climate change; provide for the constitution, functions and composition of catchment councils, sub-catchment and water user associations; provide for international and regional cooperation in, an equitable and sustainable utilization of shared water resources; provide for the domestication and implementation of the basic principles and rules of international law relating to the environment and shared water resources as specified in the treaties, conventions and agreements to which Zambia is a State Party; repeal and replace the Water Act, 1949; and provide for matters connected with, or incidental to, the foregoing

Forest Act No. 4 Of 2015 ; An Act to provide for the establishment and declaration of National Forests, Local Forests, joint forest management areas, botanical reserves, private forests and community forests; provide for the participation of local communities, local authorities, traditional institutions, non-governmental organizations and other stakeholders in sustainable forest management; provide for the conservation and use of forests and trees for the sustainable management of forests ecosystems and biological diversity; establish the Forest development Fund; provide for the implementation of the United Nations Framework Convention on

Climate Change, Convention on International Trade in Endangered Species of Wild Flora and Fauna, the Convention on Wetlands of International Importance, especially as Water Fowl Habitat, the Convention on Biological Diversity, the Convention to Combat Desertification in those Countries experiencing Serious Drought and/or Desertification, particularly in Africa and any other relevant international agreement to which Zambia is a party; repeal and replace the Forests Act, 1999; and provide for matters connected with, or incidental to, the foregoing.

3.2.5. Planning Policies for the Protection of National and Regional Interests

Zambia's settlement planning and development is governed by the Urban and Regional Planning No.3 of 2015 (Esther Nchimunya Nyemba, 2018). The Act repealed the Town Country Planning Act of 1962 and the Housing (Statutory and Improvement Areas Act of 1975 (Urban and Regional Planning Act, 2015).

The Urban and Regional Planning Act Number 3 of 2015 is the document where land use planning policies are found for the protection of national and regional interest within Zambia. This Act became law in August 2015 and applies to the entire country that is urban and rural areas. It is the law that which guides planning and development regardless of the land tenure or land administration system and provides for cooperation with traditional authorities (Ministry of Local Government, 2016). This policy document provides for development, planning and administration principles, standards and requirements for urban regional planning processes and systems. It provides a planning framework, guidelines, systems and processes for urban and regional planning for the Republic (Urban and Regional Planning Act, 2015). The current Act provides for the establishment of planning authorities, and supervises and controls land use and the preparation and implementation of development plans. A planning authority is a public body with delegated authority to under town and or regional planning (Augustine Mulolwa, 2016)

The Urban and Regional Planning Act despite not having an operational strategy provides for the following in its implementation:

- Development, planning and administration of principles, standards and requirements for urban and regional planning processes and systems.
- Provide a planning framework, guidelines, systems and processes for urban and regional planning.
- Establish a transparent, accountable, participatory and inclusive process for urban and regional development that allows for the involvement of all players in the housing sector in the planning, implementation and operation of human settlements.

- Ensure sustainable urban and rural development by promoting environmental, social and economic sustainability in development initiatives and controls at all levels of urban and regional planning; ensure uniformity of law and policy with respect to urban and regional planning.
- Establish procedures for integrated urban and regional planning in a devolved system of governance so as to ensure multi-sector cooperation, coordination and involvement of different levels of ministries, provincial administration, local authorities, traditional leaders and other stakeholders in urban and regional planning (Nyemba, 2018).

The proposed policies for the protection of national and regional interests include the policies found in the draft National Land Policy which was adopted in 2002. This proposed policy document was redrafted and the revised draft land policy was released in 2007 but dated October 2006 (Machina, 2007). This document is needed to streamline the delivery of land administration services and management in the development of the country. The policy seeks to strengthen land tenure security and enhance sustainable and productive management of land resources by upholding transparent and cost-effective systems administration (Draft National Land Policy, 2017). Another policy document which contains proposed land use policies for national interest and regional interest are the regional planning regulations. This at present, the Urban and Regional Planning Act has not attained full implementation due to lack of urban and regional planning regulations. The regulations are supposed to provide for the operationalization of the Act, but since its adoption only few parts of the Act have been operationalize while others which are new can only be operationalize through a regulatory framework. Such parts include the provision for local authorities to be planners for their areas. Section 13 of the Act stipulates that; “local authorities shall for the purpose of this Act be designated as planning authority for its area by the minister through a statutory instrument, to regulate, control and plan for development, use of land and provision of services as required by the area”. The advantage of this provision is if fully operationalized, it will foster quick delivery of development that is currently hindered by bureaucracy. Planning for land for housing, service delivery and infrastructure are done at provincial level (Zambia Institute of Policy Analysis and Research (ZIPAR, 2018). Therefore, the regional planning regulation is a supporting document to the Urban and Regional Planning Act.

3.2.6. Planning Policies for Informal Settlements

According to Marie Huchzermeyer and Aly Kara (2006), they have defined informal settlements as settlements of the urban poor developed through the unauthorised occupation of land, they are regarded by many as unhealthy and overcrowded blights on the urban landscape ‘squatter camps’ in common parlance. The Urban and Regional Planning Act No. 3 of 2015

contains land use planning policies in informal settlements. The Urban and Regional Act No. 3 of 2015 defines informal settlements as follows:

“Group of people living on land they have no legal claim to Houses of temporary, semi-permanent or permanent nature erected on land that have not formally been permitted by the planning authority and serviced by residential use under this Act or any other written law”.

Clusters of housing and other infrastructure built without the formal permission of the planning authorities under this Act, any other written law or repealed Acts: or Settlements that have only temporary permission from the planning authority to occupy the settled land (Urban and Regional Planning Act, 2015).

The Human and Social Development Strategic Development Area of the 8th National Development Plan addresses the need to develop human capacities, through investments and interventions in education and skills development, health and nutrition, water and sanitation. In addition, it aims at reducing poverty, inequalities and vulnerability. As such, the needs of informal settlements will be met through this Strategic Development Area. Therefore, the need for human and social development cannot be over emphasised as is evident from the many interventions so far put in place since the launch of the 8NDP by the New Dawn Government.

The URP Act provides that a local authority may declare or undeclared an improvement area. Meaning that an informal settlement can be declared or undeclared by a local authority. This is where an improvement area is declared, a local area plan must be approved. Once the Local Area Plan is adopted the provisions in the URP Act in relation to planning application, land use and development control will apply to the improvement area (Ministry of Local Government, 2016). In terms of appropriate building and development control of informal settlements, the local authority does not intervene but upgrading of informal settlements can be implemented when the settlements grown beyond its capacities. The URP Act under section 40(7) provides provision for the Minister to prescribe specific requirements for planning permission and development control in informal settlements where the local plan exists (Urban and Regional Planning Act, 2015). Currently, there are no proposals for land use policies within informal settlement.

PART FOUR IMPLEMENTATION PROGRAM

4.0. Implementation Program

The Implementation Programme will comprise of the following four elements:

- i. A capital investment programme indicating the major programmes and investments required to implement the development framework with the agency responsible for delivery;
- ii. A local authority's financial plan providing a budget projection for the first five years after the adoption of the plan and indicating the manner in, and extent to, which the capital and operational budgets of the local authority reflect the priorities and objectives of the integrated development plan, including the local authority's expectation of specific funding allocations from the Government;
- iii. A programme for the preparation of local area plans and other detailed planning activities to assist in the implementation of the integrated development plans; and
- iv. Proposals for monitoring and review of the integrated development plan, including key performance indicators.

4.1 Capital Investment Plan

The Capital Investment Plan will be for a period of five years only and will be directly related to the priorities, objectives and strategies identified in the IDP. The Capital Investment Plan will, however, expand to include investment in physical infrastructure and services which are identified as priorities for Ndola district. The capital investment is as tabulated in the tables below;

Identified Activities for Implementation of Strategies and Program

ECONOMIC SECTOR

Objective 1	To enhance the development of sustainable domestic and foreign market linkages by 2033.								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Strengthening the chamber of commerce	Creating market linkages	Conduct stakeholder meetings	200,000.00	4 meetings conducted	4 meetings conducted	4 meetings conducted	4 meetings conducted	4 meetings conducted	Ministry of Agriculture/ Local Authority/ Ministry of commerce/ Chamber of commerce/ ministry of information
		Advertisement of local products	500,000.00	5	5	5	5	5	
		Cluster the business segments	96,000.00	4 Clusters	5 Clusters	6 Clusters	7 Clusters	8 Clusters	
Cluster Group Support and Capacity Building	Training Programs	Conduct entrepreneurship Trainings	425,000.00	10 Cluster groups trained	10 Cluster groups trained	10 Cluster groups trained	10 Cluster groups trained	10 farmer groups trained	Ministry of Agriculture & Livestock/ Ministry /Chamber of commerce
Local product inspection and market information management	Market Information and Research	Producing marketing bulletins	520,000.00	52 weekly bulletins produced	52 weekly bulletins produced	52 weekly bulletins produced	52 weekly bulletins produced	52 weekly bulletins produced	Ministry of Agriculture & livestock/ local authority/ chamber of
	Field Visits	Inspection of	100,000.00	4 field visits	4 field visits	4 field visits	4 field visits	4 field visits	

system		district Marketing Infrastructure							commerce
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Objective 2	To promote mechanized farming methods in the district by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Promotion of technology adoption: conservation agriculture, crop diversification and mechanisation	Crop diversification	Capacity building of farmers in the production of various crops through farmer field schools	230,000.00	17 farmer field schools	17 farmer field schools	17 farmer field schools	17 farmer field schools	17 farmer field schools	Ministry of Agriculture / Local Authority
		Setting up demonstration plots of various crop varieties	250,000.00	17 demonstration plots	17 demonstration plots	17 demonstration plots	17 demonstration plots	17 demonstration plots	
		Conducting field days	500,000.00	17 field days	17 field days	17 field days	17 field days	17 field days	
		Farmer exchange visits	125,000.00	5 farmer exchange visits	5 farmer exchange visits	5 farmer exchange visits	5 farmer exchange visits	5 farmer exchange visits	
	Conservation agriculture	Training farmers in conservation	510,000.00	17 farmer field schools	17 farmer field schools	17 farmer field schools	17 farmer field schools	17 farmer field schools	

		agriculture through farmer field schools.							
	Farm power and Mechanisation	To showcase small-holder technologies used for timely land preparation and crop processing	425,000.00	17 demonstrations	17 demonstrations	17 demonstrations	17 demonstrations	17 demonstrations	
Promotion of irrigated agriculture	Irrigation development	Development of small-scale irrigation schemes	150,000.00	3 smallholder irrigation schemes	3 smallholder irrigation schemes	3 smallholder irrigation schemes	3 smallholder irrigation schemes	3 smallholder irrigation schemes	
Promotion of dairy and small livestock production	Livestock production	Capacity build farmers in small livestock production	240,000.00	8 farmer groups trained	8 farmer groups trained	8 farmer groups trained	8 farmer groups trained	8 farmer groups trained	Ministry of Livestock and Fisheries
		Establishing a breeding center for small livestock		1 centre	-	-	-	-	
		Control of Livestock Diseases	200,000.00	80 surveillance visits	80 surveillance visits	80 surveillance visits	80 surveillance visits	80 surveillance visits	
Promotion of	Climate	Capacity	350,000	25 farmers	50 farmers	75 farmers	100 farmers	125 farmers	Ministry of

climate smart Agriculture	Smart Agriculture promoted	build farmers in Smart Agriculture technologies		per camp (425 farmers)	per camp (850 farmers)	per camp (1,275 farmers)	per camp (1,700 farmers)	per camp (2,125 farmers)	Agriculture /Ministry of Green Economy and Environment
		Increase area under conservation Agriculture	250,000	100	200	300	400	500	Ministry of Agriculture
		Increase number of farmer using drought tolerant crop varieties	450,000	1000	1500	2000	4000	8000	
Infrastructure development for Improvement of extension service delivery	Improved agriculture extension services	Renovation of Camp houses	600,000.00	-	1	2	-	-	Ministry of Agriculture / Local Authority
		Construction of camp houses	11,200,000.00	2	3	3	3	3	
		Construction of one district Agriculture office Block	5,000,000.00	-	1	-	-	-	
Fish Clusters Formation and Capacity Building	Aquaculture and capture fisheries	Formulate and support 3 community	150,000.00	1 hatchery	1 hatchery	1 hatchery	-	-	Ministry of Livestock and Fisheries

	Manageme nt and Developm ent	based hatcheries							
		Training and Sensitizatio n of Fishers and Fish Farmers	200,000 .00	5 fish clusters registered and trained	5 fish clusters registered and trained	5 fish clusters registered and trained	5 fish clusters registered and trained	5 fish clusters registered and trained	
		Extension Visits to Fishers and Fish Farmers	150,000 .00	24 extension visits	24 extension visits	24 extension visits	24 extension visits	24 extension visits	
		Restocking and conservation of small water bodies	300,000 .00	10 small water bodies restocked and conserved	10 small water bodies restocked and conserved	10 small water bodies restocked and conserved	10 small water bodies restocked and conserved	10 small water bodies restocked and conserved	
		Distribution of fish seed (fingerlings) to fish farmers	250,000 .00	100 farmer beneficiaries	100 farmer beneficiaries	100 farmer beneficiaries	100 farmer beneficiaries	100 farmer beneficiaries	

Objective 3	To promote value addition on locally produced raw materials by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Promotion of value addition on local products in order to increase shelf life	Food processing and value addition	To build capacity in farmers on improved food processing and value addition	850,000.00	17 farmer group to be trained	17 farmer groups to be trained	17farmer groups to be trained	17 farmer groups trained to be	17farmer groups to be trained	Ministry of Agriculture & livestock/ chamber of commerce/ education
	Stockholder engagement (PPP)	Creating a stakeholder directory	20,000.00	1	1	1	1	1	
		Stakeholder meetings	50,000.00	4	4	4	4	4	
	Value addition on locally produced materials	Capacity building on the value addition	500,000.00	10 clusters per year	10 clusters per year	10 clusters per year	10 clusters per year	10 clusters per year	

Objective 4	To expand and strengthen the revenue base in the district by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Increased Investment in ICT	Human resource Development on E-governance	Capacity building Training	840,000.00	20 Members of staff	25 Members of staff	25 Members of staff	30 Members of staff	20 Members of staff	Local government/ Education/ ZICTA
		Purchase of IT Infrastructure	500,000.00	1					
		Installation and operationalization	1,000,000.00			1	1	1	
		Public engagement on E- governance	50,000.00	1	1	1	1	1	

Objective 5		To promote tourism in the district by 2033							
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Promotion of tourism product development and attractions	New innovative Tourism Products developed	Construction of Theme park (ZITF stand)	1,000,000		1				
	Marketing of tourism	Stakeholder engagement	47,296.00		1	1	1	1	Local Authority/ ministry of tourism and culture
		Profiling of tourism activates	35,472.00		1	1	1	1	
		Develop a Marketing strategy	94,592.00		1				
		Operationalize the strategy	189,184.00		1	1	1	1	
	Improving tourism packaging and capacity	Creation of Ndola visitor information desk	256,000.00	1					
		Creation of community heritage corridor	256,000.00	1					
Infrastructure development	Construction, rehabilitation and maintenance of tourism sites	Renovation of tourism infrastructure (Copperbelt Museum)	2,000,000.00					1	
		Renovation of the Henry Makulu	2,000,000.00				1		

		House							
		Maintenance of tourism sites (Chichele Mofu Tree & Dag Hammarskjöld)	800,000.00		1		1		
		Construction of a museum along Ndola-Kitwe dual carriageway.	4,000,000.00			1			
		Construction of recreation facility (Masala Cultural Village)	2,000,000.00				1		

Objective 6	To facilitate the establishment of a Multi- Facility Economic Zone in the district by 2033								
Strategy	Program	Activities	Budget (ZMK)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Facilitate the creation of Multi Facility Economic Zone	Promotion of Public Private Partnership in the City	Conduct stakeholder meetings	300,000.00	4 meetings	2 meetings	2 meetings	2 meetings	2 meetings	Chamber of Commerce/ ministry of land/ forestry NCC/ ministry of infrastructure
		Site identification	50,000	1					
		Drawing up of the MOU	50,000.00	1	1				
		MFEZ construction	68, 000, 000					1	

PHYSICAL SECTOR

Objective 1	To improve and maintain transport infrastructure in the district by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Formulate a deliberate policy to allocate a percentage of revenue towards the construction, rehabilitation and road maintenance	Upgrading of 122.162km on selected roads to bituminous standard	Stakeholder meeting	300,000.00	1	1	1	1	1	Ndola City Council/Road Development Agency
		Field Visit	35,000.00	2	1	1			
		Presentation to local authority	5,000.00	2	1	1			
		Full council meetings	12,000	2	1	1			
		Tendering process	93, 500.00	1	1	1	1	1	
		Upgrading of the 122.162km	1,099,458,000		50km	50km	22.162km		
	Maintenance of 43 Km of roads	Procurement of earth moving equipment	100,000,000.00			1			

		Maintenance of 43km of roads	387,000,000			20km	20km	3km	
	Rehabilitation of 344.9 km on the 5 district major roads	Rehabilitation of roads	22,073,600,000	1	1	1	1	1	
Infrastructure development	Street lighting	Installation of street lights on 230km of road	172,500,000	50km	50km	50km	50km	30km	Ndola City Council
	Provision of a modern bus terminus	Construction of a modern bus terminus	4, 000, 000				1		Ndola City Council

Objective 2	To construct and upgrade market infrastructure in the district by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Formulate a deliberate policy to allocate 5% of local revenue towards market infrastructure	Infrastructure development	Stakeholder meeting	482,110.00	4					Ndola City Council
		Conducting condition based survey	35,220.00	✓					
		Architectural Designs and developing BOQs	10,000.00	✓	✓	✓	✓	✓	
		Presentation of designs and BOQs to management	5,200.00	✓	✓	✓	✓	✓	

		Full council approval of designs and BOQs	63,080.00	✓	✓	✓	✓	✓	
		Tendering process	92,500.00	✓	✓	✓	✓	✓	
		Market construction	12,000,000.00	✓	✓	✓	✓	✓	
Engaging in PPP	Infrastructure development	Stakeholder meeting	300,000.00	4	4	4	4	4	
		Drawing up of the MOU for the construction of markets through PPP	37,200.90	1	1				

Objective 3	To Establish more health centers in the district by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Lobby for increased budgetary allocation	Health infrastructure development	Development of sector budgets from the plan	620,000	1	1	1	1	1	Ministry of Health and Ndola council
		Stakeholder engagement	41,000.00	1	1	1	1	1	
		Site identification	26,180.00	1					
		Designs and cost estimates	10,210.00	1					
		Resource mobilisation	41,250.00	1	1	1	1	1	

		Tendering process	55,210.00	1	1	1	1	1	
		Construction of project(health facilities)	4,510,030.03	2	2	2	2	2	

Objective 4	To establish more educational facilities in the district by 2033								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Lobby for increased budgetary allocation	Education infrastructure development	Development of sector budgets from the plan	620,000	1	1	1	1	1	Ministry of education and Ndola city council
		Stakeholder engagement in the implementation of School health and nutrition (SHN)	41,000.00	1	1	1	1	1	
		Construction of additional 1*3 classrooms with Desks	30, 000, 000	4	4	4	4	4	
		Construction of a complete secondary school with Desks	207, 000, 000	1		1		1	
		Ablution Blocks	6,000,000	2	2	2	2	2	
		Desk Rehabilitation	1,293,750	300	300	300	300	300	
		Purchase of New Desks (double and single seaters)	90,638,000	9,064	9064	9064	9064	9064	
	ICT	Purchase of Computers (laptops and desk tops)	2,099,500	50	50	50	50	47	

OBJECTIVE 5									
Objective: To increase the water service coverage by 2033									
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Provision of water infrastructure	Lay Water Networks	Purchase of excavator	1,000,000.00	1					Ministry of Water and Sanitation
		Conduct stakeholder meetings	1,000,000.00	9	12	4			
		Lay water network in Jacaranda	32,000,000.00	1					
		Lay water network in Dola Hill	32,000,000.00	1					
		Lay water network in Pamodzi	40,000,000.00	1					
		Lay water network in Mushili	42,000,000.00	1					
		Lay water network in Pinegrove	32,000,000.00	1					
		Lay water network in Kaloko	20,000,000.00	1					

		Lay water network in Twapia	10,000,000.00		1				
		Lay water network in Canaan	20,000,000.00		1				
		Lay water network for water Kiosks	5,000,000.00		1				
		Lay water network in Airport	20,000,000.00		1				
		Lay water network in Misundu/Mapalo	30,000,000.00	1					
		Lay water network in Bwana Mkubwa/Ndeke	20,000,000.00	1					
		Lay water network in George	4,000,000.00	1					
		Construct water network Kopaland	5,000,000.00		1				
		Construct water network in Itawa extension	5,000,000.00		1				
		Construct water network in Kamba	5,000,000.00		1				

OBJECTIVE	Objective: To increase the sewerage service coverage by 2027								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Provision of sewerage infrastructure	Lay Sewer Networks	Purchase of vacuum tanker	2,000,000.00						Ministry of Water and Sanitation
		Conduct stakeholder meetings	1,000,000.00	4	5	4	6	3	
		Lay sewer network in Mitengo	20,000,000.00	1					
		Upgrade sewer networks in Masala/Chifubu	50,000,000.00	1					
		Lay sewer network in Kopaland	20,000,000.00	1					
		Lay sewer network in Itawa extension	20,000,000.00	1					
		Lay sewer network in Twapia	40,000,000.00		1				
		Construction of sewer scheme to the airport	40,000,000.00			1			
		Lay sewer network in Misundu & Mapalo	60,000,000.00			1			
		Lay sewer network in BwanaMkubwa/N deke	40,000,000.00			1			

		Lay sewer network in George area	20,000,000.00				1	
	Construct Wastewater Treatment Plant in Ndola	Construct Wastewater treatment plant	10,000,000.00				1	
	Construct FST Treatment Plant in Ndola	Construct FST treatment plant	10,000,000.00					1

Objective 6	To promote the development of high rise housing units to maximize land use in the district by 2032								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Develop standardised storey building plans	Maximum land utilisation	Development of approved standard building plans	90,012.00	1	1	1	1	1	NCC
Entice potential Investors (PPP)	Upgrade informal settlements	Community meetings	450,000	12	12				
		Developmental Meetings	260,000	4					
		Tendering process for a private partner to conduct the process of upgrading	47,000.00	1	1	1	1	1	

Objective 7	To improve waste management infrastructure in the district by 2032								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Create platforms of key stake holders to partner with in PPP	Solid waste management	Sensitisation	51,800.00	4	4	4	4	4	Ndola city council, All public and private institutions within the district
		Procurement of waste disposal equipment	2,000,550.00	1	1	1	1	1	
		Expansion of street sweeping services to 5 new communities outside the CBD	1,977,000	12	12	12	12	12	
		Identification and engagement of CBEs in medium and low cost	750,000	4	4	4	4	4	
	Construction of engineered land fill	site identification	14,010.50	1					
		Designs and cost estimates	10,000.00	1					
		Meeting with key stakeholders	300,000.00	1	1	1	1	1	
		Sign an MOU with interested stakeholders	43,070.00	1					
		Environmental Impact Assessment	50,000	1					
		Construction of an incinerator	250,000	1					
		Construction of the weigh bridge	2,000,000	1					
		Construction of a borehole	50,000	1					
		Construction of the boundary fence	4,000,000	1	1	1	1	1	

		Construction of an ablution block	300,000	1	1				
	Keep Zambia clean and health campaign	Sensitisation to the public	97,060.00	4	4	4	4	4	
		Procurement of cleaning items	101,900.00	2	2	2	2	2	
		Transportation (purchase of vehicles and fuel)	109,161,003.30	12	12	12	12	12	

SOCIAL SECTOR

Objective 1	To increase access to quality education in the district by 2032.								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Provision of human resource & materials	Staff recruitment and deployment	Recruitment of qualified teachers	205,120.02	1	1	1	1	1	Ministry of general education
	Staff development	Support in-service capacity building	280,12.02	1	1	1	1	1	
	Provision of teaching and learning materials	Procurement of teaching and learning materials	21,150,000.01	1	1	1	1	1	

	Lobbying for colleges and universities to have branches in the district	Stakeholder engagement	30,000.00	1					
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Objective 2	To increase access to improved sanitation and safe drinking water for all residents by 2030								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Increase safe clean healthy water point	Awareness campaigns on WASH	School shows	24,400.00	3	3	3	3	3	Ndola city council /Kafubu water and sewerage/ WARMA and other utility companies
		Road shows	30,050.00	2	2	2	2	2	
		Talk show on both radio and television	47,000.00	4	4	4	4	4	
	Advocating for lower connecting fees	Meetings with utility companies	7,640.00	1					
		Review meetings	7,640.00		2				
	Lobby for the construction of water kiosk	Meetings with different stakeholders	8,400.00		2				

Objective 3									
To Promote gender mainstreaming in all sectors by 2030									
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Gender mainstreaming	Promotion of gender work place policy	Sensitisation	383,427.70	✓	✓	✓	✓	✓	Ministry of gender NCC District administration
		Formulation of gender work place policy	500,060.00	1	1	1	1	1	
		Reviewing of gender work place policy	82,195.40	1				1	

Objective 4									
To increase access to quality primary and secondary health care in all wards by 2032.									
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Lobby for more human resource in the health sector	Development of a human resource assessment framework	identification of vacancies	17,320.00	1	1	1	1	1	Ministry of health
		Recruitment of human resource	109,800.00	1	1	1	1	1	
		Training of existing human resource	2,870,080.00	1	1	1	1	1	

Objective 5									
To mitigate the impact of climate change in the district by 2032									
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Promotion of enhanced forestry extension services at schools and	Enhanced Forestry Extension Services	Carrying out Environmental awareness campaigns in schools and at farmer level on the	336,000.00	20 farmers and 5 schools	20 farmers and 6 schools	20 farmers and 7 schools	20 farmer field and 7schools	20 farmer field and 7 schools	Ministry of Green Economy and Environment

farm level		importance of conserving forests							
		Establishment community forest nursery	80,000.00	5 nurseries	5 nurseries	5 nurseries	5 nurseries	5 nurseries	
		Establishment of nursery in schools	50,000.00	4 schools	5 schools	6 schools	7 schools	8 schools	
	Promotion of modern beekeeping	Training farmers in modern beekeeping and processing of honey and beeswax	80,000.00	10 farmers	10 farmers	10 farmers	10 farmers	10 farmers 5 schools and 20 farmers	
Promotion of plantation/ woodlot establishment and management	Enhanced sensitization of farmers and schools on tree planting and management in order to private climate change	Training farmers and school children on how to plant and manage trees	80,000.00	20 farmer field and 5 schools	20 farmer field and 5 schools	20 farmer field and 5 schools	20 farmer field and 5 schools	20 farmer field and 5 schools	
		Management of established plantations through engaging casual workers (Job creation) to carryout weeding, pruning, controlled early burning and fire break maintenance	288,000.00	123 ha	123 ha	123 ha	123 ha	123 ha	

Establishment of forestry farmer support program	Registration of farmers in sustainable forest conservation and utilization	Capacity build farmers in plantation/woodlot establishment and management	240,000.00	10 farmers	10 farmers	10 farmers	10farmers	10 farmers	Ministry of Green Economy and Environment
		Distribution of assorted seedlings to farmers for plantation/woodlot establishment and management	24,000.00	10,000 seedlings / 20 farmers	10,000 seedlings /20 farmers	10,000 seedlings /20 farmers	10,000 seedlings /20 farmers	10,000 seedlings /20 farmers Once every month	
Promotion of forest protection and management	Recruitment of honorary forestry officers	Training of honorary forestry officers in forestry law and policy	120,000.00	10 honorary forestry officers	10 honorary forestry officers	10 honorary forestry officers	10 honorary forestry officers	10 honorary forestry officers	Ministry of Green Economy and Environment
		Conducting	650,000.00	48 blitz	48 blitz	48 blitz	48 blitz	48 blitz	

		routine/blitz patrols in order to promote sustainable utilization of forest resources		patrols and 48 routine patrols	patrols and 48 routine patrols	patrols and 48 routine patrol	patrols and 48 routine patrol	patrols and 48 routine patrol	
Procurement of office tools and equipment	Procuring tools and equipment in order to facilitate enhanced Office administration and service delivery	Procuring one 4 doors van (landcruiser) in order to improving service delivery to the general public in promoting sustainable forest utilization and conservation	1,200,000.00	4 doors van (land cruiser)	-	-	-	-	
		Procuring one modern desk top computer, one laptop and one printer for efficient and effective production of activities report for planning purposes	50,000.00	One modern desktop, laptop and printer	-	-	-	-	

Objective 6	To increase social protection programs for the vulnerable groups in the district by 2030								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Increase the coverage of social protection	Increase in access point	Stackholder engagement	250, 000.00	2	2	2	2	2	NCC
		Sensitization	380,522.09	4	4	4	4	4	
		Identification of service access points	18,300.50	1			1		
		Training of volunteers	1,402,600.00	1	✓	✓	1	✓	
Lobby for increased budgetary allocation	Increase social protection program	Stake holder engagement	45,000.00	1	✓	✓	1	✓	
Engage in PPP	Drawing up the MoU	Hold stakeholder meeting	35,000.00	1	✓	✓	1	✓	

Institutional

Objective 1	To increase qualified human resource in public institutions by 2032								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Institution assessment of human resource needs	Development of a human resource assessment framework	identification of vacancies within the institution	7,320.00	1	1	1	1	1	-Human resource personal
		Recruitment of human resource	109,800.00	1	1	1	1	1	Service commission
		Training of existing human resource	2,870,080.00	1	1	1	1	1	

		Submission of monthly returns to the employer	7,320.00	4	4	4	4	4	-administration department
		Staff appraisal	7,320.00	1	1	1	1	1	
		Recruitment of qualified human resource	15,000.00	1	1	1	1	1	

Objective 2	To develop a smart city by linking all sectors to the district data bank by 2030								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Development of a district data bank	Linking all sectors	Stakeholder meeting	100,000	1					-IDP district Committee -Department of Finance (IT Section)
		Creation of a district website	500,000	1					
		Creation of district database	500,000	1					
		orientation workshop	100,000	1					
		stakeholder data bank updates	250,000	1	1	1	1	1	
		District Development coordination	60,000	1	1	1	1	1	

Objective 3	To strengthen the legal and policy framework of various sectors to enhance service delivery in public sector by 2032								
Strategy	Program	Activities	Budget (zmk)	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Responsibility
Formulation and enforcing by-laws	Framing of existing and	1. Stakeholder meeting	192,000.00	1	1	1	1		Local authority

	new laws	by-	Reviewing of existing by-laws and new submissions from various stakeholders	72,800.00				1		
			Presentation of proposed new by-laws to full council	6,320.00					1	
			Publishing of by-laws	63,000					1	
			submission of document to the Ministry of Local Government	7,140.00					1	
			monitoring and evaluation of adherence and compliance levels in the district	10,000.00	4	4	4	4	4	

4.2 Local Authority's Financial Plan

The financial plan is directly related to the the **priorities, objectives and strategies** identified in the IDP and will run for a period of **5 years**. While the IDP will incorporate inputs from government sectors outside of the local authority, the financial plan will relate wholly to the functions of and services provided by the local authority. In addition to incorporating operational budgets for identified objectives and strategies, the Financial Plan will also reflect standard operational budgets which the Local Authority will incur such as salaries and emoluments.

Revenue Collection Efficiency of Ndola City Council Assessment of the Revenue Base 2018 -2020

ITEM	2018	2019	2020
Budgeted Revenue	112,417,790.00	108,190,272.00	135,273,521.00
Actual Revenue Collected	88,631,459.00	88,964,180.00	104,876,568.30
Collection Efficiency %	79	74	78

Source: Ndola City Council Finance Department (2021)

There has been a minimal decrease in revenues collected in the past three (3) years. This can be attributed to the issues set below.

Issues Impeding Revenue Collection

1. These include lack of transport to effectively conduct debt collection
2. Some households are headed by elderly people with no income hence fail to property rates
3. Closed buildings whose owners cannot be traced making debt on such properties difficult to collect.
4. Manual distribution of bills makes it difficult to deliver bills on time.

Strategies to Improve Revenue Collection

1. Procurement of Vehicles to ease transport challenges
2. Implementation of e-billing system to ensure bills are sent on time.
3. Forming task forces in revenue collection activities as a way of ensuring compliance.

Types of Sources of Revenue

1. Own Sources of Revenue
 - (a) Property Rates
 - (b) Personal Levy
 - (c) Fees and Charges
 - (d) Land Charges
 - (e) Licences
 - (f) Permits

2. External Sources of Revenue
 - a. Government Grants
 - b. Constituency Development Fund (CDF)

4.3. Programme for the preparation of Local Area Plans

For effective implementation of the Ndola District Integrated Development Plan, detailed Local Area Plans need to be specified and prepared. The table below highlights the list of the proposed Local Area Plans and the period of preparation for these plans.

No.	Name of Plan	Period of Preparation	Responsible institution
1	Misaka 5 Local Area Plan – Planning of the Excised part of Misaka Forest no. 5	January 2023 to December 2023	Local Planning Authority, Utility Companies/Providers
2	Kavu Local Area Plan – Unplanned Settlement Upgrading	January 2023 to December 2023	Local Planning Authority, Utility Companies/Providers
3	Mwenye Extension Local Area Plan – Unplanned Settlement Upgrading	January 2024 to December 2024	Local Planning Authority, Utility Companies/Providers
4	Chipulukusu Local Area Plan – Unplanned Settlement Upgrading	January 2024 to December 2024	Local Planning Authority, Utility Companies/Providers
5	Twapia George Local Area Plan – Unplanned Settlement Upgrading	January 2025 to December 2025	Local Planning Authority, Utility Companies/Providers

6	Kantolomba Local Area Plan – Unplanned Settlement Upgrading	January 2026 to June 2026	Local Planning Authority, Utility Companies/Providers
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Financial Plan

Revenue and Expenditure Projections for the Five-Year Period from 2021 to 2025

Revenue						
Details of Revenue	2020	2021	2022	2023	2024	2025
	Approved Estimates	Approved Estimates	Approved Estimates	Estimate	Estimate	Estimate
Local Revenue	ZMK	ZMK	ZMK	ZMK	ZMK	ZMK
Local Taxes	57,285,350	40,896,706	39,534,257	87,918,684	87,918,684	87,918,684
Fees and Charges	34,089,714	39,932,340	47,048,711	34,664,839	36,000,000	37000000
Licences	3,578,487	5,294,298	5,992,750	5,450,091	5,900,000	5950000
Permits	7,840,132	11,046,500	20,678,184	20,130,418	21,200,000	21500000
National Support	32,479,838	33,595,810	131,045,810	140,595,810	141,000,000	142,000,000
	135,273,521	130,765,655	244,299,712	288,759,842	292,018,684	294,368,684

Expenditure						
Expenditure Details	2020	2021	2022	2023	2024	2025
	Approved Estimates	Approved Estimates	Approved Estimates	Estimate	Estimate	Estimate
	ZMK	ZMK	ZMK	ZMK	ZMK	ZMK
Personal Emoluments	48,500,910	57,802,698	63,797,094	87,893,738	87,893,738.00	88,893,738
Office Costs & Other Administrative costs	6,209,960	2,743,503	5,257,246	10,875,421	10,975,421.00	12,000,000
Building Repairs & Maintenance Costs	7,891,731	7,220,262	4,204,198	3,786,422	4,000,000	3,500,000
Plant, Machinery & Vehicle Running Costs	6,364,558	6,597,217	8,283,346	3,841,555	3,986,422	4,000,000

Requisites	4,844,663	4,313,954	4,109,941	5,706,629	5,807,629	5,000,000
Services	30,437,558	31,885,739	68,134,072	74,884,543	78,742,474	79,000,000
Travel Expenses	1,844,588	1,638,941	835,999	2,062,613	2,500,000.00	2,500,000
Staff Training	1,119,610	724,758	1,029,567	2,272,200	2,272,200.00	2,474,946
Legal Costs	682,166	512,477	1,500,000	2,438,800	2,450,800.00	1,000,000
Non-Financial Assets	9,763,077	9,151,803	79,817,529	82,197,921	83,000,000.00	87,000,000
Staff Creditors	6,797,500	5,534,303	5,850,720	8,400,000	6,400,000.00	5,000,000
Other Creditors	5,619,700	750000	650,000	2,000,000	990,000.00	1,000,000
Statutory Contributions (Obligations)	5,197,500	1,890,000	830,000	2400000	3,000,000.00	3,000,000
Total	135,273,521	130,765,655	244,299,712	288,759,842	292,018,684.00	294,368,684

4.4 Monitoring and Evaluation plan

The implementation of the Integrated Development Plan (IDP) for Ndola district will involve the public and private sector including non-governmental organisations and every leadership at community level structures in the district. Standard reporting formats will be used to track progress of implementation. It will entail developing and implementing departmental plans in support of the IDP. It will also heavily rely on advocacy in many respects requiring policy changes at national level. For the Plan to effectively meet its objectives, a Monitoring and Evaluation (M&E) system has been put in place.

The implementation process of this Plan will be monitored and evaluated by tracking inputs, activities and outputs using the standard report system to District Development Coordinating Committee (DDCC)/Council. This will form the basis for determining whether plan implementation is on course. Monitoring reports will be periodically prepared by the Technical Committee of the Cluster Advisory Groups of each Strategic area for discussion at DDCC level. Consolidated district reports will then be tabled before the Provincial Development Coordinating Committee for further discussion and action. Issues requiring national attention will then be forwarded to relevant national level authorities for guidance and action.

The Monitoring and Evaluation system will assess the impact of the plan on diversification, job creation and poverty reduction. It will also provide feedback for stakeholders on results and lessons learned during the implementation of the plan.

Monitoring and evaluation plan-ECONOMIC-SECTOR

Objective 1		To enhance the development of sustainable domestic and foreign market linkages by 2032.										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verifications	Responsibility
Strengthening the chamber of commerce	Creating market linkages	Conduct stakeholder meetings	4	4	4	4	4	Number of meetings conducted	Ministry of Agriculture/ Local Authority/ Ministry of commerce/ Chamber of commerce/ ministry of information	Quarterly	Minutes/ reports/ attendance register/ pictures	Ministry of Agriculture/ Local Authority/ Ministry of commerce/ Chamber of commerce/ ministry of information
		Advertise ment of local products	5	5	5	5	5	Number of adverts done	Ministry of Agriculture/ Local Authority/ Ministry of commerce/ Chamber of commerce/ ministry of information	Quarterly	Piece of advert from the newspapers, record of the advert from the radio, video clip of the advert, proof of payment to the media houses	

		Cluster the business segments	4	5	6	7	8	Number of clusters formed	Ministry of Agriculture/ Local Authority/ Ministry of commerce/ Chamber of commerce/ ministry of information	Quarterly	List of members of the cluster, minutes of the meetings of the cluster, reports	
Cluster Group Support and Capacity Building	Training Programs	Conduct entrepreneurship Trainings	10	10	10	10	10	Number of clusters trained	Local authority/ agriculture/ chamber of commerce / ministry of commerce	Quarterly	Minutes/ reports/ attendance register/ pictures	Local authority/ agriculture/ chamber of commerce / ministry of commerce
Local product inspection and market information management system	Market Information and Research	Producing marketing bulletins	52	52	52	52	52	Number of bulletins produced	Local authority/ agriculture/ chamber of commerce / ministry of commerce	Quarterly	The bulletins	Local authority/ agriculture/ chamber of commerce / ministry of commerce
		Inspection of district Marketing Infrastructure	4	4	4	4	4	Number of inspection conducted	Local authority/ agriculture/ chamber of commerce / ministry of commerce	Quarterly	Reports, pictures,	

	public engagement on E- governance		1	1	1	1	1	Number of public engagement s conducted	Local government/ Education/ ZICTA	Annually	Attendance registers, reports, pictures	Local government/ Education/ ZICTA
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Objective 2		To promote mechanized farming methods in the district by 2030										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verifications	Responsibility
Promotion of technology adoption: conservation agriculture, crop diversification and mechanisation	Crop diversification	Capacity building of farmers in the production of various crops through farmer field schools	17	17	17	17	17	Number of capacity building programs conducted	Ministry of Agriculture/ Local Authority	Quarterly	Reports, attendance registers, pictures	Ministry of Agriculture/ Local Authority
		Setting up demonstration plots of various crop varieties	17	17	17	17	17	Number of demonstration plots conducted	Ministry of Agriculture/ Local Authority	Bi-annual	Reports, pictures	
		Conducting field days	17	17	17	17	17	Number of field days conducted	Ministry of Agriculture/ Local Authority	Quarterly	Reports, pictures	
		Farmer exchange visits	5	5	5	5	5	Number of exchange visits done	Ministry of Agriculture/ Local Authority	Quarterly	Reports, pictures, number of areas visited	
	Conservation agriculture	Training farmers in conservation agriculture through farmer field schools.	17	17	17	17	17	Number of farmers trained	Ministry of Agriculture/ Local Authority	Quarterly	Training materials, reports, pictures, attendance register	Ministry of Agriculture/ Local Authority

	Farm power and Mechanisation	To showcase small-holder technologies used for timely land preparation and crop processing	17	17	17	17	17	Number of demonstration conducted	Ministry of Agriculture/ Local Authority	Quarterly	Pictures, reports, list of participates	Ministry of Agriculture/ Local Authority
Promotion of irrigated agriculture	Irrigation development	Development of small-scale irrigation schemes	3	3	3	3	3	Number of irrigation schemes	Ministry of Agriculture/ Local Authority	Quarterly	Reports, pictures,	
Promotion of dairy and small livestock production	Livestock production	Capacity build farmers in small livestock production	8	8	8	8	8	Number of farmer groups trained	Ministry of Agriculture/ Local Authority	Quarterly	Reports, pictures, training materials, attendance registers,	Ministry of Agriculture/ livestock/ Local Authority
		Establishing a breeding centre for small livestock	1	-	-	-	-	Number of canters established	Ministry of Agriculture/ Local Authority	Annually	Pictures, the actual canter,	Ministry of Agriculture/ livestock/ Local Authority

		Control of Livestock Diseases	80	80	80	80	80	Number of surveillance visits conducted	Ministry of Agriculture/ Local Authority	Quarterly	Reports, pictures	Ministry of Agriculture/ livestock/ Local Authority
Promotion of climate smart Agriculture	Climate Smart Agriculture promoted	Capacity build farmers in Smart Agriculture technologies	425	850	1,275	1,700	2,125	Number of capacity building programs conducted	Ministry of Agriculture/ Ministry of Green Economy and Environment	Quarterly	Reports, attendance registers, pictures	Ministry of Agriculture/ Ministry of Green Economy and Environment
		Increase area under conservation Agriculture	100	200	300	400	500	No. of farming blocks	Ministry of Agriculture	Annually	Reports, pictures	Ministry of Agriculture
		Increase number of farmers using drought tolerant crop varieties	1000	1500	2000	4000	8000	Number of farmer groups	Ministry of Agriculture	Bi-annually	Reports, pictures	Ministry of Agriculture
Infrastructure development for Improvement of extension service delivery	Improved agriculture extension services	Renovation of Camp houses	600,000.00	-	1	2	-	-	Ministry of Agriculture	Bi - Annually	Reports, pictures	Ministry of Agriculture
		Construction of camp houses	11,200,000.00	2	3	3	3	3	Ministry of Agriculture	Quarterly	Reports, pictures	Ministry of Agriculture

		Construction of one district Agriculture Office Block	5,000,000.00	-	1	-	-	-	Ministry of Agriculture	Annually	Reports, pictures	Ministry of Agriculture
Fish Clusters Formation and Capacity Building	Aquaculture and capture fisheries Management and Development	Formulate and support 3 community based hatcheries	1	1	1	-	-	Number of hatcheries formulated	Ministry of Agriculture/ Local Authority	Annually	The Hatcheries, reports	Ministry of Agriculture/ livestock/ Local Authority
		Training and Sensitization of Fishers and Fish Farmers	5	5	5	5	5	Number of fish clusters registered and trained	Ministry of Agriculture/ livestock/ Local Authority	Quarterly	Attendance register, pictures, reports	Ministry of Agriculture/ livestock/ Local Authority
		Extension Visits to Fishers and Fish Farmers	24	24	24	24	24	Number of extension visits	Ministry of Agriculture/ livestock/ Local Authority	Quarterly	Pictures, reports	Ministry of Agriculture/ livestock/ Local Authority
		Restocking and conservation small water bodies	10	10	10	10	10	Number of small water bodies restocked and conserved	Ministry of Agriculture/ livestock/ Local Authority	Quarterly	Pictures, report	Ministry of Agriculture/ livestock/ Local Authority
		Distribution of fish seed (fingerlings) to fish farmers	100	100	100	100	100	Number of farmer	Ministry of Agriculture/ livestock/	Quarterly	Pictures, reports	Ministry of Agriculture/ livestock/

								beneficiaries	Local Authority			Local Authority
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Objective 3	To promote value addition on locally produced raw materials by 2032											
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Indicators	Source of information	Frequency	Means of verifications	Responsibility
Promotion of value addition on local products in order to increase shelf life	Food processing and value addition	To build capacity in farmers on improved food processing and value addition	17	17	17	17	17	Number of farmers participated in capacity building	Ministry of Agriculture/ livestock/ Local Authority	Quarterly	Reports, attendance registers, pictures	Ministry of Agriculture/ livestock/ Local Authority
	Stockholder engagement (PPP)	Creating a stakeholder directory	1	1	1	1	1	Number of stockholder directorate	Ministry of Agriculture/ livestock/ Local Authority	Annually	Reports, attendance registers,	Ministry of Agriculture/ livestock/ Local Authority

		Stakeholder meetings	4	4	4	4	4	Number of stakeholder meeting conducted	Ministry of Agriculture/ livestock/ Local Authority	Quarterly	Reports, attendance registers, pictures	Ministry of Agriculture/ livestock/ Local Authority
	Value addition on locally produced materials	Capacity building on the value addition	10	10	10	10	10	Number of capacity building workshop conducted	Ministry of Agriculture/ livestock/ Local Authority	Quarterly	Reports, attendance registers, pictures	Ministry of Agriculture/ livestock/ Local Authority

Objective 4		To expand and strengthen the revenue base in the district by 2032										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verifications	Responsibility
Increased Investment in ICT	Capacity building Training		20	25	25	30	20	Number of capacity building conducted	Local government/ Education/ ZICTA	Quarterly	Attendance register, reports, pictures	Local government/ Education/ ZICTA

	Purchase of IT Infrastructure		1					Number of ICT infrastructure purchased	Local government/ Education/ ZICTA	Annually	Pictures, proof of payments	Local government/ Education/ ZICTA
	Installation and operationalization				1	1	1	Number of ICT installed	Local government/ Education/ ZICTA	Annually	Pictures, proof of payments	Local government/ Education/ ZICTA

Objective 5	To promote tourism in the district by 2032											
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verifications	Responsibility
Promotion of tourism sites and activates	Marketing of tourism	Stakeholder engagement		1	1	1	1	Number of stakeholder engagements conducted	Local Authority/ ministry of tourism and culture	Annually	Attendance registers , reports, pictures	Local Authority/ ministry of tourism and culture
		Profiling of tourism adverts		1	1	1	1	Number of tourism adverts conducted	Local Authority/ ministry of tourism and culture	Annually	Attendance registers , reports, pictures	Attendance registers, reports, pictures
		Develop a Marketing strategy		1				Number of market strategies developed	Local Authority/ ministry of tourism and culture	Annually	Attendance registers , reports, pictures	Local Authority/ ministry of tourism and culture
		Operationalize the strategy		1	1	1	1	Number of strategies operationalize	Local Authority/ ministry of tourism and culture	Annually	Attendance registers , reports, pictures	Local Authority/ ministry of tourism and culture

	Improving tourism packaging and capacity	Creation of Ndola visitor information Centre	1					Number of information centres created	Local Authority/ ministry of tourism and culture	Annually	Information centre/number of visitors	
		To participate in the creation of community heritage corridor	1					Number of community heritage corridors created	Local Authority/ ministry of tourism and culture	Annually	Community heritage/minute, reports, attendance registers	
Infrastructure development	Construction and rehabilitation of tourism sites	Renovation of tourism infrastructure	1	1	1	1	1	Number of tourism infrastructure renovated	Local Authority/ ministry of tourism and culture	Annually	tourism infrastructure renovated	
		Maintenance of tourism sites	1	1	1	1	1	Number of tourism sites maintained	Local Authority/ ministry of tourism and culture	Annually	Maintained tourism sites	

		Constructi on of a museum	1	1	1			Number of museums constructe d	Local Authority/ ministry of tourism and culture	Annually	Museum	
		Constructi on of recreationa l facilities	1	1				Number of recreation al facilities constructe d	Local Authority/ ministry of tourism and culture	Annually		

Objectives 6		To facilitate the establishment of a Multi- Facility Economic Zone in the district by 2032										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verifications	Responsibility
Facilitate the creation of Multi Facility Economic Zone	Promotion of Public Private Partnership in the City	Conduct stakeholder meetings	4	2	2	2	2	Number of stakeholder meeting conducted	Ministry of lands, local authority, ministry of commas	Bi-annually	reports, attendance register	Ministry of lands, local authority, ministry of commas
		Site identification	1					Number of sites identified	Ministry of lands, local authority, ministry of commas	Annually	reports, attendance register	Ministry of lands, local authority, ministry of commas
		Drawing up of the MOU	1	1				Number of MOU drawn	Ministry of lands, local authority, ministry of commas	Annually	reports	Ministry of lands, local authority, ministry of commas

Monitoring and Evaluation Plan Tool-Physical Sector

Objective 1. To improve and maintain transport infrastructure in the district by 2032												
Strategy	Programme	Activity	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Indicator	Source of Information	Frequency	Means of Verification	By Whom
Formulate a deliberate policy to allocate a % of revenue towards the construction, rehabilitation and road maintenance	Upgrading of 122.162km on selected roads to bituminous standard	Stakeholder meetings	1	1	1	1	1	Number of meetings held	-Ndola City Council -Road Development Agency	Quarterly	-Reports - Attendance registers -Pictures	-Ndola City Council (Engineering Department) -RDA

		Field visit	2	1	1			Number of visits conducted	-Ndola city council - Road development Agency	Quarterly	-Pictures -Reports	NCC/RDA
		Upgrading of roads to bituminous standard	22	22	22	22	22	Number of roads upgraded	Ndola city council - Road development Agency	Quarterly	Upgraded roads	
		Develop resources mobilisation strategy	1					Number of strategies formulated	Ndola City Council	Annually	-Reports - Attendance list	Ndola City Council (Engineering Department) -RDA
	Maintenance of 43km of feeder roads	Procurement of earth moving equipment	1					Earth moving equipment procured	Ndola city council - Road development Agency	Annually	Earth moving equipment	

	Rehabilitation of 344.9km on the 5 district major roads	Rehabilitation of roads	51.5	62.3	69.8	125	36.3	344.9km of 5 district major roads rehabilitated	Ndola city council - Road development Agency	Quarterly	Rehabilitated roads	
Infrastructure development	Street lighting	Installation of street lights on 230km of road		50km		50km	30km	Number of kms of roads installed with street lights	Ndola city council	Quarterly	Reports, pictures	Ndola City Council (Engineering Department)
	Provision of a modern bus terminus	Construction of a modern bus terminus				1		Number of bus terminus constructed	Ndola city council	Annually	Reports, pictures	Ndola City Council (Engineering Department)

Objective 2. To construct and upgrade market infrastructure in the district by 2032

Strategy	Programme	Activity	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Indicator	Source of Information	Frequency	Means of Verification	By Whom
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Formulate a policy to allocate a percentage of revenue towards market infrastructure	Infrastructure development	Stakeholder meeting	12	12	12	12	12	Number of meetings held	Ndola City Council	Quarterly	Reports -Attendance list of Meetings	Ndola city council Office of the District Commissioner The office of the Mayor
		Conducting condition based survey	1					Number of surveys conducted	Ndola City Council	Annually	Minutes/ attendance register	Ndola city council Office of the District Commissioner The office of the Mayor
		Architectural design and developing BoQs	1					Number of architectural	Ndola City Council Ndola City Council	Bi-annually	Architectural designs	

								designs create				
		Presentation of the BoQs to management	1					Number of BoQs presented	Ndola City Council	Bi-Annually	BoQs	
		Full council approval designs and BoQs	1					Full council meeting	Ndola city council	Bi-Annually	Minutes/attendance register	
		Tendering process	1					Number of tenders processed	Ndola city council	Quarterly	Tenders/minutes	
		Market construction	1					Number of market constructed	Ndola city council	Annually	market	
Engage in PPP	infrastructure development	Stakeholder meeting		4	4		4	number of meeting held	Ndola City Council	Quarterly	Minute Books/-Reports	Ndola City Council
		Drawing up the MoU for the construction of markets through PPP	1					MoU created	Ndola City Council	Annually	MoU	

Objective 3: To establish more health centres in the district by 2032												
Strategy	Programme	Activity	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Indicator	Source of Information	Frequency	Means of Verification	By Whom
Lobby for increased budgetary allocation	Health infrastructure development	Development of sector budget from the plan	1					Number of budgets formulated	Ndola City Council	Annually	Minutes of meetings/ Reports	Ndola City Council
		Stakeholder engagement	1					Number of meetings conducted	Ndola City Council	Quarterly	Means of Verification Minutes of meetings/ Reports	
		Site identification	1					Number of sites identified	Ndola City Council	Annually	Identified sites	
		Designs and cost estimates	1					Number of designs and cost estimates created	Ndola City Council	Annually	Designs and cost estimates	
		Resource mobilisation	1	1	1	1	1	Number of resources mobilised	Ndola City Council	Annually	Resources mobilised	
		Tendering process										
		Construction of the project (health centres)										

Objective 4		Objective: To increase the water service coverage by 2033										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verification s	Responsibility
Provision of water infrastructure	Lay Water Networks	• Purchase of excavator	1					Number of vacuum tankers purchased	Ministry of Water and Sanitation	1	Reports/pictures	Ministry of Water and Sanitation
		• Conduct stakeholder meetings	9	12	4			No of meetings	Ministry of Water and Sanitation	1	Minutes/reports/attendance register/pictures	
		• Lay water network in Jacaranda	1					Number of water pipe laid	KWSC	1	Reports/pictures	
		• Lay water network in Dola Hill	1					Number of water pipe laid	KWSC	1	Reports	
		• Lay water network in Pamodzi	1					Number of water pipe laid	KWSC	1	Reports	
		• Lay water network in Mushili	1					Number of water pipe laid	KWSC	1	Reports	

		Lay water network in Pinegrove	1					Number of water pipe laid	KWSC	1	Reports	
		Lay water network in Kaloko	1					Number of water pipe laid	KWSC	1	Reports	
		Lay water network in Twapia		1				Number of water pipe laid	KWSC	1	Reports	
		Lay water network in Canaan		1				Number of water pipe laid	KWSC	1	Reports	
		Lay water network for water Kiosks		1				Number of water pipe laid	KWSC	1	Reports	
		Lay water network in Airport		1				Number of water pipe laid	KWSC	1	Reports	
		Lay water network in Misundu/Map alo	1					Number of water pipe laid	KWSC	1	Reports	

		Lay water network in Bwana Mkubwa/Ndeke	1					Number of water pipe laid	KWSC	1	Reports	
		Lay water network in George	1					Number of water pipe laid	KWSC	25	Reports	
		Construct water network Kopaland		1				Number of water pipe laid	KWSC	1	Reports	Number of water pipe laid
		Construct water network in Itawa extension		1				Number of water pipe laid	KWSC	1	Reports	Number of water pipe laid
		Construct water network in Kamba		1				Number of water pipe laid	KWSC	1	Reports	Number of water pipe laid

Objective		To establish more educational facilities in the district by 2032										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Indicator	Source of Information	Frequency	Means of Verification	Responsibility
Lobby for increased budgetary allocation	Education infrastructure development	Development of sector budgets from the plan	1	1	1	1	1					Ministry of education and Ndola city council
		Stakeholder engagement in the implementation of School health and nutrition (SHN)	1	1	1	1	1					
		Construction of additional 1*3 classrooms with Desks	4	4	4	4	4					
		Construction of a complete secondary school with Desks	1		1		1					
		Ablution Blocks	2	2	2	2	2					
		Desk Rehabilitation	300	300	300	300	300					

		Purchase of New Desks (double and single seaters)	9,064	9064	9064	9064	9064					
	ICT	Purchase of Computers (laptops and desk tops)	50	50	50	50	47					

Objective 5: To Promote high-rise housing units to maximise land use in the district by 2032													
Strategy	Program me	Activity	Targ et Year 1	Targ et Year 2	Targ et Year 3	Targ et Year 4	Targ et Year 5	responsibil ity	Indicato r	Source of Informati on	Frequen cy	Means of Verificati on	By Whom
Develop standardis ed storey building plans	Maximu m land utilisatio n	Developm ent of approved standard building plans	1	1	1	1	1	NCC	- Number of site inspectio ns	Ndola City Council	Bi-annually	-Reports	Ndola City Council -Ministry of Local Governme nt and Housing -Ministry of Infrastruct ure and Developm ent

		Production of cost estimates		4				NCC	Number of bill of quantities produced	Ndola City Council	Quarterly	-bill of quantities produced -reports	-Ndola City Council -Ministry of Infrastructure and Development -Ministry of Local Government and Housing
Entice potential Investors by engaging in (PPP)	Upgrade informal settlements	Communit y meetings	12	12				NCC	Number of stakehold er meeting	Ndola City Council	Quarterly	-Reports - attendance list	-Ndola City Council -NGOs -Ministry of Communit y Developm ent and Social Welfare
		Tendering process for a private partner to conduct the process of upgrading	1	1	1	1	1	NCC	Number of tender processe d	Ndola City Council	Annually	Tender processed	

Objective		Objective: To increase the sewerage service coverage by 2027										
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicators	Source of information	Frequency	Means of verifications	Responsibility
Provision of sewerage infrastructure	Lay Sewer Networks	Purchase of vacuum tanker										
		Conduct stakeholder meetings	4	5	4	6	3	No of meetings	Ministry of Water and Sanitation	1	Minutes/ reports/ attendance register/ pictures	Ministry of Water and Sanitation
		Lay sewer network in Mitengo	1					Number of water pipe laid	KWSC	1	Reports	
		Upgrade sewer networks in Masala/Chifubu	1					Number of water pipe laid	KWSC	1	Reports	
		Lay sewer network in Kopaland	1					Number of water pipe laid	KWSC	1	Reports	
		Lay sewer network in Itawa extension	1					Number of water pipe laid	KWSC	1	Reports	

		Lay sewer network in Twapia		1				Number of water pipe laid	KWSC	25	Reports	
		Construction of sewer scheme to the airport			1			Number of water pipe laid	KWSC	25	Reports	
		Lay sewer network in Misundu & Mapalo			1			Number of water pipe laid	KWSC	25	Reports	
		Lay sewer network in BwanaMkubwa/Ndeke			1			Number of water pipe laid	KWSC	1	Reports	
		Lay sewer network in George area				1		Number of water pipe laid	KWSC	1	Reports	
	Construct Wastewater Treatment Plant in Ndola	Construct Wastewater treatment plant				1		No of treatment plants constructed	KWSC	1	Operational Plant	
	Construct FST Treatment Plant in Ndola	Construct FST treatment plant					1	No of treatment plants constructed	KWSC	1	Operational Plant	

Objective 6: To improve waste management infrastructure in the district by 2032												
Strategy	Programme	Activity	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	Indicator	Source of Information	Frequency	Means of Verification	By Whom
Identify platforms for key stakeholders to partner with	Solid waste management	sensitization	10	10	10	10	10	Amount of wastes managed	Ndola City Council	Quarterly	-Reports	Ndola City Council
		Procurement of waste disposal equipment	2	2	2	2	2	Number of waste disposal equipment procured	Ndola city council	Bi-annually	Procured waste disposal equipment	
		Expansion of street sweeping services to 5 new communities outside the CBD	12	12	12	12	12	Number of communities serviced	Ndola city council	Quarterly	Reports	
		Identification and engagement of CBEs in medium and low cost	4	4	4	4	4	No. of CBEs engaged	Ndola city council	Quarterly	Reports	

	Constructi on of engineere d land fill	Site identificat ion	4	4	4	4	4	Number of sites identifie d	Ndola City Council	Bi- annually	Reports	-Ndola City Council
		Designed and costs	1					Number of designs and costs drawn	Ndola City Council	Quarterl y	Designs drawn	
		Meetings with stakehold ers	1			1	1	Number of meeting s held	Ndola city council	Quarterl y	Attendanc e registers, reports. Minutes	
		Signing an MoU	1					Number of MoU signed	Ndola city council	Annually	- Drawings produce	
		Environm ental Impact Assessme nt	1					Number of Docume nts produce d	Ndola city council	Annually	Reports	
		Constructi on of an incinerato r	1					Number of incinerat ors construc ted	Ndola city council	Quarterl y	Reports, incinerato r, pictures	

		Constructi on of the weigh bridge	1					Number of weigh bridges construc ted	Ndola city council	Quarterl y	Reports, weigh bridge, pictures	
		Constructi on of a borehole	1					Number of borehole s construc ted	Ndola city council	Quarterl y	Reports, boreholes , pictures	
		Constructi on of the boundary fence	1	1			1	Boundar y fence construc ted	Ndola city council	Quarterl y	Reports, boundary fence, pictures	
		Constructi on of an ablution block	1	1				Number of ablution blocks construc ted	Ndola city council	Quarterl y	Reports, ablution block, pictures	
	Keep Zambia clean and health campaign	Sensitizati on to the public		5					Ndola City Council	Quarterl y	d -Reports	
		Procurem ent of cleaning items	1	1	1	1	1	Number of cleaning items procured	Ndola city council	Quarterl y	Cleaning items procured	
		Transport ation (purchase	1	1	1	1	1	Transpo rt provided	Ndola city council	Quarterl y	Transport provided	

		of vehicles and fuel)										
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Objective 1	Increase access to quality education in the district by 2032											
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 3	Target Year 5	Indicators	Source of Information	Frequency	Means of Verification	By Whom
Provision of human resource & materials	Staff recruitment and deployment	Recruitment of qualified teacher	1	1	1	1	1	Number of teachers recruited	Ministry of general education	Annually	Staff returns	Ministry of general education
	Staff development	Support in- service capacity building	1	1	1	1	1	Number of trainings held	Ministry of general education	Bo- annually	Attendance list, training materials and reports	Ministry of general education
	Provision of teaching and learning materials	Procurement of teaching and learning materials	1	1	1	1	1	Number of teaching materials procured	Ministry of general education	Quarterly	Proof of payments,	Ministry of general education

Social Sector

To Promote gender mainstreaming in all sectors by 2032												
Objective 2												
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 3	Target Year 5	Indicators	Source of Information	Frequency	Means of Verification	By Whom
Gender mainstreaming	Promotion of gender work place policy	Sensitisation	1	1	1	1	1	Number of gender work place policy produced	Ministry of gender NCC	Bi-annually	Gender work place policy	Ministry of gender, NCC
		Formulation of gender work place policy	1	1	1	1	1	Number of formulated gender work place policy	Ministry of gender NCC	Annually	Gender work place policy	Ministry of gender NCC
		Reviewing of gender work place policy	1				1	Number of reviewed	Ministry of gender NCC and district administration	Annually	Gender work place policy	Ministry of gender

								gender work place policy				NCC

Objective 3: To mitigate the impact of climate change in the district by 2032

Strategy	Programme	Activity	Target Year 1	Target Year 2	Target Year 3	Targ et Year 4	Target Year 5	Indicato r	Source of Information	Frequen cy	Means of Verification	By Whom
Promoti on of enhance d forestry extensio n service in schools and at farm level	Enhanced Forestry Extension Services	Carryin g out Environ mental awarene ss campaig n meeting s in schools and at farmer level on the importa nce of conservi	4	4	4	4	4	Number of meetings held	Forestry Department Agriculture	Quarterly	-Reports -Attendance registers -Pictures	Provincial Forestry Department Agriculture Education

		ng forests										
		Establishment of community forest nurseries	5	5	5	5	5	Number of nurseries established and number of seedlings raised	Forestry Department Agriculture Interested Partners	Quarterly	-Pictures -Reports	Forestry Department Agriculture NCC Education
		Establishment of nursery in schools	5	5	5	5	5	Number of nurseries and seedlings raised	Forestry Department Education	Quarterly	-Pictures -Reports	

Promotion of plantation/ woodlot establishment and management		Training farmers in modern beekeeping and processing of honey and beeswax	10					Number of farmers trained in modern beekeeping and processing of honey and beeswax	Forestry Department Agriculture	Annually	-Reports -Attendance list	Forestry Department Agriculture
	Enhanced sensitization of farmers and schools on tree planting and management in order to private climate change	Training farmers and school children on how to plant and manage trees	12					Number of farmers and school children	Forestry Department Education	Bi-annually	Reports -Attendance registers -Pictures	Provincial Forestry Department Agriculture Education
	Management of established plantations through engaging casual	Weeding, pruning, Firebreak maintenance	2					2 Number of hectares covered	Forestry Department	Bi-annually	Reports -Pictures	Provincial Forestry Department

Establishment of forestry farmer support program	workers (job creation)	and controlled early burning										
	Registration of farmers in sustainable forest conservation and utilization	Capacity build farmers in plantation/woodlot establishment and management	5	5	5	5	5	Number of farmers showing interest in sustainable forest conservation and utilization	Forestry Department	Quarterly	Reports -Attendance registers -Pictures	Provincial Forestry Department Agriculture
		Distribution of assorted seedlings to farmers for plantation/woodlot establishment	2	2	2	2	2	Number of seedlings distributed to farmers	Forestry Department Education	Quarterly	Reports -Pictures	Provincial Forestry Department Agriculture

Promotion of forest protection and management	Recruitment of honorary forestry officers	Training of honorary forestry officers in forestry law and policy	5	5	5	5	5	Number of honorary forestry officers trained	Forestry Department Agriculture	Quarterly	Reports -Attendance registers	Provincial Forestry Department
		Conducting blitz/routine patrols in order to promote sustainable utilization of forest resources	96	96	96	96	96	Number of routine/blitz patrols conducted	Forestry Department Police service	Quarterly	-Reports -Pictures	Provincial Forestry Department Zambia Police Service
Procurement of office tools and equipment	Office administration	Procuring one 4 doors van in order to improving service	1							Annually	-Reports -Pictures	Ndola district commissioner

		delivery to the general public in promoting sustainable forest utilization and conservation										
		Procuring of one modern desk top computer, one laptop and one printer for efficient and effective production of activities report for planning	1							Annually	-Reports -Pictures	

		purpose										
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Objective 4	To increase social protection programs for the vulnerable groups in the district by 2032											
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 3	Target Year 5	Indicators	Source of Information	Frequency	Means of Verification	By Whom
Increase the coverage of social protection	Increase in access point	Stockholder engagement	2	2	2	2	2	Number of meetings held	Community development and social welfare	Bi-annually	Minutes and attendance list	Community development and social welfare
		Sensitization	4	4	4	4	4	Number of sensitization meetings held	Community development and social welfare	Quarterly	Reports and attendance list	Community development and social welfare
		Identification of service access points	1	1	1	1	1	Number of service points sites identified	Community development and social welfare	Bi-annually	Reports	Community development and social welfare

		Training of volunteers	1	1	1	1	1	Number of volunteers trained	Community development and social welfare	Annully	Attendance list, training manuals and reports	Community development and social welfare
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Institutional Sector

Objective 1: To increase staff levels through recruitment of qualified human resource in public institutions according to institution establishments by 2030.

Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicator	Source of information	frequency	Means of Verification	By whom
Institution assessment of human resource needs	Development of a human resource assessment framework	identification of vacancies within the institution	1	1	1	1	1	Number of vacancies identified	Administration Department	Quarterly	-Reports	Departmental Heads
		advertisement of vacancies within the media	1	1	1	1	1	Number of adverts conducted	Public Relations	Quarterly	-Reports	Public Relations
		Requesting for qualified human resource from the employer.	1	1	1	1	1	Number of employees requested	Human Resource and Administration Department	Annually	-Reports	Human Resource
		Submission of monthly returns to the employer	12	12	12	12	12	Number of returns submitted	Human Resource	Quarterly	-Reports	Human Resource
		Staff appraisal	1	1	1	1	1	Number of staff appraised	Head of Departments	Annually	-Reports	-Head of Departments

												-Human Resources
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Objective 2	Increase access to quality primary and secondary health care in all wards by 2032											
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 3	Target Year 5	Indicators	Source of Information	Frequency	Means of Verification	By Whom
Lobby for more human resource in the health sector	Development of the human resource assessment framework	Identification of vacancies	1					Number of vacancies identified	Ministry of health	Annually	Identified vacancies	Ministry of gender, NCC
		Recruitment of human resource	1	1	1	1	1	Number of human resource recruited	Ministry of health	Bi-annually	Recruiter human resource	Ministry of health
		Training of existing human resource	1	1	1	1	1	Number of human resource trained	Ministry of health	Annually	Trained human resource	Ministry of health

Objective 3: To strengthen the legal and policy framework for various sectors to enhance service delivery in public sectors by 2030.												
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 4	Target Year 5	indicator	Source of information	frequency	Means of Verification	By whom
Strengthen legal and policy framework Enhancement of service delivery	legal and policy framework	Stakeholder meeting	1					Number of meetings held	Local authority	Quarterly	-reports -attendance list	Local authority
		Reviewing legal and policy framework	1					Number of meetings	Local government	Annually	-reports	Local government
		Submission of policy document						Policy document submitted	Local government	Annually	-delivery note	Local government
		monitoring and evaluation of the implementation of the policy	25 participants - annually	25 participants - annually	25 participants - annually	25 participants - annually	25 participants - annually	Number of M&E conducted	Local authority	Quarterly	-reports -photos	Local authority

Objective 4	To increase access to improved sanitation and safe drinking water for all residents by 2032											
Strategy	Program	Activities	Target Year 1	Target Year 2	Target Year 3	Target Year 3	Target Year 5	Indicators	Source of Information	Frequency	Means of Verification	By Whom
Increase safe clean healthy water point	Awareness campaigns on WASH	School shows	3	3	3	3	3	Number of school shows held	Ndola city council ,ministry of education and ministry of health	Quarterly	reports	Ndola city council ,ministry of education and ministry of health
		Road shows	2	2	2	2	2	Number of road shows held	Ndola city council ,ministry of education and ministry of health	Bi- annually	reports	Ndola city council ,ministry of education and ministry of health
		Talk show on both radio and television	4	4	4	4	4	Number of talk shows and both radio and television done	Ndola city council ,ministry of education and ministry of health	Quarterly	reports	Ndola city council ,ministry of education and

												ministry of health
	Advocating for lower connecting fees	Meetings with utility companies	1					Number of meetings held	Ndola city council	Bi-annually	Minutes, reports attendance list	
		Review meetings		2				Number of review meetings held	Local authority	Bi-annually	minutes	Local authority
	Lobby for the construction of water kiosk	Meetings with different stakeholders		2				Number of meetings	Local authority and Kafubu	Bi-annually	minutes	Local authority and Kafubu

PART FIVE CONCLUSION

5.0 Conclusion

The process of formulating the Ndola District Integrated Development Plan was done in accordance with the provisions of the Urban and Regional Planning Act No. 3 of 2015 (URP Act. No. 3 of 2015), with further guidance from the Ministry of Local Government and Rural Development at various levels. Development of this important document ensured the full participation of all relevant stakeholders across the District of Ndola. Correctively, the broader goal of the District was formulated to help anchor all development sectors in an effort to align and coordinate development in the District to ensure prudent use of resources (natural, human, government/private and financial).

The long term vision of the Ndola District IDP has been aligned to various sectoral plans with the aim of working towards feeding into the National Plans through shared interfaces amongst stakeholders and decision makers. It is anticipated that such interactions will lead to the realization of the Nation's vision 2030, whose aim is to see Zambia attaining a prosperous middle-income nation status by 2030. Further, the Ndola District IDP has all its investment plans aligned to the 8NDP whose goal is to ensure **economic transformation** through **industrialization and diversification of the economy** by way of **enhancing the participation of citizens and the private** sector while **ensuring that human development** is enhanced to **ensure environmentally sustainable use of natural resources** through **improved policy and governance environment**.

This IDP will run for a period of 10 years from 2022 through to 2032. There will be a midterm review of the performance of the IDP after 5 years (2027), while the final review will be after 10 years. During the review period, all relevant stakeholders in the district will be invited to contribute as was the case during formulation.

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Appendixes

Public Participation Reports

1. Ndola Central constituency
2. Chifubu Constituency
3. Bwana Mkubwa Constituency
4. Kabushi Constituency